ABCD ADHD menarche analysis: Output for cross-sectional supplementary/sensitivity models

Model 2 ADHD present output

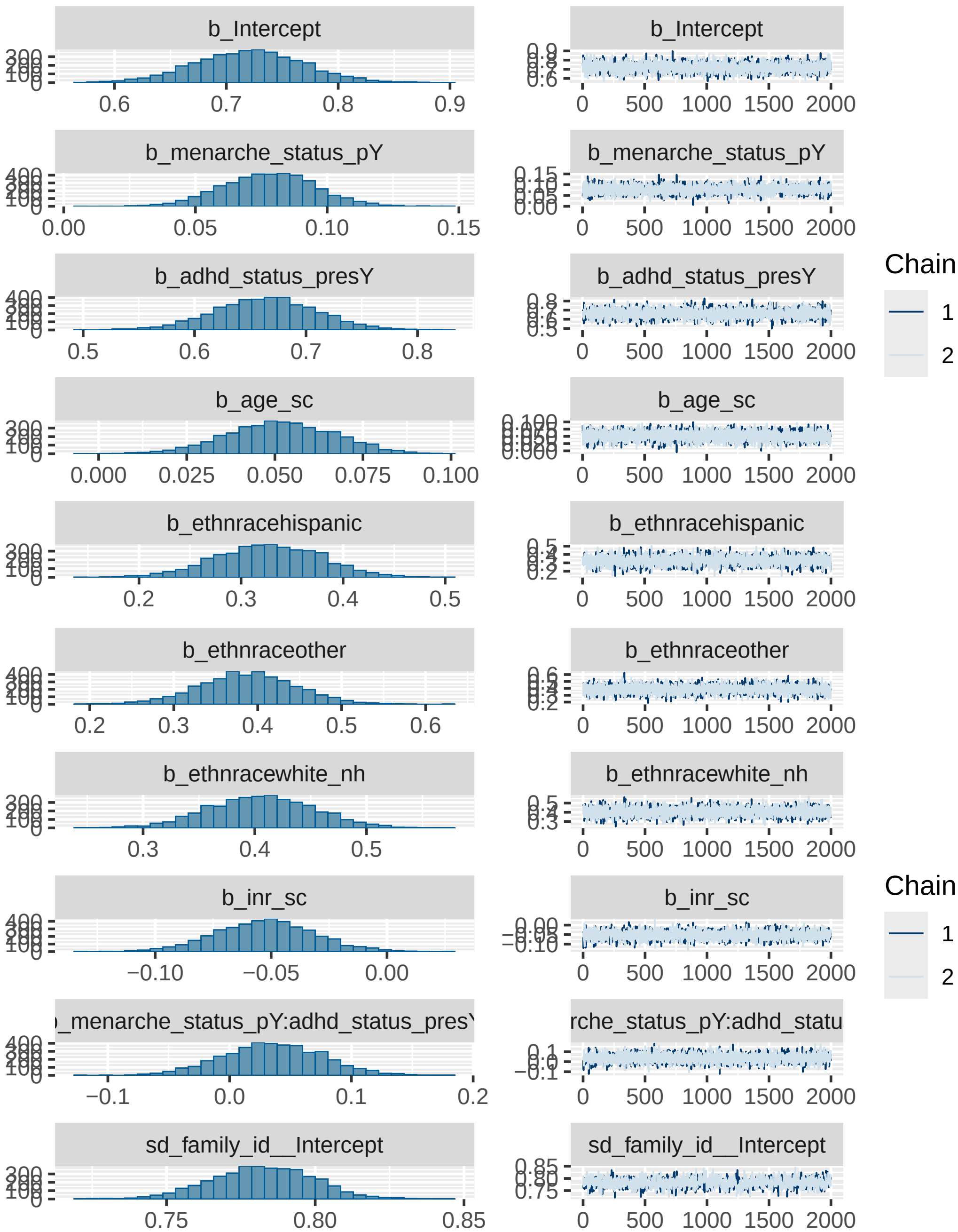
```
    Estimate  Est.Error      Q2.5     Q97.5
R2 0.8386163 0.00251197 0.8336968 0.8435774


 Family: poisson
  Links: mu = log
Formula: internalising ~ menarche_status_p * adhd_status_pres + age_sc + ethnrace + inr_sc +
   Data: analytic_df (Number of observations: 19375)
  Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~family_id (Number of levels: 4620)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.782     0.017    0.749    0.815 1.001      828     1996

~obs_id (Number of levels: 19375)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.447     0.006    0.435    0.460 1.005     1453     2237

~participant_id (Number of levels: 5291)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.443     0.021    0.402    0.485 1.002      486      937

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.121     0.026    0.078    0.180 1.000     2586     2684
```

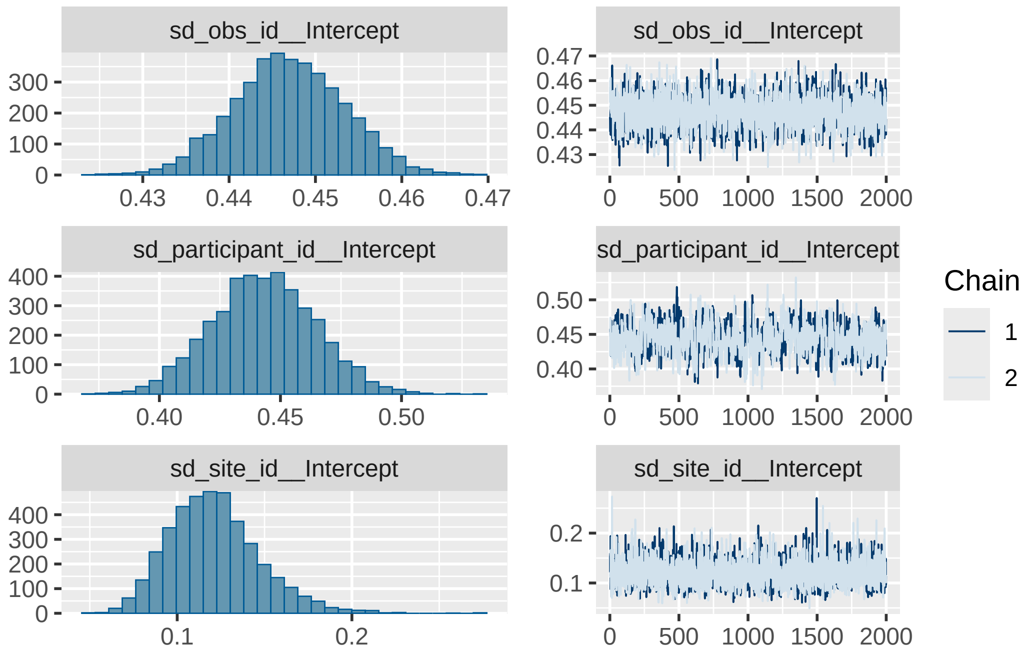
```
Regression Coefficients:
                                     Estimate Est.Error l-95% CI u-95% CI  Rhat
Intercept                               0.721     0.049    0.625    0.817 1.003
menarche_status_pY                      0.078     0.018    0.044    0.113 1.001
adhd_status_presY                       0.664     0.046    0.574    0.754 1.001
age_sc                                  0.052     0.015    0.022    0.080 1.000
ethnracehispanic                        0.327     0.053    0.222    0.427 1.000
ethnraceother                           0.388     0.057    0.274    0.500 1.000
ethnracewhite_nh                        0.406     0.047    0.315    0.497 1.003
inr_sc                                 -0.053     0.021   -0.095   -0.010 1.000
menarche_status_pY:adhd_status_presY    0.035     0.041   -0.044    0.115 1.000
                                     Bulk_ESS Tail_ESS
Intercept                                3192     2857
menarche_status_pY                       4909     3187
adhd_status_presY                        3783     2732
age_sc                                   4421     3299
ethnracehispanic                         3369     2884
ethnraceother                            3503     3004
ethnracewhite_nh                         3314     3022
inr_sc                                   5340     2998
menarche_status_pY:adhd_status_presY     4866     2961

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

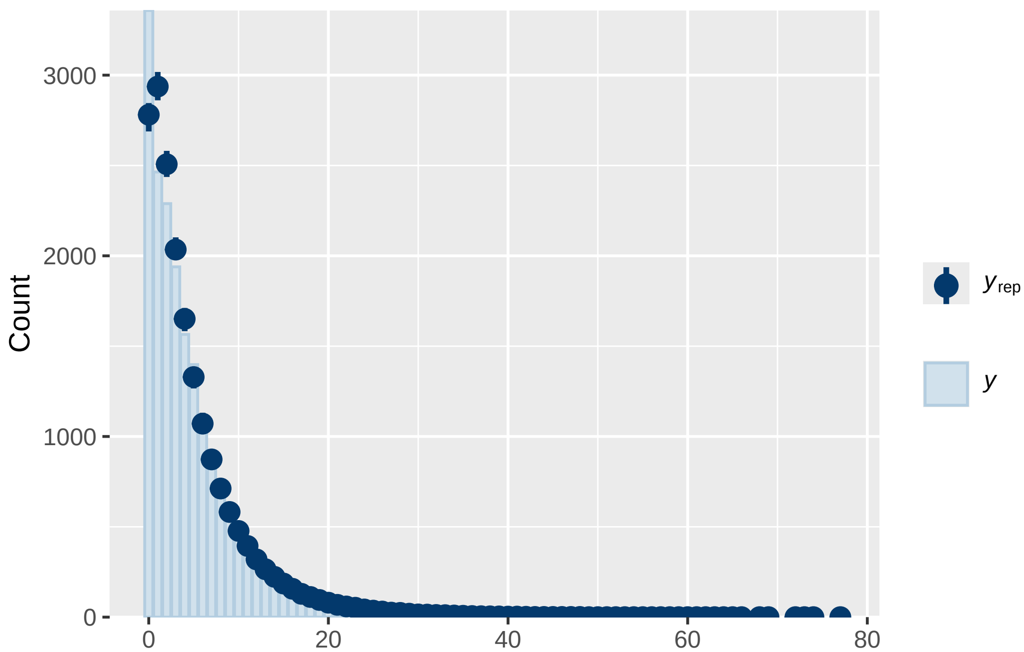
Model 2 ADHD present diagnostics

```
                prior     class                                 coef
         normal(0, 1)         b
         normal(0, 1)         b                    adhd_status_presY
         normal(0, 1)         b                               age_sc
         normal(0, 1)         b                     ethnracehispanic
         normal(0, 1)         b                        ethnraceother
         normal(0, 1)         b                     ethnracewhite_nh
         normal(0, 1)         b                               inr_sc
         normal(0, 1)         b                   menarche_status_pY
         normal(0, 1)         b menarche_status_pY:adhd_status_presY
 student_t(3, 1.1, 2.5) Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd
```

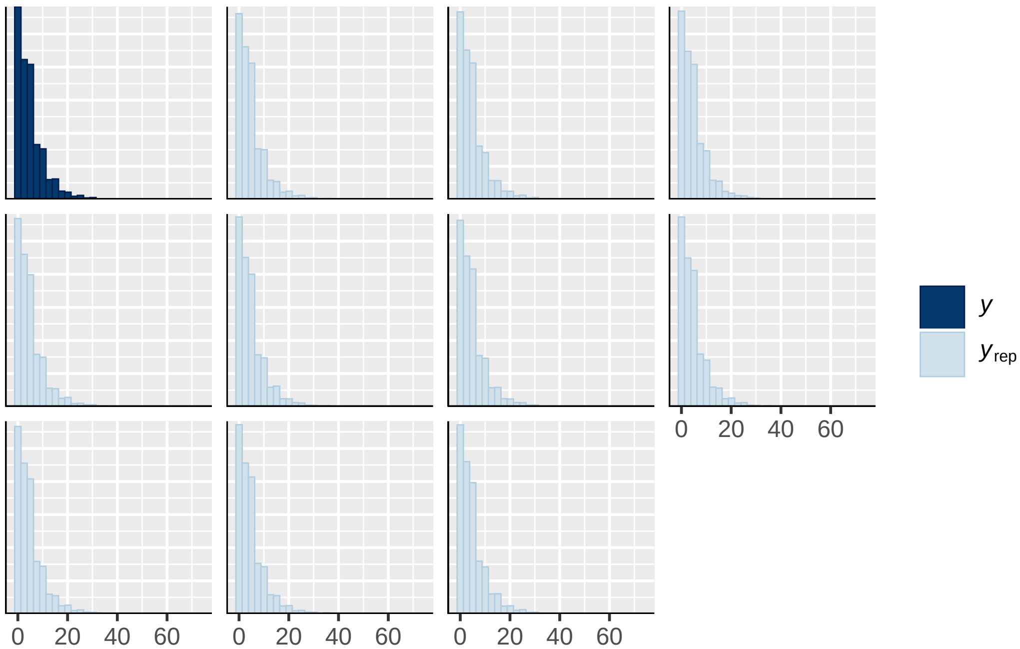
```
  student_t(3, 0, 2.5)        sd                              Intercept
  student_t(3, 0, 2.5)        sd
  student_t(3, 0, 2.5)        sd                              Intercept
  student_t(3, 0, 2.5)        sd
  student_t(3, 0, 2.5)        sd                              Intercept
  student_t(3, 0, 2.5)        sd
  student_t(3, 0, 2.5)        sd                              Intercept
         group resp dpar nlpar lb ub tag       source
                                                 user
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                              default
                                 0            default
     family_id                   0       (vectorized)
     family_id                   0       (vectorized)
        obs_id                   0       (vectorized)
        obs_id                   0       (vectorized)
participant_id                   0       (vectorized)
participant_id                   0       (vectorized)
       site_id                   0       (vectorized)
       site_id                   0       (vectorized)
```


b_Intercept
b_menarche_status_pY
b_adhd_status_presY
b_age_sc
b_ethnracehispanic
b_ethnraceother
b_ethnracewhite_nh
b_inr_sc
_menarche_status_pY:adhd_status_pres
rche_status_pY:adhd_statu
sd_family_id__Intercept
Chain
1
2

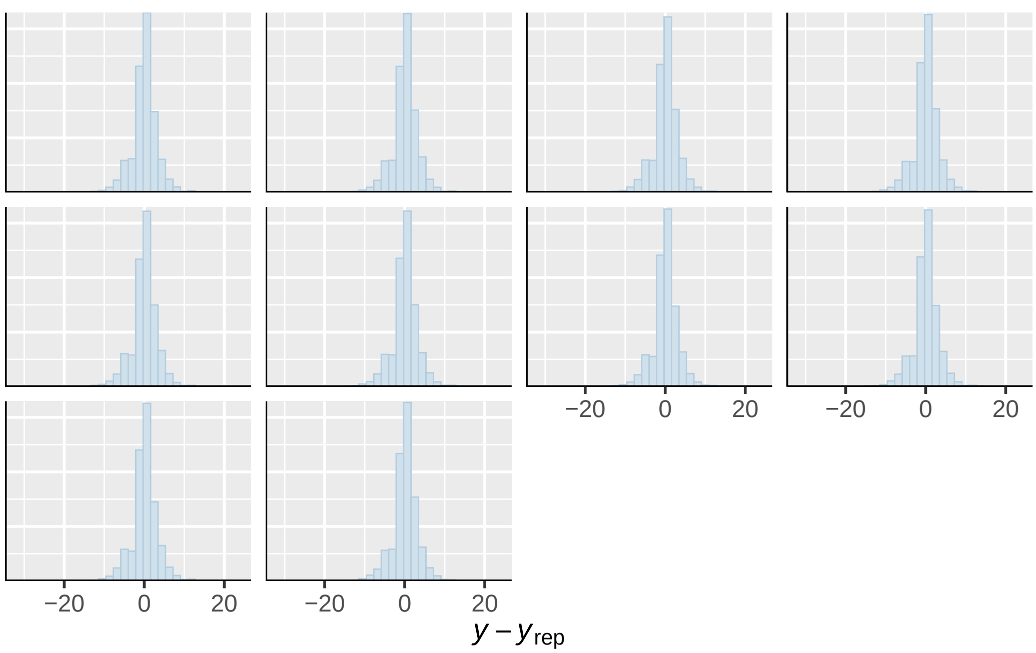





```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

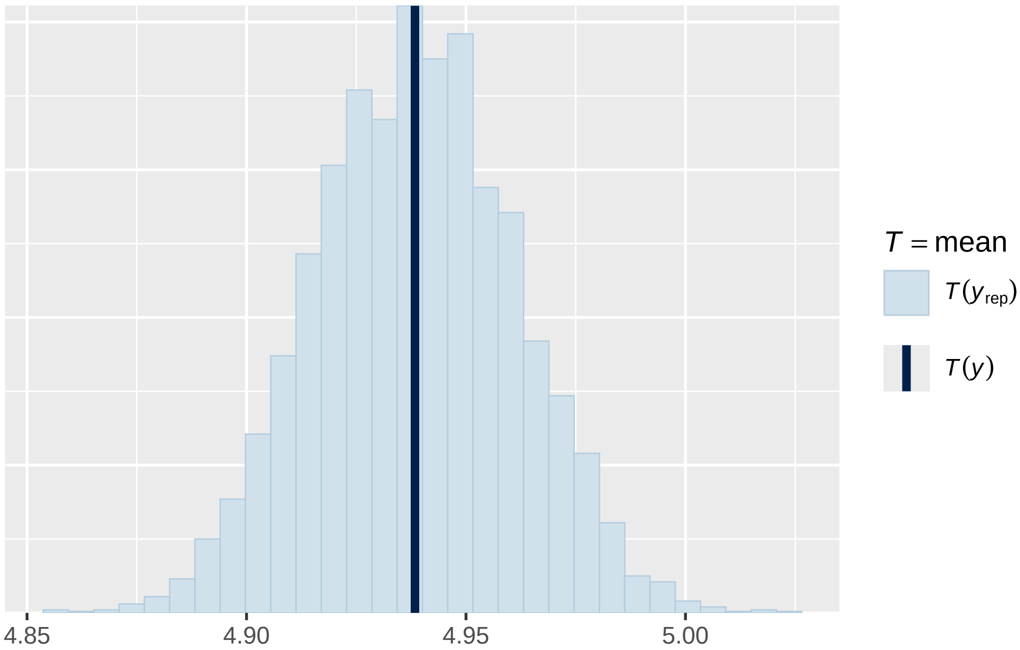



```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

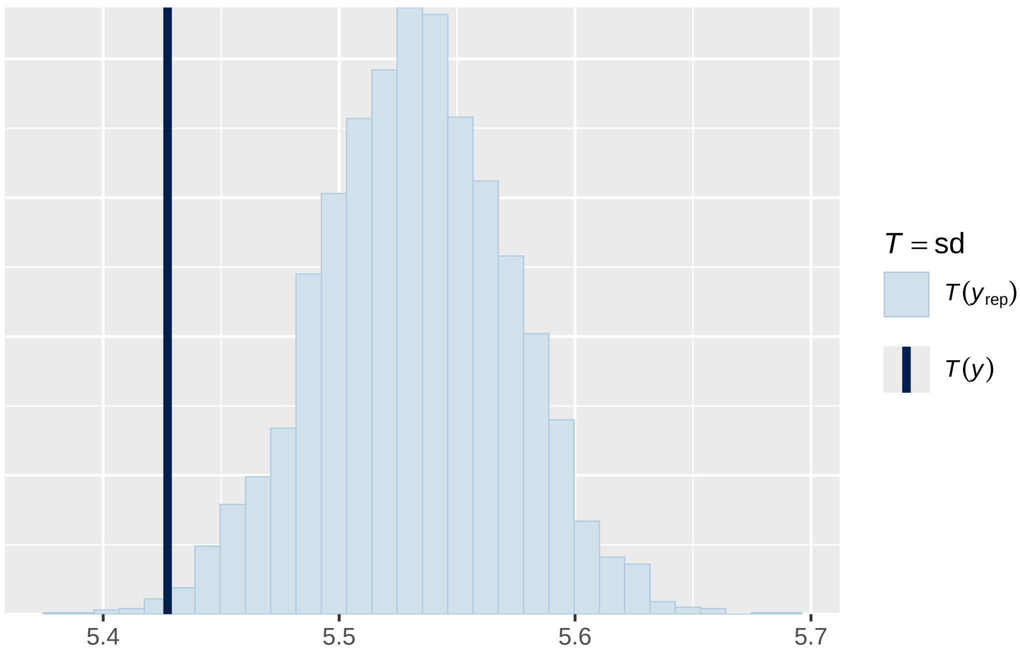



```
Using all posterior draws for ppc type 'stat' by default.
```

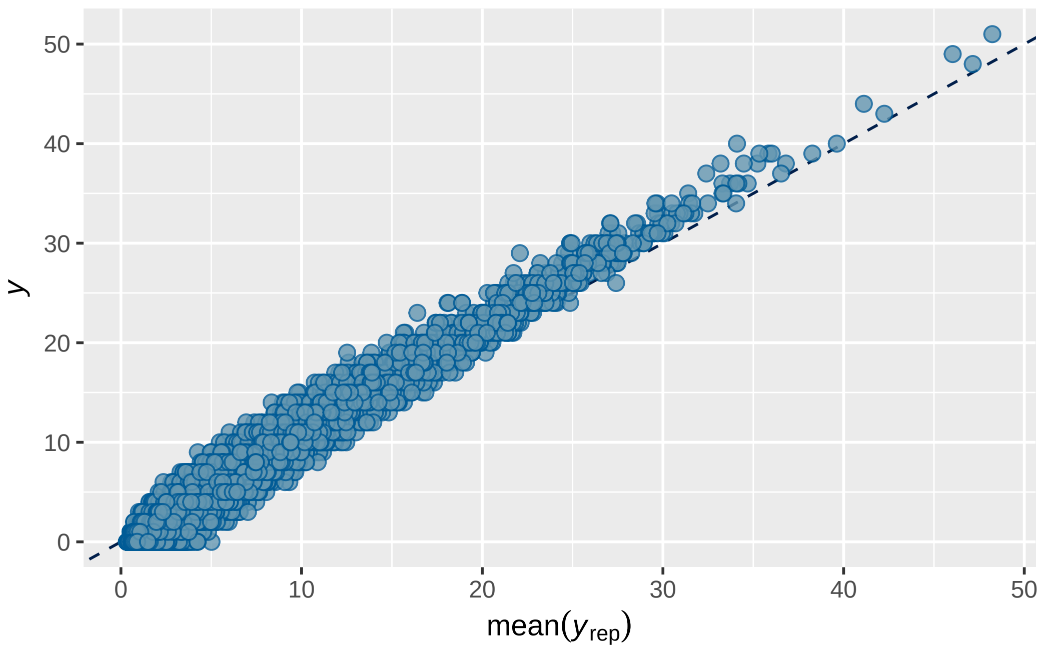
```
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 2b ADHD present output

```
   Estimate   Est.Error      Q2.5     Q97.5
R2 0.838637 0.002536781 0.8336032 0.8435009
```

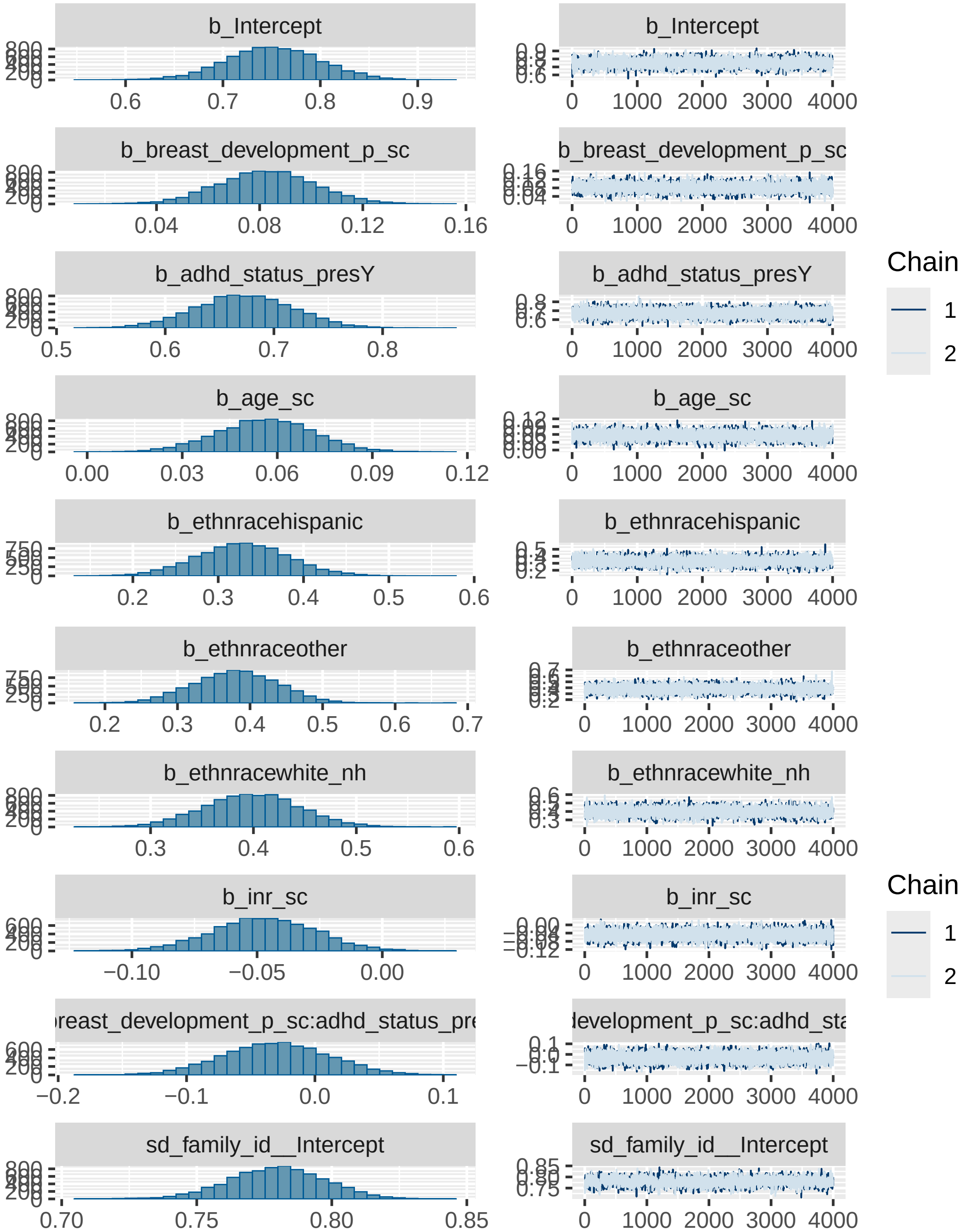
```
 Family: poisson
  Links: mu = log
Formula: internalising ~ breast_development_p_sc * adhd_status_pres + age_sc + ethnrace + in
   Data: analytic_df (Number of observations: 19358)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 4613)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.780     0.018    0.744    0.813 1.000     1663     2948

~obs_id (Number of levels: 19358)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.447     0.006    0.434    0.459 1.000     2610     4731
```

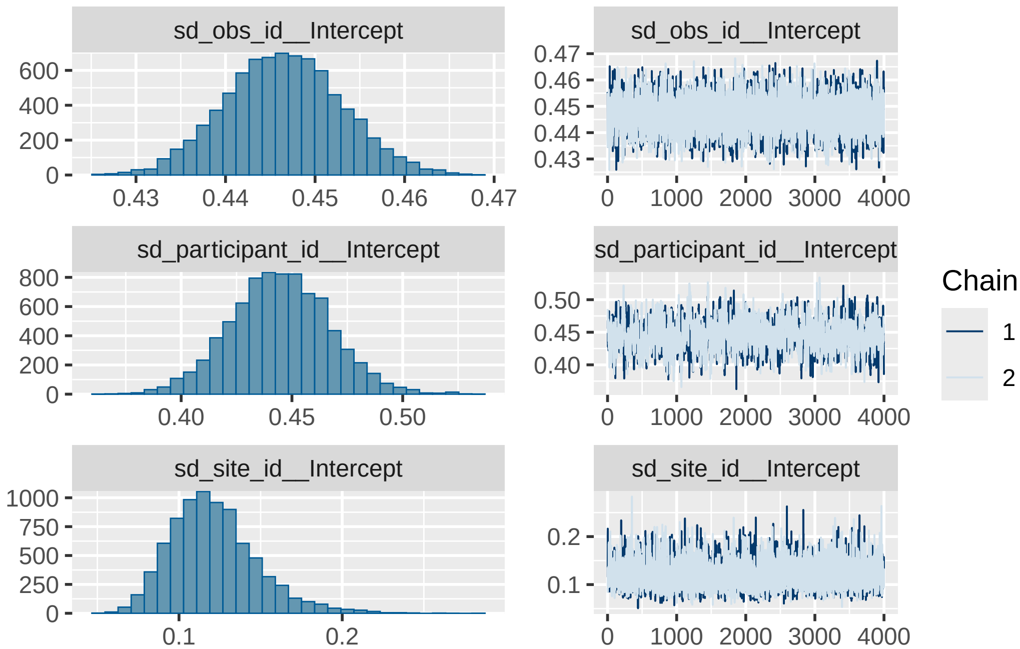
```
~participant_id (Number of levels: 5283)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.445     0.022    0.401    0.489 1.002      826     1340

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.121     0.027    0.078    0.185 1.000     6098     5868

Regression Coefficients:
                                          Estimate Est.Error l-95% CI u-
95% CI
Intercept                                    0.752     0.049    0.655    0.847
breast_development_p_sc                      0.083     0.019    0.046    0.120
adhd_status_presY                            0.673     0.044    0.585    0.759
age_sc                                       0.056     0.015    0.026    0.086
ethnracehispanic                             0.329     0.053    0.226    0.435
ethnraceother                                0.380     0.058    0.267    0.491
ethnracewhite_nh                             0.398     0.047    0.308    0.489
inr_sc                                      -0.047     0.021   -0.088    -
0.007
breast_development_p_sc:adhd_status_presY   -0.030     0.041   -0.110    0.051
                                           Rhat Bulk_ESS Tail_ESS
Intercept                                 1.000     6696     5637
breast_development_p_sc                   1.000    10014     5667
adhd_status_presY                         1.000     7843     6310
age_sc                                    1.000    10683     6416
ethnracehispanic                          1.000     6091     5937
ethnraceother                             1.002     6606     5780
ethnracewhite_nh                          1.000     6194     5650
inr_sc                                    1.001    12246     6174
breast_development_p_sc:adhd_status_presY 1.001    10809     6736

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

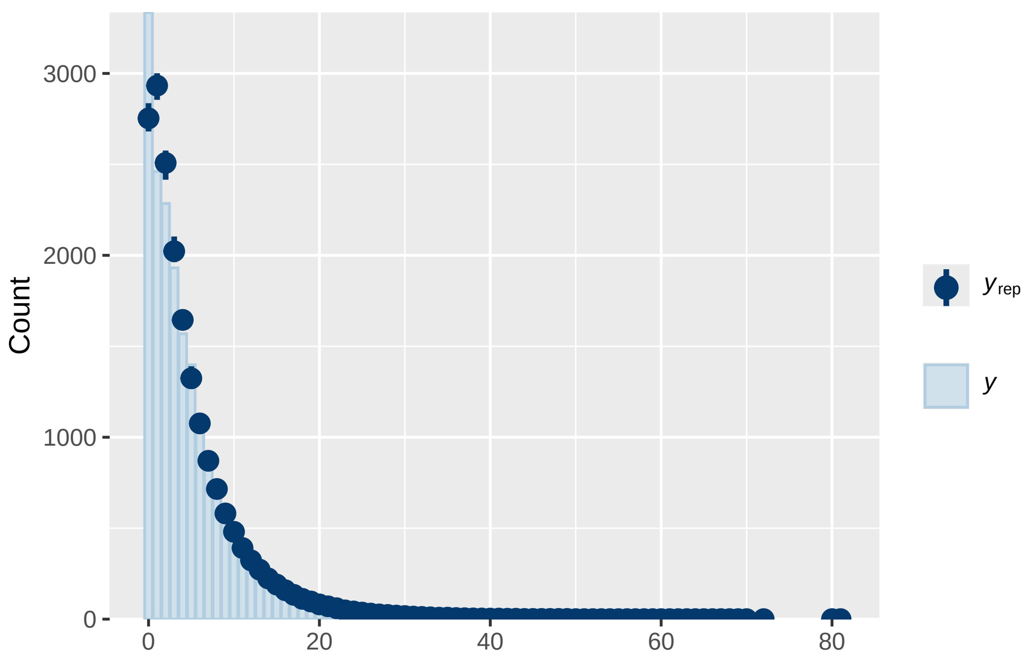
Model 2b diagnostics

```
       prior     class                          coef
normal(0, 1)         b
normal(0, 1)         b             adhd_status_presY
```

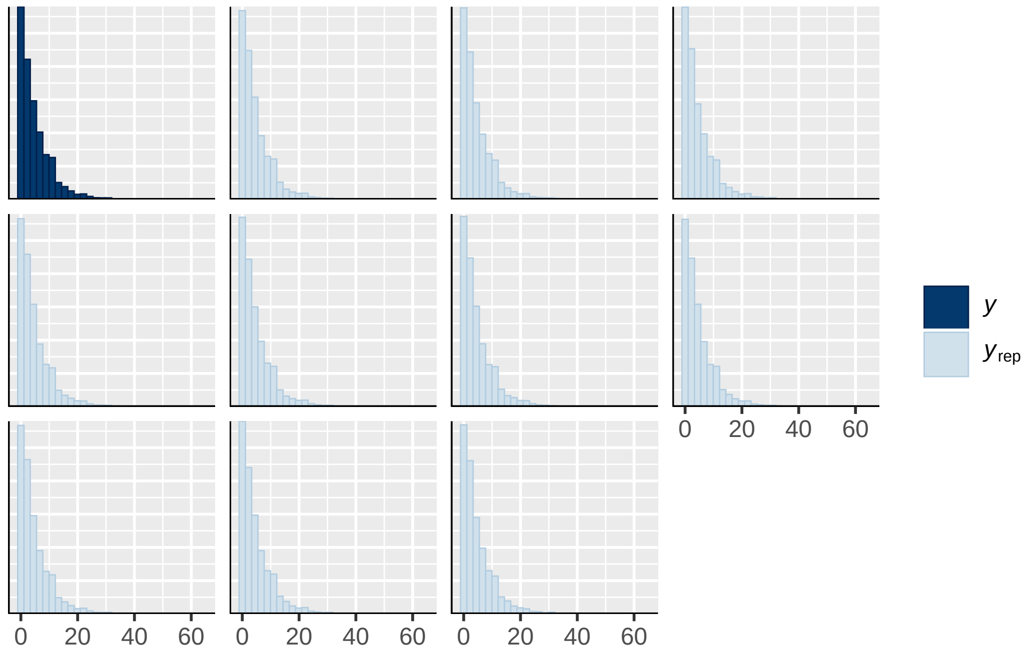
```
          normal(0, 1)         b                                    age_sc
          normal(0, 1)         b                   breast_development_p_sc
          normal(0, 1)         b breast_development_p_sc:adhd_status_presY
          normal(0, 1)         b                          ethnracehispanic
          normal(0, 1)         b                             ethnraceother
          normal(0, 1)         b                          ethnracewhite_nh
          normal(0, 1)         b                                    inr_sc
student_t(3, 1.1, 2.5) Intercept
  student_t(3, 0, 2.5)        sd
  student_t(3, 0, 2.5)        sd
  student_t(3, 0, 2.5)        sd                                 Intercept
  student_t(3, 0, 2.5)        sd
  student_t(3, 0, 2.5)        sd                                 Intercept
  student_t(3, 0, 2.5)        sd
  student_t(3, 0, 2.5)        sd                                 Intercept
  student_t(3, 0, 2.5)        sd
  student_t(3, 0, 2.5)        sd                                 Intercept
         group resp dpar nlpar lb ub tag       source
                                                 user
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                              default
                                0             default
     family_id                  0        (vectorized)
     family_id                  0        (vectorized)
        obs_id                  0        (vectorized)
        obs_id                  0        (vectorized)
participant_id                  0        (vectorized)
participant_id                  0        (vectorized)
       site_id                  0        (vectorized)
       site_id                  0        (vectorized)
```


b_Intercept
0.6 0.7 0.8 0.9
b_Intercept
0 1000 2000 3000 4000
b_breast_development_p_sc
0.04 0.08 0.12 0.16
b_breast_development_p_sc
0 1000 2000 3000 4000
b_adhd_status_presY
0.5 0.6 0.7 0.8
b_adhd_status_presY
0 1000 2000 3000 4000
b_age_sc
0.00 0.03 0.06 0.09 0.12
b_age_sc
0 1000 2000 3000 4000
b_ethnracehispanic
0.2 0.3 0.4 0.5 0.6
b_ethnracehispanic
0 1000 2000 3000 4000
b_ethnraceother
0.2 0.3 0.4 0.5 0.6 0.7
b_ethnraceother
0 1000 2000 3000 4000
b_ethnracewhite_nh
0.3 0.4 0.5 0.6
b_ethnracewhite_nh
0 1000 2000 3000 4000
b_inr_sc
−0.10 −0.05 0.00
b_inr_sc
0 1000 2000 3000 4000
reast_development_p_sc:adhd_status_pre
−0.2 −0.1 0.0 0.1
development_p_sc:adhd_sta
0 1000 2000 3000 4000
sd_family_id__Intercept
0.70 0.75 0.80 0.85
sd_family_id__Intercept
0 1000 2000 3000 4000
Chain
1
2
Chain
1
2

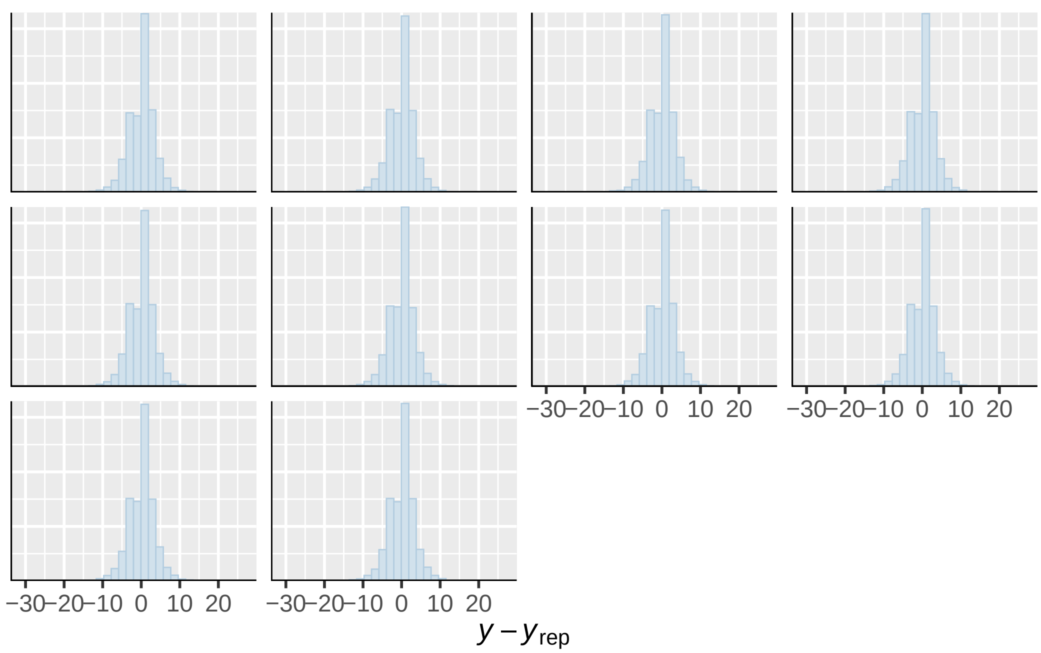




`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

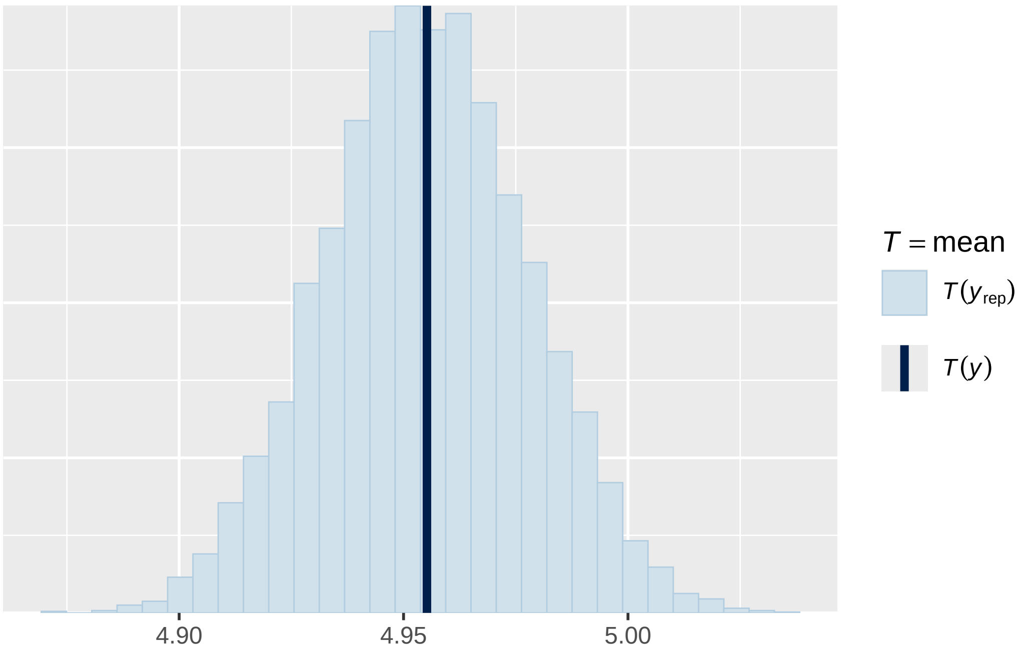



```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

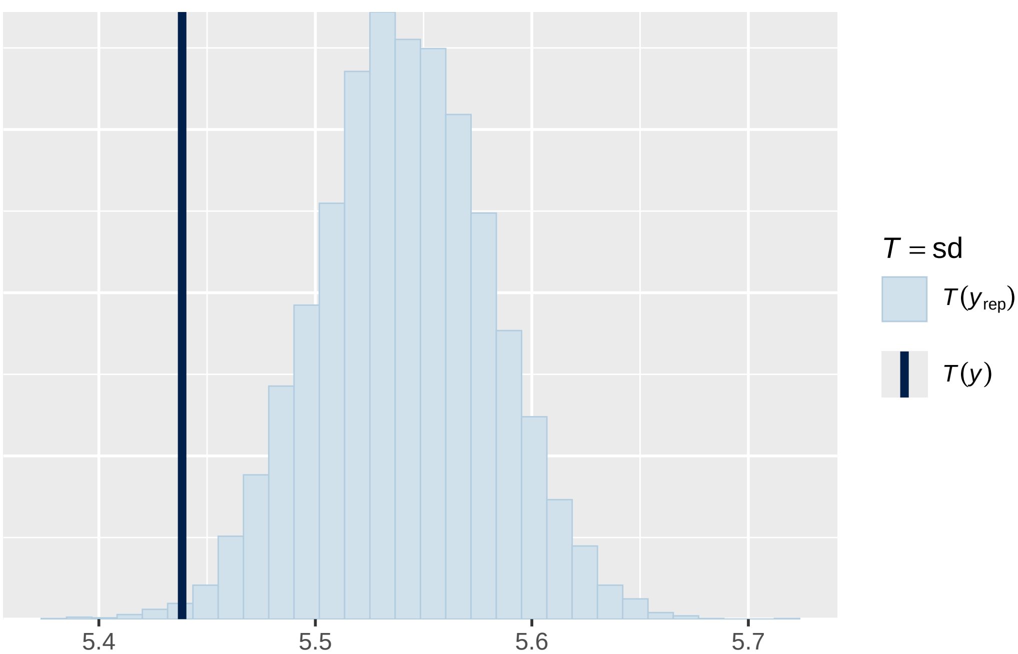



```
Using all posterior draws for ppc type 'stat' by default.
```

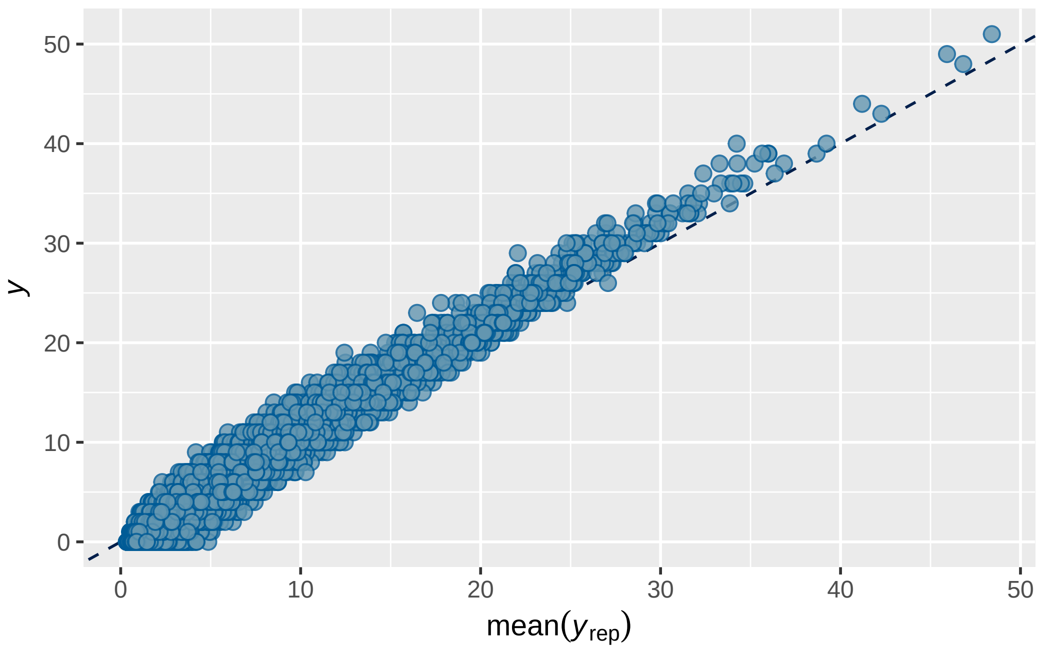
```
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



ABCD ADHD menarche analysis: Output for cross-sectional supplementary/sensitivity models

Model 3 ADHD present output

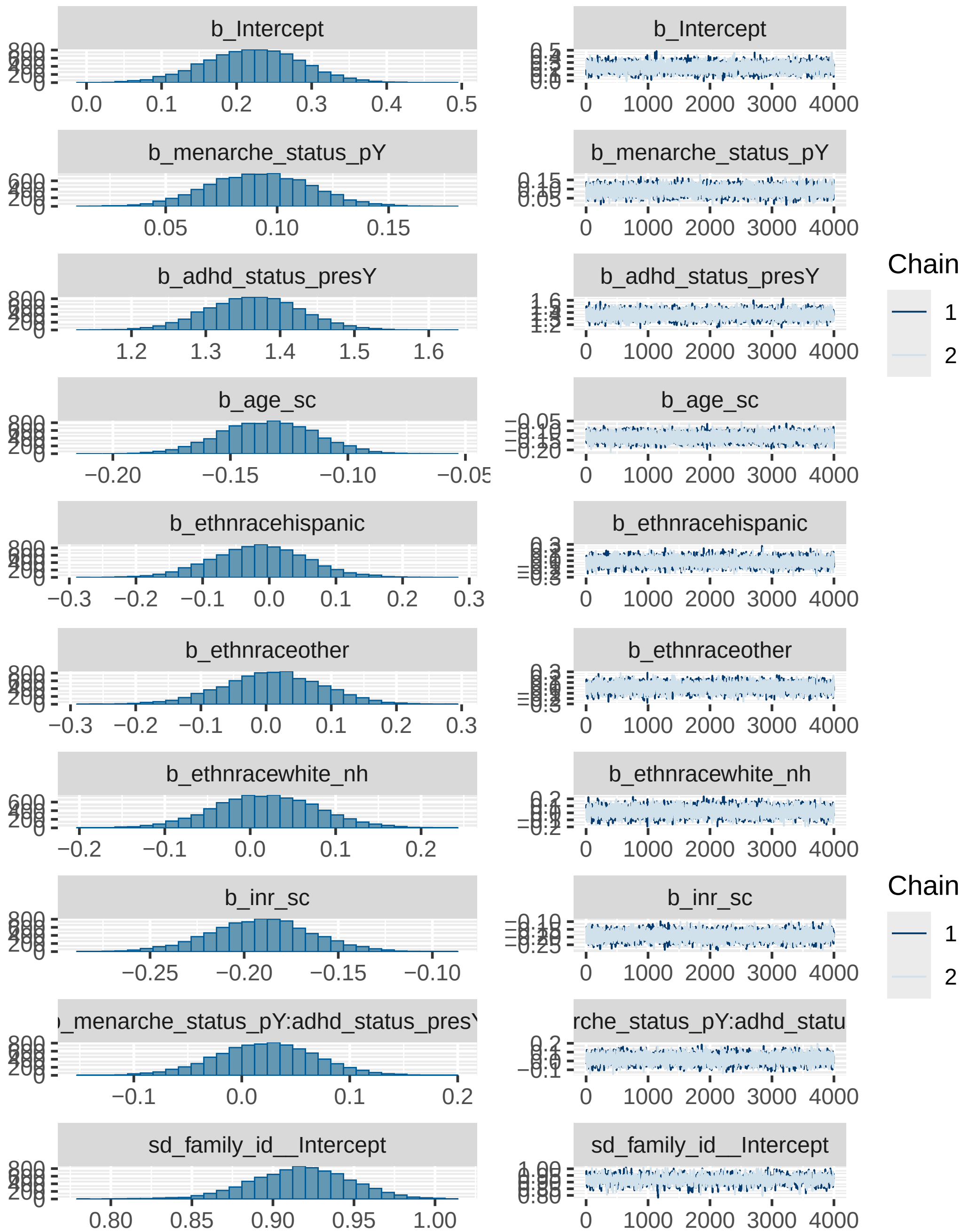
```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.8594888 0.002517475 0.8544448 0.8643727


 Family: poisson
  Links: mu = log
Formula: externalising ~ menarche_status_p * adhd_status_pres + age_sc + ethnrace + inr_sc +
   Data: analytic_df (Number of observations: 19375)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 4620)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.9174    0.0307   0.8554   0.9758 1.0007     1105     1889

~obs_id (Number of levels: 19375)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.4976    0.0082   0.4816   0.5138 0.9999     3394     5376

~participant_id (Number of levels: 5291)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.7946    0.0304   0.7369   0.8553 1.0015     1003     1929

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.1637    0.0370   0.1028   0.2470 0.9999     5110     5214
```

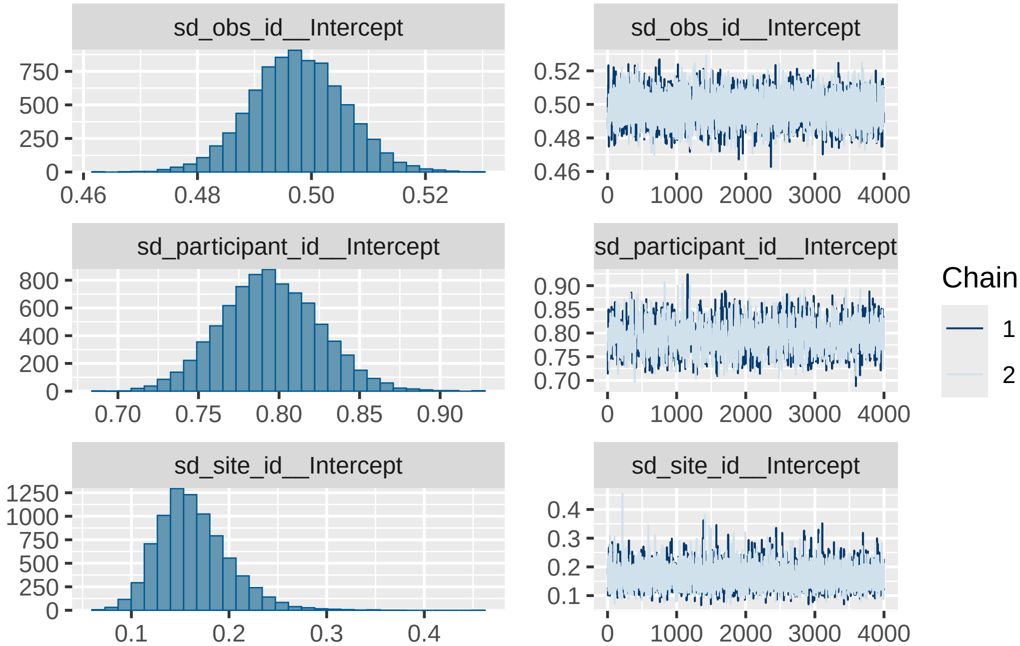
```
Regression Coefficients:
                                    Estimate Est.Error l-95% CI u-95% CI
Intercept                             0.2221    0.0646   0.0947   0.3482
menarche_status_pY                    0.0922    0.0228   0.0486   0.1373
adhd_status_presY                     1.3650    0.0628   1.2431   1.4883
age_sc                               -0.1341    0.0198  -0.1725  -0.0955
ethnracehispanic                     -0.0140    0.0705  -0.1518   0.1282
ethnraceother                         0.0098    0.0764  -0.1425   0.1573
ethnracewhite_nh                      0.0187    0.0610  -0.1001   0.1396
inr_sc                               -0.1909    0.0272  -0.2454  -0.1370
menarche_status_pY:adhd_status_presY  0.0222    0.0464  -0.0701   0.1117
                                      Rhat Bulk_ESS Tail_ESS
Intercept                           1.0003     4884     5563
menarche_status_pY                  1.0001     7328     6301
adhd_status_presY                   1.0007     6716     5644
age_sc                              1.0000     7548     5866
ethnracehispanic                    1.0001     5502     5287
ethnraceother                       1.0002     5858     5479
ethnracewhite_nh                    1.0006     5412     6021
inr_sc                              1.0000     9143     6193
menarche_status_pY:adhd_status_presY 1.0004     6681     5597

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

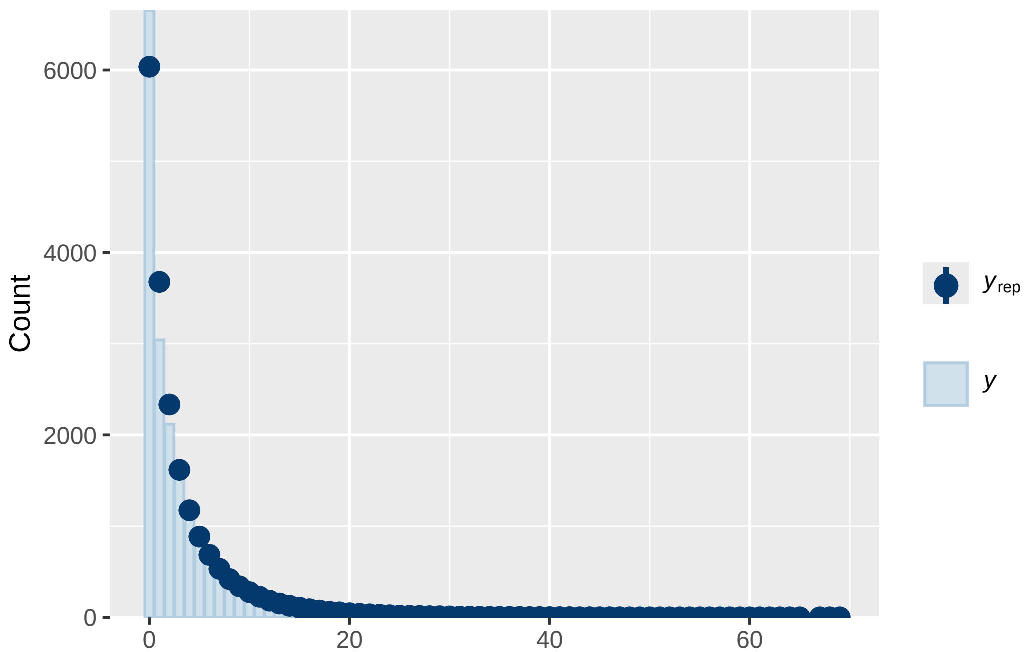
Model 3 ADHD present diagnostics

```
               prior     class                                 coef
        normal(0, 1)         b
        normal(0, 1)         b                    adhd_status_presY
        normal(0, 1)         b                               age_sc
        normal(0, 1)         b                     ethnracehispanic
        normal(0, 1)         b                        ethnraceother
        normal(0, 1)         b                     ethnracewhite_nh
        normal(0, 1)         b                               inr_sc
        normal(0, 1)         b                   menarche_status_pY
        normal(0, 1)         b menarche_status_pY:adhd_status_presY
 student_t(3, 0, 2.9) Intercept
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd
```

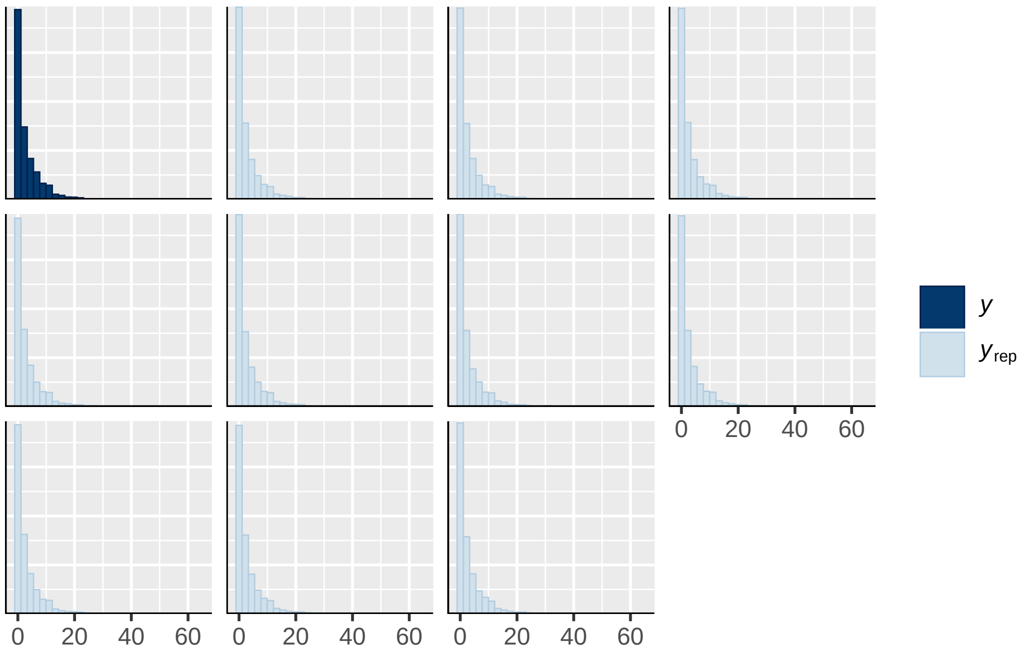
```
student_t(3, 0, 2.9)        sd                              Intercept
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd                              Intercept
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd                              Intercept
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd                              Intercept
         group resp dpar nlpar lb ub tag       source
                                                 user
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                              default
                                  0           default
     family_id                    0      (vectorized)
     family_id                    0      (vectorized)
        obs_id                    0      (vectorized)
        obs_id                    0      (vectorized)
participant_id                    0      (vectorized)
participant_id                    0      (vectorized)
       site_id                    0      (vectorized)
       site_id                    0      (vectorized)
```


b_Intercept
b_menarche_status_pY
b_adhd_status_presY
b_age_sc
b_ethnracehispanic
b_ethnraceother
b_ethnracewhite_nh
b_inr_sc
_menarche_status_pY:adhd_status_presY
sd_family_id__Intercept
Chain
1
2

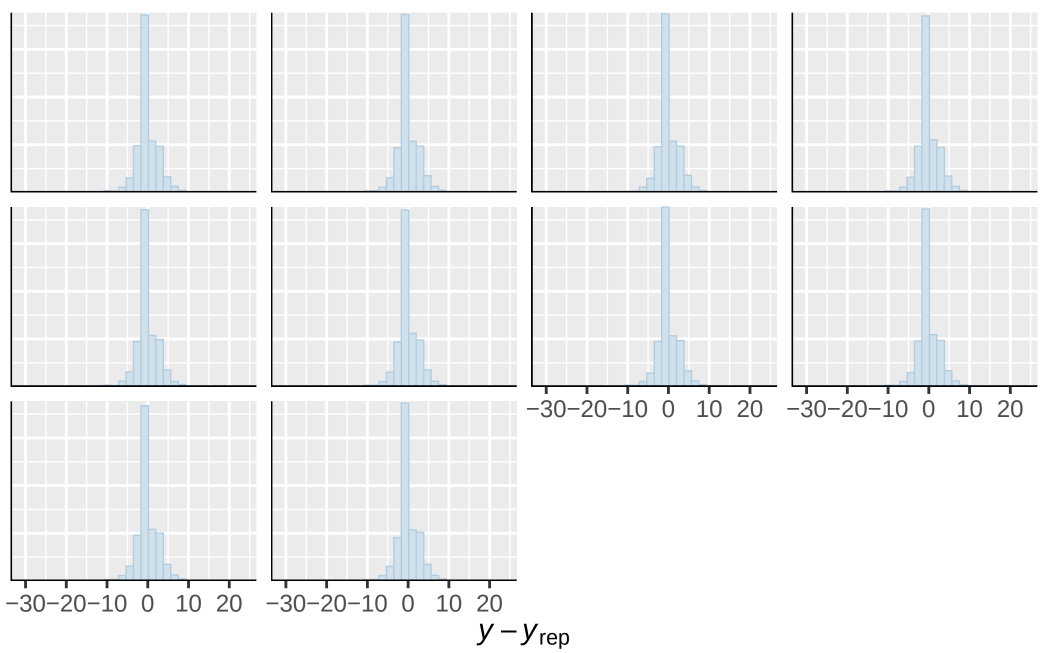




`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

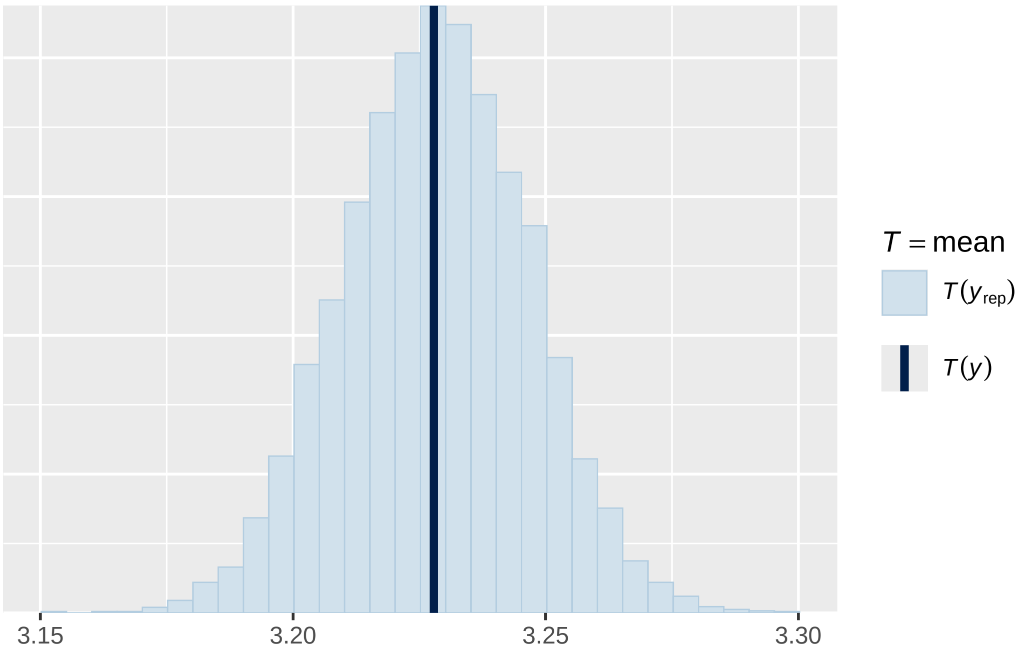



```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

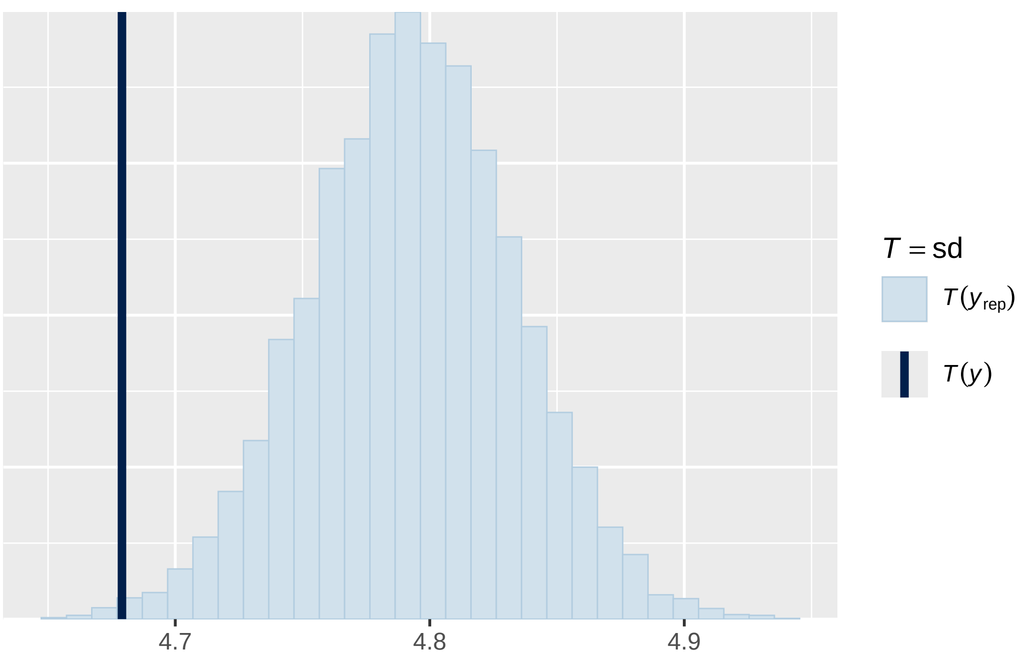



```
Using all posterior draws for ppc type 'stat' by default.
```

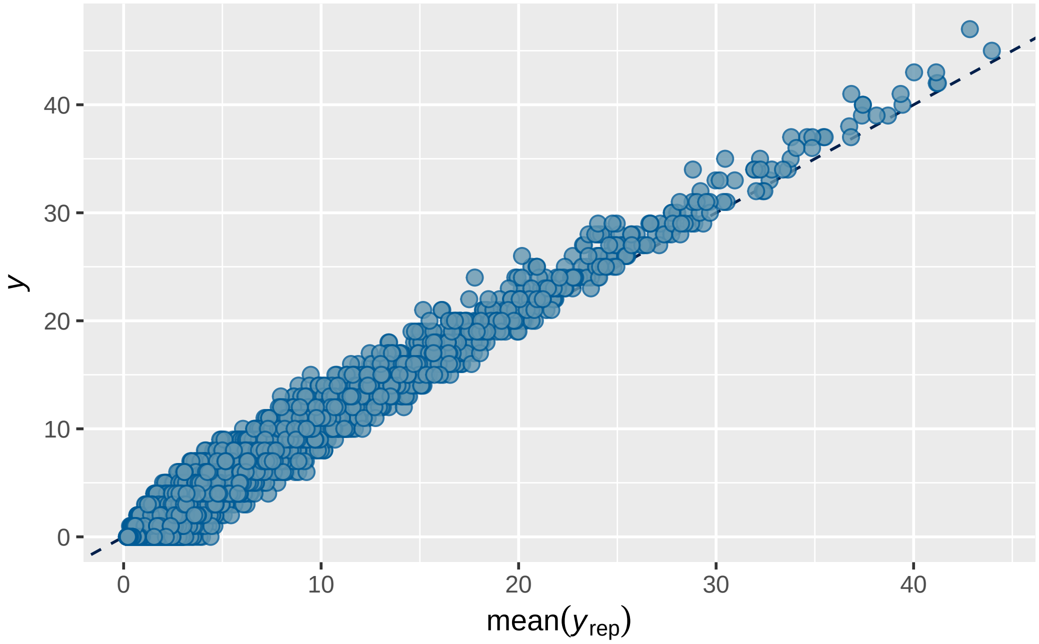
```
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 3b ADHD present output

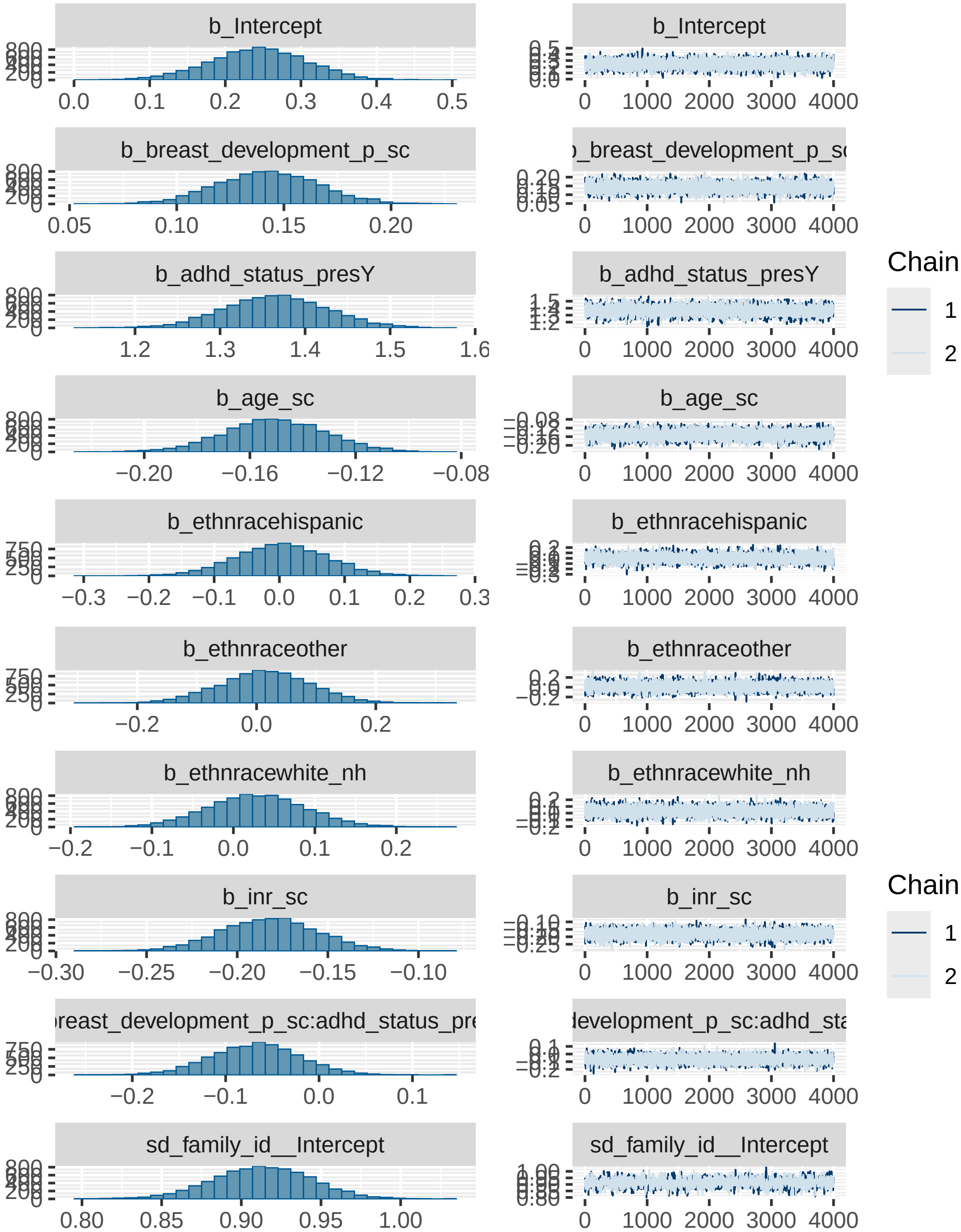
```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.8591862 0.002558196 0.8539733 0.8641035


 Family: poisson
  Links: mu = log
Formula: externalising ~ breast_development_p_sc * adhd_status_pres + age_sc + ethnrace + in
   Data: analytic_df (Number of observations: 19358)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 4613)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.9131    0.0309   0.8513   0.9722 1.0011      992     1909

~obs_id (Number of levels: 19358)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.4946    0.0082   0.4784   0.5109 0.9999     3245     4878
```

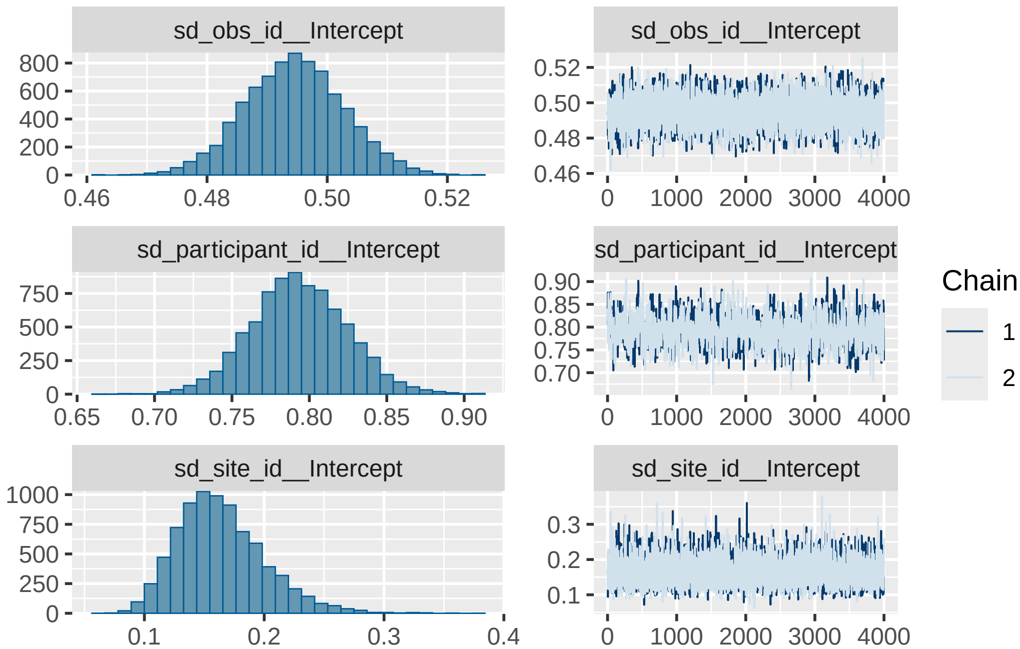
```
~participant_id (Number of levels: 5283)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.7934    0.0310   0.7337   0.8552 1.0027      815     1577

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.1639    0.0365   0.1042   0.2469 1.0007     5087     5193

Regression Coefficients:
                                          Estimate Est.Error l-95% CI u-
95% CI
Intercept                                   0.2436    0.0647   0.1152   0.3672
breast_development_p_sc                     0.1425    0.0236   0.0964   0.1897
adhd_status_presY                           1.3657    0.0610   1.2480   1.4873
age_sc                                     -0.1505    0.0192  -0.1884  -
0.1130
ethnracehispanic                            0.0035    0.0706  -0.1353   0.1416
ethnraceother                               0.0149    0.0768  -0.1346   0.1639
ethnracewhite_nh                            0.0288    0.0599  -0.0871   0.1469
inr_sc                                     -0.1820    0.0268  -0.2347  -
0.1288
breast_development_p_sc:adhd_status_presY  -0.0681    0.0474  -0.1623   0.0245
                                            Rhat Bulk_ESS Tail_ESS
Intercept                                 1.0003     4593     5273
breast_development_p_sc                   1.0002     7326     6100
adhd_status_presY                         1.0000     6339     5356
age_sc                                    1.0000     7751     6087
ethnracehispanic                          0.9999     5349     4801
ethnraceother                             0.9999     5558     5534
ethnracewhite_nh                          0.9999     5138     5343
inr_sc                                    1.0000     8338     6266
breast_development_p_sc:adhd_status_presY 1.0002     7297     6428

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

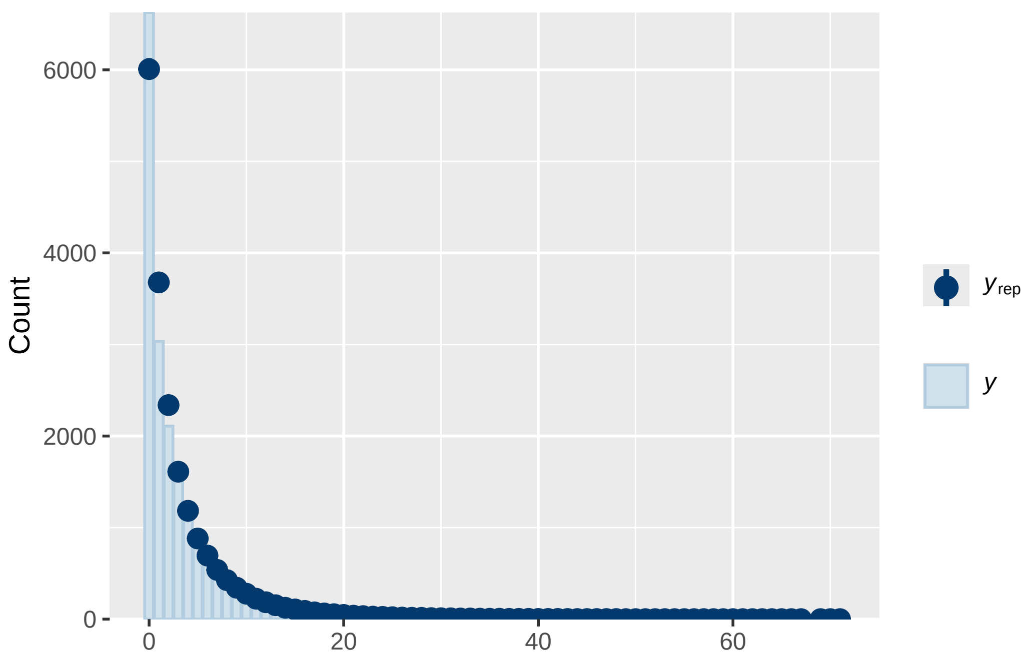
Model 3B ADHD present diagnostics

```
                 prior     class                                      coef
           normal(0, 1)        b
```

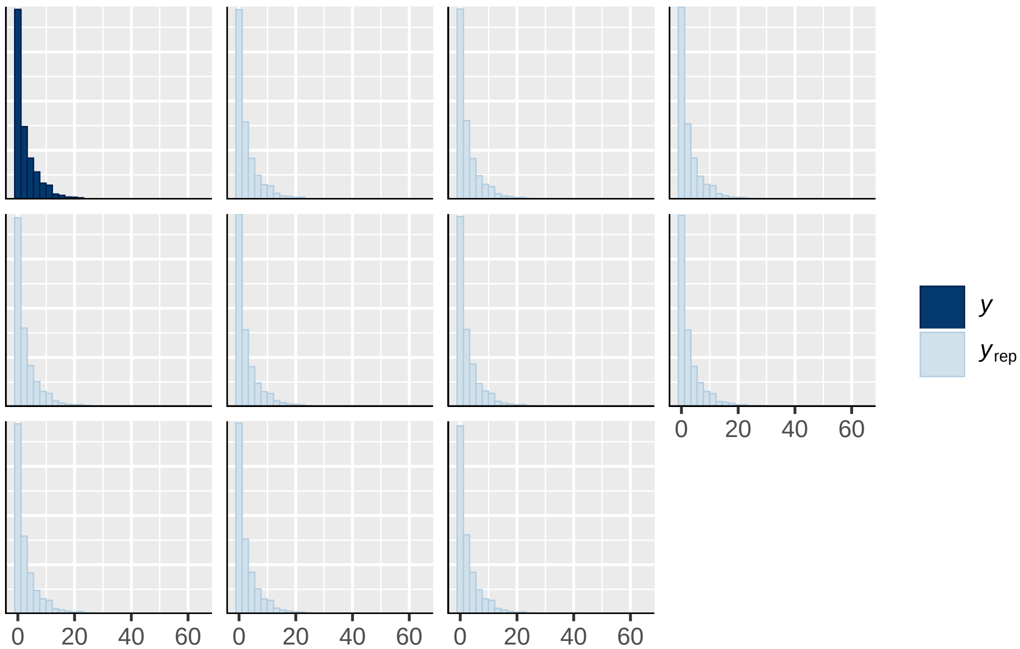
```
          normal(0, 1)         b                            adhd_status_presY
          normal(0, 1)         b                                       age_sc
          normal(0, 1)         b                      breast_development_p_sc
          normal(0, 1)         b breast_development_p_sc:adhd_status_presY
          normal(0, 1)         b                           ethnracehispanic
          normal(0, 1)         b                              ethnraceother
          normal(0, 1)         b                           ethnracewhite_nh
          normal(0, 1)         b                                       inr_sc
student_t(3, 0.7, 2.5) Intercept
  student_t(3, 0, 2.5)        sd
  student_t(3, 0, 2.5)        sd
  student_t(3, 0, 2.5)        sd                                    Intercept
  student_t(3, 0, 2.5)        sd
  student_t(3, 0, 2.5)        sd                                    Intercept
  student_t(3, 0, 2.5)        sd
  student_t(3, 0, 2.5)        sd                                    Intercept
  student_t(3, 0, 2.5)        sd
  student_t(3, 0, 2.5)        sd                                    Intercept
         group resp dpar nlpar lb ub tag       source
                                                 user
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                              default
                                0             default
     family_id                  0        (vectorized)
     family_id                  0        (vectorized)
        obs_id                  0        (vectorized)
        obs_id                  0        (vectorized)
participant_id                  0        (vectorized)
participant_id                  0        (vectorized)
       site_id                  0        (vectorized)
       site_id                  0        (vectorized)
```


b_Intercept
b_breast_development_p_sc
b_adhd_status_presY
b_age_sc
b_ethnracehispanic
b_ethnraceother
b_ethnracewhite_nh
b_inr_sc
sd_family_id__Intercept
Chain
1
2

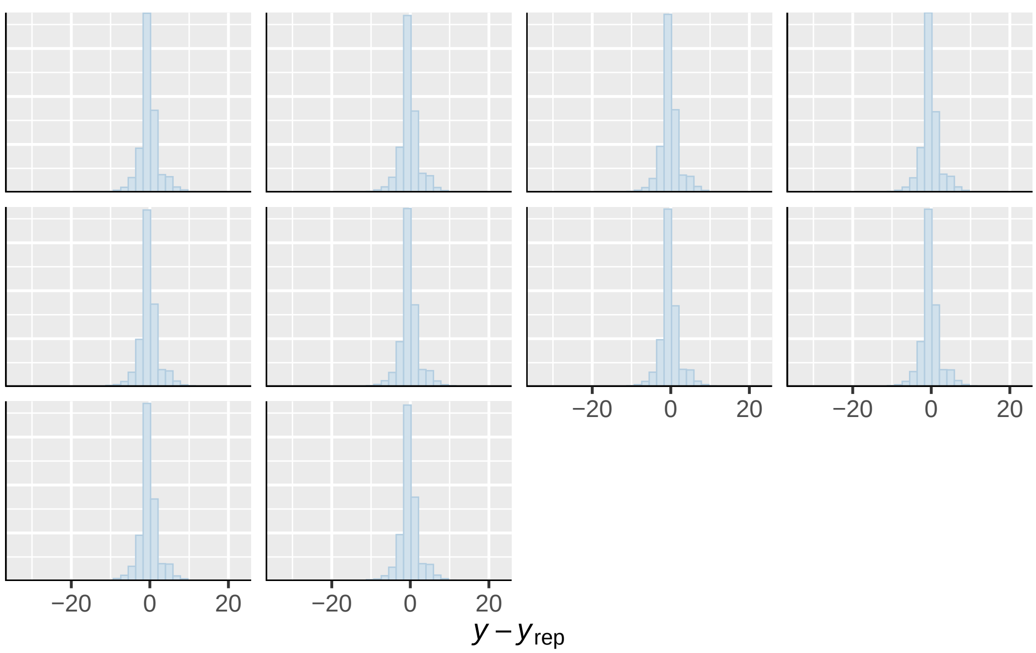




`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

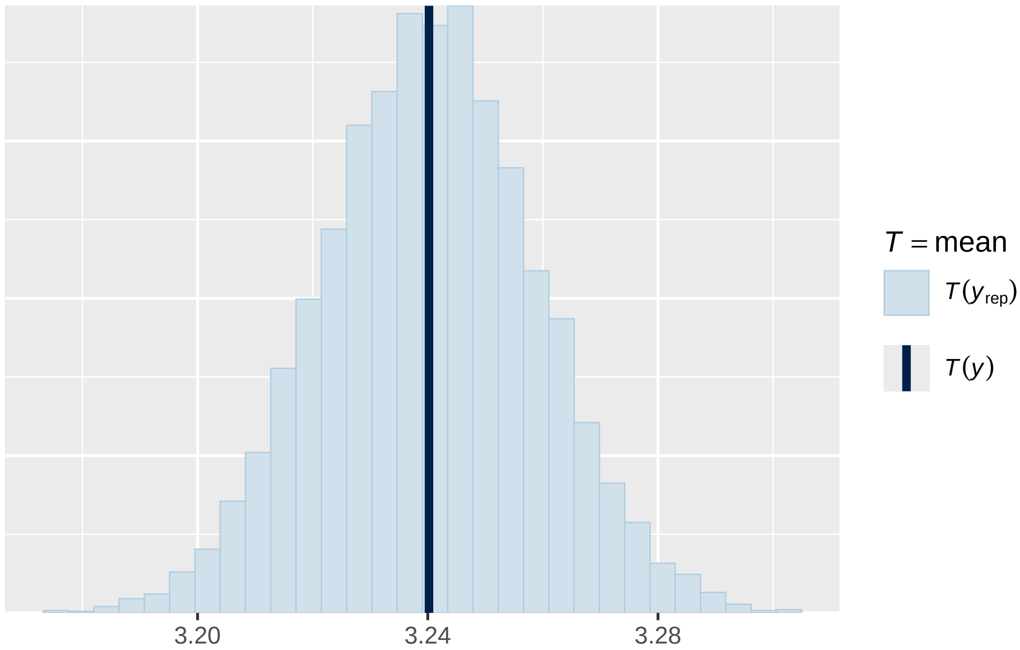



```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

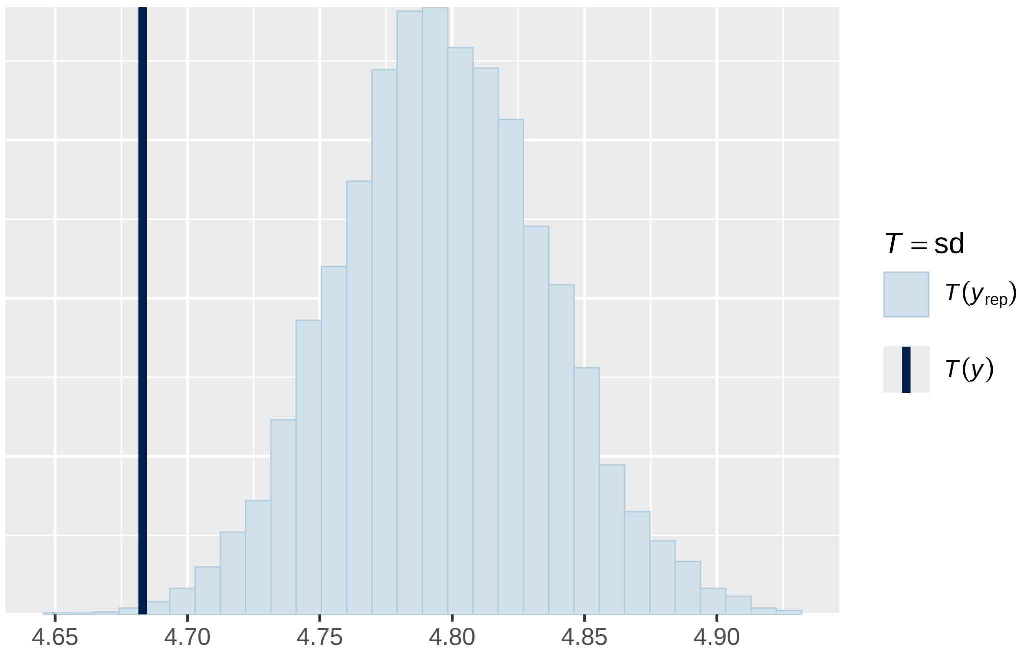



```
Using all posterior draws for ppc type 'stat' by default.
```

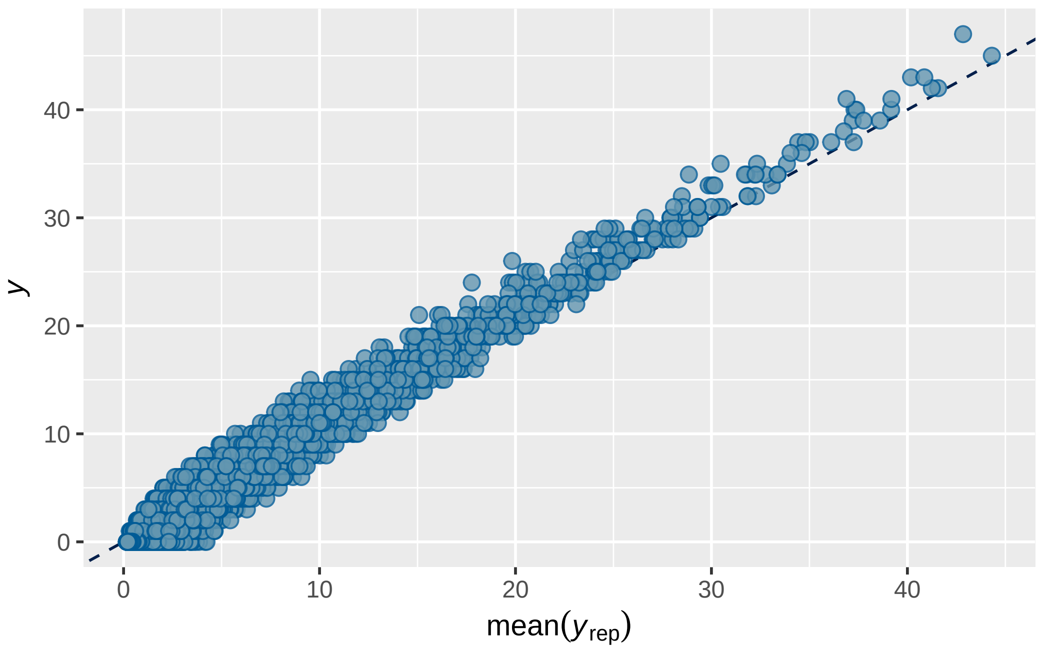
```
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



ABCD ADHD menarche analysis: Output for cross-sectional supplementary/sensitivity models

Model 1b BMI output

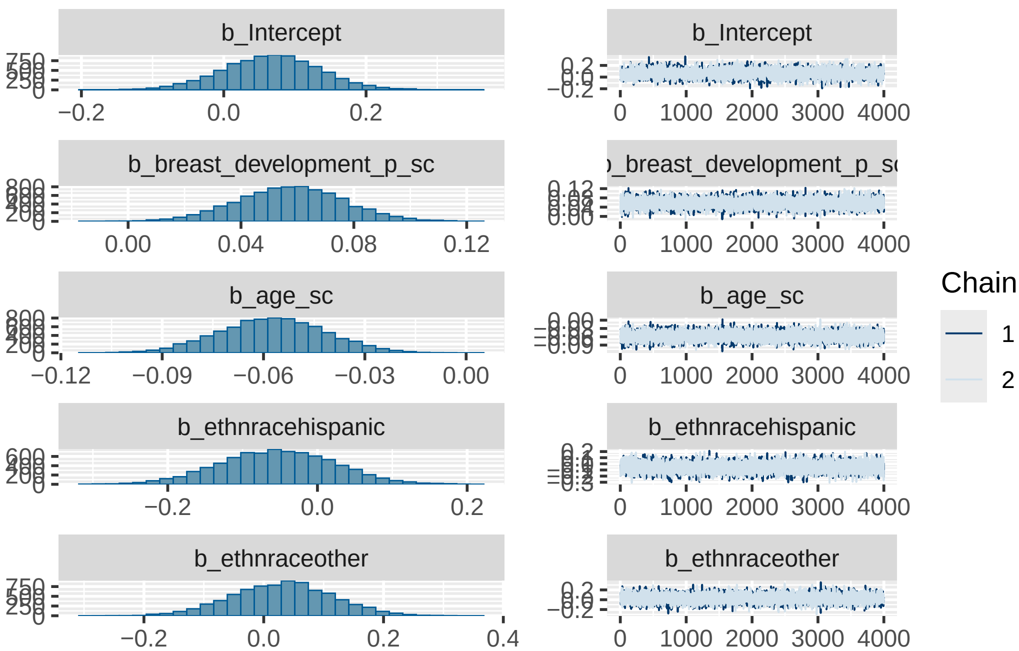
```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.7026613 0.003306821 0.6961907 0.7091034


 Family: poisson
  Links: mu = log
Formula: adhd_traits ~ breast_development_p_sc + age_sc + ethnrace + inr_sc + bmi_wave_0_sc +
   Data: analytic_df (Number of observations: 19364)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 4619)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.834     0.046    0.743    0.921 1.003      625     1546

~participant_id (Number of levels: 5289)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    1.037     0.036    0.968    1.108 1.002      811     2026

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.175     0.039    0.111    0.266 1.000     6635     5310

Regression Coefficients:
                        Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS
Intercept                  0.067     0.068   -0.068    0.202 1.000     6608
```

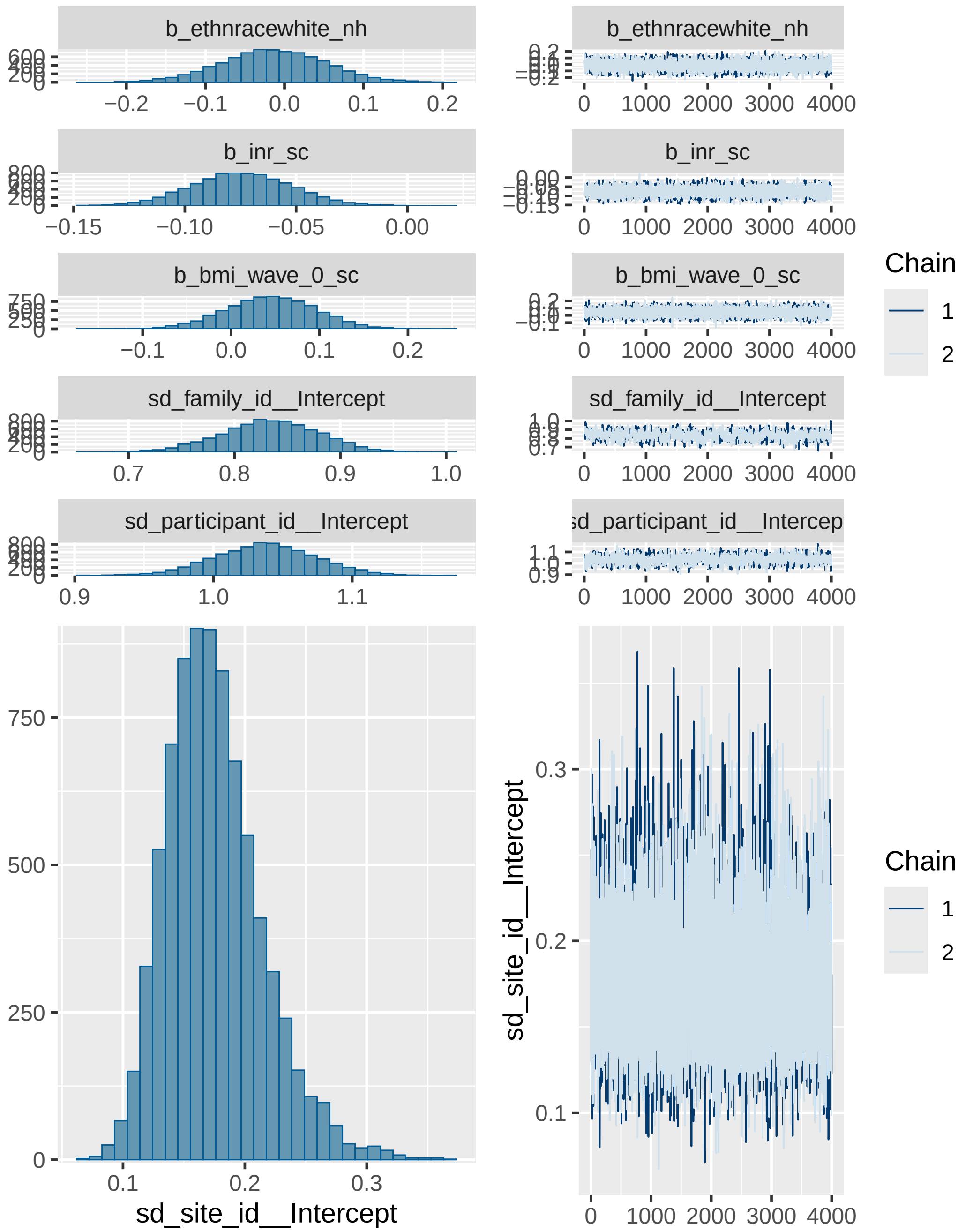
```
breast_development_p_sc    0.058     0.019    0.021    0.095 1.001    15140
age_sc                    -0.057     0.016   -0.087   -0.027 1.000    14562
ethnracehispanic          -0.053     0.077   -0.202    0.095 1.000     8340
ethnraceother              0.031     0.084   -0.133    0.194 1.000     8533
ethnracewhite_nh          -0.014     0.066   -0.144    0.117 1.000     8330
inr_sc                    -0.075     0.022   -0.118   -0.031 1.000    17133
bmi_wave_0_sc              0.045     0.050   -0.054    0.141 1.000    10095
                        Tail_ESS
Intercept                   5759
breast_development_p_sc     5962
age_sc                      5881
ethnracehispanic            6247
ethnraceother               6433
ethnracewhite_nh            6235
inr_sc                      6196
bmi_wave_0_sc               5863

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

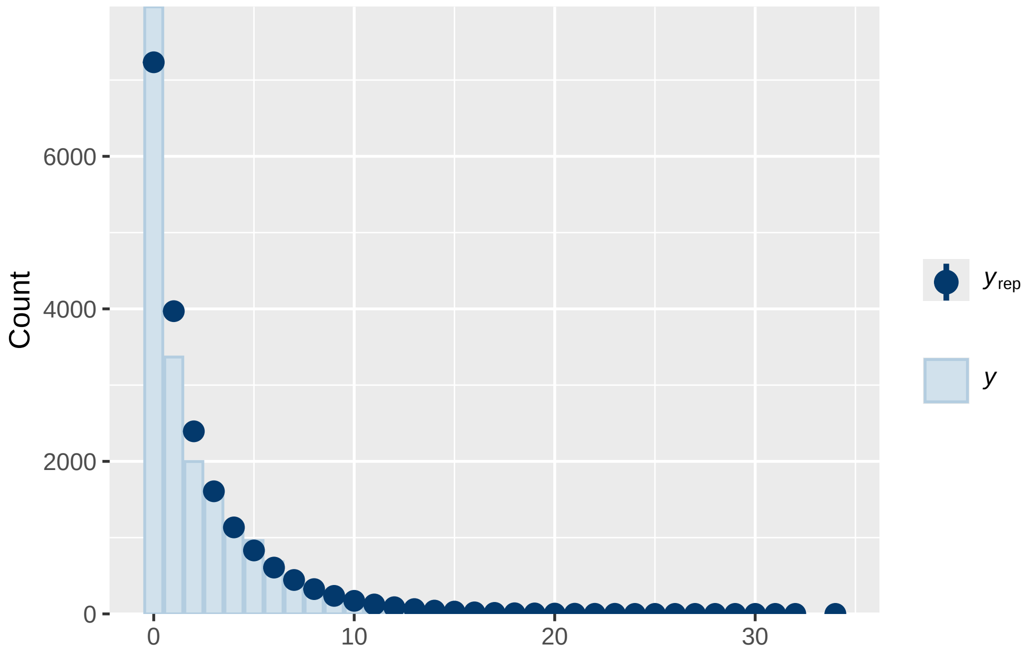
Model 1b BMI diagnostics

```
                prior     class                    coef          group resp
         normal(0, 1)         b
         normal(0, 1)         b                  age_sc
         normal(0, 1)         b           bmi_wave_0_sc
         normal(0, 1)         b breast_development_p_sc
         normal(0, 1)         b        ethnracehispanic
         normal(0, 1)         b           ethnraceother
         normal(0, 1)         b        ethnracewhite_nh
         normal(0, 1)         b                  inr_sc
 student_t(3, 0, 2.7) Intercept
 student_t(3, 0, 2.7)        sd
 student_t(3, 0, 2.7)        sd                              family_id
 student_t(3, 0, 2.7)        sd               Intercept      family_id
 student_t(3, 0, 2.7)        sd                         participant_id
 student_t(3, 0, 2.7)        sd               Intercept participant_id
 student_t(3, 0, 2.7)        sd                                site_id
 student_t(3, 0, 2.7)        sd               Intercept        site_id
 dpar nlpar lb ub tag       source
                              user
```

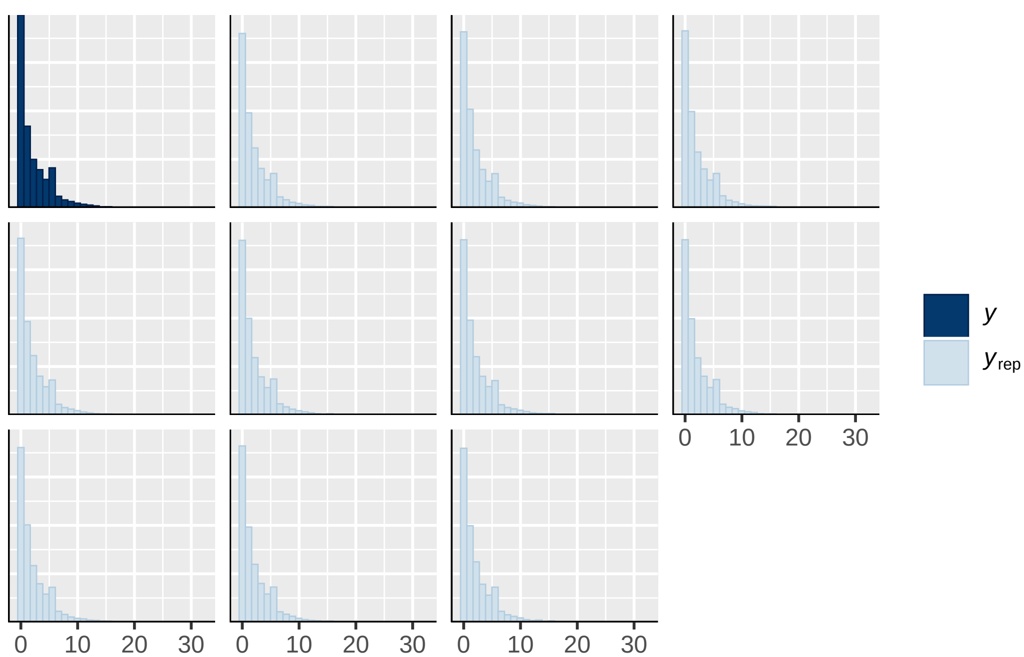
```
                    (vectorized)
                    (vectorized)
                    (vectorized)
                    (vectorized)
                    (vectorized)
                    (vectorized)
                    (vectorized)
                         default
0                        default
0                   (vectorized)
0                   (vectorized)
0                   (vectorized)
0                   (vectorized)
0                   (vectorized)
0                   (vectorized)
```



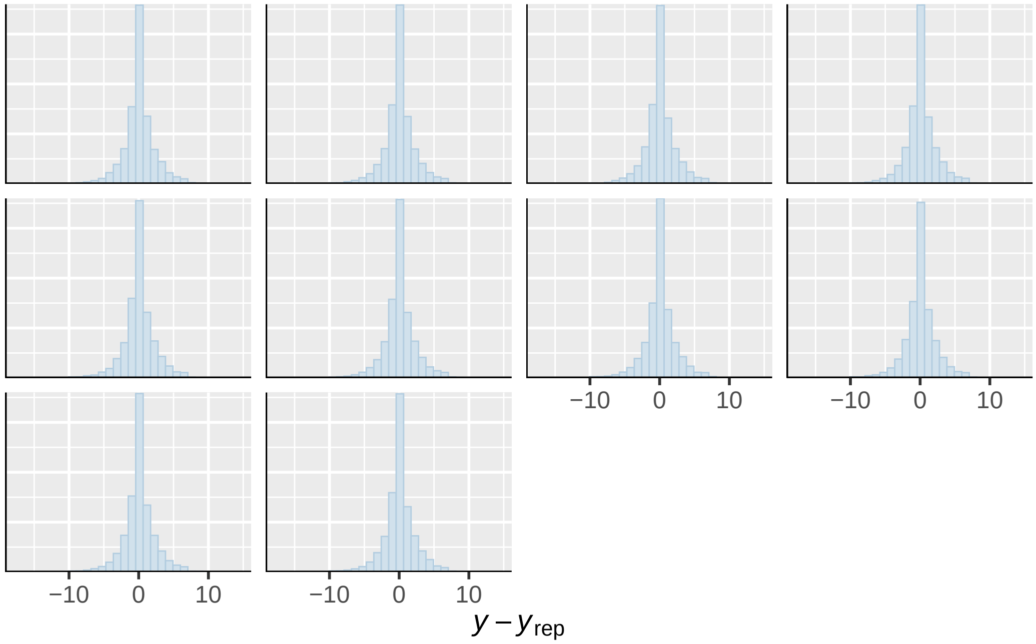

b_ethnracewhite_nh
b_inr_sc
b_bmi_wave_0_sc
sd_family_id__Intercept
sd_participant_id__Intercept
sd_site_id__Intercept
Chain
1
2



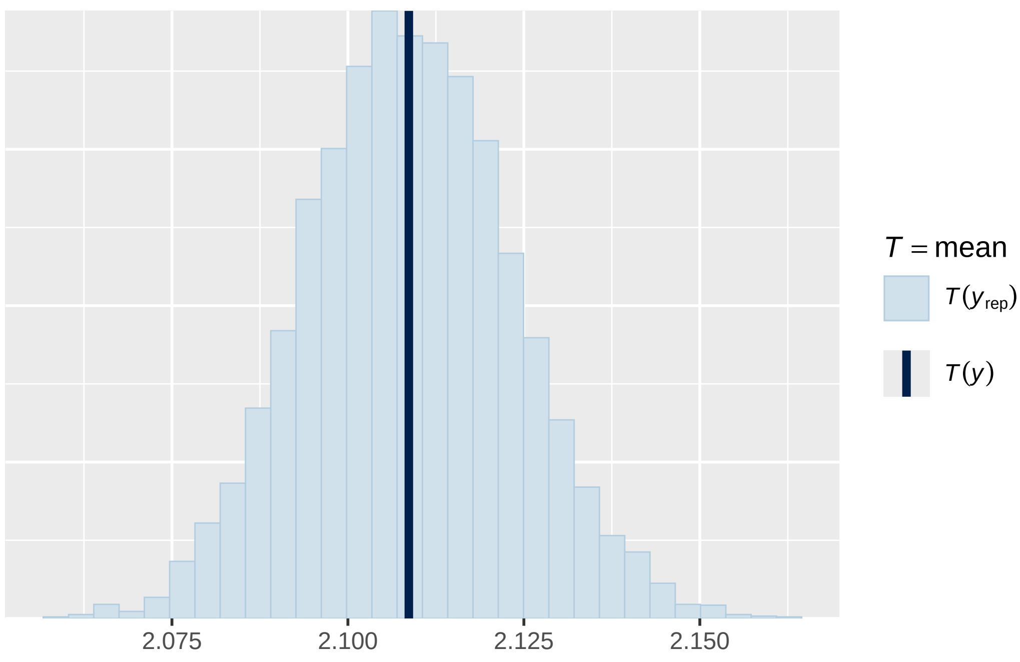

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



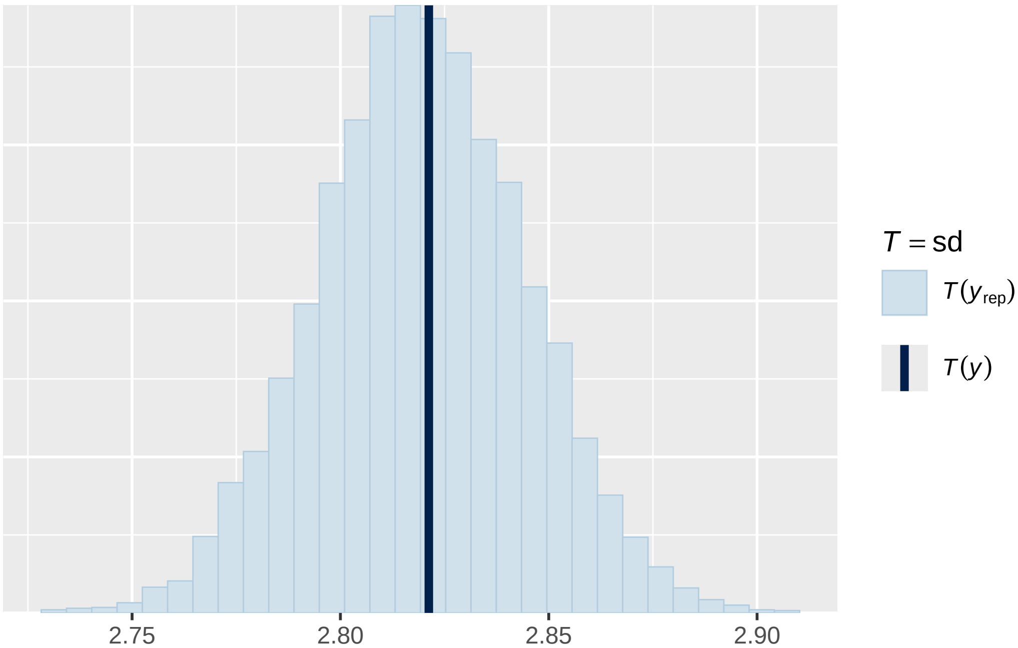

```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



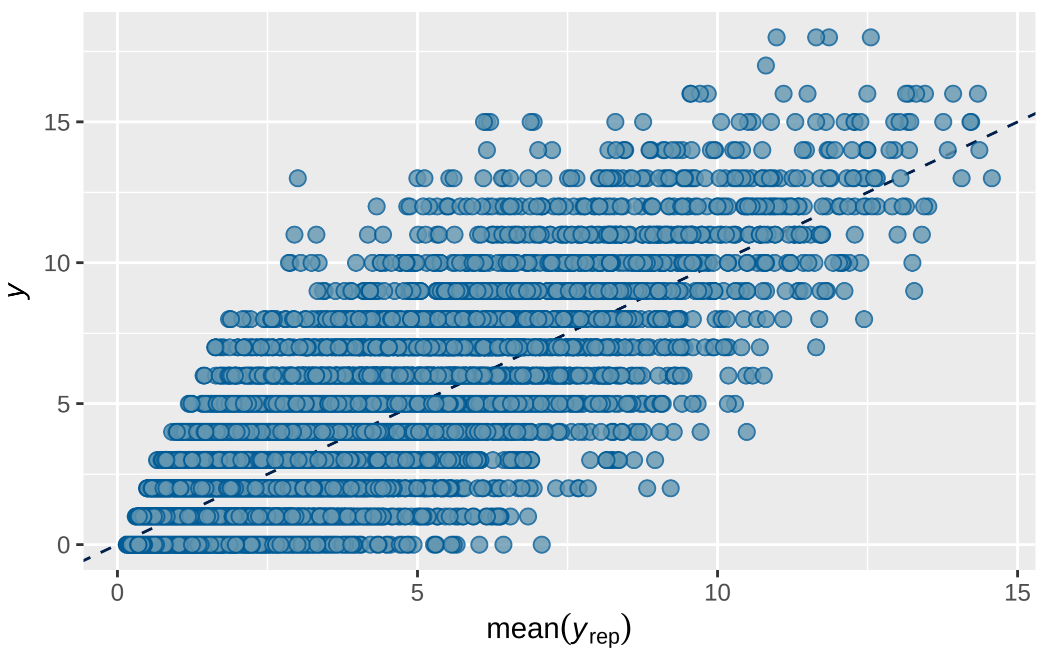

```
Using all posterior draws for ppc type 'stat' by default.
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 4 BMI output

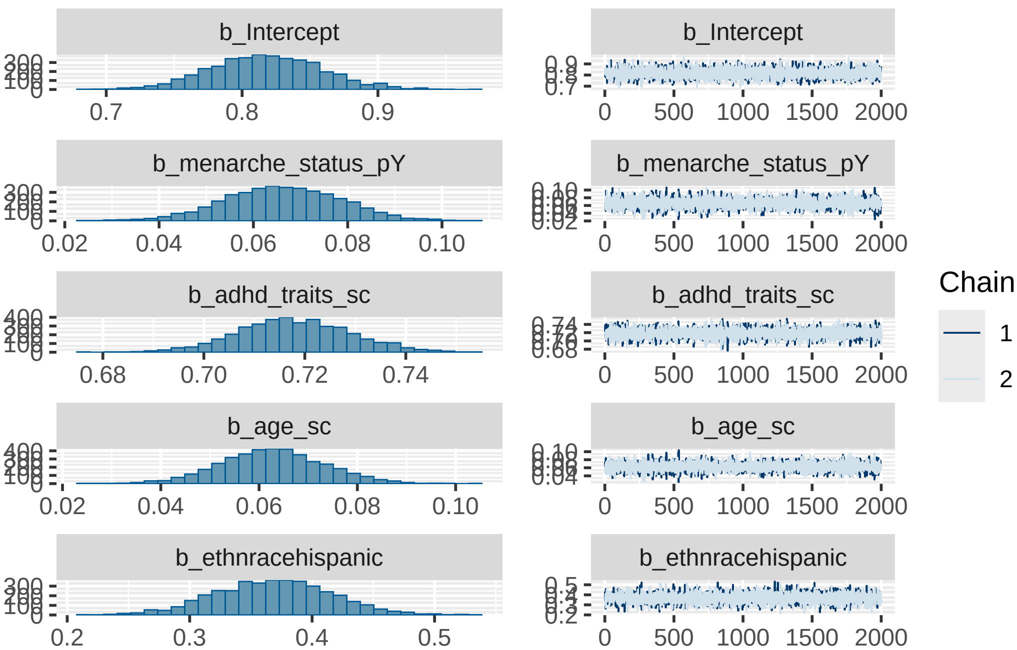
```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.7363026 0.002180899 0.7318509 0.7405877


 Family: poisson
  Links: mu = log
Formula: internalising ~ menarche_status_p * adhd_traits_sc + age_sc + ethnrace + inr_sc + bm
   Data: analytic_df (Number of observations: 19381)
  Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~family_id (Number of levels: 4626)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.662     0.017    0.627    0.695 1.006      754     1681

~participant_id (Number of levels: 5297)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.484     0.019    0.448    0.522 1.008      467     1244

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.088     0.022    0.052    0.135 1.000     3813     3076

Regression Coefficients:
                                  Estimate Est.Error l-95% CI u-95% CI  Rhat
Intercept                            0.818     0.041    0.740    0.902 1.001
menarche_status_pY                   0.066     0.012    0.042    0.090 1.001
adhd_traits_sc                       0.718     0.011    0.696    0.740 1.000
age_sc                               0.063     0.011    0.042    0.084 1.000
ethnracehispanic                     0.369     0.047    0.274    0.460 1.000
ethnraceother                        0.416     0.051    0.317    0.513 1.001
ethnracewhite_nh                     0.473     0.041    0.389    0.553 1.001
inr_sc                              -0.017     0.015   -0.047    0.012 1.000
bmi_wave_0_sc                        0.154     0.030    0.095    0.213 1.000
menarche_status_pY:adhd_traits_sc    0.007     0.013   -0.020    0.032 1.001
                                  Bulk_ESS Tail_ESS
Intercept                             4128     3106
menarche_status_pY                    7952     3100
adhd_traits_sc                        8880     3066
age_sc                                8701     2562
```

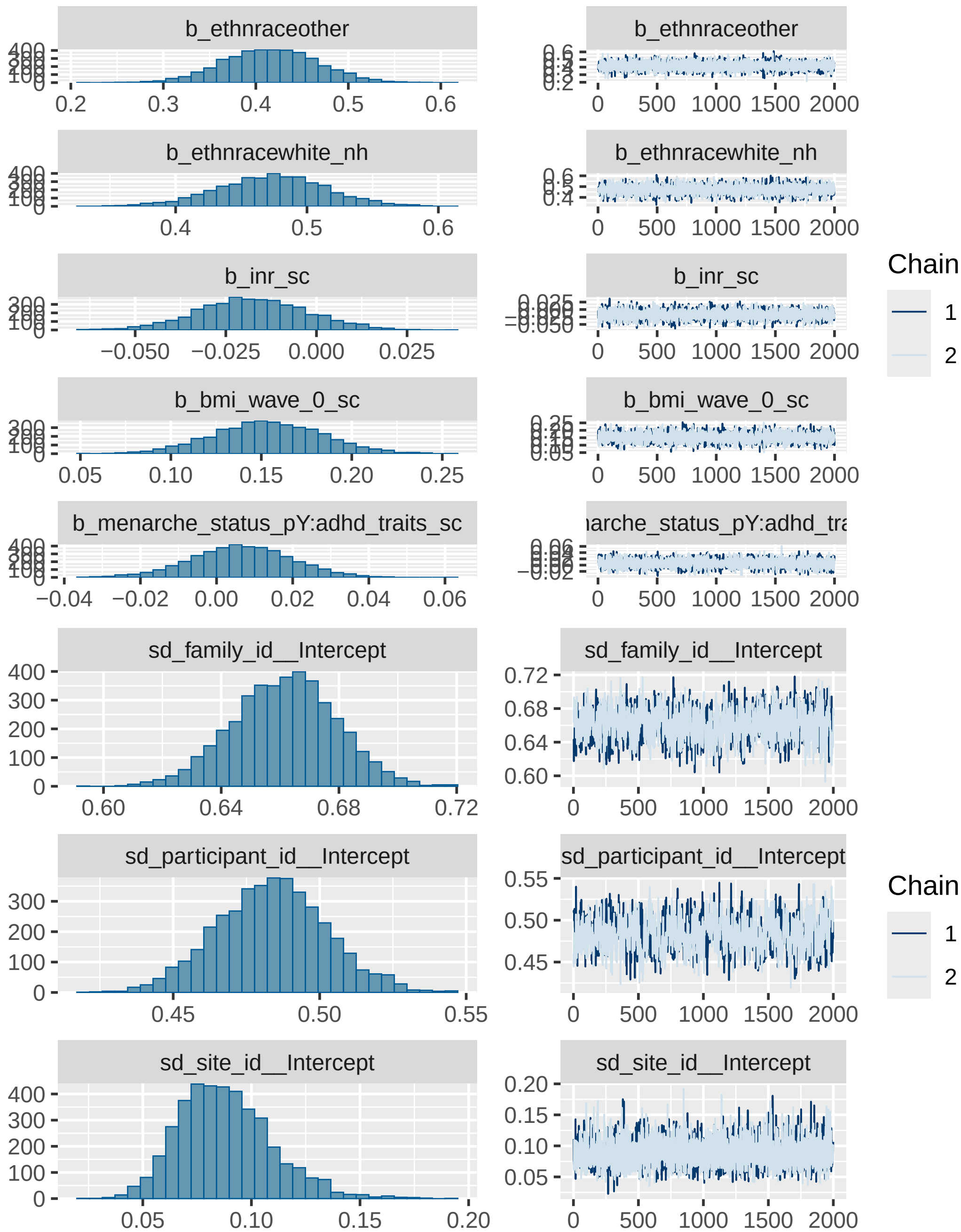
```
ethnracehispanic                  4528     3001
ethnraceother                     4620     3349
ethnracewhite_nh                  4280     2841
inr_sc                            8757     2663
bmi_wave_0_sc                     5774     3311
menarche_status_pY:adhd_traits_sc     7736     3353

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Model 4 BMI diagnostics

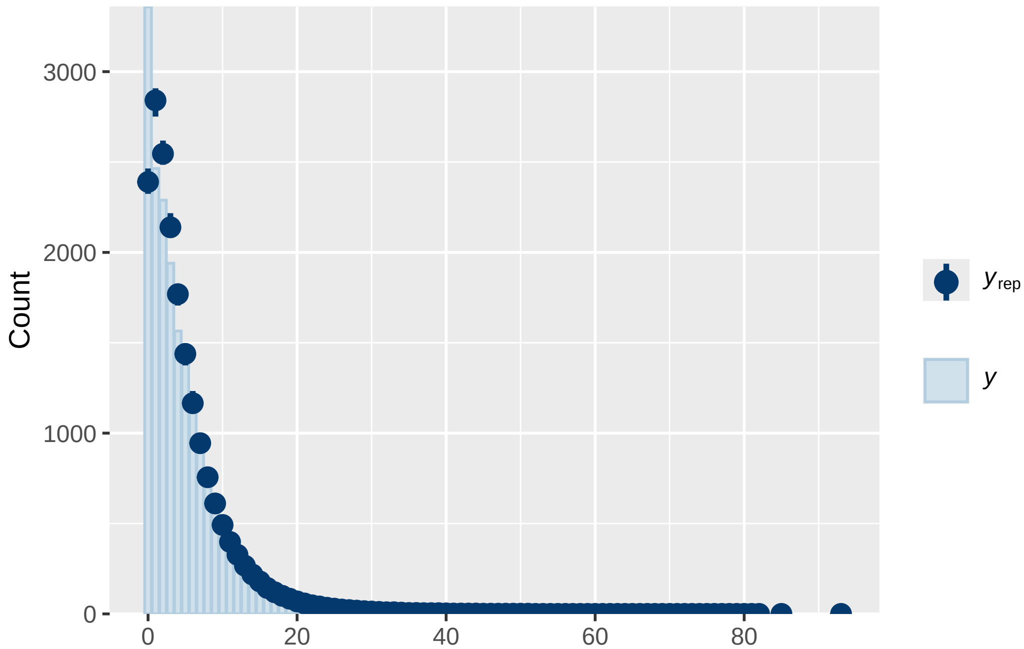
```
                  prior     class                              coef
           normal(0, 1)         b
           normal(0, 1)         b                    adhd_traits_sc
           normal(0, 1)         b                            age_sc
           normal(0, 1)         b                     bmi_wave_0_sc
           normal(0, 1)         b                  ethnracehispanic
           normal(0, 1)         b                     ethnraceother
           normal(0, 1)         b                  ethnracewhite_nh
           normal(0, 1)         b                            inr_sc
           normal(0, 1)         b                menarche_status_pY
           normal(0, 1)         b menarche_status_pY:adhd_traits_sc
 student_t(3, 1.1, 2.5) Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd                         Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd                         Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd                         Intercept
          group resp dpar nlpar lb ub tag       source
                                                  user
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
```

		(vectorized)
		default
	0	default
family_id	0	(vectorized)
family_id	0	(vectorized)
participant_id	0	(vectorized)
participant_id	0	(vectorized)
site_id	0	(vectorized)
site_id	0	(vectorized)

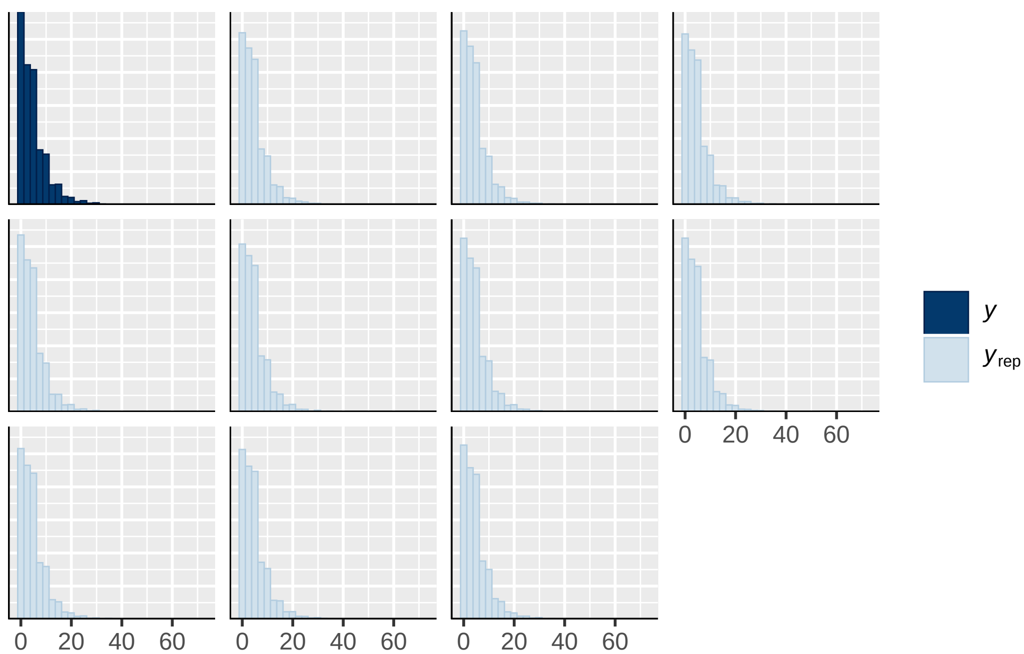




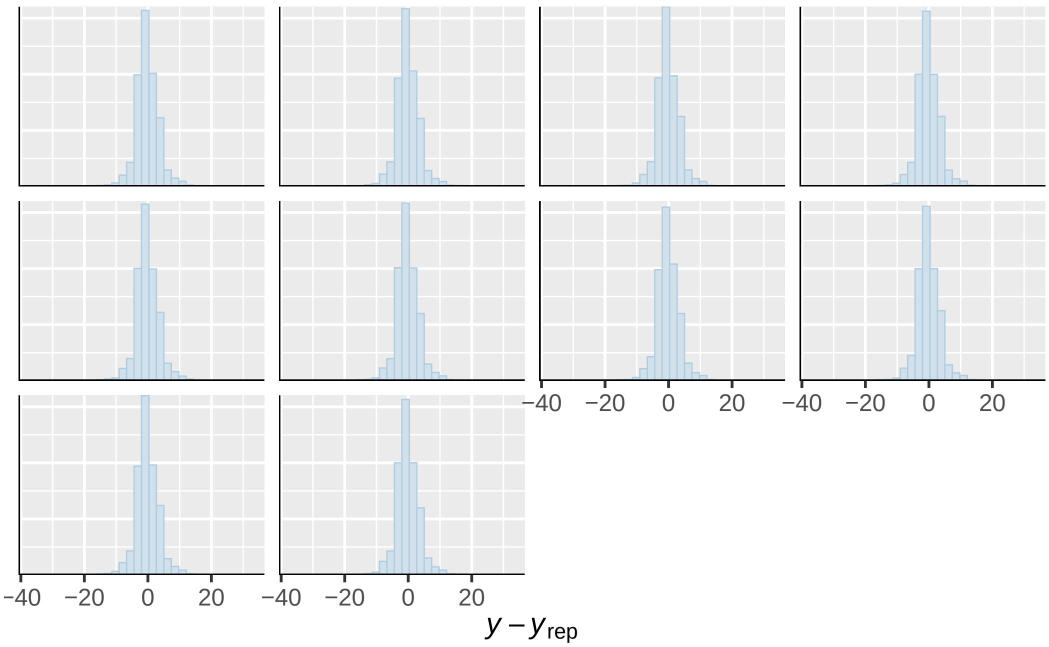
b_ethnraceother
b_ethnracewhite_nh
b_inr_sc
b_bmi_wave_0_sc
b_menarche_status_pY:adhd_traits_sc
sd_family_id__Intercept
sd_participant_id__Intercept
sd_site_id__Intercept
Chain
1
2



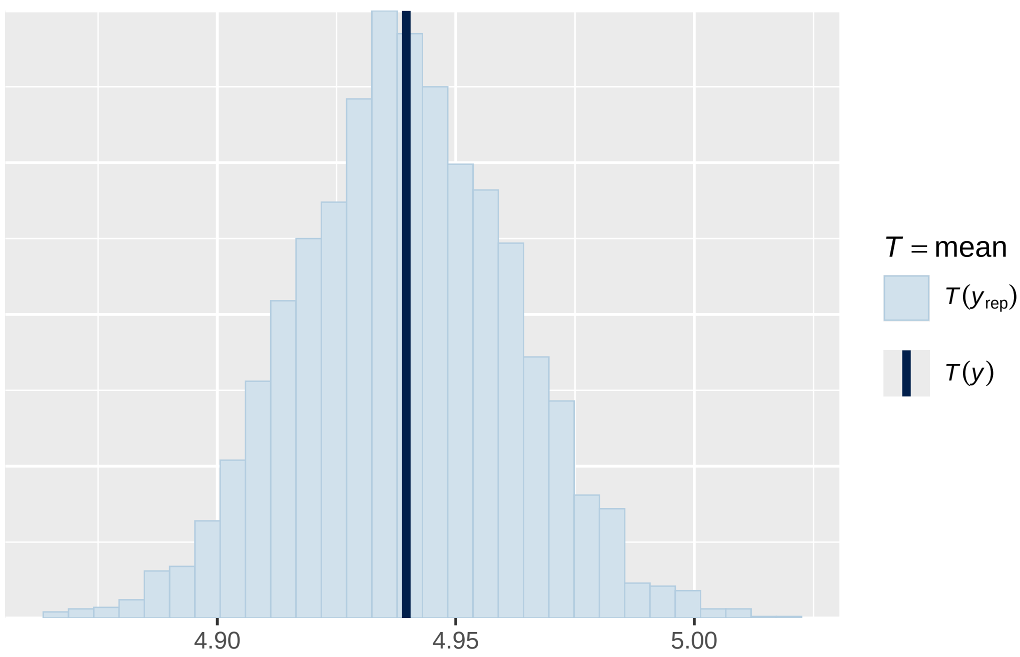

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



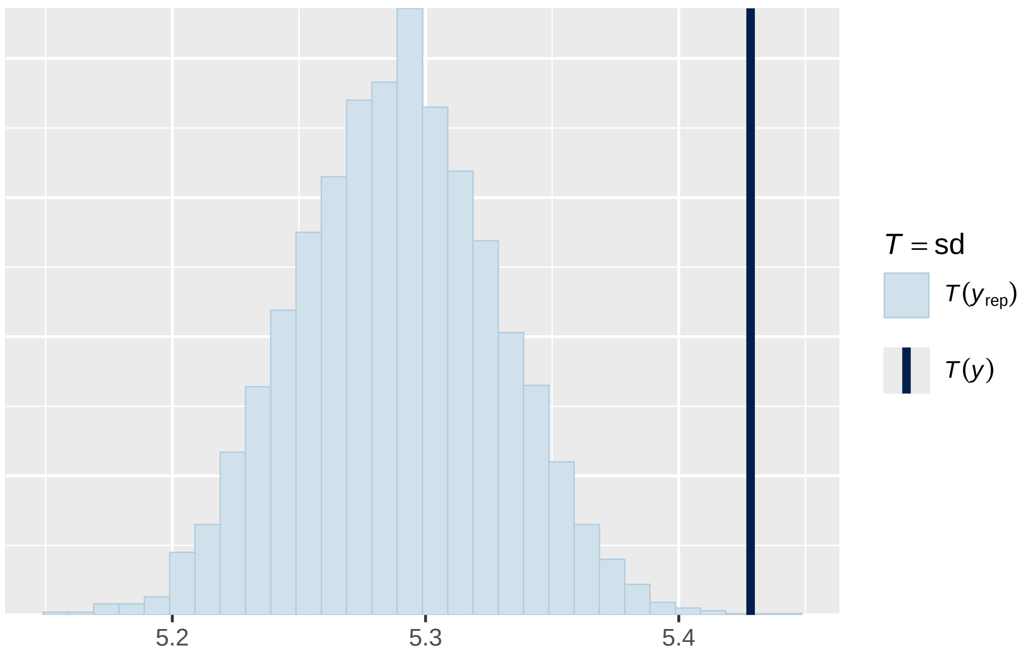

```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



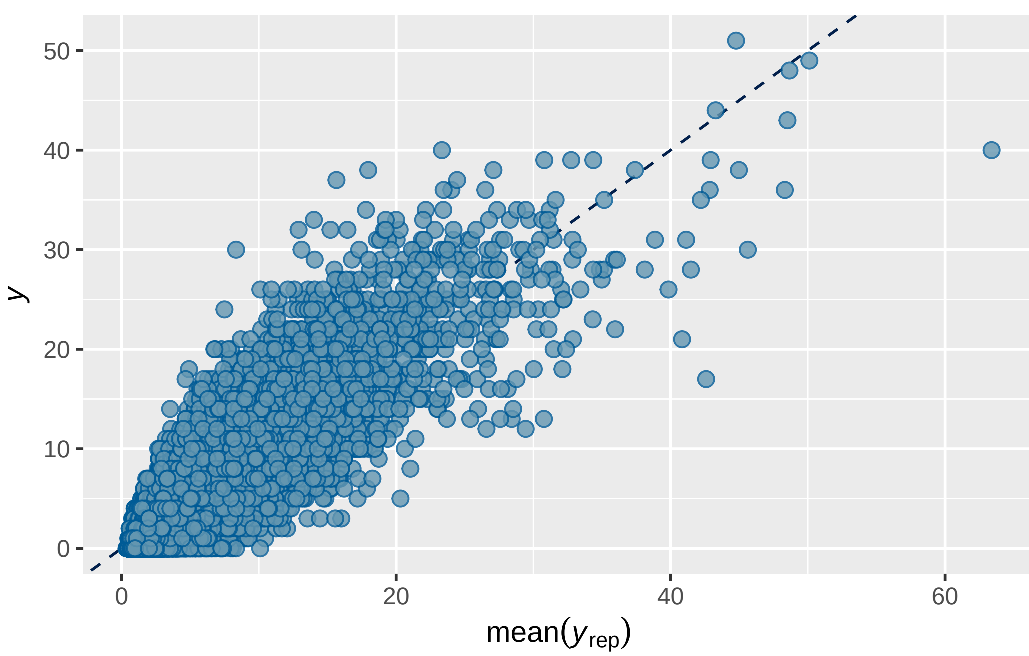

```
Using all posterior draws for ppc type 'stat' by default.
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 5 BMI output

```
   Estimate   Est.Error      Q2.5     Q97.5
R2  0.78834 0.002047103 0.7842391 0.7922249


 Family: poisson
  Links: mu = log
Formula: externalising ~ menarche_status_p * adhd_traits_sc + age_sc + ethnrace + inr_sc + bm
   Data: analytic_df (Number of observations: 19381)
  Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~family_id (Number of levels: 4626)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.745     0.027    0.690    0.797 1.004      541     1366

~participant_id (Number of levels: 5297)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.749     0.025    0.699    0.797 1.002      541     1497

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.120     0.028    0.073    0.184 1.002     3907     2856

Regression Coefficients:
                                  Estimate Est.Error l-95% CI u-95% CI  Rhat
Intercept                            0.405     0.054    0.295    0.512 1.000
menarche_status_pY                   0.065     0.016    0.034    0.097 1.001
adhd_traits_sc                       0.894     0.014    0.868    0.921 1.000
age_sc                              -0.109     0.013   -0.134   -0.083 1.000
ethnracehispanic                     0.044     0.062   -0.078    0.165 1.000
ethnraceother                        0.058     0.067   -0.076    0.193 1.001
ethnracewhite_nh                     0.119     0.054    0.016    0.224 1.000
inr_sc                              -0.124     0.019   -0.162   -0.088 1.000
bmi_wave_0_sc                        0.234     0.040    0.158    0.312 1.000
menarche_status_pY:adhd_traits_sc    0.033     0.016    0.002    0.064 1.000
                                  Bulk_ESS Tail_ESS
Intercept                             3991     2702
menarche_status_pY                    7524     2612
adhd_traits_sc                        8392     3308
age_sc                                7264     3247
```

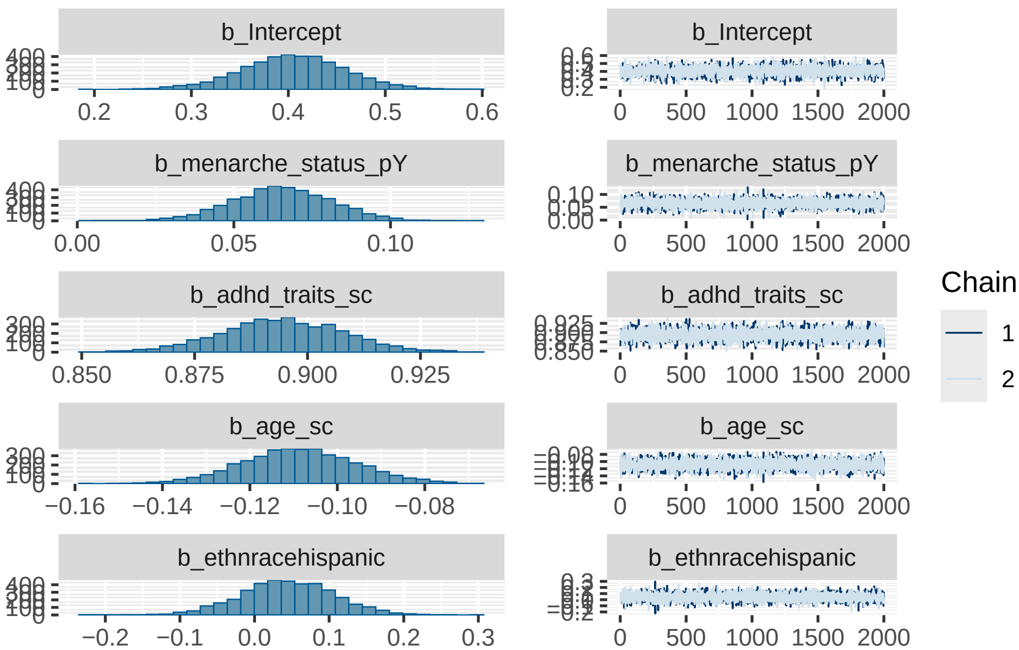
```
ethnracehispanic                    4483     2995
ethnraceother                       4514     3300
ethnracewhite_nh                    4286     2951
inr_sc                              9765     3232
bmi_wave_0_sc                       5004     3114
menarche_status_pY:adhd_traits_sc   7925     3236
```

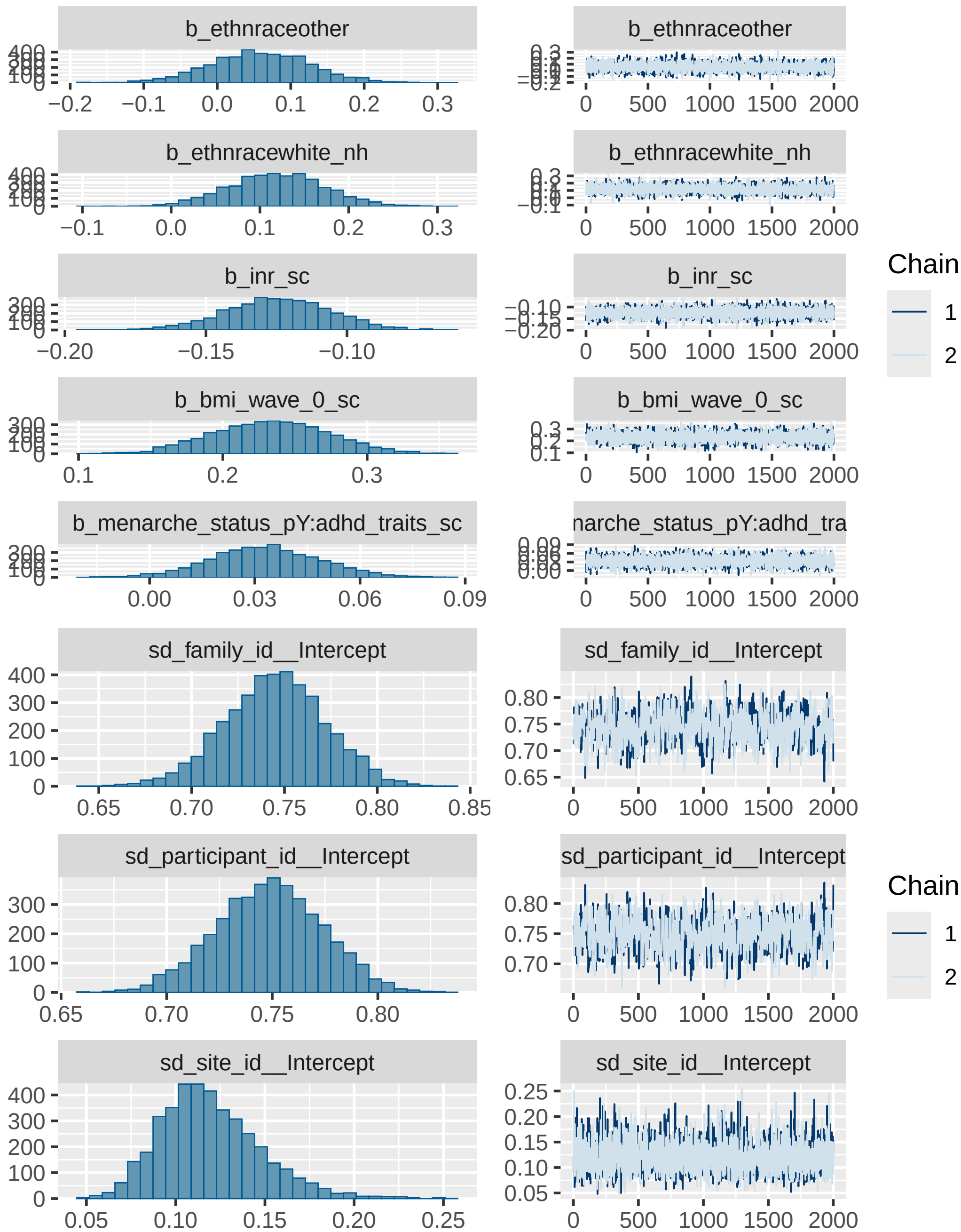
Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).

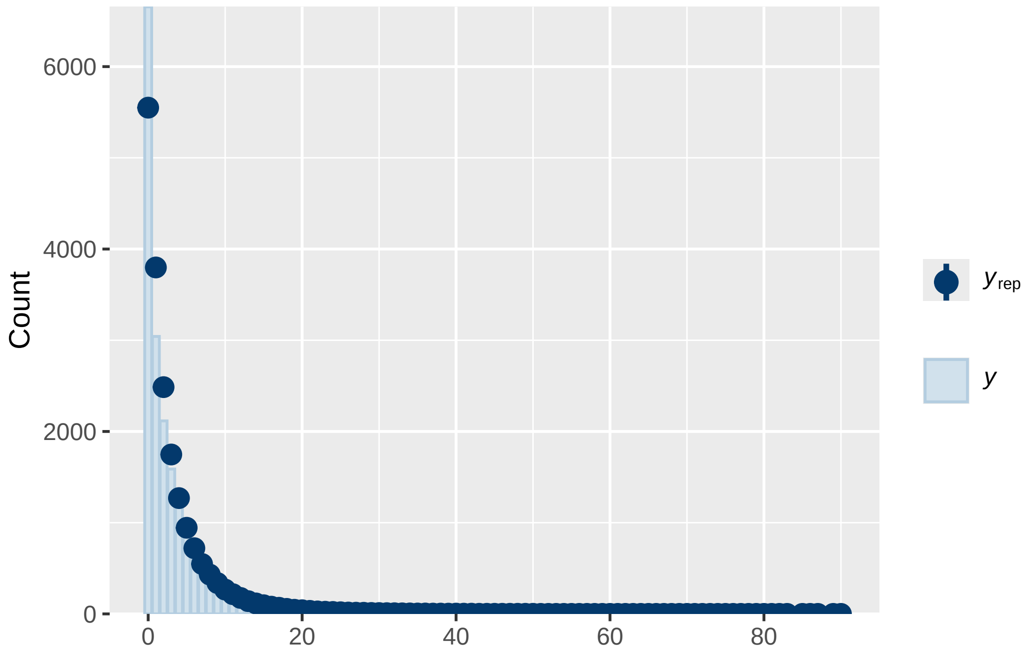
Model 5 BMI diagnostics

```
                prior     class                               coef
         normal(0, 1)         b
         normal(0, 1)         b                     adhd_traits_sc
         normal(0, 1)         b                             age_sc
         normal(0, 1)         b                      bmi_wave_0_sc
         normal(0, 1)         b                   ethnracehispanic
         normal(0, 1)         b                      ethnraceother
         normal(0, 1)         b                   ethnracewhite_nh
         normal(0, 1)         b                             inr_sc
         normal(0, 1)         b                 menarche_status_pY
         normal(0, 1)         b menarche_status_pY:adhd_traits_sc
 student_t(3, 0, 2.9) Intercept
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd                          Intercept
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd                          Intercept
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd                          Intercept
          group resp dpar nlpar lb ub tag       source
                                                  user
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
```

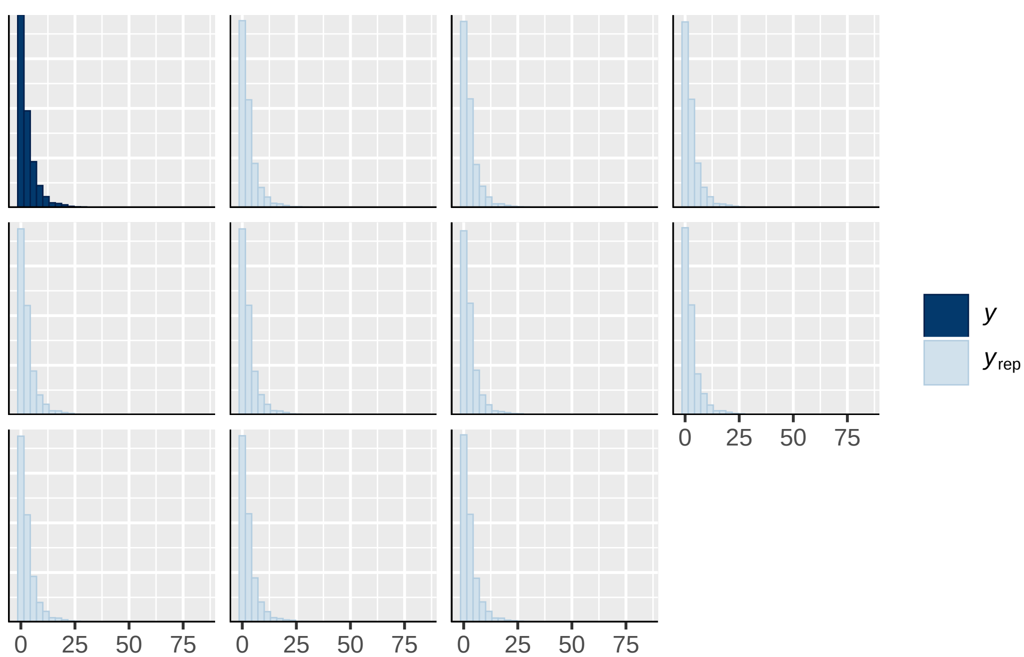
```
                                       (vectorized)
                                            default
                            0               default
    family_id               0          (vectorized)
    family_id               0          (vectorized)
participant_id              0          (vectorized)
participant_id              0          (vectorized)
      site_id               0          (vectorized)
      site_id               0          (vectorized)
```



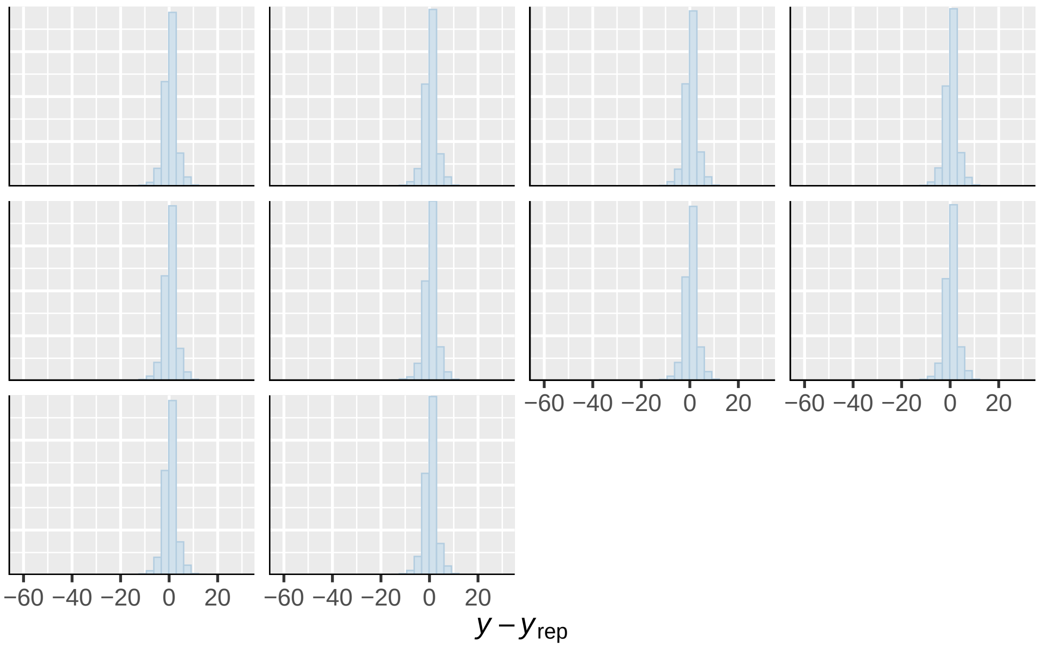

b_ethnraceother
b_ethnracewhite_nh
b_inr_sc
b_bmi_wave_0_sc
b_menarche_status_pY:adhd_traits_sc
narche_status_pY:adhd_tra
sd_family_id__Intercept
sd_participant_id__Intercept
sd_site_id__Intercept
Chain
1
2
0 500 1000 1500 2000



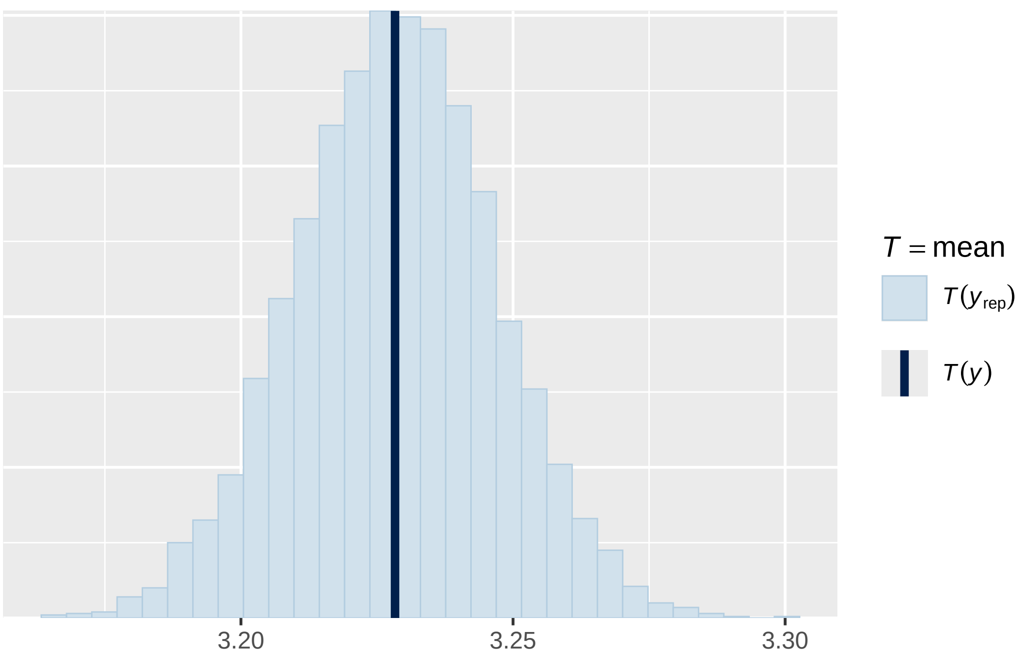

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



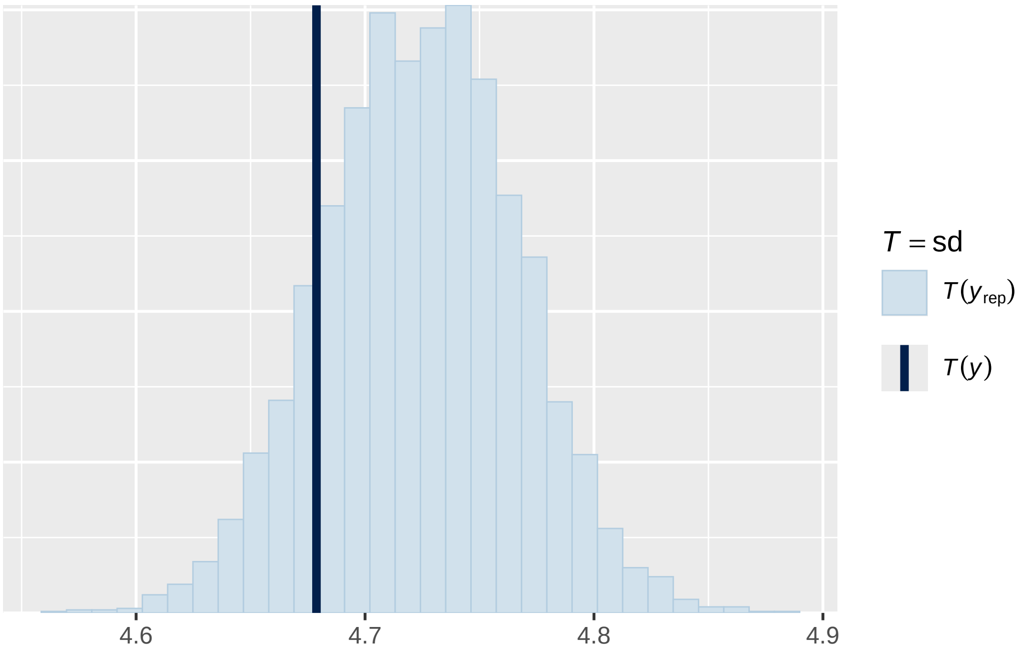

```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



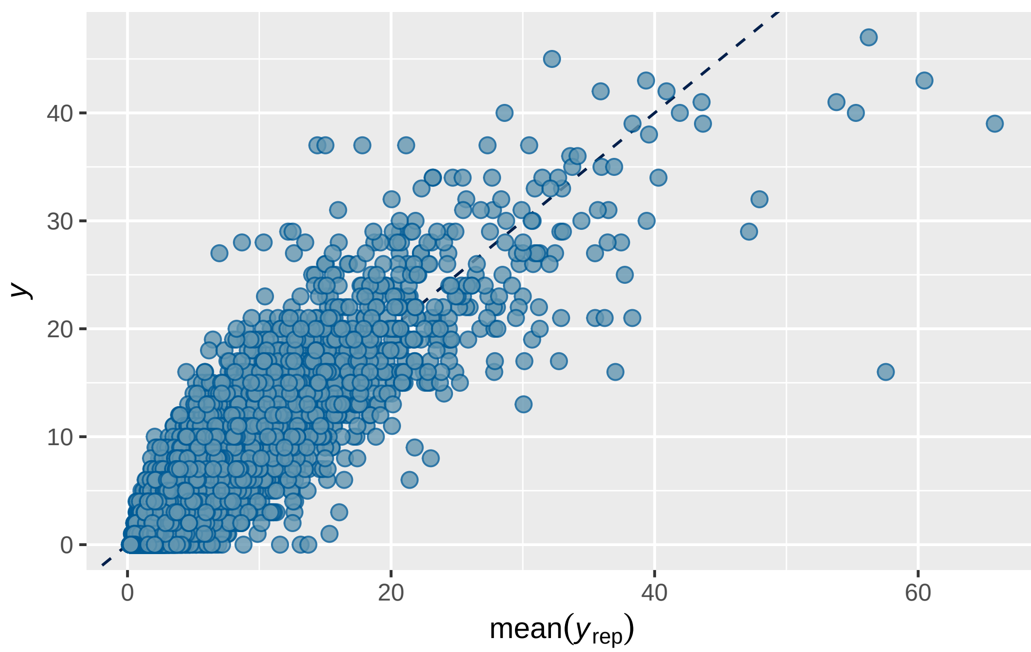

```
Using all posterior draws for ppc type 'stat' by default.
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 1b stimulant output

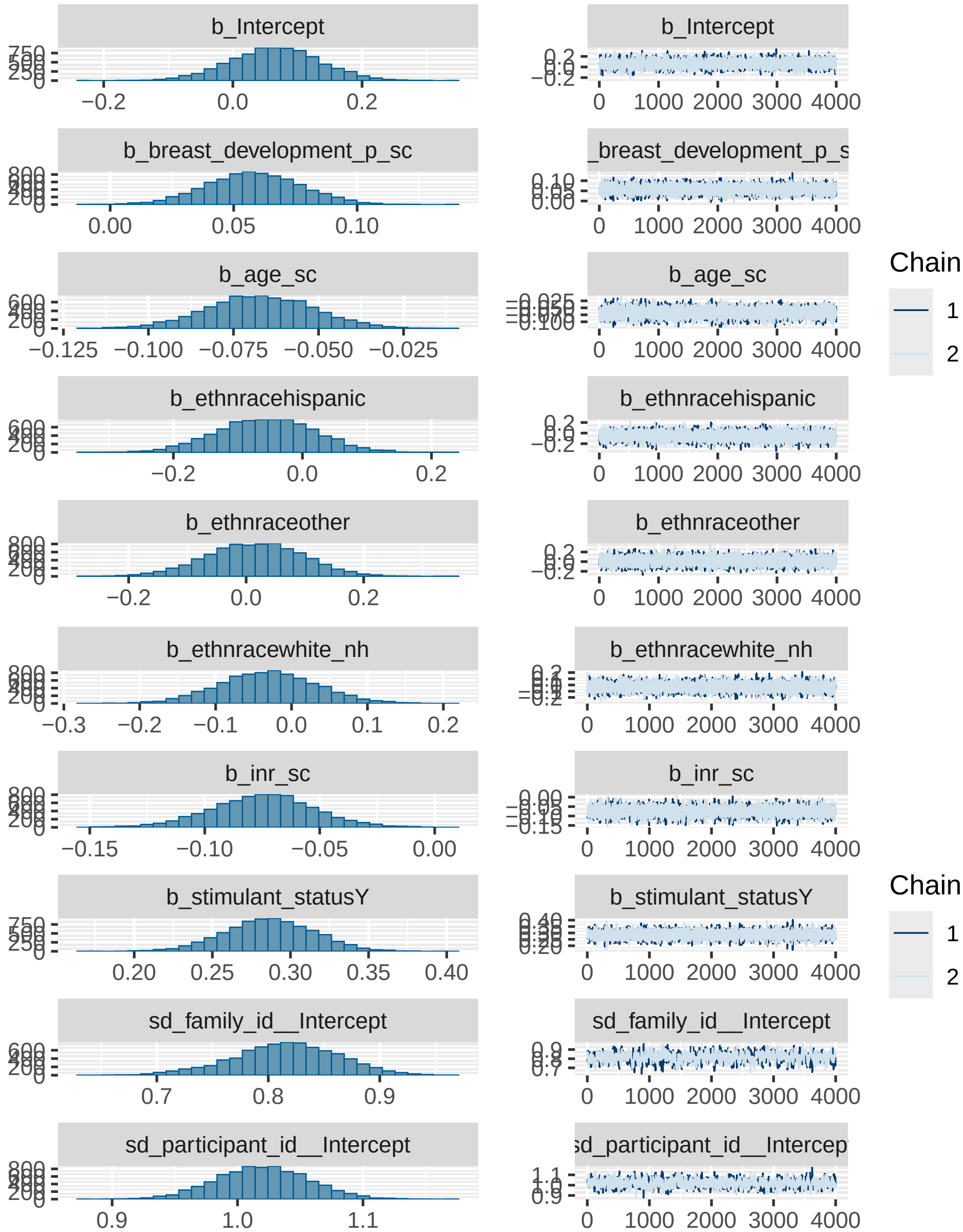
```
    Estimate  Est.Error      Q2.5     Q97.5
R2 0.7027165 0.00328854 0.6961135 0.7090334


 Family: poisson
  Links: mu = log
Formula: adhd_traits ~ breast_development_p_sc + age_sc + ethnrace + inr_sc + stimulant_statu
   Data: analytic_df (Number of observations: 19364)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 4619)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.816     0.047    0.718    0.904 1.003      586     1036

~participant_id (Number of levels: 5289)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    1.020     0.036    0.950    1.092 1.002      755     1459

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.172     0.038    0.110    0.257 1.000     6338     5466

Regression Coefficients:
                        Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS
Intercept                  0.066     0.069   -0.071    0.199 1.000     5903
breast_development_p_sc    0.058     0.019    0.022    0.095 1.000    14639
age_sc                    -0.067     0.016   -0.098   -0.036 1.000    13765
ethnracehispanic          -0.056     0.076   -0.204    0.090 1.001     6701
ethnraceother              0.015     0.081   -0.148    0.173 1.000     8090
ethnracewhite_nh          -0.031     0.064   -0.155    0.096 1.001     7314
inr_sc                    -0.075     0.022   -0.120   -0.031 1.000    18245
stimulant_statusY          0.287     0.028    0.232    0.342 1.000    18222
                        Tail_ESS
Intercept                   5924
breast_development_p_sc     6463
age_sc                      5792
ethnracehispanic            5827
ethnraceother               6369
ethnracewhite_nh            5977
```

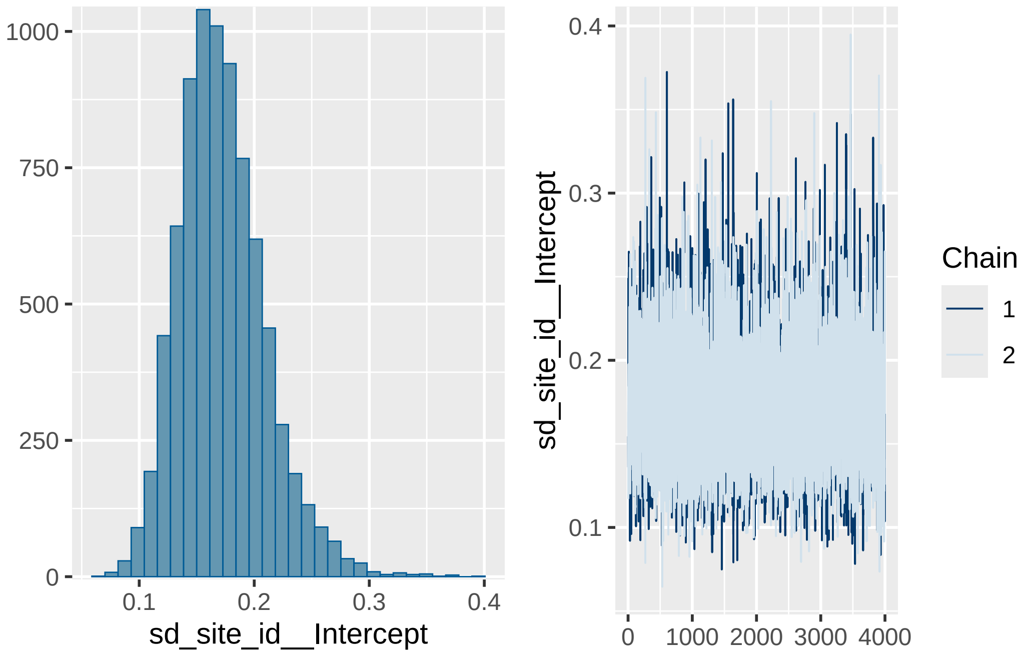
```
inr_sc                       4999
stimulant_statusY            5488

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

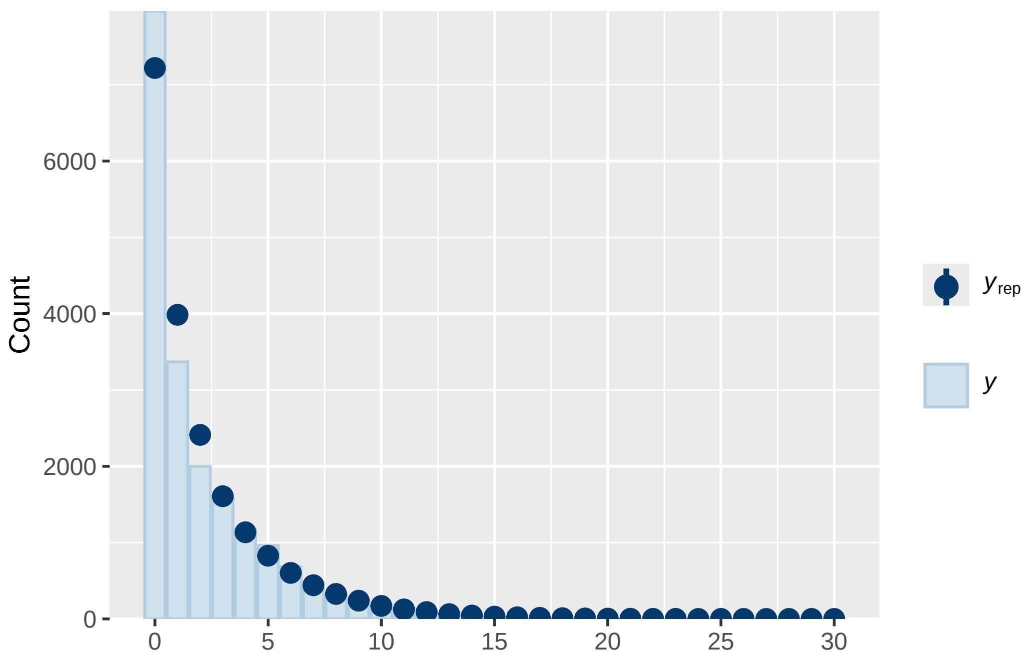
Model 1b stimulant diagnostics

```
                prior     class                     coef          group resp
         normal(0, 1)         b                                             
         normal(0, 1)         b                   age_sc                    
         normal(0, 1)         b breast_development_p_sc                    
         normal(0, 1)         b        ethnracehispanic                    
         normal(0, 1)         b           ethnraceother                    
         normal(0, 1)         b        ethnracewhite_nh                    
         normal(0, 1)         b                   inr_sc                    
         normal(0, 1)         b       stimulant_statusY                    
 student_t(3, 0, 2.7) Intercept                                             
 student_t(3, 0, 2.7)        sd                                             
 student_t(3, 0, 2.7)        sd                              family_id     
 student_t(3, 0, 2.7)        sd                Intercept      family_id     
 student_t(3, 0, 2.7)        sd                         participant_id     
 student_t(3, 0, 2.7)        sd                Intercept participant_id     
 student_t(3, 0, 2.7)        sd                                site_id     
 student_t(3, 0, 2.7)        sd                Intercept        site_id     
 dpar nlpar lb ub tag       source
                              user
                      (vectorized)
                      (vectorized)
                      (vectorized)
                      (vectorized)
                      (vectorized)
                      (vectorized)
                      (vectorized)
                           default
             0             default
             0        (vectorized)
             0        (vectorized)
             0        (vectorized)
             0        (vectorized)
             0        (vectorized)
```

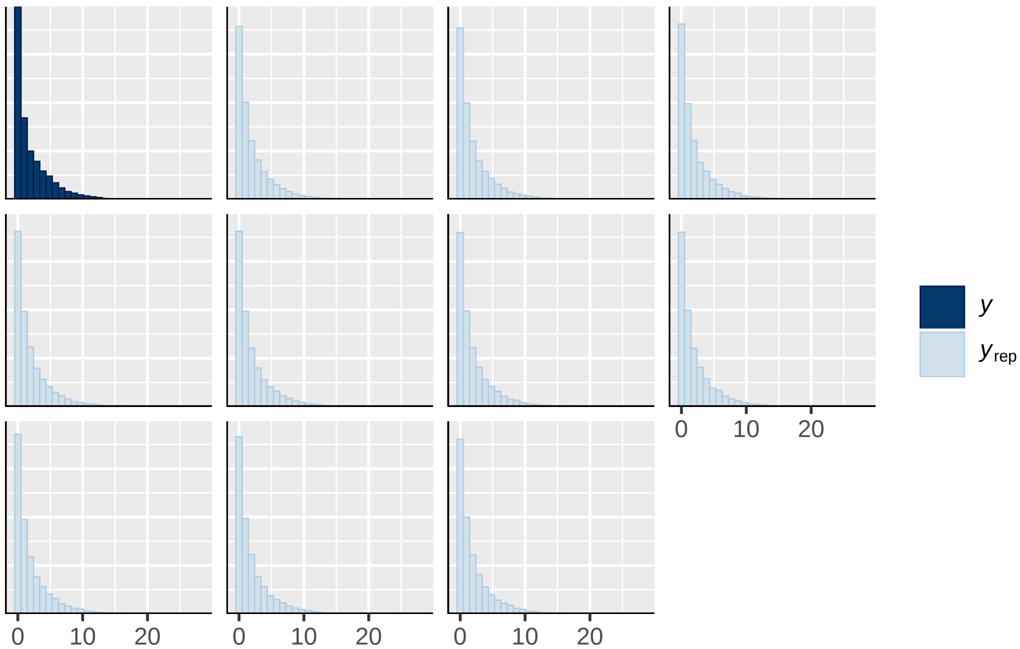
0 (vectorized)



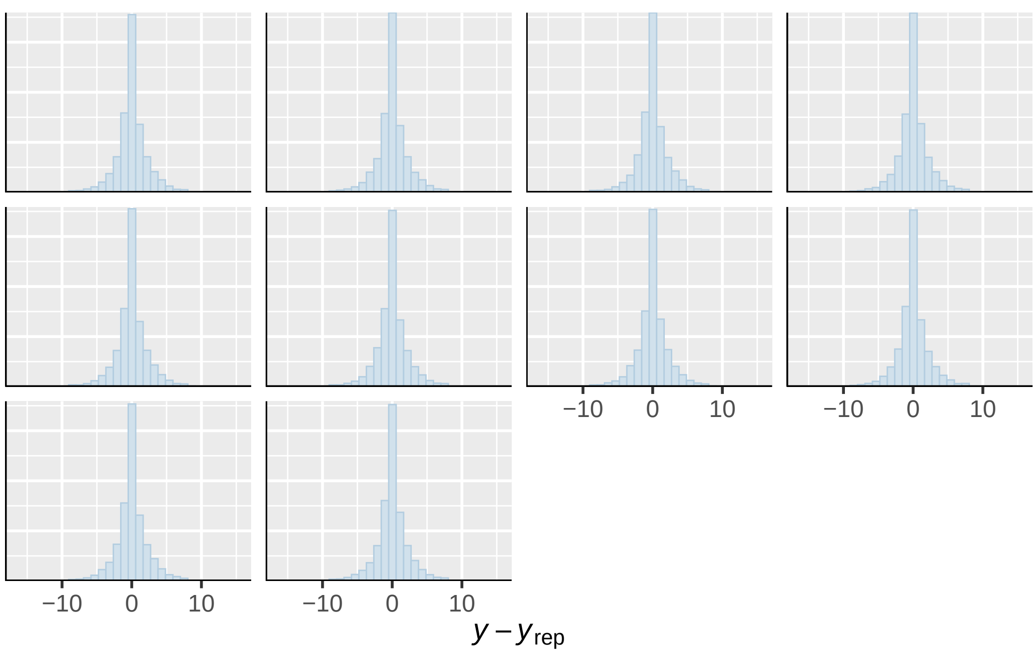





```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

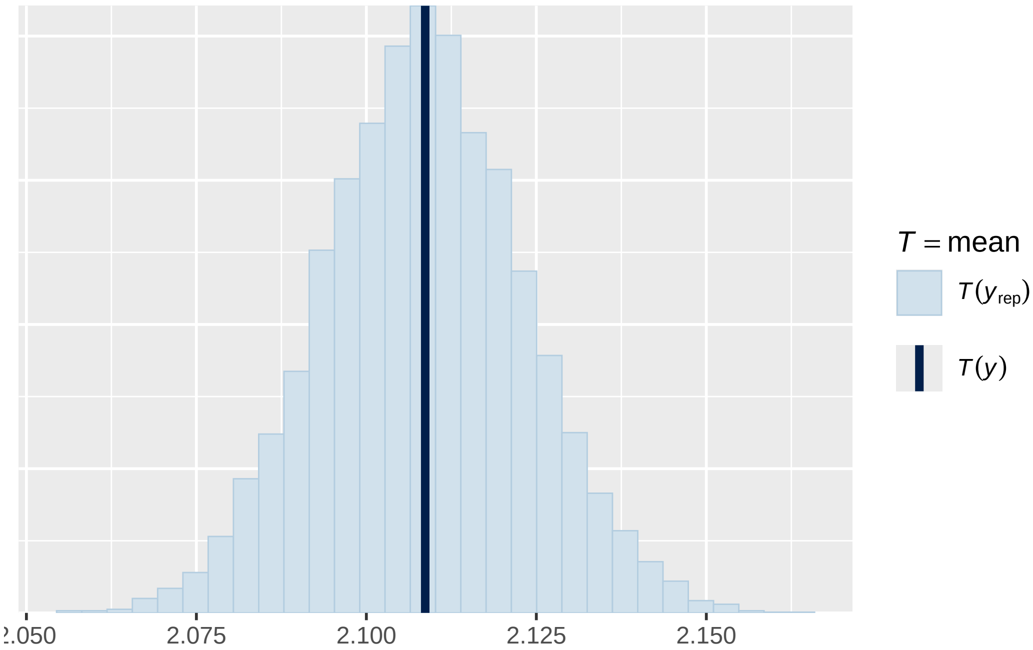



```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

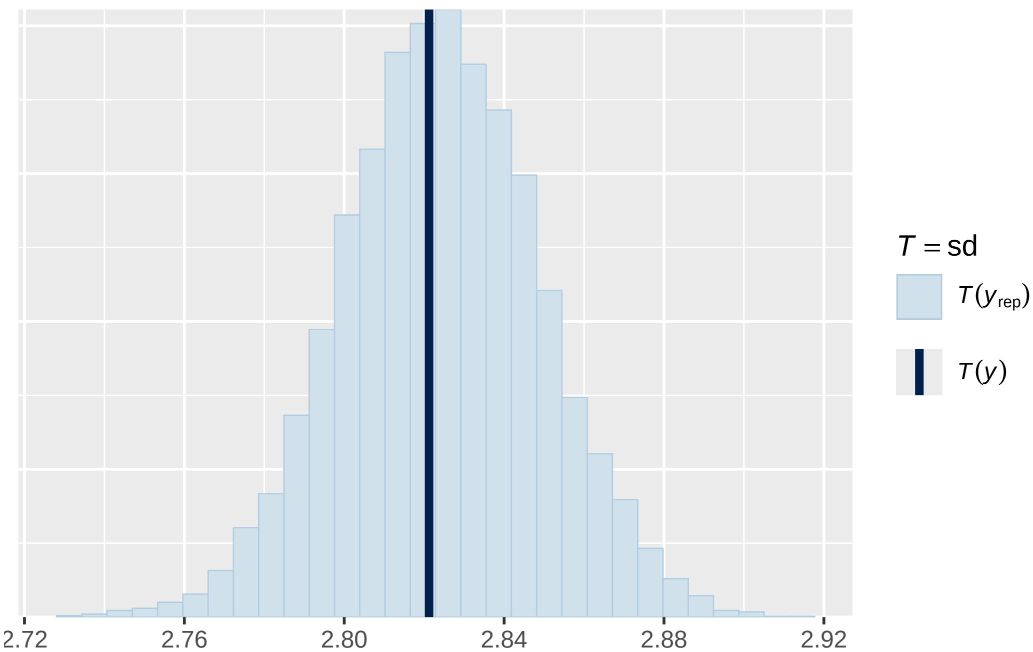



```
Using all posterior draws for ppc type 'stat' by default.
```

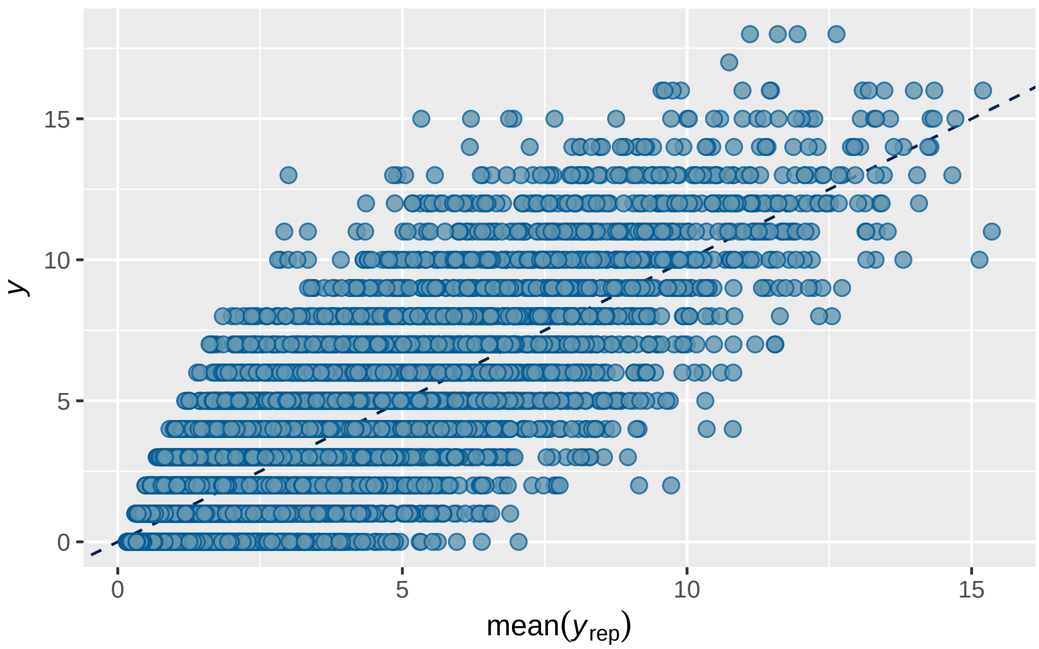
```
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 4 stimulant output

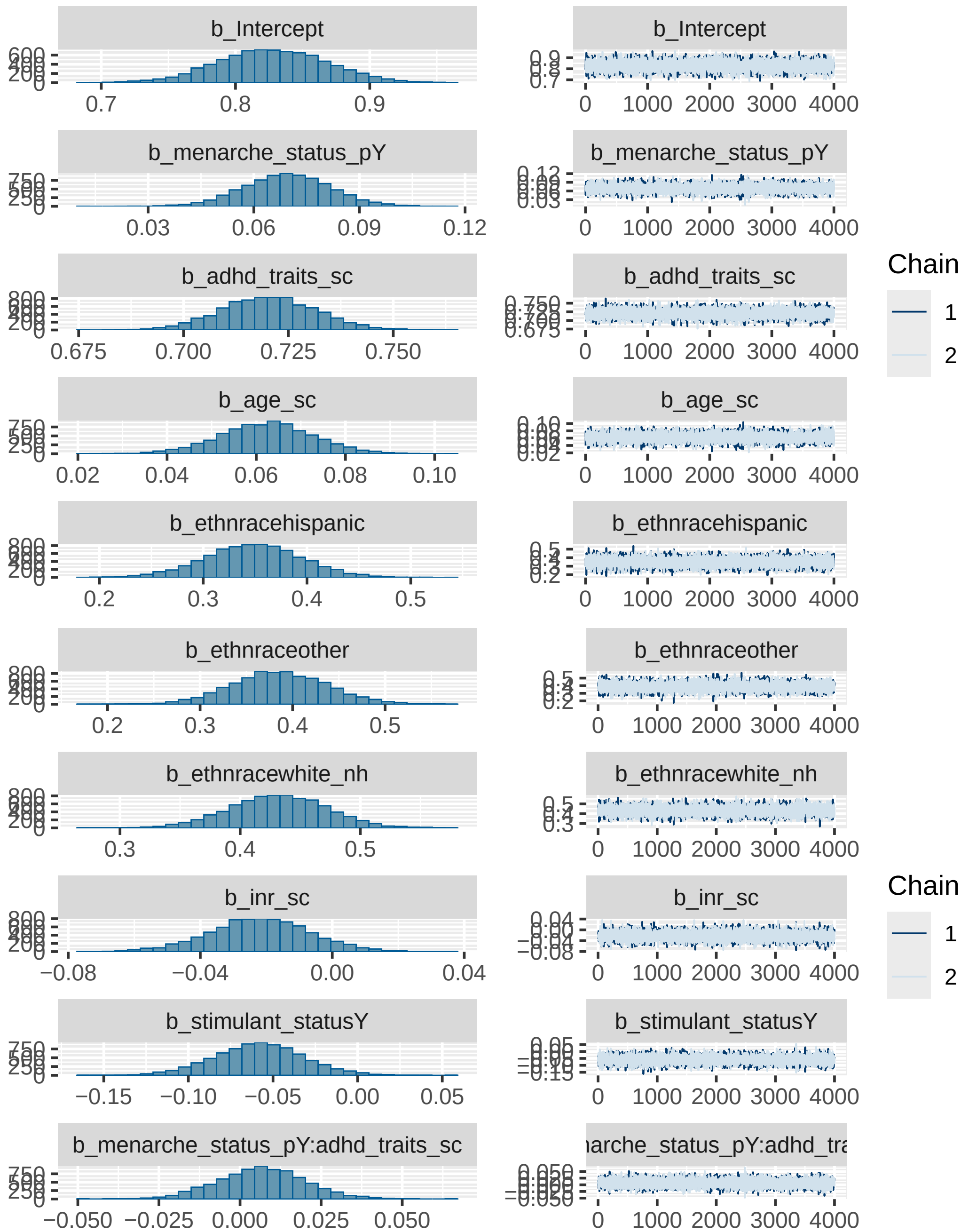
```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.7362039 0.002200039 0.7317853 0.7404623


 Family: poisson
  Links: mu = log
Formula: internalising ~ menarche_status_p * adhd_traits_sc + age_sc + ethnrace + inr_sc + st
   Data: analytic_df (Number of observations: 19381)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 4626)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.660     0.018    0.624    0.694 1.001     1819     3114

~participant_id (Number of levels: 5297)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.487     0.019    0.451    0.524 1.001     1165     2344
```

```
~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.089     0.022    0.053    0.137 1.000     6525     6311

Regression Coefficients:
                                  Estimate Est.Error l-95% CI u-95% CI  Rhat
Intercept                            0.828     0.041    0.749    0.906 1.000
menarche_status_pY                   0.069     0.012    0.046    0.094 1.000
adhd_traits_sc                       0.720     0.011    0.698    0.743 1.000
age_sc                               0.062     0.010    0.041    0.083 1.000
ethnracehispanic                     0.348     0.047    0.256    0.439 1.001
ethnraceother                        0.386     0.051    0.287    0.487 1.000
ethnracewhite_nh                     0.432     0.040    0.352    0.509 1.000
inr_sc                              -0.022     0.015   -0.052    0.008 1.000
stimulant_statusY                   -0.058     0.026   -0.112   -0.006 1.001
menarche_status_pY:adhd_traits_sc    0.007     0.013   -0.019    0.033 1.000
                                  Bulk_ESS Tail_ESS
Intercept                             9148     6292
menarche_status_pY                   16104     6097
adhd_traits_sc                       16702     6559
age_sc                               15947     5806
ethnracehispanic                      9922     6155
ethnraceother                         9329     6174
ethnracewhite_nh                      9419     5808
inr_sc                               15357     5580
stimulant_statusY                    15745     5596
menarche_status_pY:adhd_traits_sc    14922     6436

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

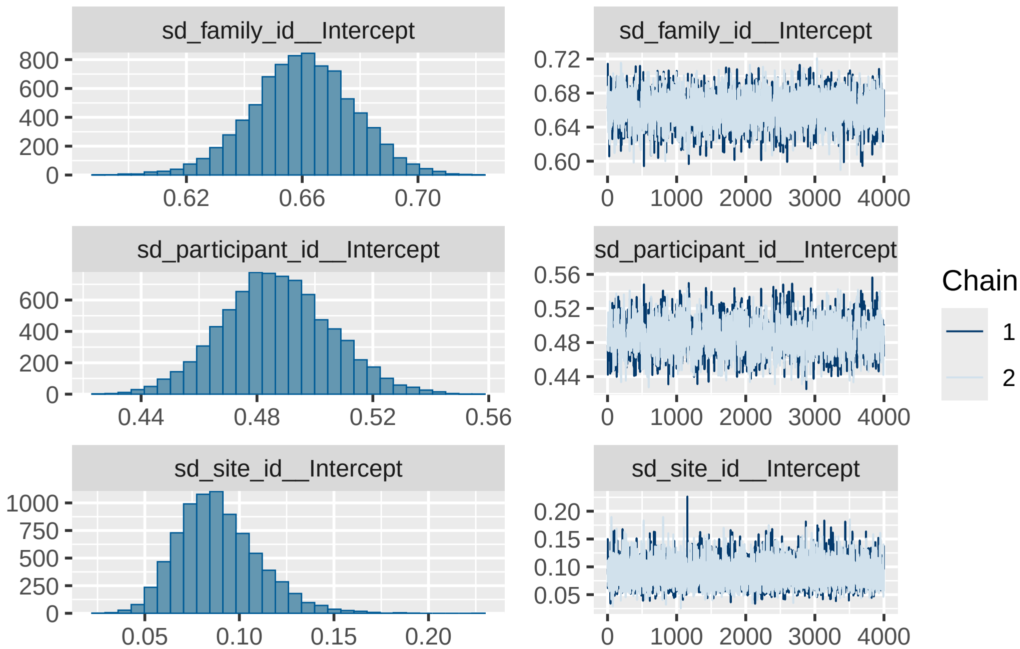
Model 4 stimulant diagnostics

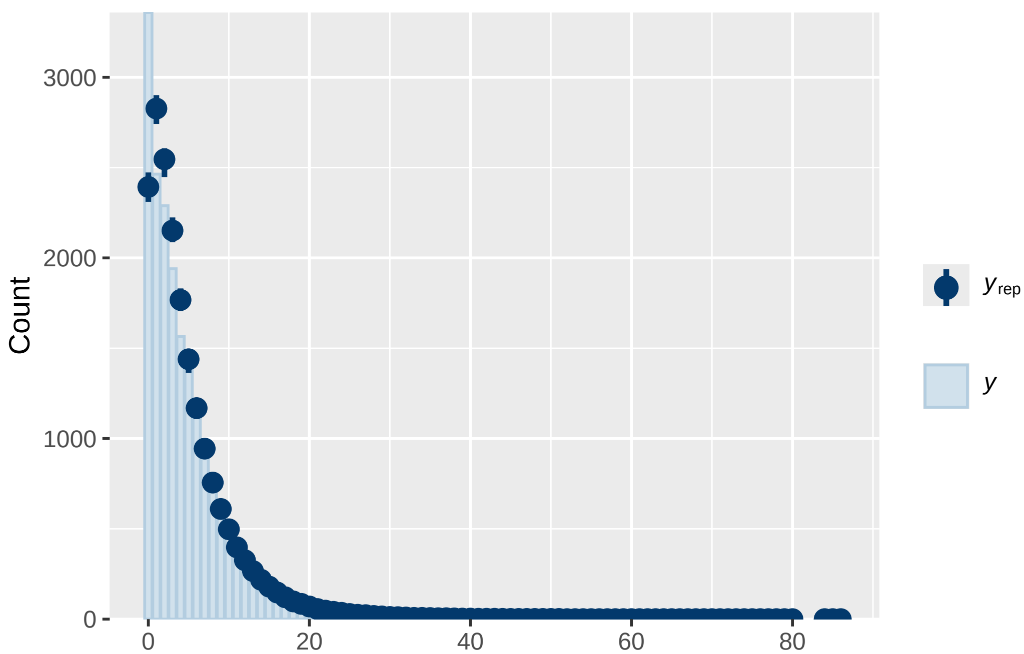
```
        prior     class                              coef
 normal(0, 1)         b
 normal(0, 1)         b                    adhd_traits_sc
 normal(0, 1)         b                            age_sc
 normal(0, 1)         b                  ethnracehispanic
 normal(0, 1)         b                     ethnraceother
 normal(0, 1)         b                  ethnracewhite_nh
```

```
          normal(0, 1)         b                              inr_sc
          normal(0, 1)         b                  menarche_status_pY
          normal(0, 1)         b menarche_status_pY:adhd_traits_sc
          normal(0, 1)         b                   stimulant_statusY
student_t(3, 1.1, 2.5) Intercept
  student_t(3, 0, 2.5)        sd
  student_t(3, 0, 2.5)        sd
  student_t(3, 0, 2.5)        sd                           Intercept
  student_t(3, 0, 2.5)        sd
  student_t(3, 0, 2.5)        sd                           Intercept
  student_t(3, 0, 2.5)        sd
  student_t(3, 0, 2.5)        sd                           Intercept
         group resp dpar nlpar lb ub tag       source
                                                 user
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                              default
                                0             default
     family_id                  0        (vectorized)
     family_id                  0        (vectorized)
participant_id                  0        (vectorized)
participant_id                  0        (vectorized)
       site_id                  0        (vectorized)
       site_id                  0        (vectorized)
```

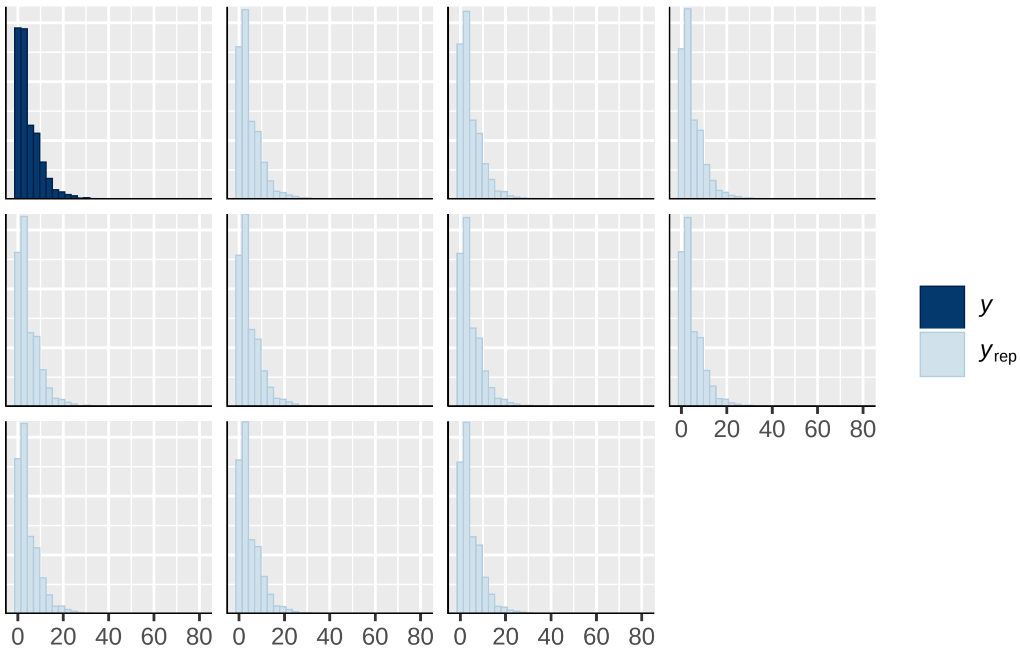


b_Intercept
0.7
0.8
0.9
b_Intercept
0 1000 2000 3000 4000
b_menarche_status_pY
0.03
0.06
0.09
0.12
b_menarche_status_pY
0 1000 2000 3000 4000
b_adhd_traits_sc
0.675
0.700
0.725
0.750
b_adhd_traits_sc
0 1000 2000 3000 4000
b_age_sc
0.02
0.04
0.06
0.08
0.10
b_age_sc
0 1000 2000 3000 4000
b_ethnracehispanic
0.2
0.3
0.4
0.5
b_ethnracehispanic
0 1000 2000 3000 4000
b_ethnraceother
0.2
0.3
0.4
0.5
b_ethnraceother
0 1000 2000 3000 4000
b_ethnracewhite_nh
0.3
0.4
0.5
b_ethnracewhite_nh
0 1000 2000 3000 4000
b_inr_sc
−0.08
−0.04
0.00
0.04
b_inr_sc
0 1000 2000 3000 4000
b_stimulant_statusY
−0.15
−0.10
−0.05
0.00
0.05
b_stimulant_statusY
0 1000 2000 3000 4000
b_menarche_status_pY:adhd_traits_sc
−0.050
−0.025
0.000
0.025
0.050
0 1000 2000 3000 4000
Chain
1
2
Chain
1
2

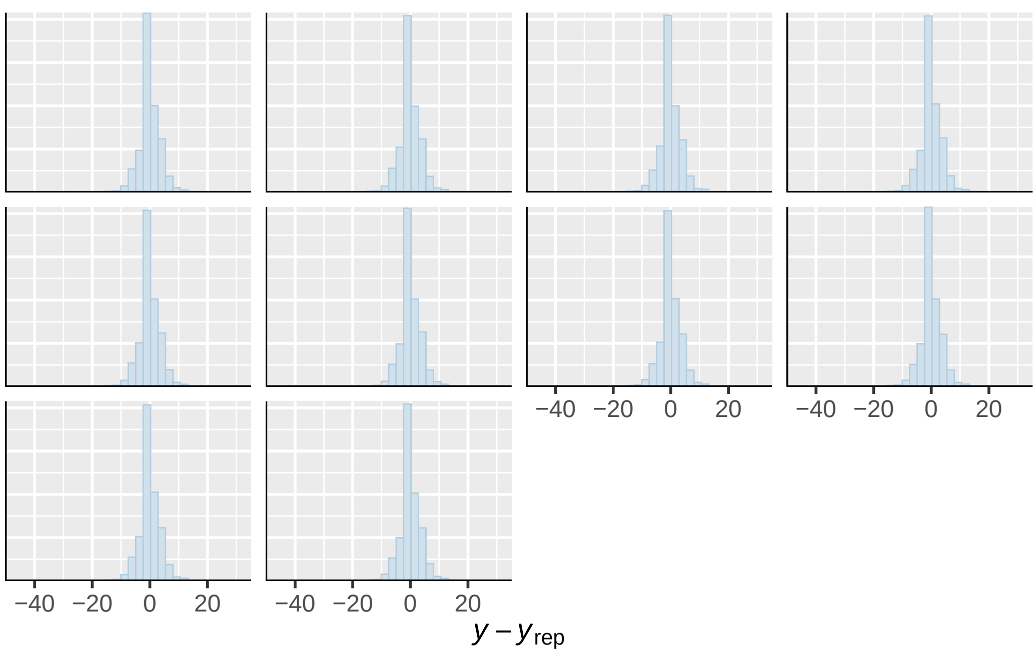




`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

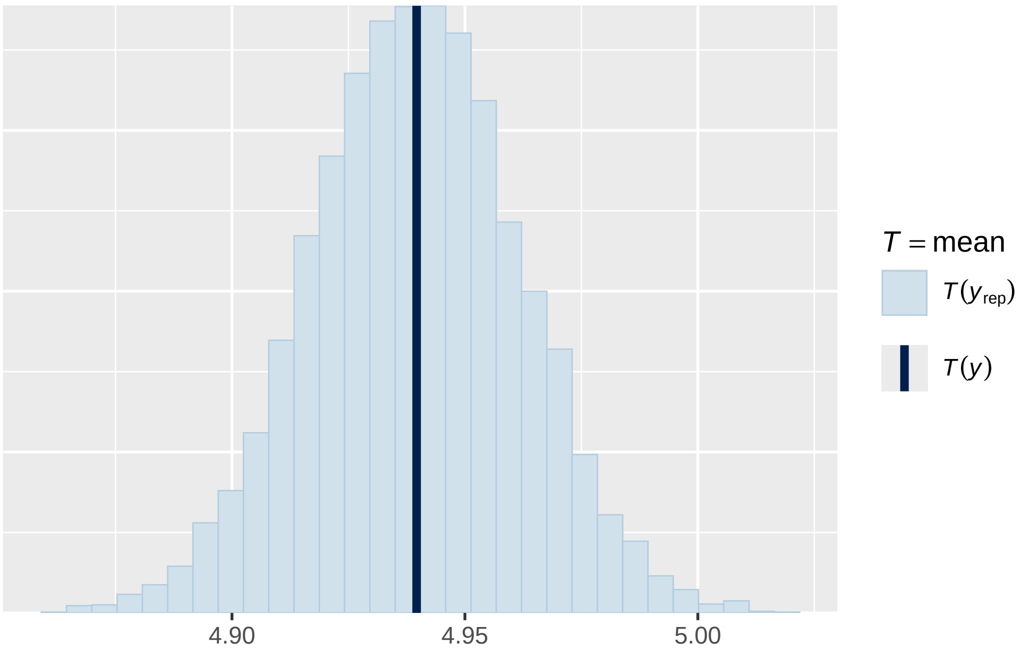



```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

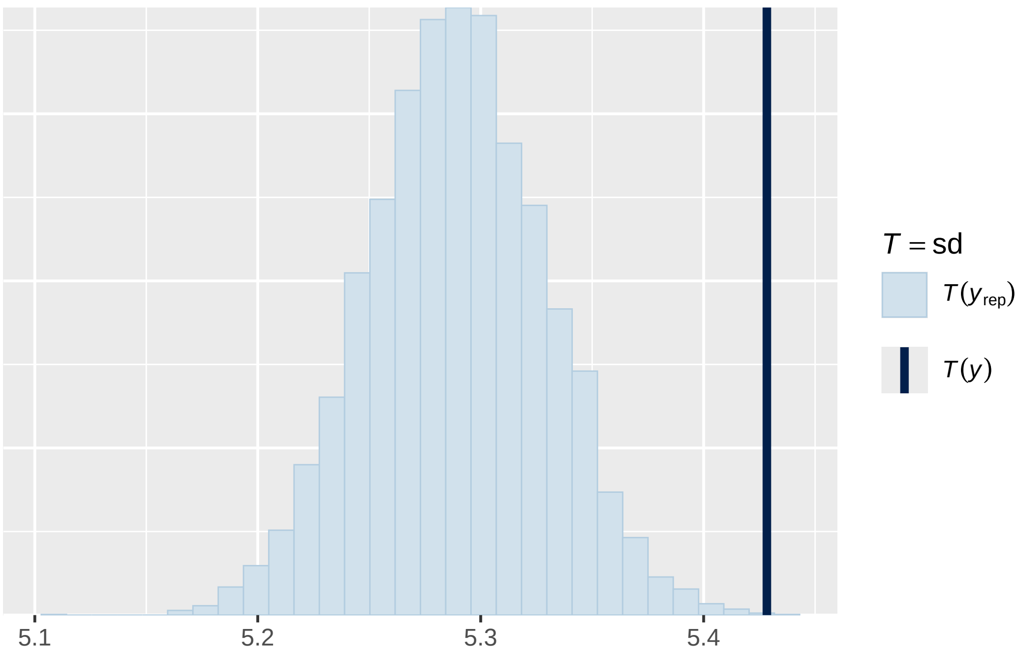



```
Using all posterior draws for ppc type 'stat' by default.
```

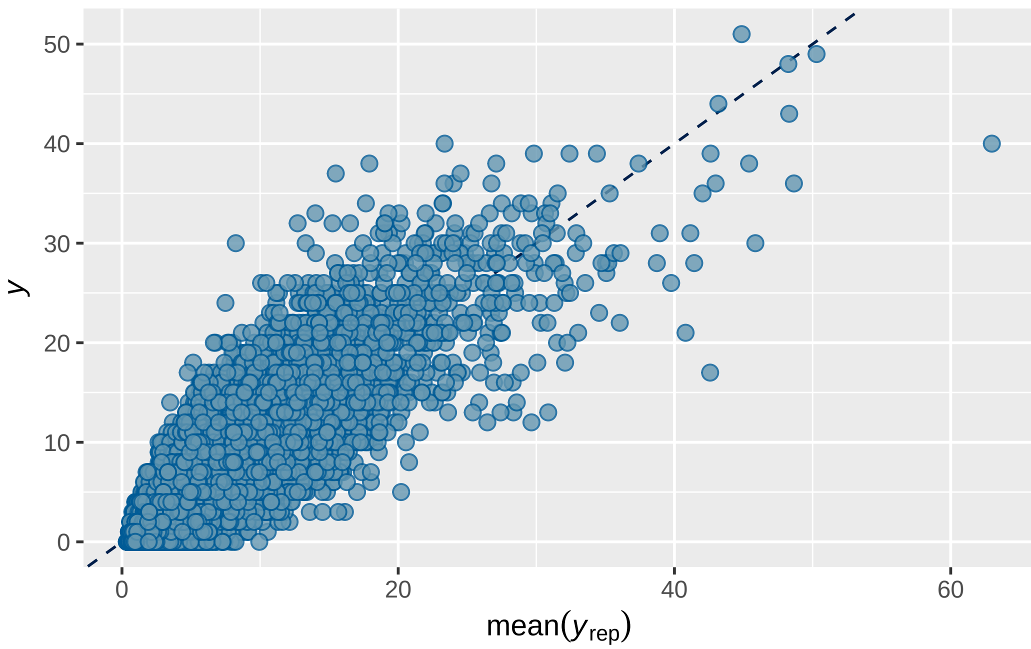
```
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 5 stimulant output

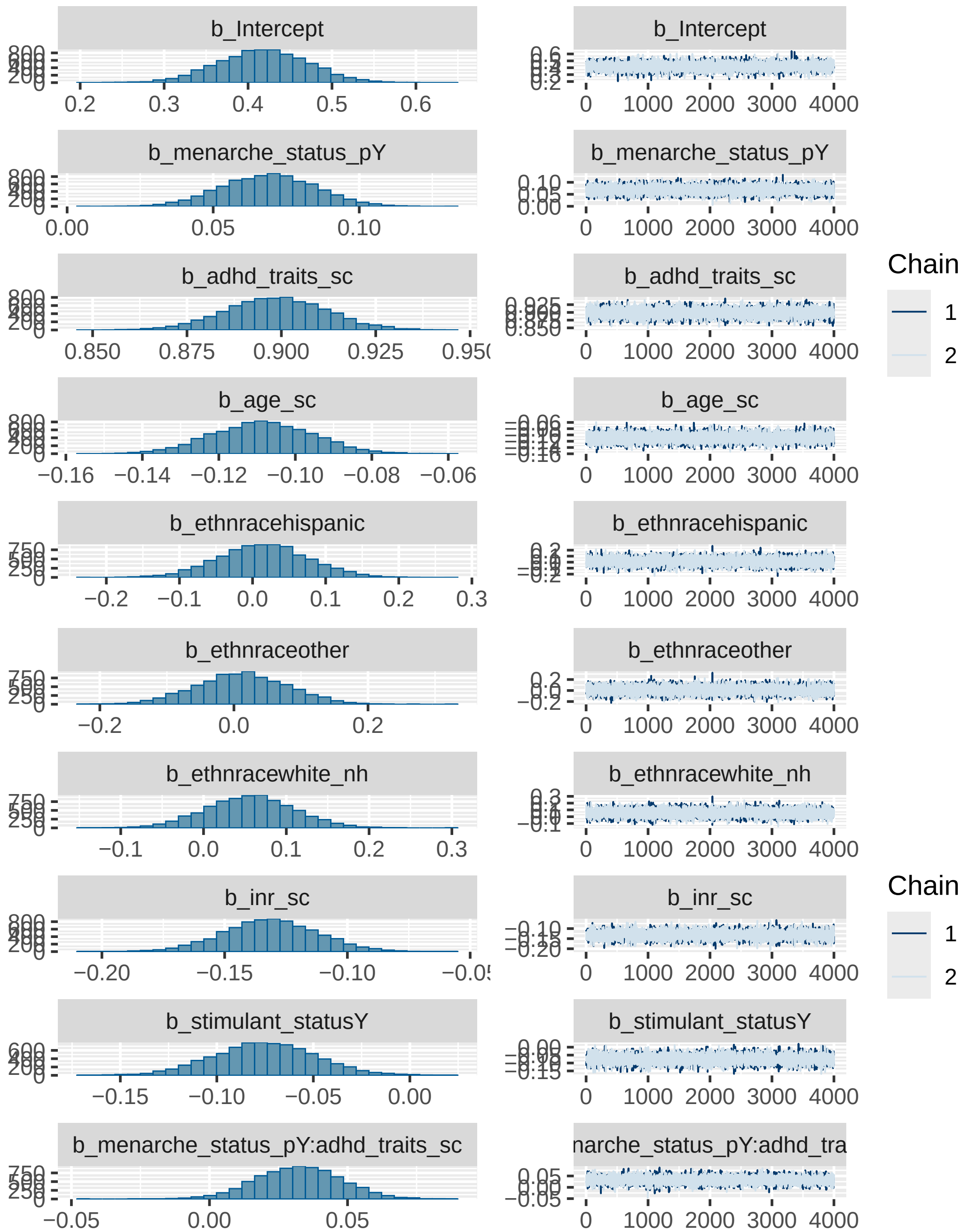
```
    Estimate  Est.Error      Q2.5     Q97.5
R2 0.7888422 0.00203214 0.7847808 0.7926695


 Family: poisson
  Links: mu = log
Formula: externalising ~ menarche_status_p * adhd_traits_sc + age_sc + ethnrace + inr_sc + st
   Data: analytic_df (Number of observations: 19381)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 4626)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.742     0.030    0.683    0.799 1.001      997     1777

~participant_id (Number of levels: 5297)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.756     0.027    0.704    0.811 1.003     1026     1704
```

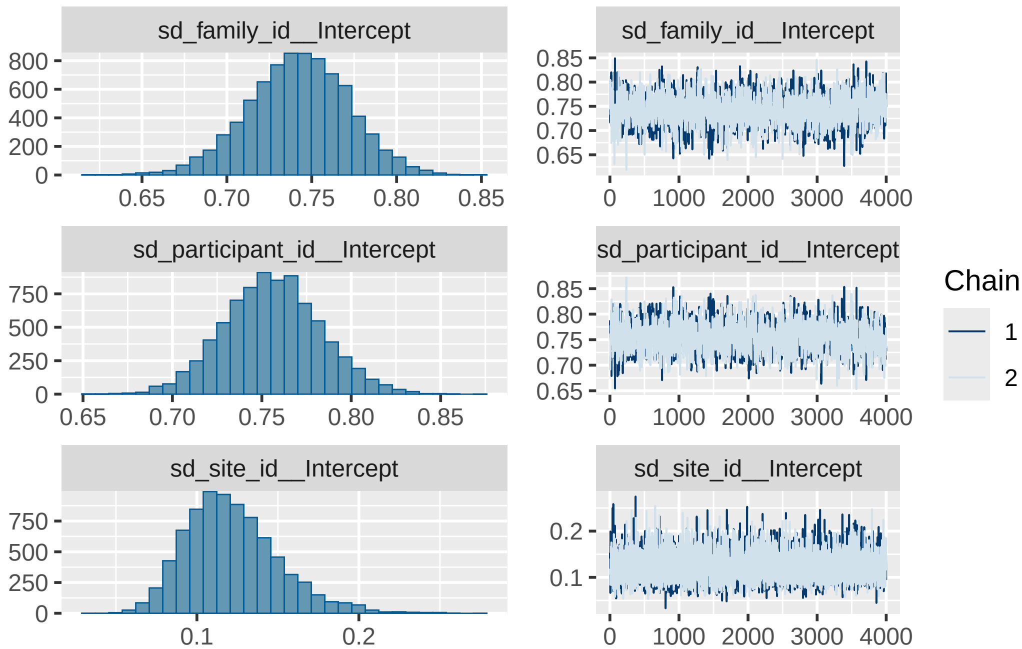
```
~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.122     0.029    0.075    0.189 1.000     7734     6102

Regression Coefficients:
                                  Estimate Est.Error l-95% CI u-95% CI  Rhat
Intercept                            0.418     0.054    0.312    0.523 1.000
menarche_status_pY                   0.069     0.016    0.038    0.100 1.000
adhd_traits_sc                       0.899     0.013    0.873    0.925 1.000
age_sc                              -0.108     0.013   -0.134   -0.083 1.000
ethnracehispanic                     0.015     0.061   -0.103    0.135 1.000
ethnraceother                        0.012     0.068   -0.120    0.143 1.000
ethnracewhite_nh                     0.056     0.053   -0.048    0.160 1.000
inr_sc                              -0.131     0.019   -0.169   -0.094 1.001
stimulant_statusY                   -0.075     0.027   -0.127   -0.023 1.000
menarche_status_pY:adhd_traits_sc    0.032     0.015    0.002    0.062 1.001
                                  Bulk_ESS Tail_ESS
Intercept                             6988     5541
menarche_status_pY                   14966     5962
adhd_traits_sc                       14865     5274
age_sc                               15313     6322
ethnracehispanic                      8316     6193
ethnraceother                         8483     6398
ethnracewhite_nh                      8022     6231
inr_sc                               18023     5441
stimulant_statusY                    17814     4897
menarche_status_pY:adhd_traits_sc    17067     6170

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

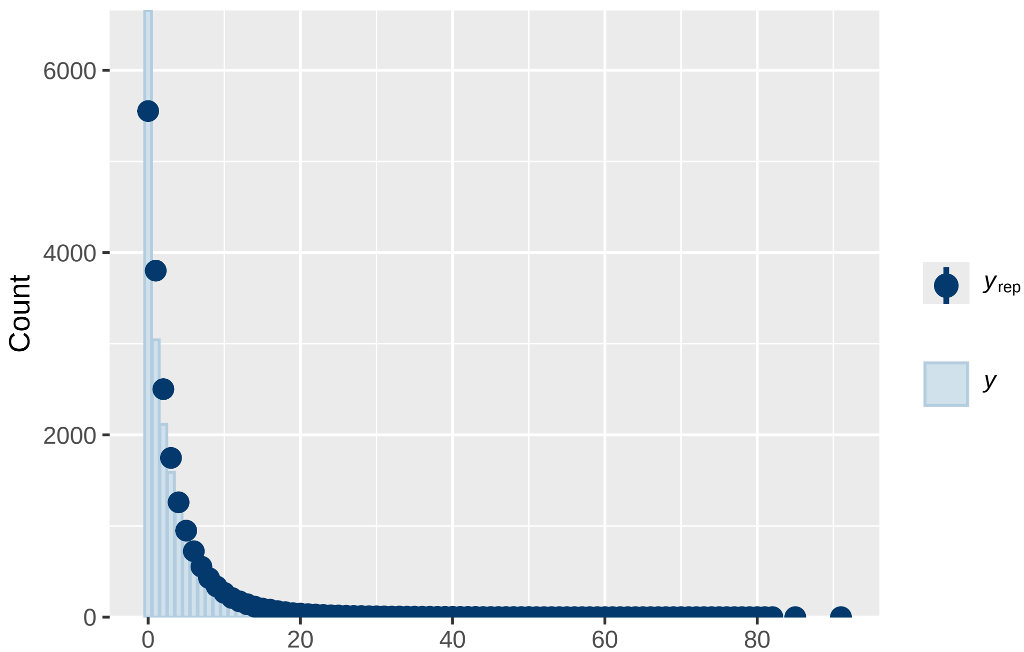
Model 5 stimulant diagnostics

```
       prior     class                              coef
normal(0, 1)         b
normal(0, 1)         b                    adhd_traits_sc
normal(0, 1)         b                            age_sc
normal(0, 1)         b                  ethnracehispanic
normal(0, 1)         b                     ethnraceother
normal(0, 1)         b                  ethnracewhite_nh
```

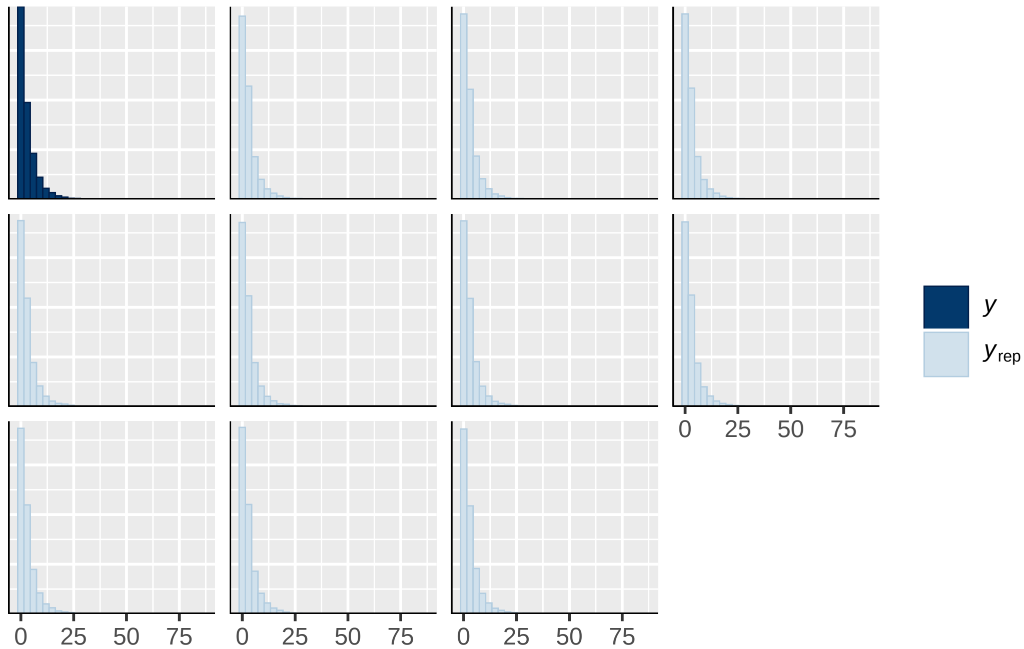
```
         normal(0, 1)         b                              inr_sc
         normal(0, 1)         b                menarche_status_pY
         normal(0, 1)         b menarche_status_pY:adhd_traits_sc
         normal(0, 1)         b                 stimulant_statusY
 student_t(3, 0, 2.9) Intercept
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd                          Intercept
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd                          Intercept
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd                          Intercept
          group resp dpar nlpar lb ub tag       source
                                                  user
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                               default
                                 0             default
      family_id                  0        (vectorized)
      family_id                  0        (vectorized)
 participant_id                  0        (vectorized)
 participant_id                  0        (vectorized)
        site_id                  0        (vectorized)
        site_id                  0        (vectorized)
```


b_Intercept
b_menarche_status_pY
b_adhd_traits_sc
b_age_sc
b_ethnracehispanic
b_ethnraceother
b_ethnracewhite_nh
b_inr_sc
b_stimulant_statusY
b_menarche_status_pY:adhd_traits_sc
Chain
1
2
Chain
1
2

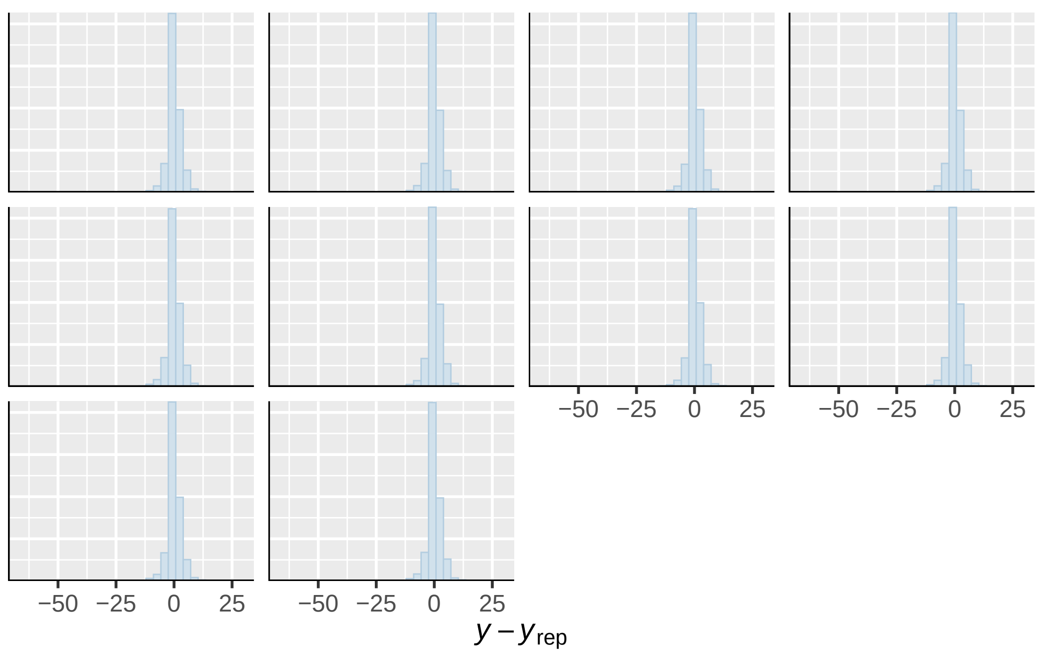




`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

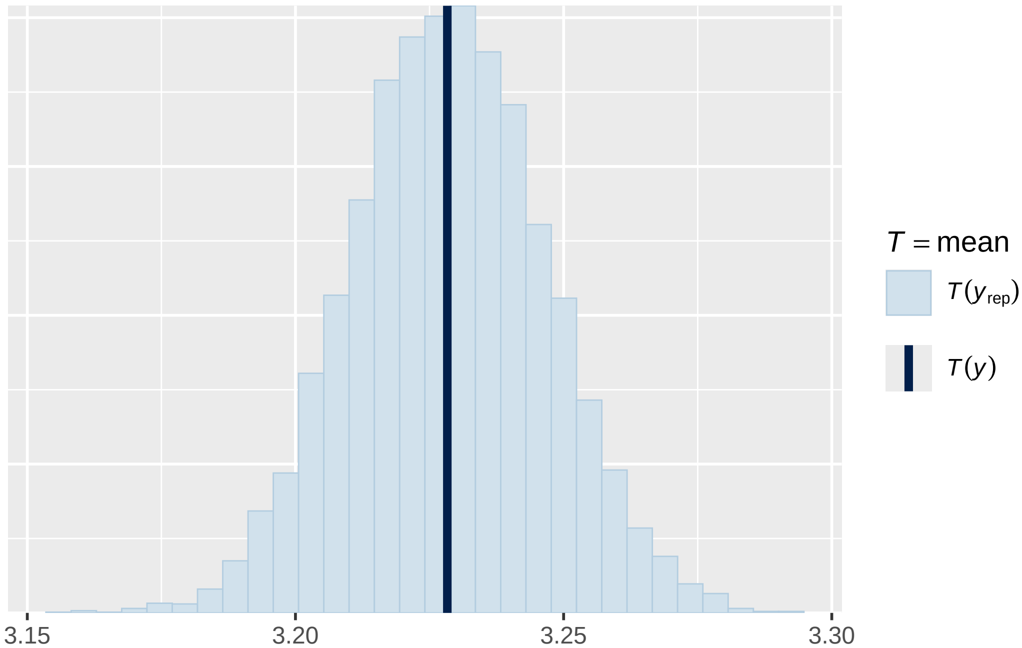



```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

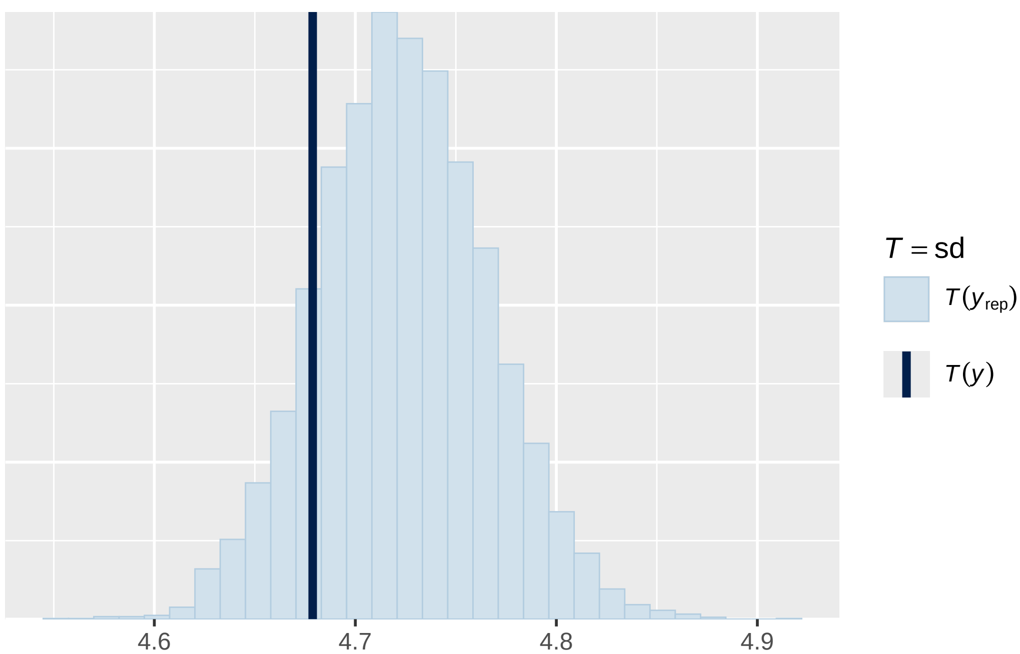



```
Using all posterior draws for ppc type 'stat' by default.
```

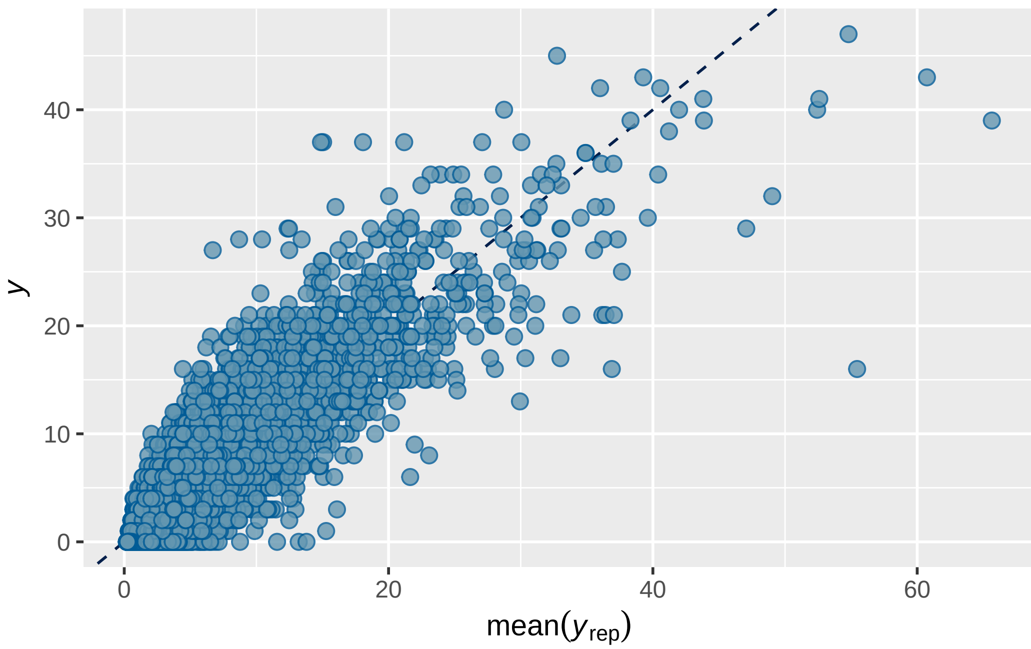
```
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



ABCD ADHD menarche analysis: Output for cross-sectional supplementary/sensitivity models

Model 1b BMI output

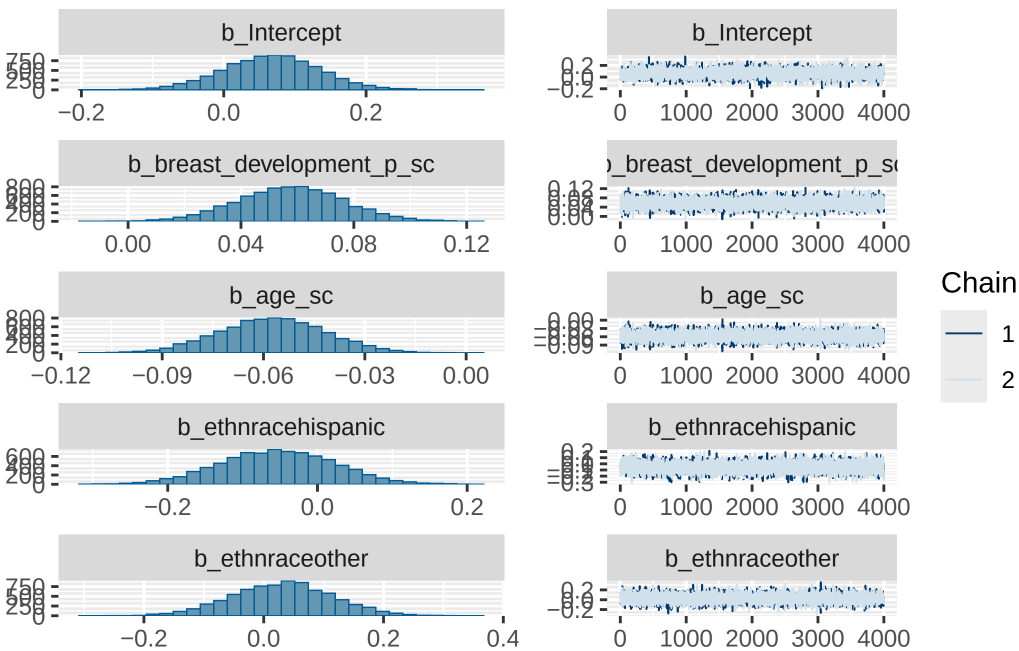
```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.7026613 0.003306821 0.6961907 0.7091034


 Family: poisson
  Links: mu = log
Formula: adhd_traits ~ breast_development_p_sc + age_sc + ethnrace + inr_sc + bmi_wave_0_sc +
   Data: analytic_df (Number of observations: 19364)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 4619)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.834     0.046    0.743    0.921 1.003      625     1546

~participant_id (Number of levels: 5289)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    1.037     0.036    0.968    1.108 1.002      811     2026

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.175     0.039    0.111    0.266 1.000     6635     5310

Regression Coefficients:
                        Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS
Intercept                  0.067     0.068   -0.068    0.202 1.000     6608
```

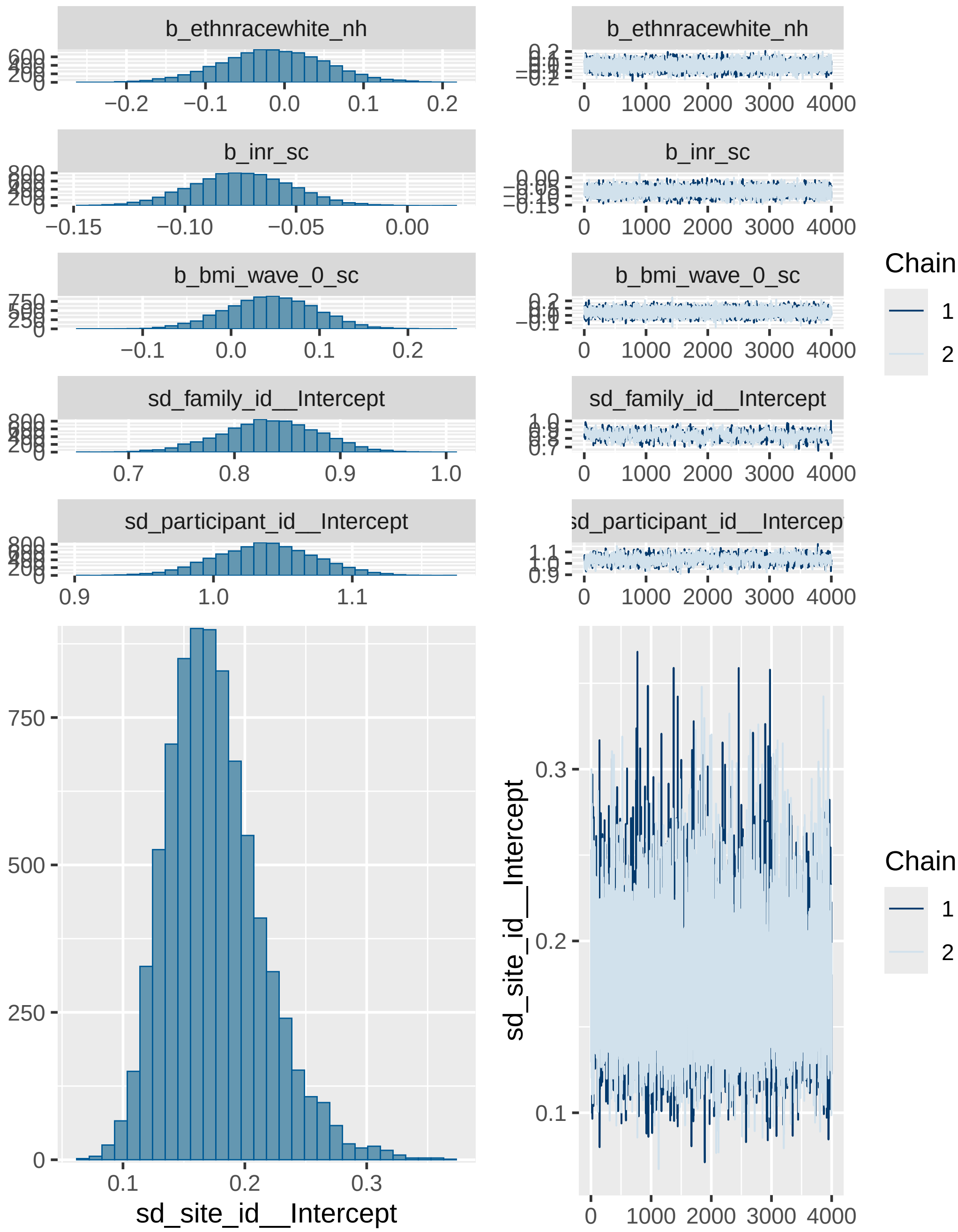
```
breast_development_p_sc    0.058     0.019    0.021    0.095 1.001    15140
age_sc                    -0.057     0.016   -0.087   -0.027 1.000    14562
ethnracehispanic          -0.053     0.077   -0.202    0.095 1.000     8340
ethnraceother              0.031     0.084   -0.133    0.194 1.000     8533
ethnracewhite_nh          -0.014     0.066   -0.144    0.117 1.000     8330
inr_sc                    -0.075     0.022   -0.118   -0.031 1.000    17133
bmi_wave_0_sc              0.045     0.050   -0.054    0.141 1.000    10095
                        Tail_ESS
Intercept                   5759
breast_development_p_sc     5962
age_sc                      5881
ethnracehispanic            6247
ethnraceother               6433
ethnracewhite_nh            6235
inr_sc                      6196
bmi_wave_0_sc               5863

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Model 1b BMI diagnostics

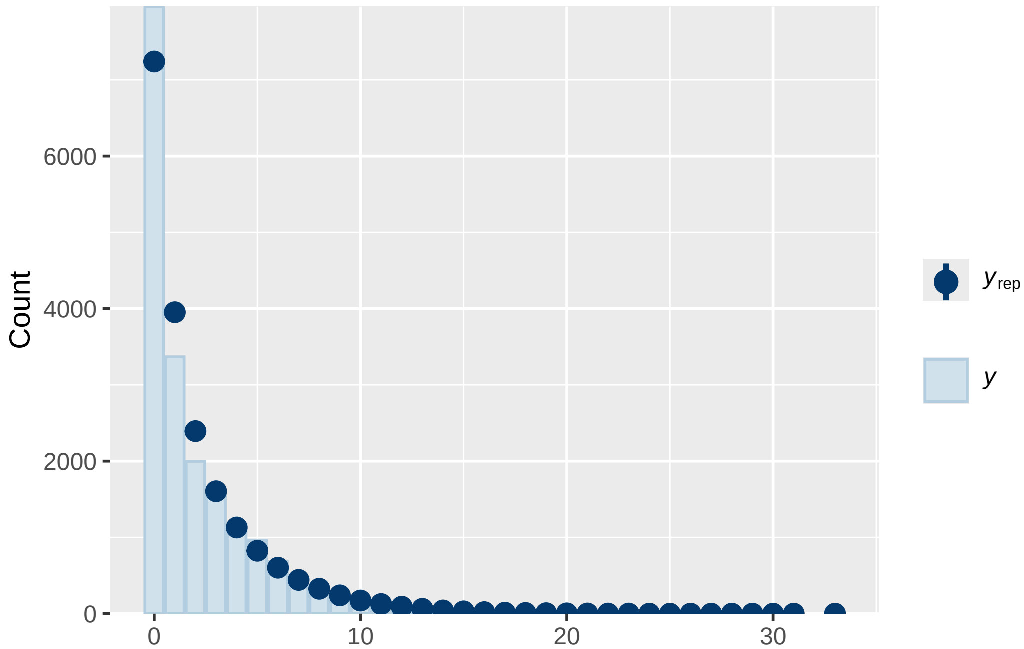
```
                prior     class                    coef          group resp
         normal(0, 1)         b
         normal(0, 1)         b                  age_sc
         normal(0, 1)         b           bmi_wave_0_sc
         normal(0, 1)         b breast_development_p_sc
         normal(0, 1)         b        ethnracehispanic
         normal(0, 1)         b           ethnraceother
         normal(0, 1)         b        ethnracewhite_nh
         normal(0, 1)         b                  inr_sc
 student_t(3, 0, 2.7) Intercept
 student_t(3, 0, 2.7)        sd
 student_t(3, 0, 2.7)        sd                              family_id
 student_t(3, 0, 2.7)        sd               Intercept      family_id
 student_t(3, 0, 2.7)        sd                         participant_id
 student_t(3, 0, 2.7)        sd               Intercept participant_id
 student_t(3, 0, 2.7)        sd                                site_id
 student_t(3, 0, 2.7)        sd               Intercept        site_id
 dpar nlpar lb ub tag       source
                              user
```

	(vectorized)
	(vectorized)
	(vectorized)
	(vectorized)
	(vectorized)
	(vectorized)
	(vectorized)
	default
0	default
0	(vectorized)
0	(vectorized)
0	(vectorized)
0	(vectorized)
0	(vectorized)
0	(vectorized)

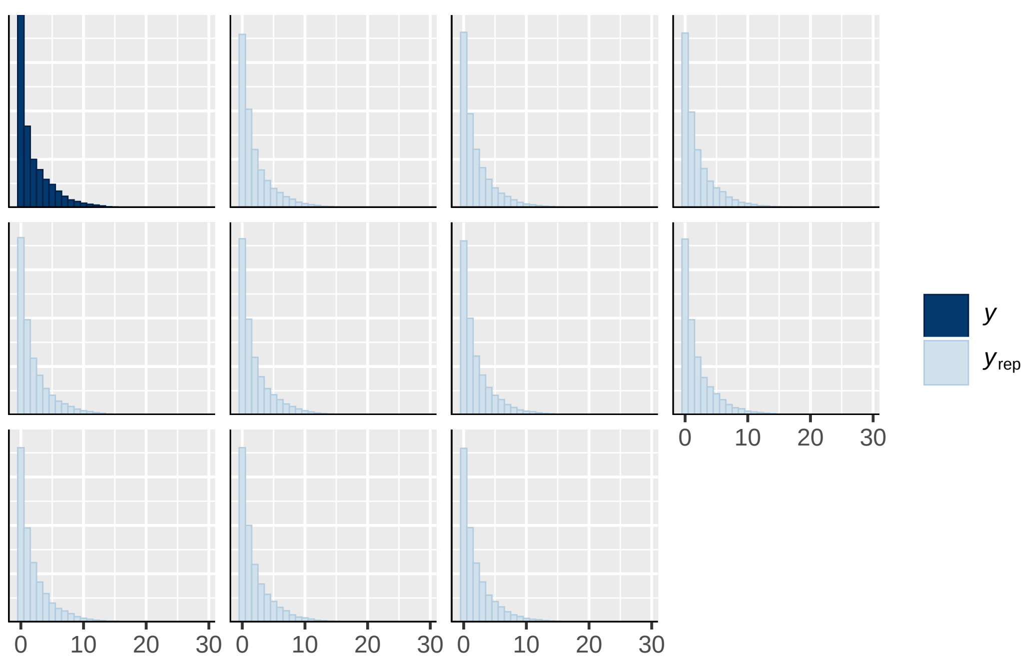




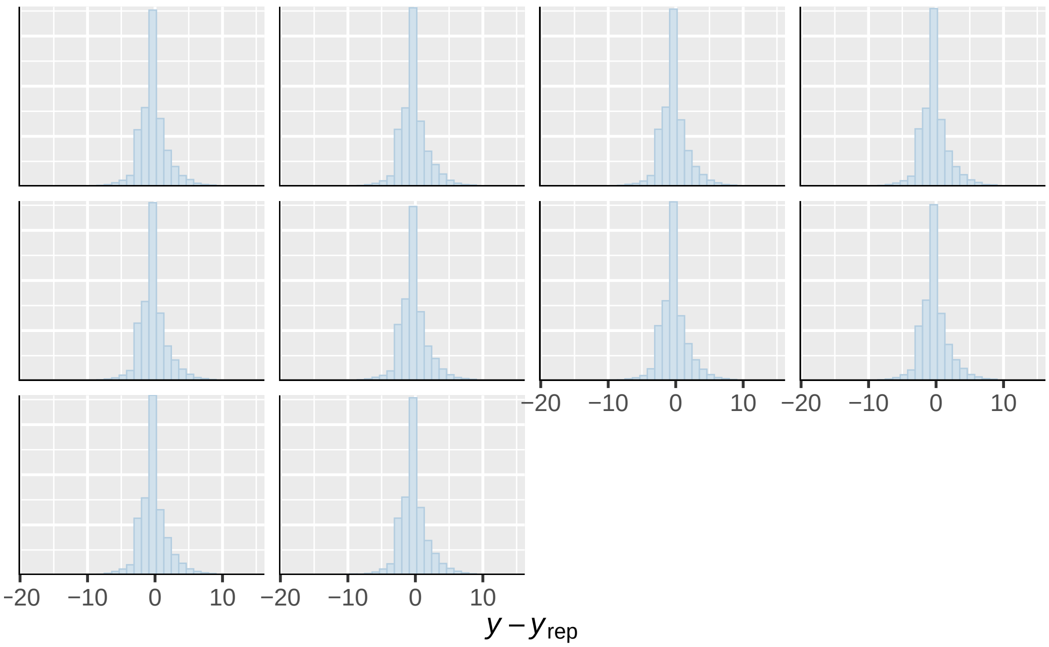
b_ethnracewhite_nh
b_inr_sc
b_bmi_wave_0_sc
sd_family_id__Intercept
sd_participant_id__Intercept
sd_site_id__Intercept
Chain
1
2



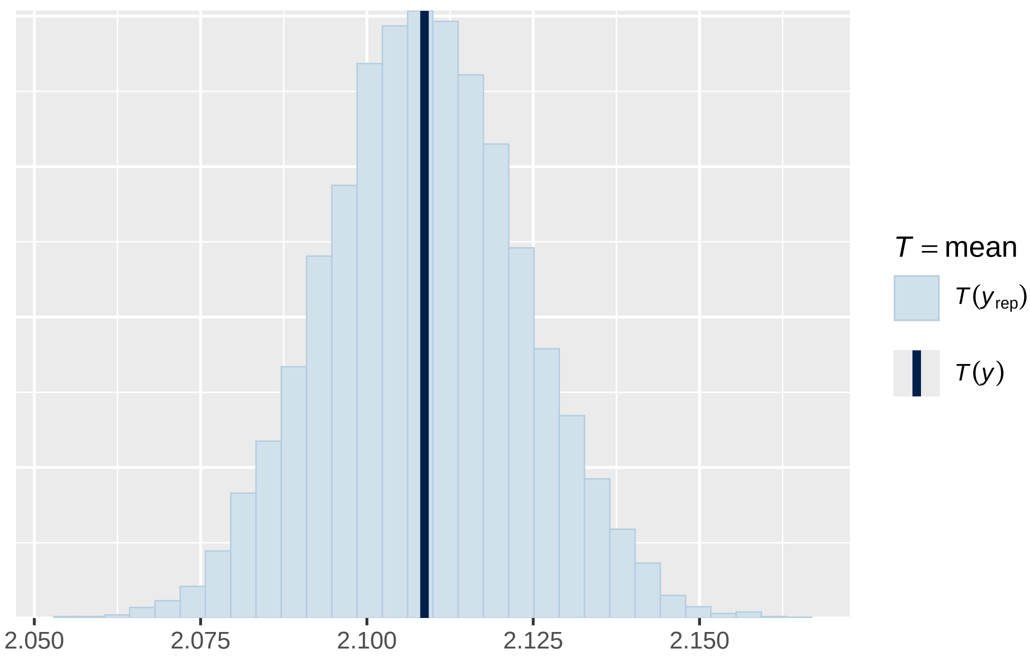
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.



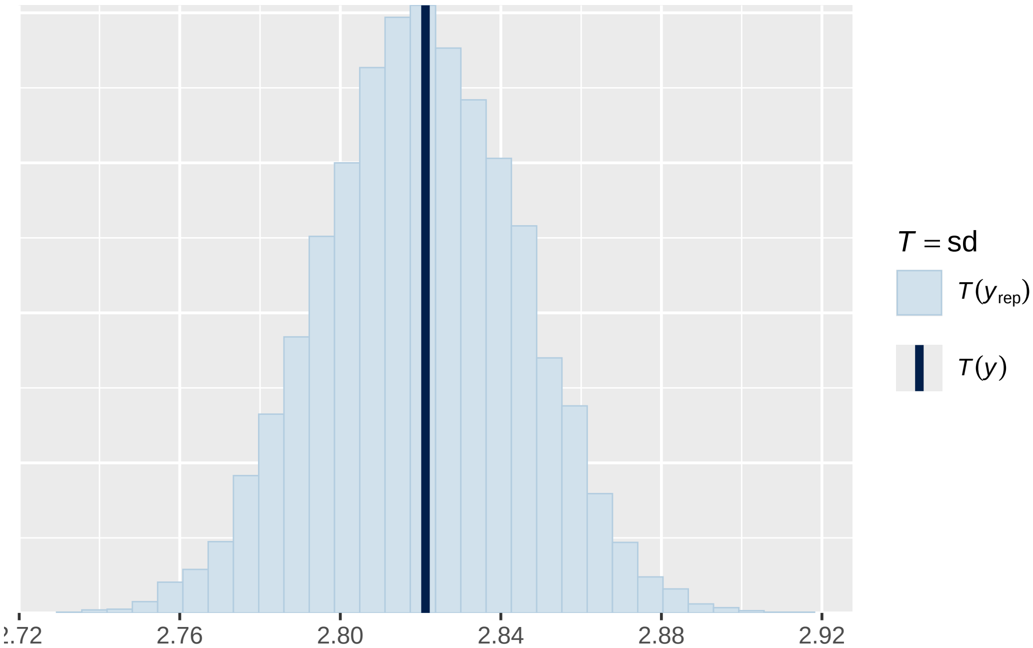
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.



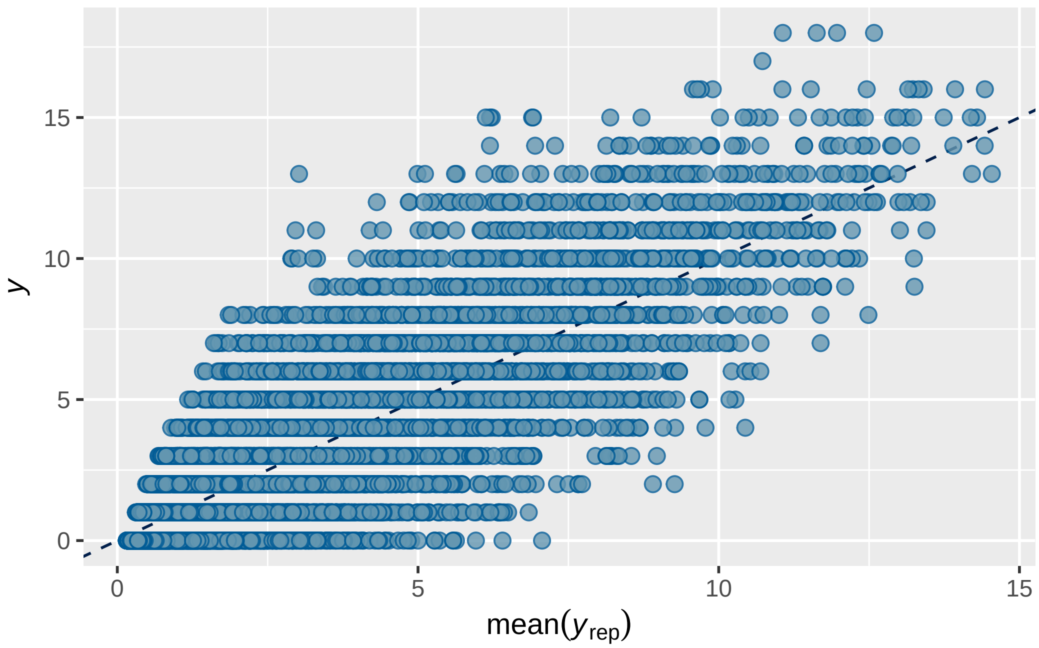

```
Using all posterior draws for ppc type 'stat' by default.
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 4 BMI output

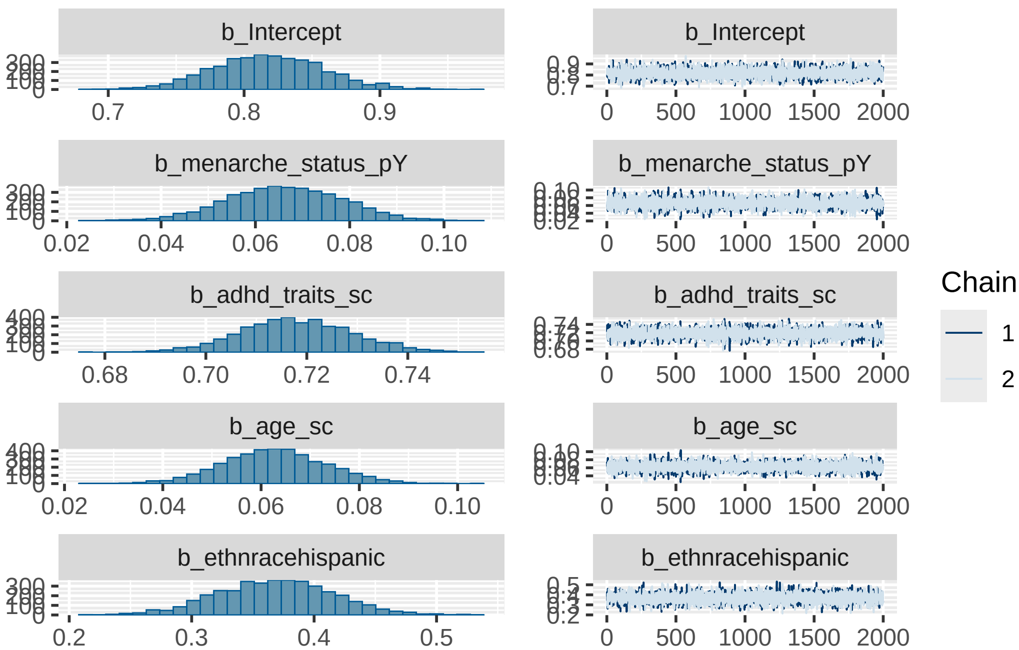
```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.7363026 0.002180899 0.7318509 0.7405877


 Family: poisson
  Links: mu = log
Formula: internalising ~ menarche_status_p * adhd_traits_sc + age_sc + ethnrace + inr_sc + bm
   Data: analytic_df (Number of observations: 19381)
  Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~family_id (Number of levels: 4626)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.662     0.017    0.627    0.695 1.006      754     1681

~participant_id (Number of levels: 5297)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.484     0.019    0.448    0.522 1.008      467     1244

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.088     0.022    0.052    0.135 1.000     3813     3076

Regression Coefficients:
                                  Estimate Est.Error l-95% CI u-95% CI  Rhat
Intercept                            0.818     0.041    0.740    0.902 1.001
menarche_status_pY                   0.066     0.012    0.042    0.090 1.001
adhd_traits_sc                       0.718     0.011    0.696    0.740 1.000
age_sc                               0.063     0.011    0.042    0.084 1.000
ethnracehispanic                     0.369     0.047    0.274    0.460 1.000
ethnraceother                        0.416     0.051    0.317    0.513 1.001
ethnracewhite_nh                     0.473     0.041    0.389    0.553 1.001
inr_sc                              -0.017     0.015   -0.047    0.012 1.000
bmi_wave_0_sc                        0.154     0.030    0.095    0.213 1.000
menarche_status_pY:adhd_traits_sc    0.007     0.013   -0.020    0.032 1.001
                                  Bulk_ESS Tail_ESS
Intercept                             4128     3106
menarche_status_pY                    7952     3100
adhd_traits_sc                        8880     3066
age_sc                                8701     2562
```

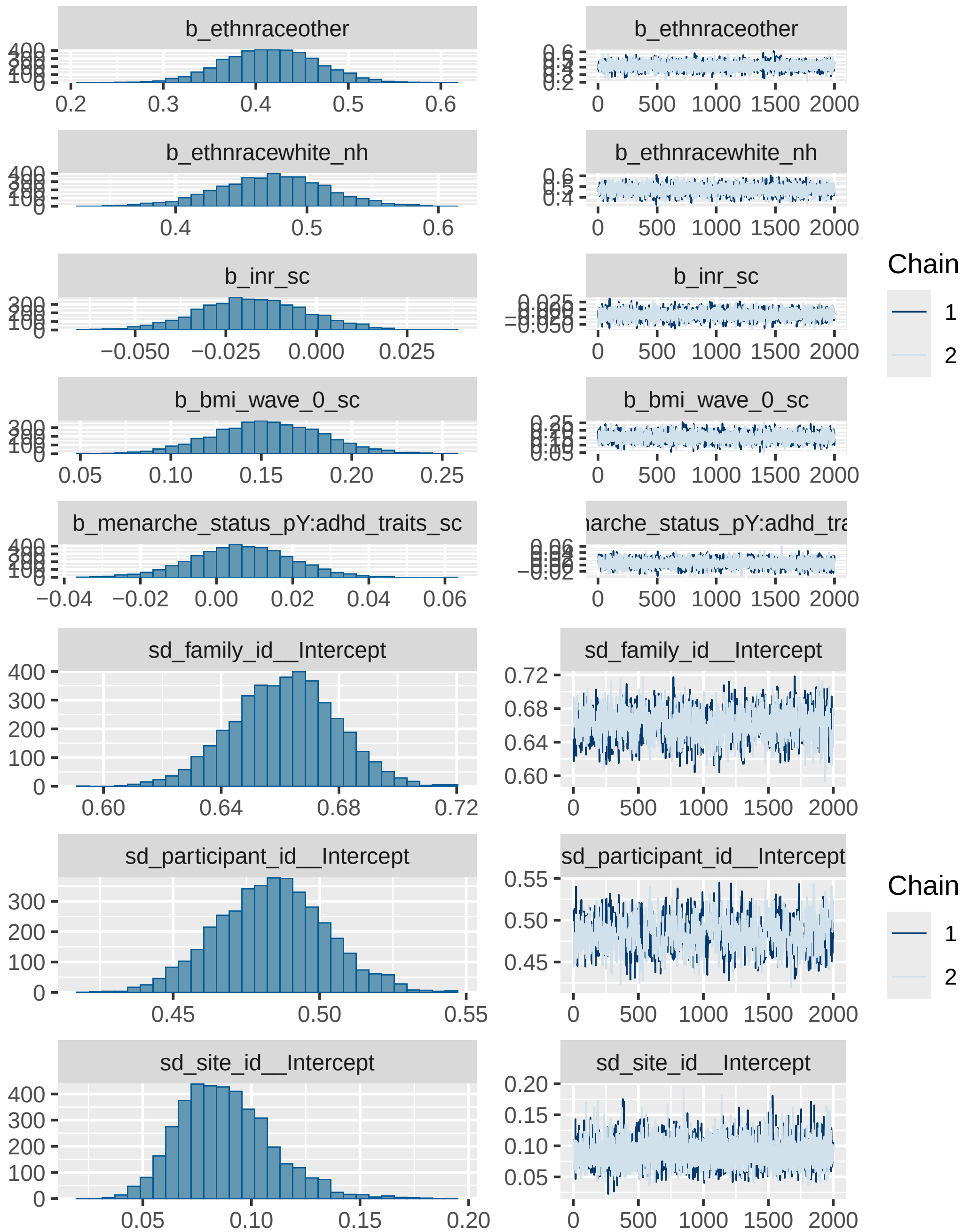
```
ethnracehispanic                      4528     3001
ethnraceother                         4620     3349
ethnracewhite_nh                      4280     2841
inr_sc                                8757     2663
bmi_wave_0_sc                         5774     3311
menarche_status_pY:adhd_traits_sc     7736     3353

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Model 4 BMI diagnostics

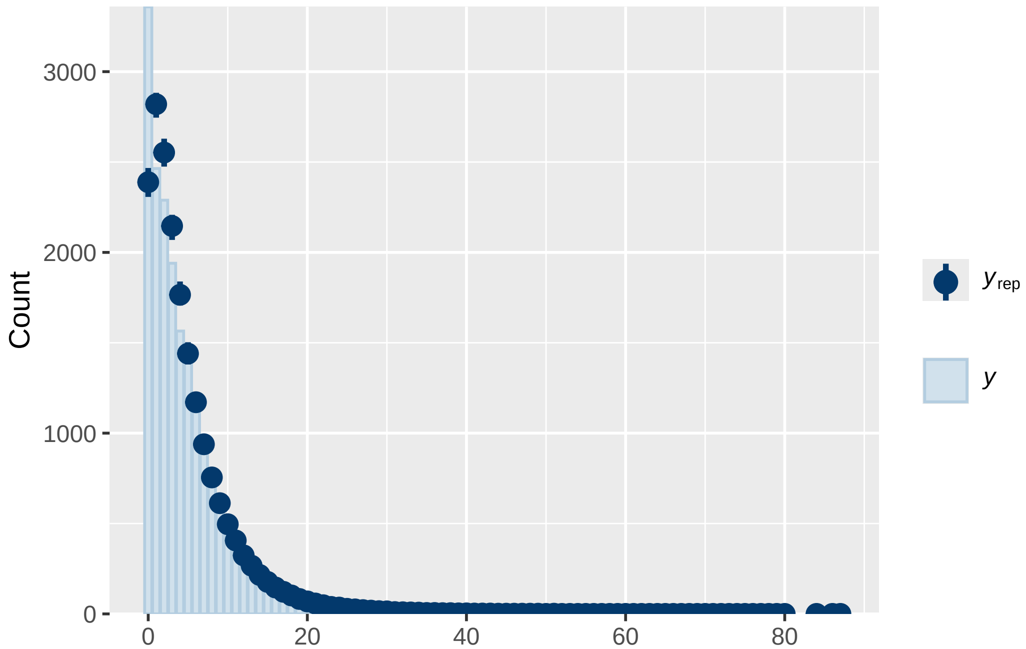
```
                 prior     class                              coef
           normal(0, 1)         b
           normal(0, 1)         b                    adhd_traits_sc
           normal(0, 1)         b                            age_sc
           normal(0, 1)         b                     bmi_wave_0_sc
           normal(0, 1)         b                  ethnracehispanic
           normal(0, 1)         b                     ethnraceother
           normal(0, 1)         b                  ethnracewhite_nh
           normal(0, 1)         b                            inr_sc
           normal(0, 1)         b                menarche_status_pY
           normal(0, 1)         b menarche_status_pY:adhd_traits_sc
 student_t(3, 1.1, 2.5) Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd                         Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd                         Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd                         Intercept
          group resp dpar nlpar lb ub tag       source
                                                  user
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
```

		(vectorized)
		default
	0	default
family_id	0	(vectorized)
family_id	0	(vectorized)
participant_id	0	(vectorized)
participant_id	0	(vectorized)
site_id	0	(vectorized)
site_id	0	(vectorized)

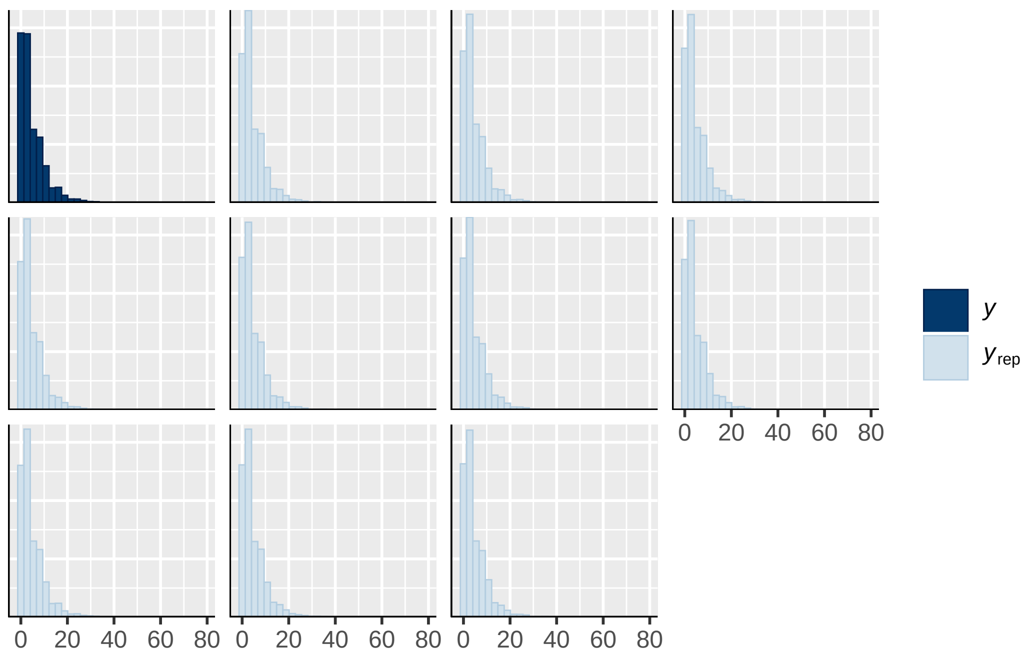




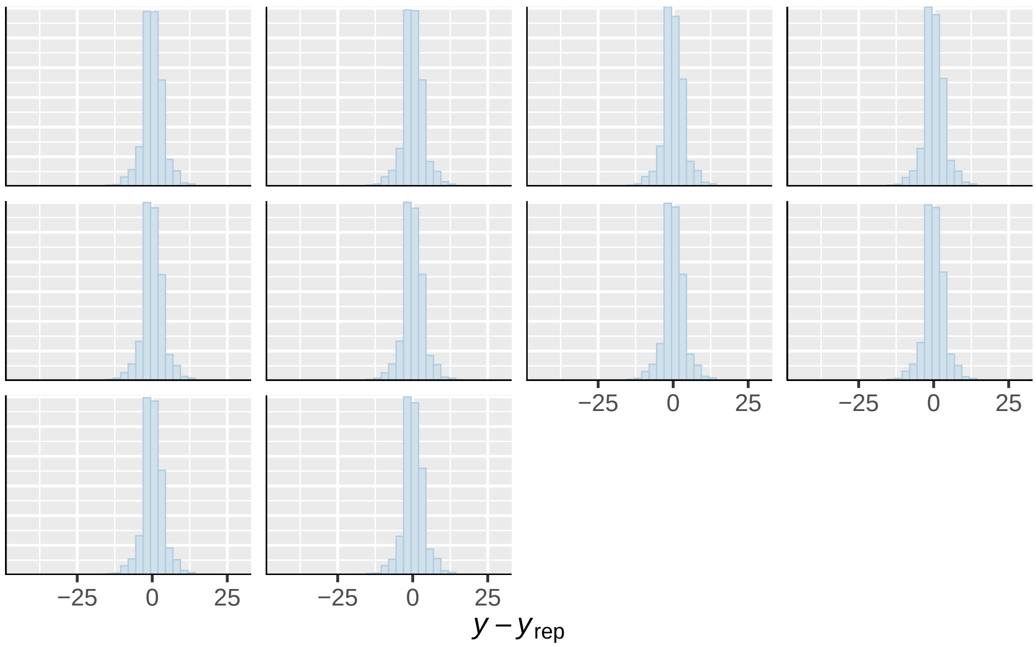
b_ethnraceother
b_ethnracewhite_nh
b_inr_sc
b_bmi_wave_0_sc
b_menarche_status_pY:adhd_traits_sc
sd_family_id__Intercept
sd_participant_id__Intercept
sd_site_id__Intercept
Chain
1
2
0
500
1000
1500
2000



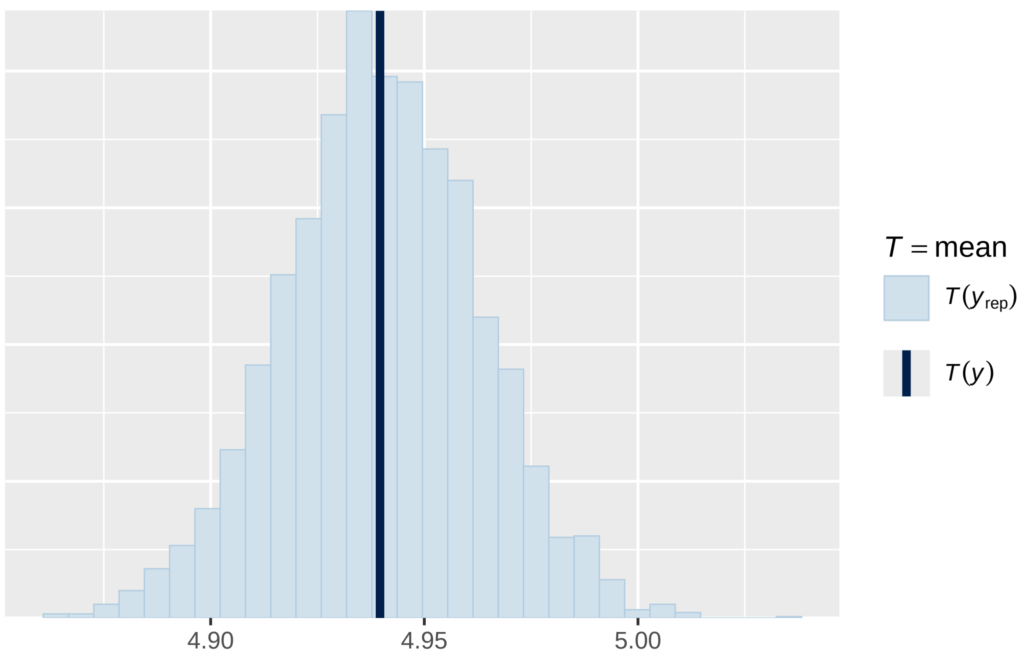

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



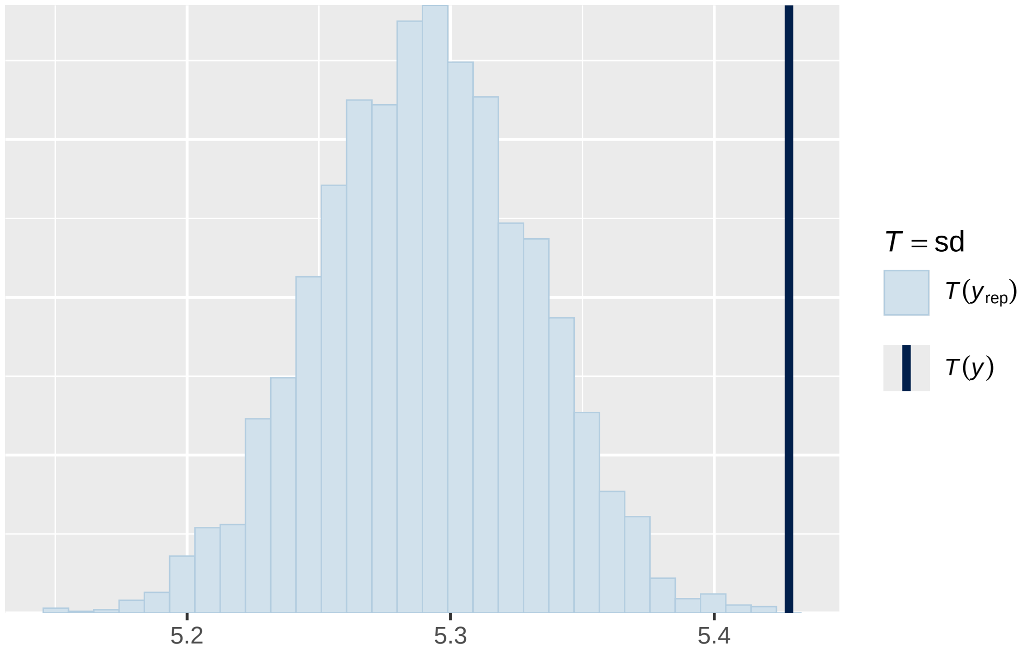

```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



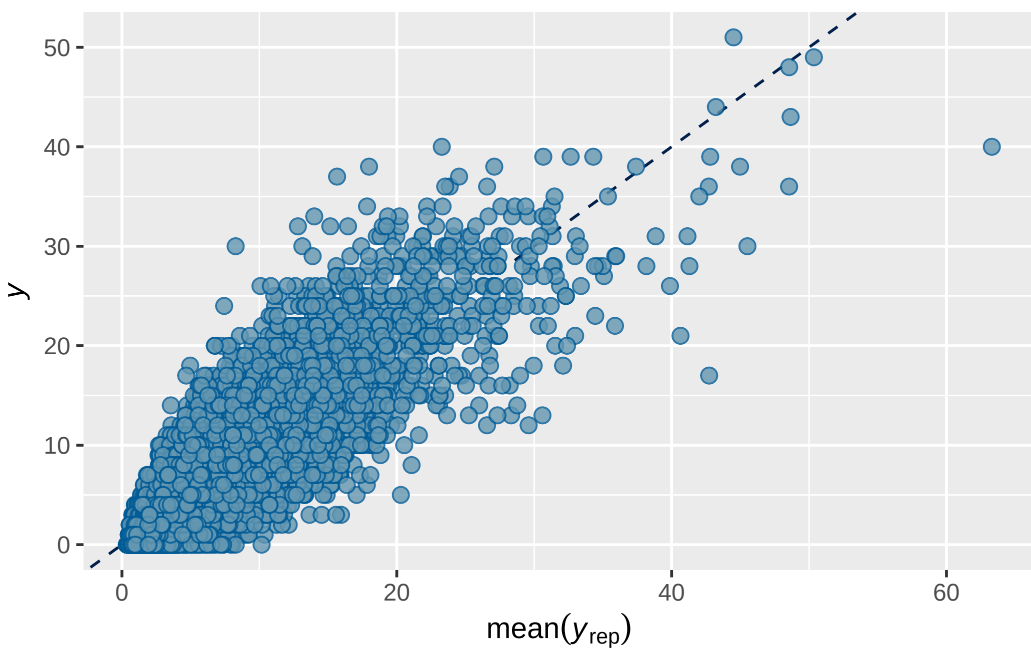

```
Using all posterior draws for ppc type 'stat' by default.
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 5 BMI output

```
   Estimate   Est.Error      Q2.5     Q97.5
R2  0.78834 0.002047103 0.7842391 0.7922249


 Family: poisson
  Links: mu = log
Formula: externalising ~ menarche_status_p * adhd_traits_sc + age_sc + ethnrace + inr_sc + br
   Data: analytic_df (Number of observations: 19381)
  Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~family_id (Number of levels: 4626)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.745     0.027    0.690    0.797 1.004      541     1366

~participant_id (Number of levels: 5297)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.749     0.025    0.699    0.797 1.002      541     1497

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.120     0.028    0.073    0.184 1.002     3907     2856

Regression Coefficients:
                                  Estimate Est.Error l-95% CI u-95% CI  Rhat
Intercept                            0.405     0.054    0.295    0.512 1.000
menarche_status_pY                   0.065     0.016    0.034    0.097 1.001
adhd_traits_sc                       0.894     0.014    0.868    0.921 1.000
age_sc                              -0.109     0.013   -0.134   -0.083 1.000
ethnracehispanic                     0.044     0.062   -0.078    0.165 1.000
ethnraceother                        0.058     0.067   -0.076    0.193 1.001
ethnracewhite_nh                     0.119     0.054    0.016    0.224 1.000
inr_sc                              -0.124     0.019   -0.162   -0.088 1.000
bmi_wave_0_sc                        0.234     0.040    0.158    0.312 1.000
menarche_status_pY:adhd_traits_sc    0.033     0.016    0.002    0.064 1.000
                                  Bulk_ESS Tail_ESS
Intercept                             3991     2702
menarche_status_pY                    7524     2612
adhd_traits_sc                        8392     3308
age_sc                                7264     3247
```

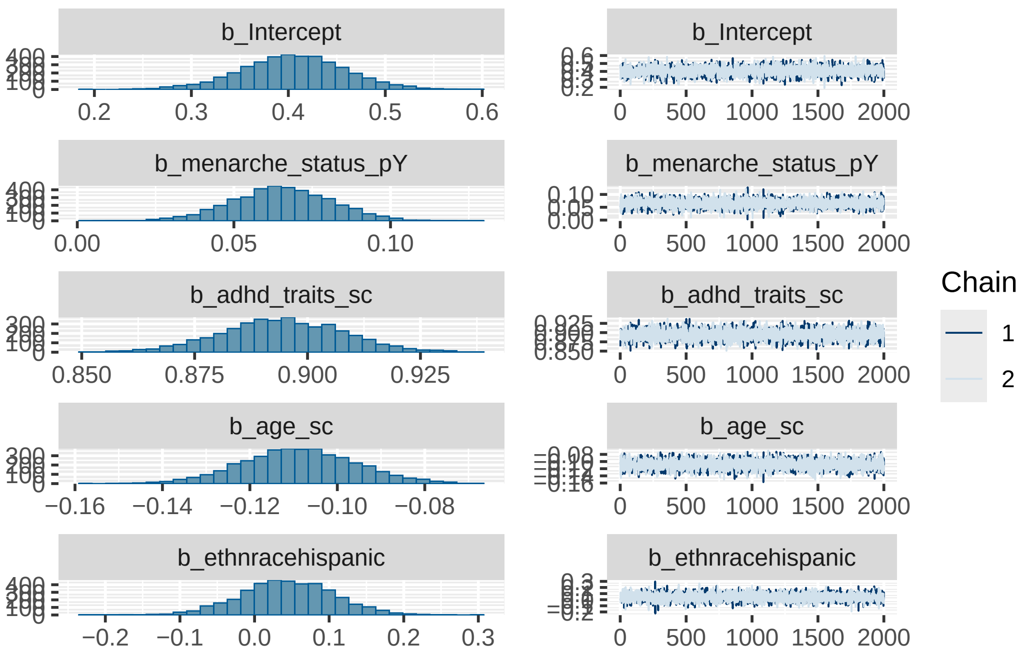
```
ethnracehispanic                       4483     2995
ethnraceother                          4514     3300
ethnracewhite_nh                       4286     2951
inr_sc                                 9765     3232
bmi_wave_0_sc                          5004     3114
menarche_status_pY:adhd_traits_sc      7925     3236
```

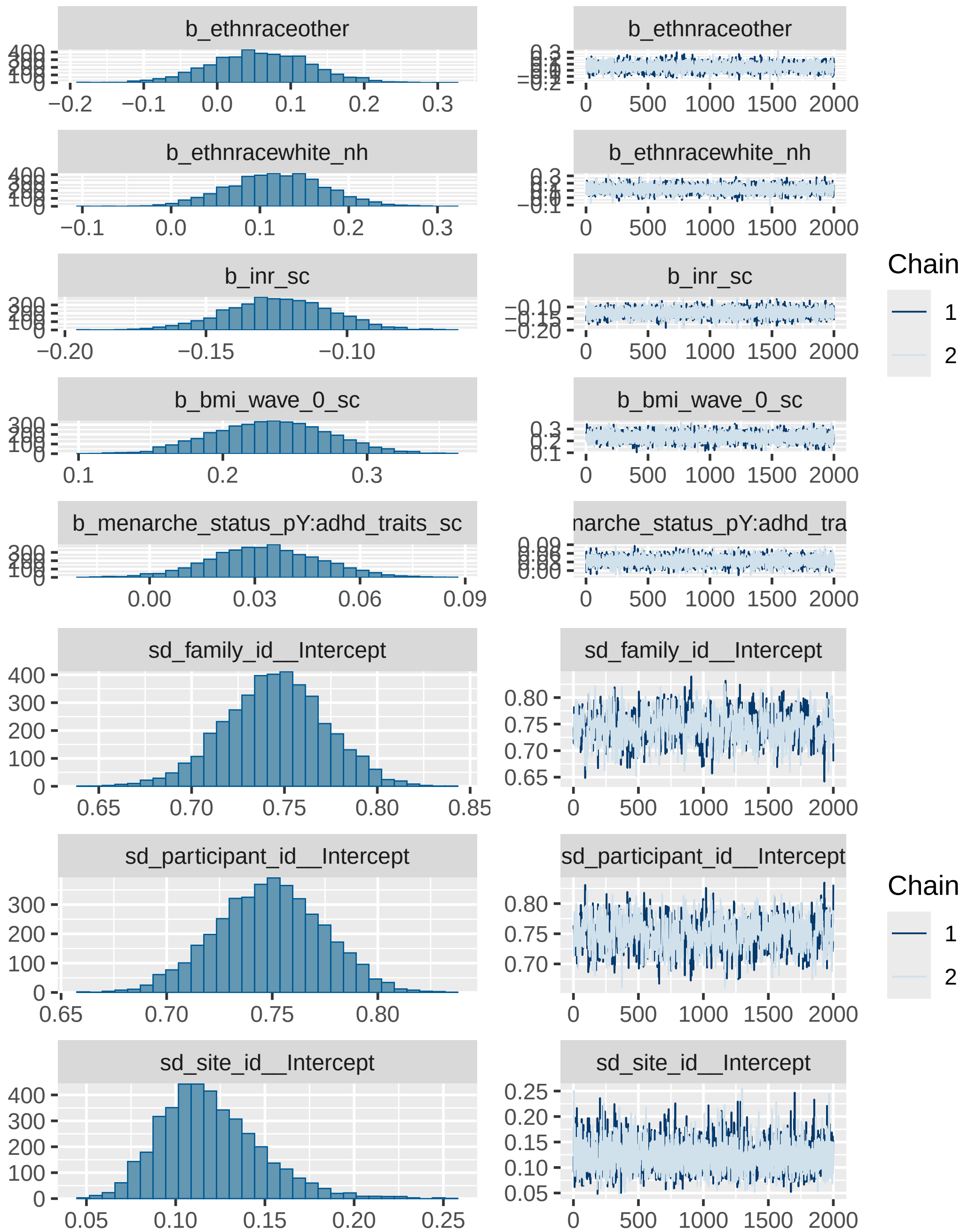
```
Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

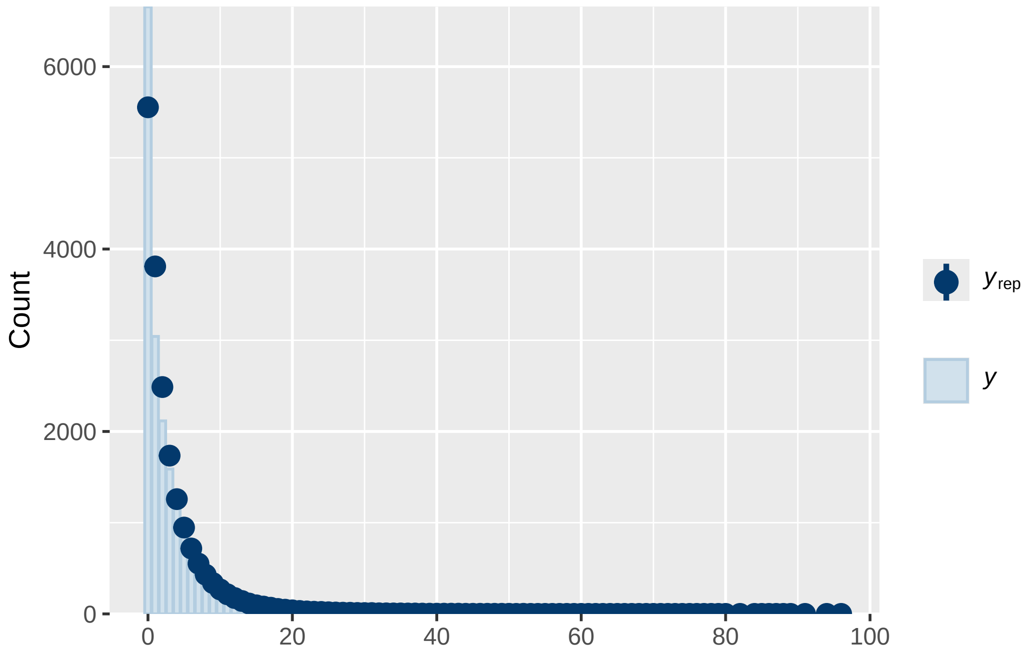
Model 5 BMI diagnostics

```
                prior     class                                coef
         normal(0, 1)         b
         normal(0, 1)         b                      adhd_traits_sc
         normal(0, 1)         b                              age_sc
         normal(0, 1)         b                       bmi_wave_0_sc
         normal(0, 1)         b                    ethnracehispanic
         normal(0, 1)         b                       ethnraceother
         normal(0, 1)         b                    ethnracewhite_nh
         normal(0, 1)         b                              inr_sc
         normal(0, 1)         b                  menarche_status_pY
         normal(0, 1)         b menarche_status_pY:adhd_traits_sc
 student_t(3, 0, 2.9) Intercept
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd                           Intercept
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd                           Intercept
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd                           Intercept
          group resp dpar nlpar lb ub tag       source
                                                  user
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
```

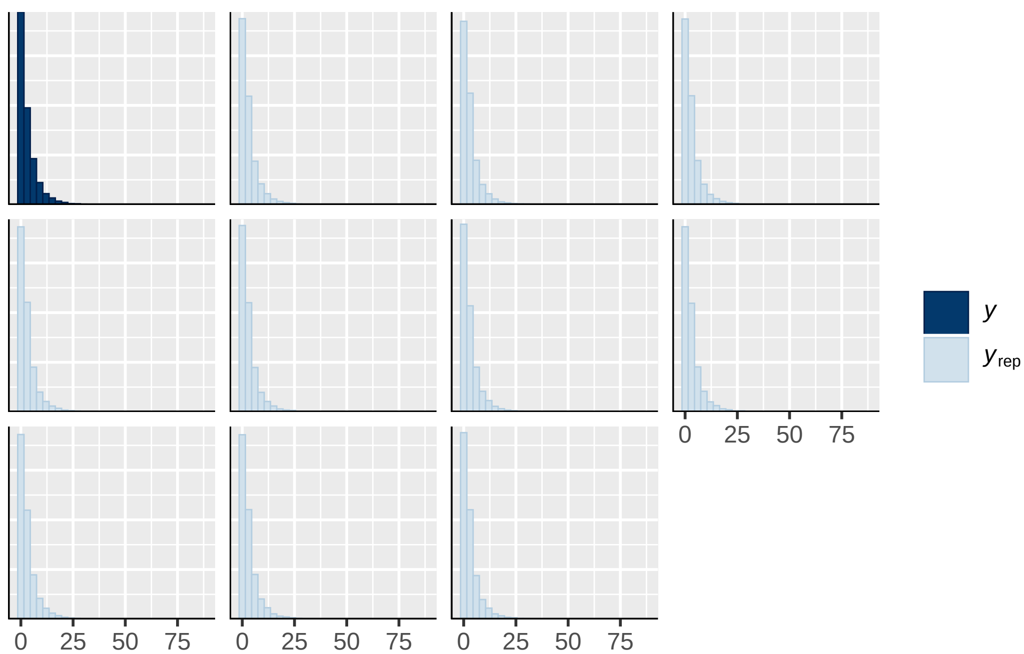
		(vectorized)
		default
	0	default
family_id	0	(vectorized)
family_id	0	(vectorized)
participant_id	0	(vectorized)
participant_id	0	(vectorized)
site_id	0	(vectorized)
site_id	0	(vectorized)



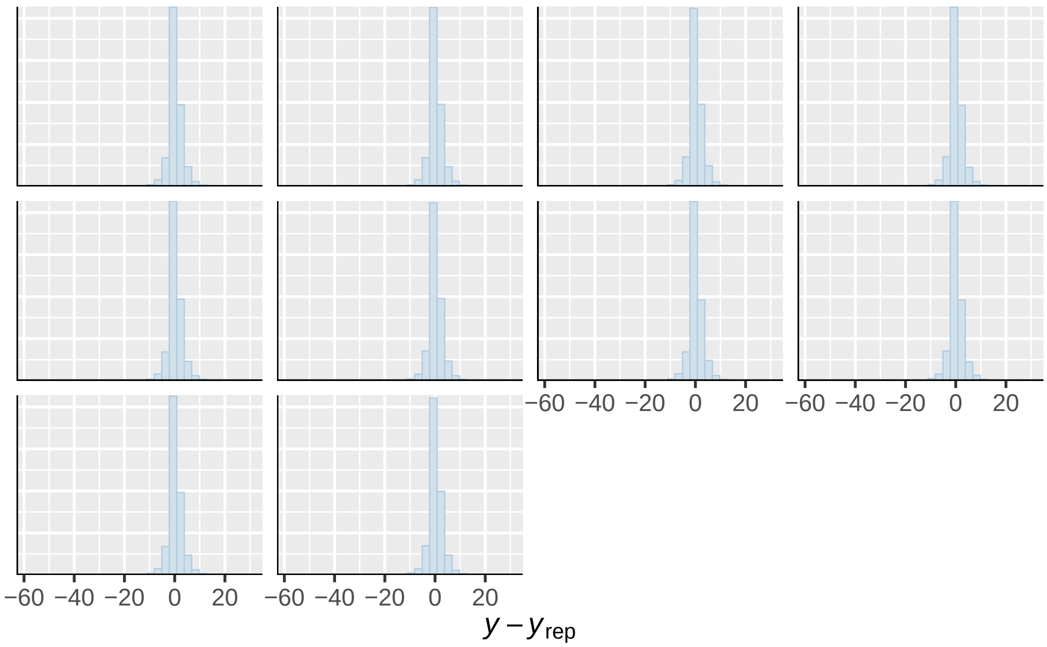

b_ethnraceother
−0.2 −0.1 0.0 0.1 0.2 0.3
b_ethnraceother
0 500 1000 1500 2000
b_ethnracewhite_nh
−0.1 0.0 0.1 0.2 0.3
b_ethnracewhite_nh
0 500 1000 1500 2000
b_inr_sc
−0.20 −0.15 −0.10
b_inr_sc
0 500 1000 1500 2000
b_bmi_wave_0_sc
0.1 0.2 0.3
b_bmi_wave_0_sc
0 500 1000 1500 2000
b_menarche_status_pY:adhd_traits_sc
0.00 0.03 0.06 0.09
narche_status_pY:adhd_tra
0 500 1000 1500 2000
Chain
1
2
sd_family_id__Intercept
400 300 200 100 0
0.65 0.70 0.75 0.80 0.85
sd_family_id__Intercept
0.80 0.75 0.70 0.65
0 500 1000 1500 2000
sd_participant_id__Intercept
300 200 100 0
0.65 0.70 0.75 0.80
sd_participant_id__Intercept
0.80 0.75 0.70
0 500 1000 1500 2000
Chain
1
2
sd_site_id__Intercept
400 300 200 100 0
0.05 0.10 0.15 0.20 0.25
sd_site_id__Intercept
0.25 0.20 0.15 0.10 0.05
0 500 1000 1500 2000



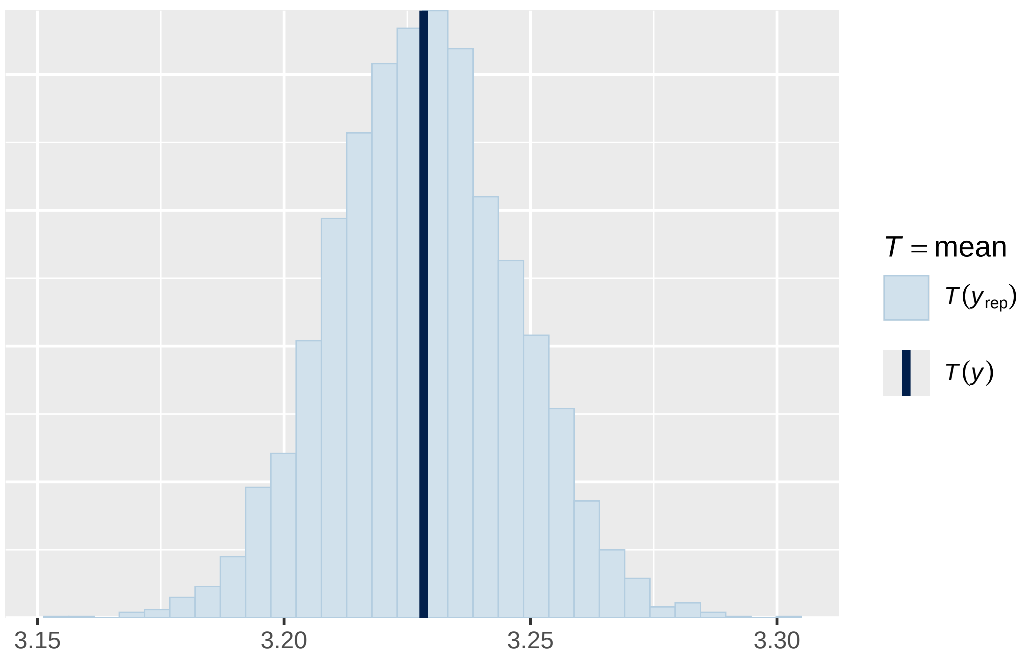

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



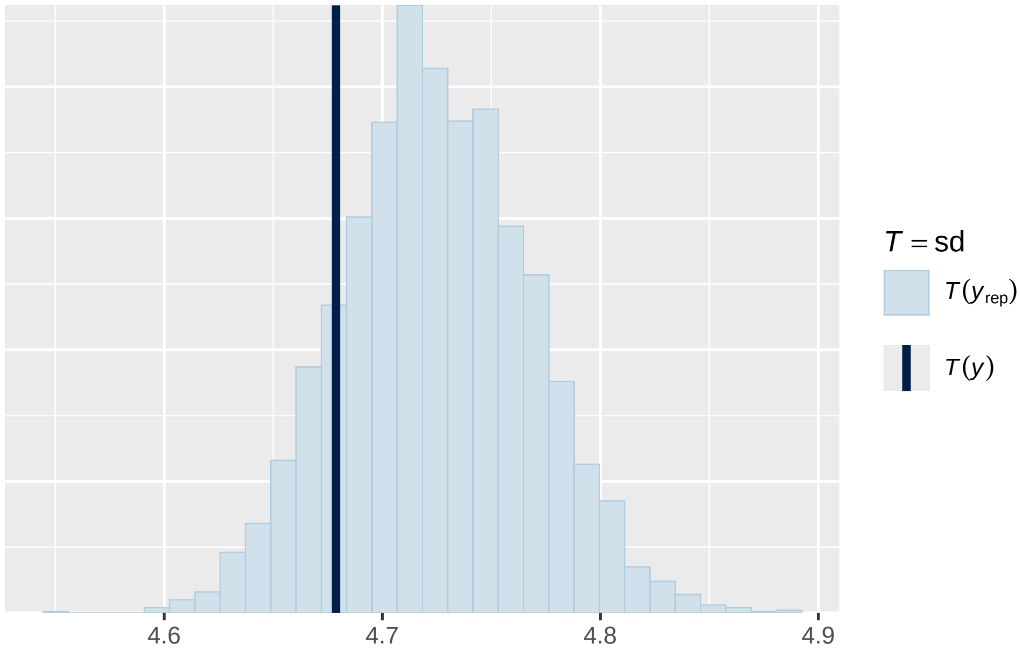

```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



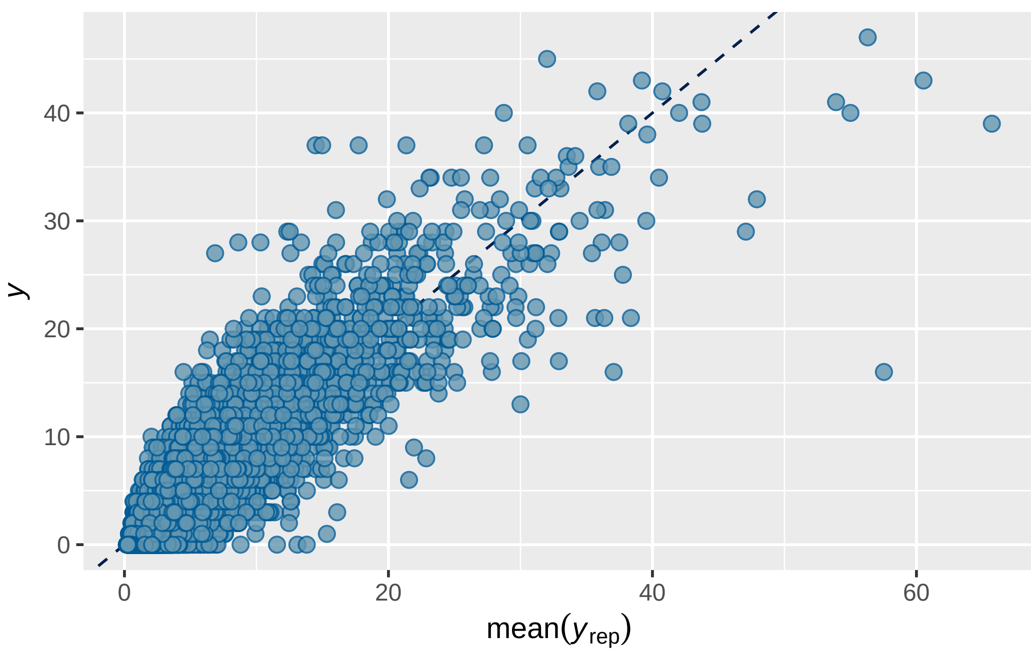

```
Using all posterior draws for ppc type 'stat' by default.
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 1b stimulant output

```
    Estimate  Est.Error      Q2.5     Q97.5
R2 0.7027165 0.00328854 0.6961135 0.7090334


 Family: poisson
  Links: mu = log
Formula: adhd_traits ~ breast_development_p_sc + age_sc + ethnrace + inr_sc + stimulant_statu
   Data: analytic_df (Number of observations: 19364)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 4619)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.816     0.047    0.718    0.904 1.003      586     1036

~participant_id (Number of levels: 5289)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    1.020     0.036    0.950    1.092 1.002      755     1459

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.172     0.038    0.110    0.257 1.000     6338     5466

Regression Coefficients:
                        Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS
Intercept                  0.066     0.069   -0.071    0.199 1.000     5903
breast_development_p_sc    0.058     0.019    0.022    0.095 1.000    14639
age_sc                    -0.067     0.016   -0.098   -0.036 1.000    13765
ethnracehispanic          -0.056     0.076   -0.204    0.090 1.001     6701
ethnraceother              0.015     0.081   -0.148    0.173 1.000     8090
ethnracewhite_nh          -0.031     0.064   -0.155    0.096 1.001     7314
inr_sc                    -0.075     0.022   -0.120   -0.031 1.000    18245
stimulant_statusY          0.287     0.028    0.232    0.342 1.000    18222
                        Tail_ESS
Intercept                   5924
breast_development_p_sc     6463
age_sc                      5792
ethnracehispanic            5827
ethnraceother               6369
ethnracewhite_nh            5977
```

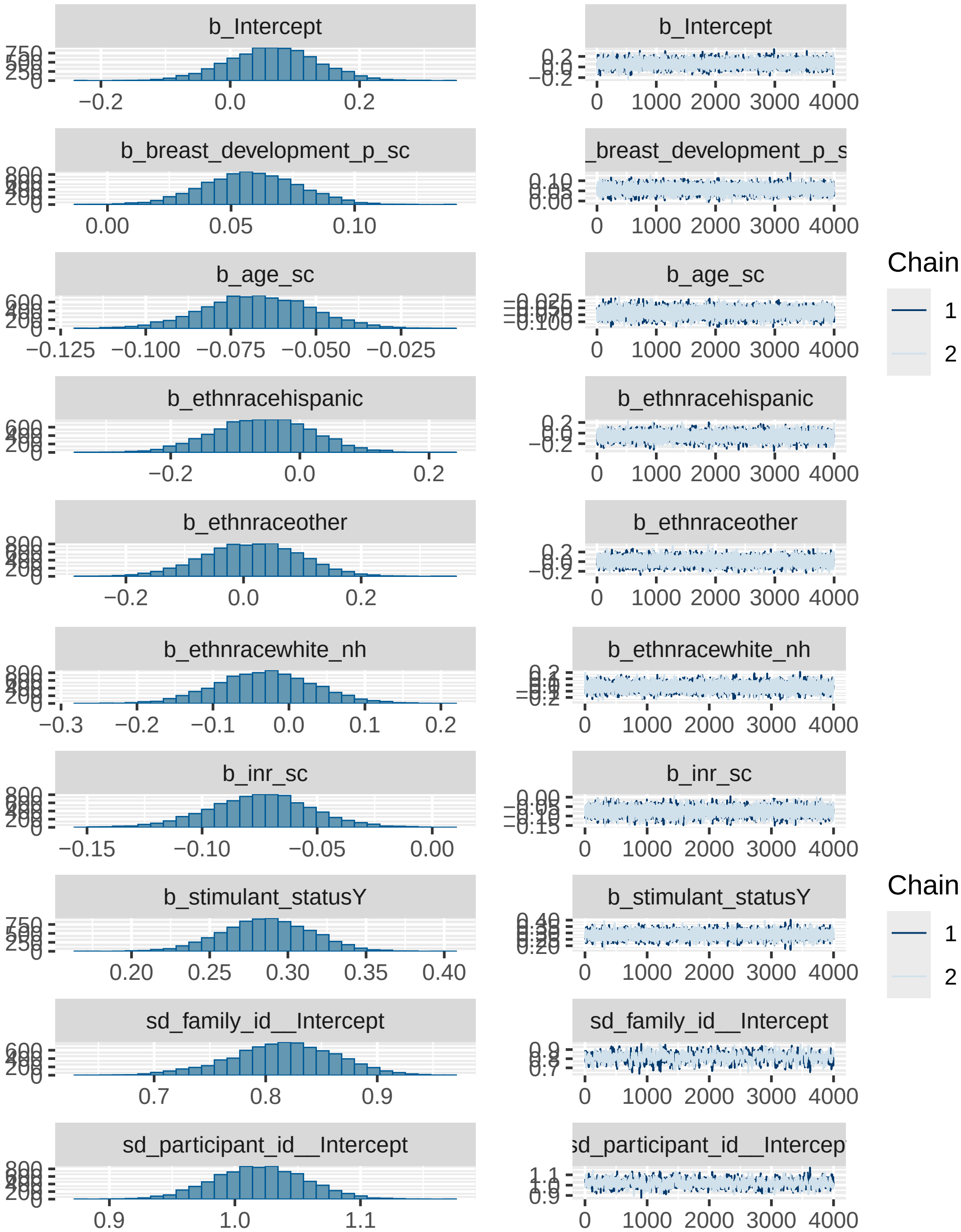
```
inr_sc                      4999
stimulant_statusY           5488
```

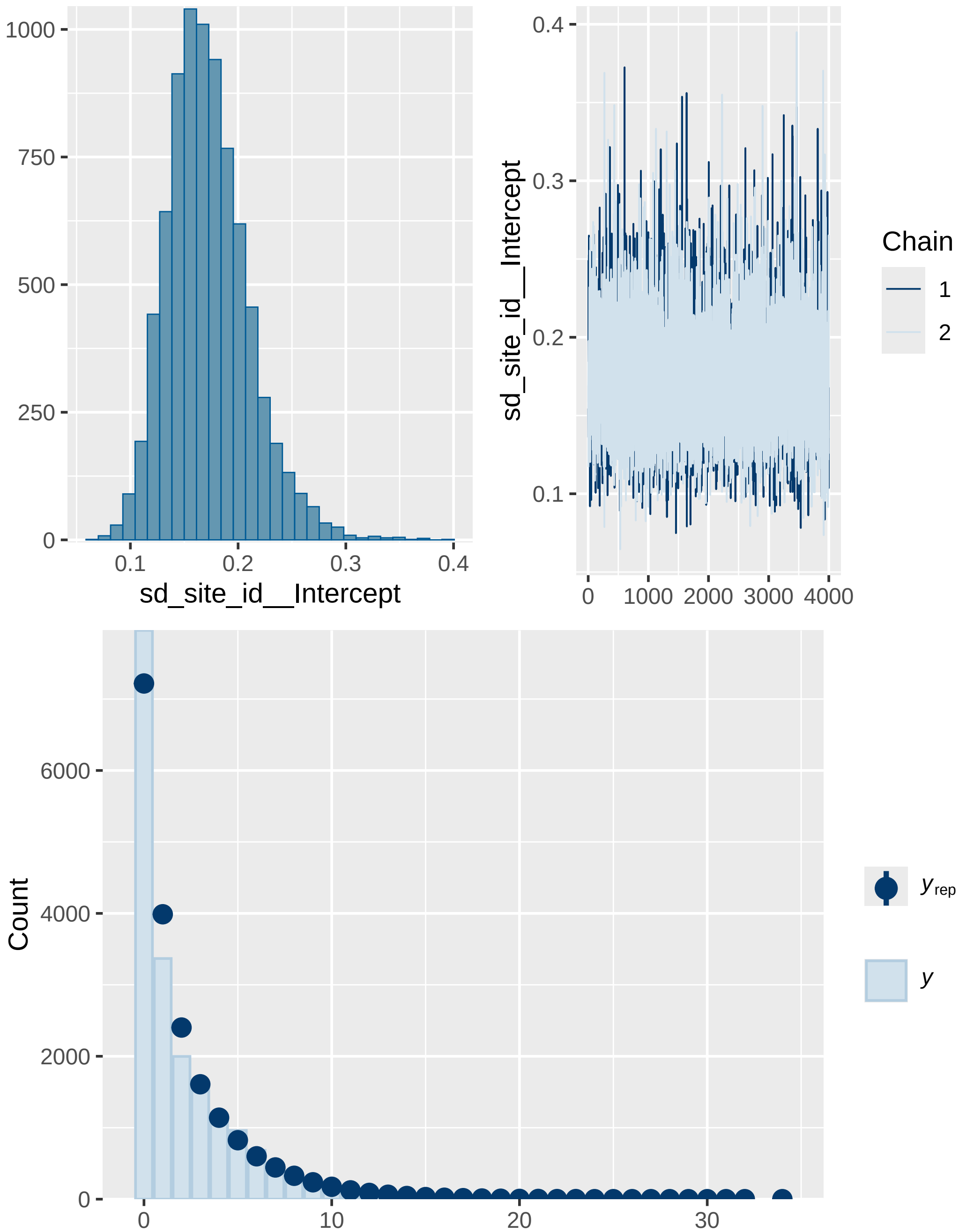
Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Model 1b stimulant diagnostics

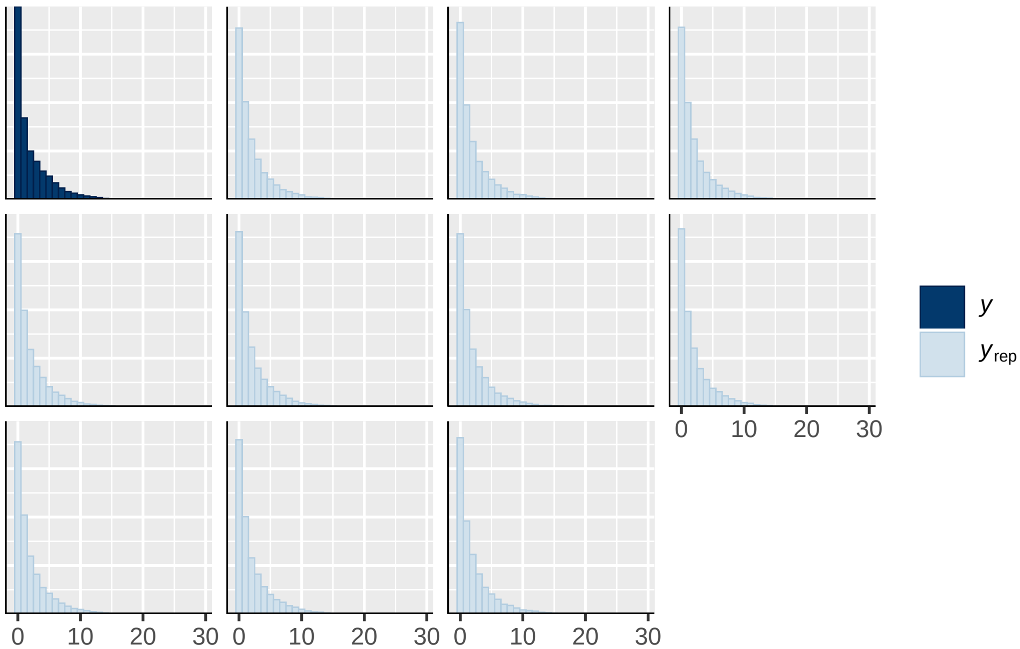
```
                prior     class                    coef          group resp
         normal(0, 1)         b                                            
         normal(0, 1)         b                  age_sc                    
         normal(0, 1)         b breast_development_p_sc                    
         normal(0, 1)         b        ethnracehispanic                    
         normal(0, 1)         b           ethnraceother                    
         normal(0, 1)         b        ethnracewhite_nh                    
         normal(0, 1)         b                  inr_sc                    
         normal(0, 1)         b       stimulant_statusY                    
 student_t(3, 0, 2.7) Intercept                                            
 student_t(3, 0, 2.7)        sd                                            
 student_t(3, 0, 2.7)        sd                              family_id     
 student_t(3, 0, 2.7)        sd               Intercept      family_id     
 student_t(3, 0, 2.7)        sd                         participant_id     
 student_t(3, 0, 2.7)        sd               Intercept participant_id     
 student_t(3, 0, 2.7)        sd                                site_id     
 student_t(3, 0, 2.7)        sd               Intercept        site_id     
 dpar nlpar lb ub tag       source
                              user
                      (vectorized)
                      (vectorized)
                      (vectorized)
                      (vectorized)
                      (vectorized)
                      (vectorized)
                      (vectorized)
                           default
              0            default
              0       (vectorized)
              0       (vectorized)
              0       (vectorized)
              0       (vectorized)
              0       (vectorized)
```

0 (vectorized)

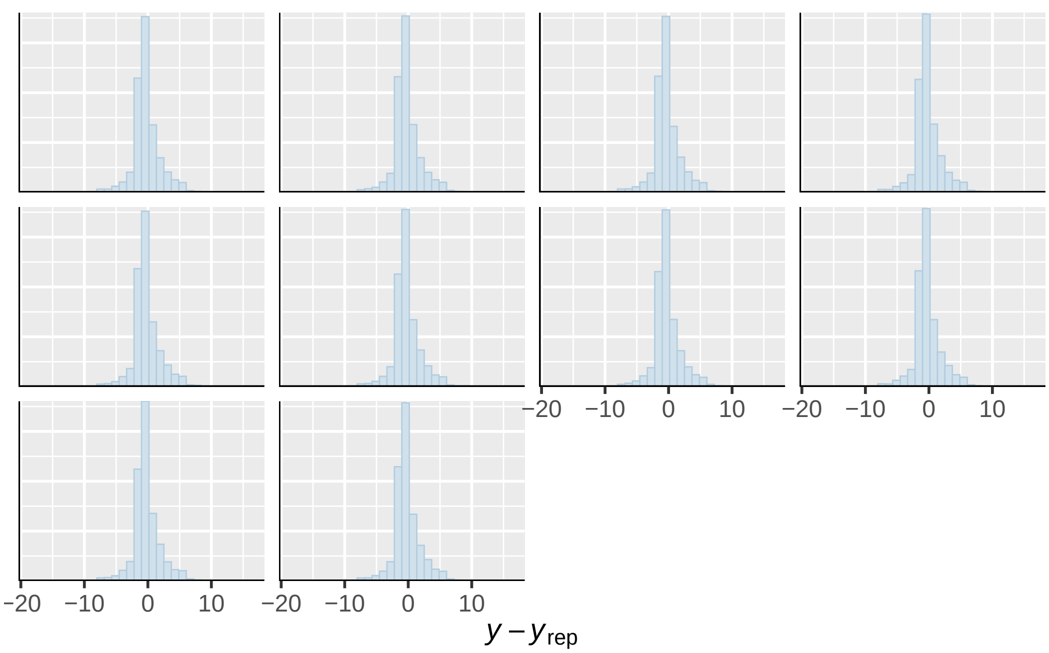





```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

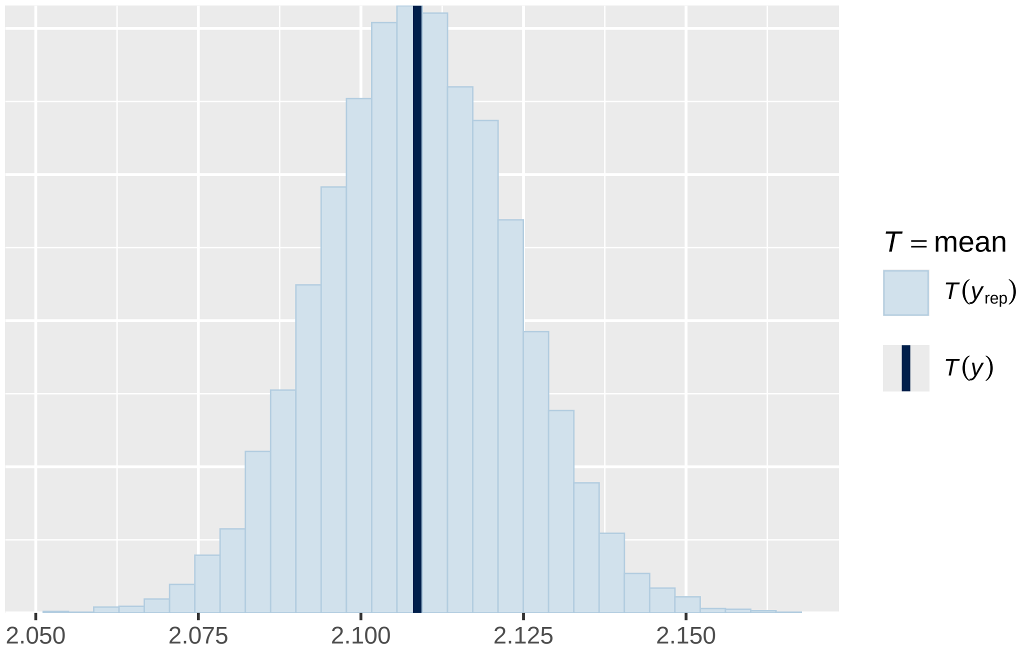


Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

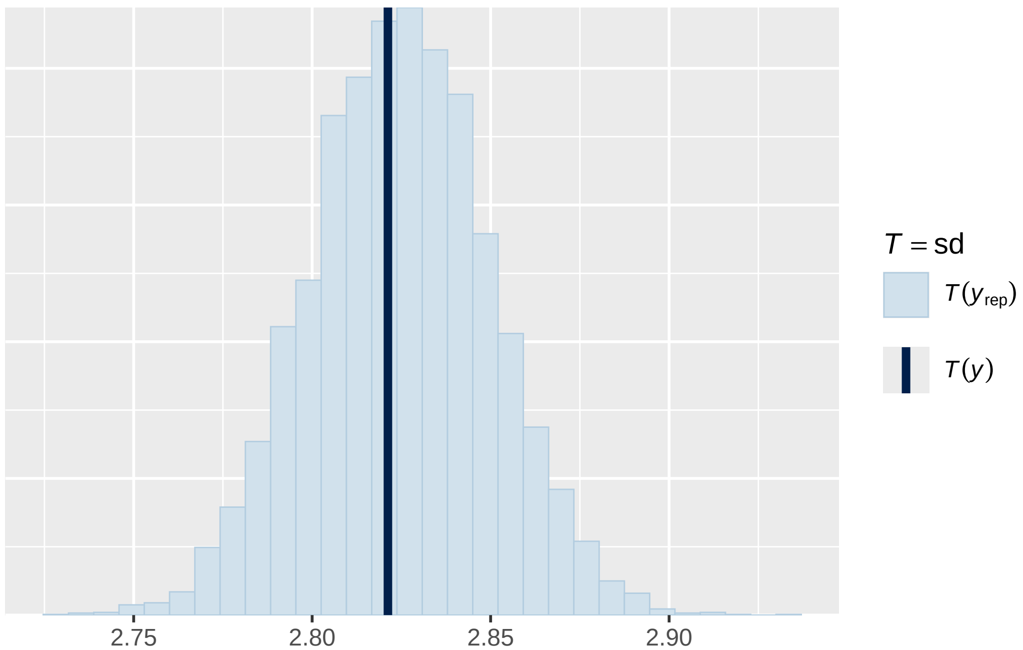


Using all posterior draws for ppc type 'stat' by default.

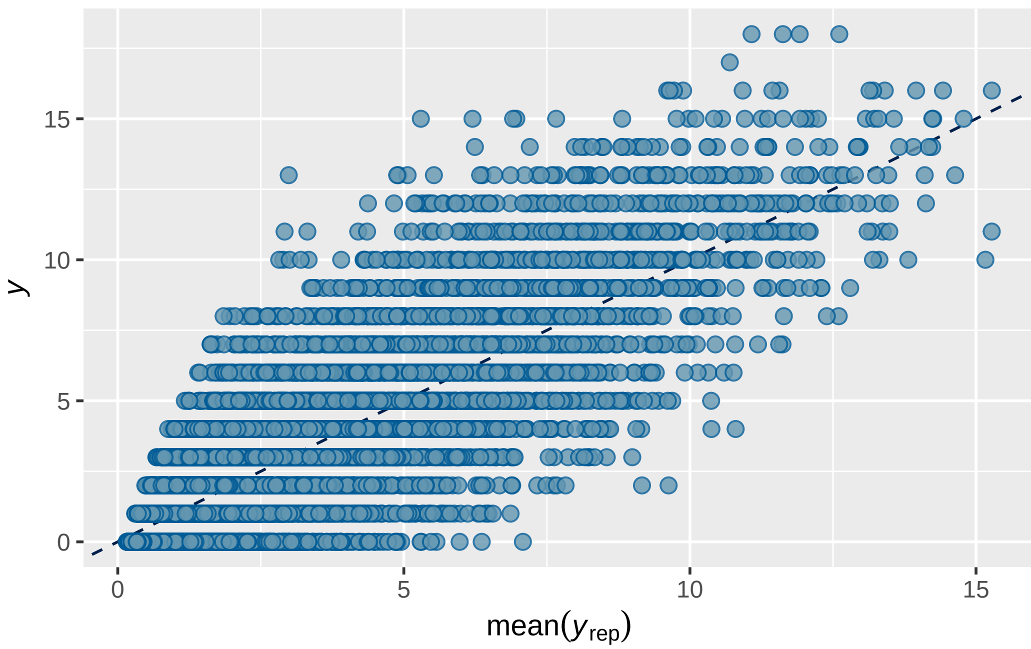
```
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 4 stimulant output

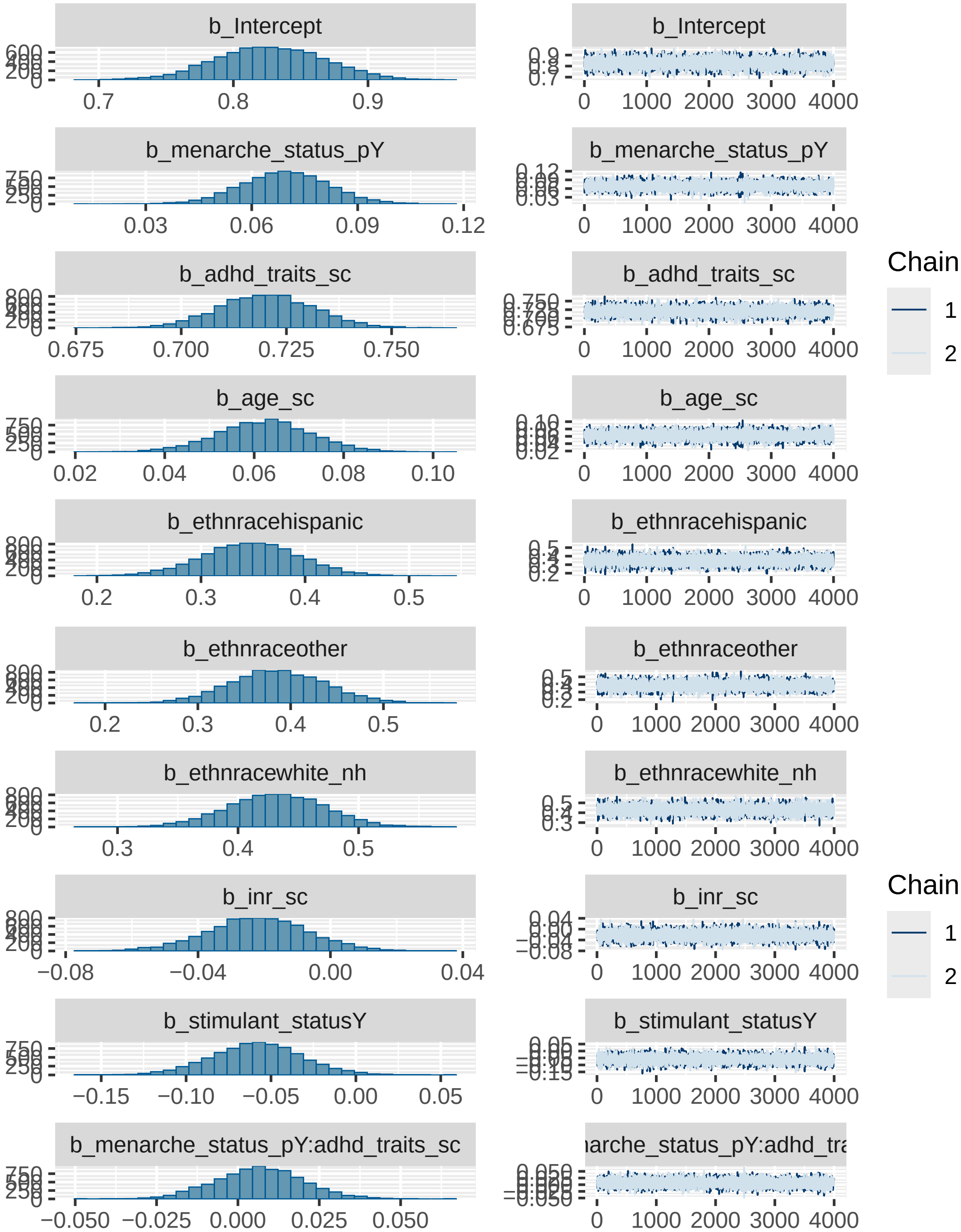
```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.7362039 0.002200039 0.7317853 0.7404623


 Family: poisson
  Links: mu = log
Formula: internalising ~ menarche_status_p * adhd_traits_sc + age_sc + ethnrace + inr_sc + st
   Data: analytic_df (Number of observations: 19381)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 4626)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.660     0.018    0.624    0.694 1.001     1819     3114

~participant_id (Number of levels: 5297)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.487     0.019    0.451    0.524 1.001     1165     2344
```

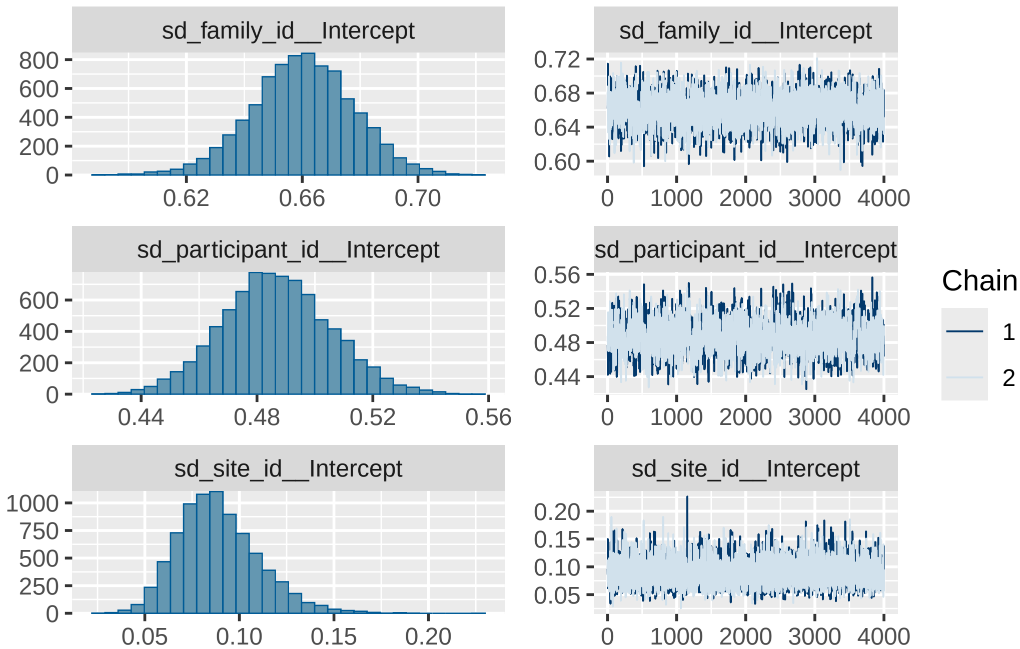
```
~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.089     0.022    0.053    0.137 1.000     6525     6311

Regression Coefficients:
                                  Estimate Est.Error l-95% CI u-95% CI  Rhat
Intercept                            0.828     0.041    0.749    0.906 1.000
menarche_status_pY                   0.069     0.012    0.046    0.094 1.000
adhd_traits_sc                       0.720     0.011    0.698    0.743 1.000
age_sc                               0.062     0.010    0.041    0.083 1.000
ethnracehispanic                     0.348     0.047    0.256    0.439 1.001
ethnraceother                        0.386     0.051    0.287    0.487 1.000
ethnracewhite_nh                     0.432     0.040    0.352    0.509 1.000
inr_sc                              -0.022     0.015   -0.052    0.008 1.000
stimulant_statusY                   -0.058     0.026   -0.112   -0.006 1.001
menarche_status_pY:adhd_traits_sc    0.007     0.013   -0.019    0.033 1.000
                                  Bulk_ESS Tail_ESS
Intercept                             9148     6292
menarche_status_pY                   16104     6097
adhd_traits_sc                       16702     6559
age_sc                               15947     5806
ethnracehispanic                      9922     6155
ethnraceother                         9329     6174
ethnracewhite_nh                      9419     5808
inr_sc                               15357     5580
stimulant_statusY                    15745     5596
menarche_status_pY:adhd_traits_sc    14922     6436

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

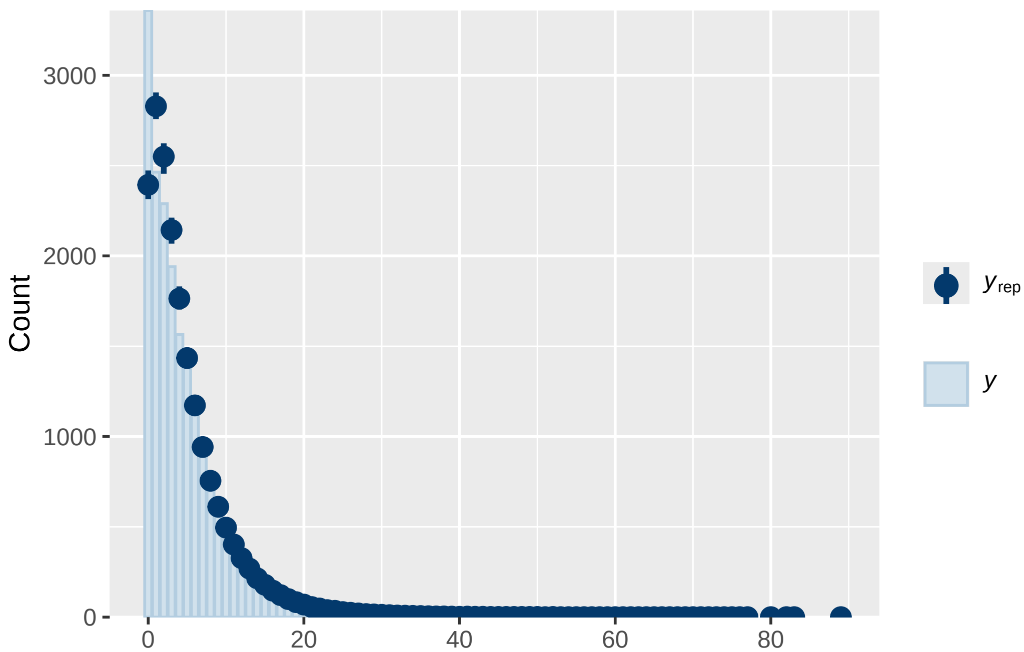
Model 4 stimulant diagnostics

```
          prior     class                               coef
   normal(0, 1)         b
   normal(0, 1)         b                     adhd_traits_sc
   normal(0, 1)         b                             age_sc
   normal(0, 1)         b                   ethnracehispanic
   normal(0, 1)         b                      ethnraceother
   normal(0, 1)         b                   ethnracewhite_nh
```

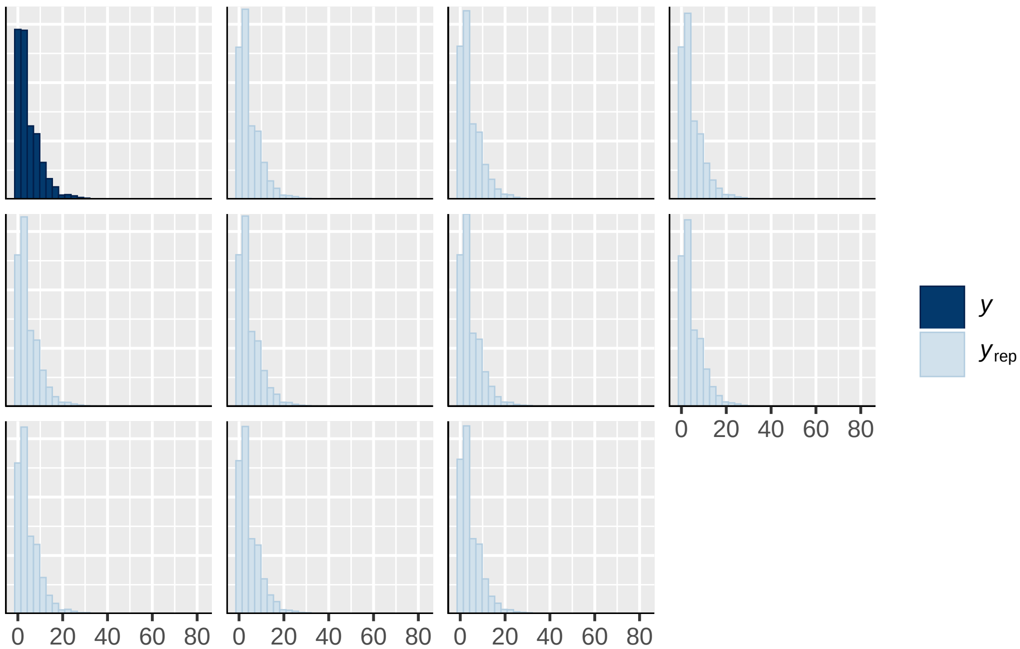
```
          normal(0, 1)         b                              inr_sc
          normal(0, 1)         b                  menarche_status_pY
          normal(0, 1)         b menarche_status_pY:adhd_traits_sc
          normal(0, 1)         b                   stimulant_statusY
student_t(3, 1.1, 2.5) Intercept
  student_t(3, 0, 2.5)        sd
  student_t(3, 0, 2.5)        sd
  student_t(3, 0, 2.5)        sd                           Intercept
  student_t(3, 0, 2.5)        sd
  student_t(3, 0, 2.5)        sd                           Intercept
  student_t(3, 0, 2.5)        sd
  student_t(3, 0, 2.5)        sd                           Intercept
         group resp dpar nlpar lb ub tag       source
                                                 user
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                              default
                                0             default
     family_id                  0        (vectorized)
     family_id                  0        (vectorized)
participant_id                  0        (vectorized)
participant_id                  0        (vectorized)
       site_id                  0        (vectorized)
       site_id                  0        (vectorized)
```


b_Intercept
b_menarche_status_pY
b_adhd_traits_sc
b_age_sc
b_ethnracehispanic
b_ethnraceother
b_ethnracewhite_nh
b_inr_sc
b_stimulant_statusY
b_menarche_status_pY:adhd_traits_sc
Chain
1
2

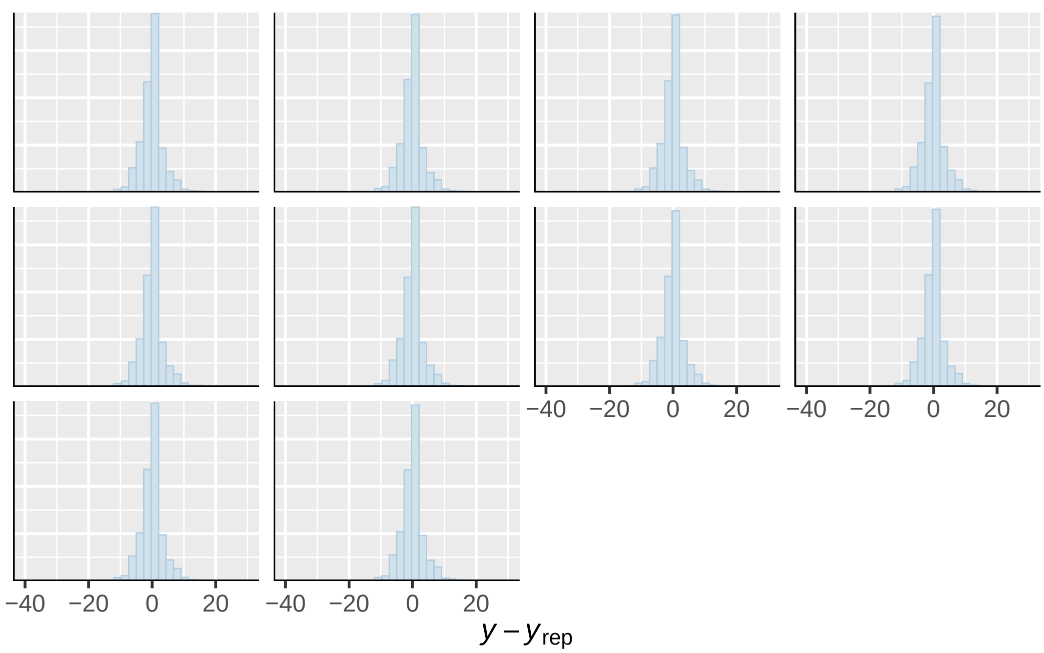





```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

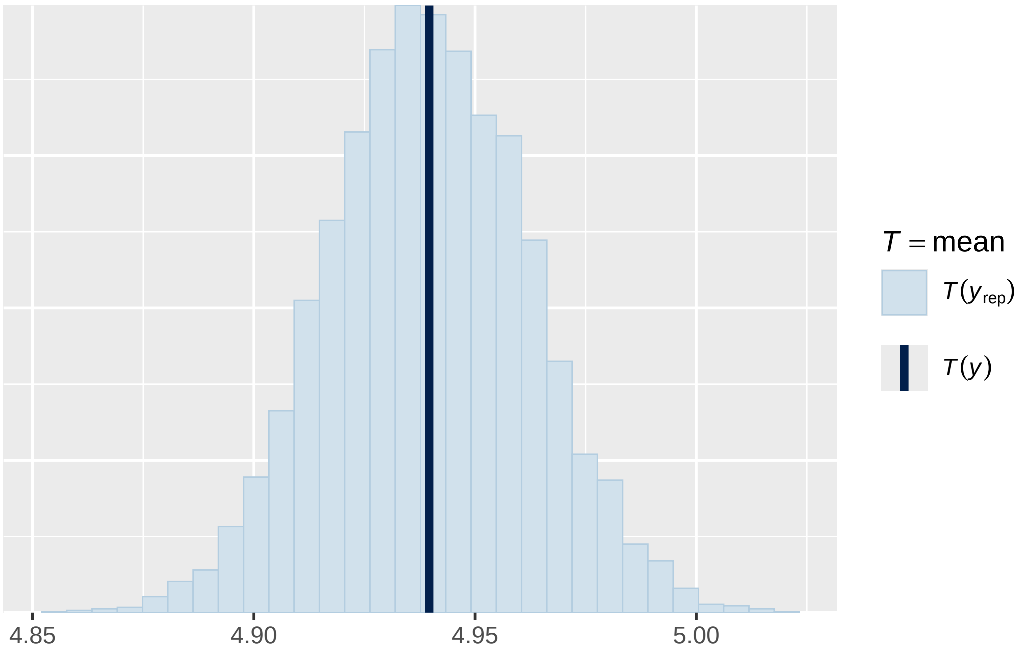



```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

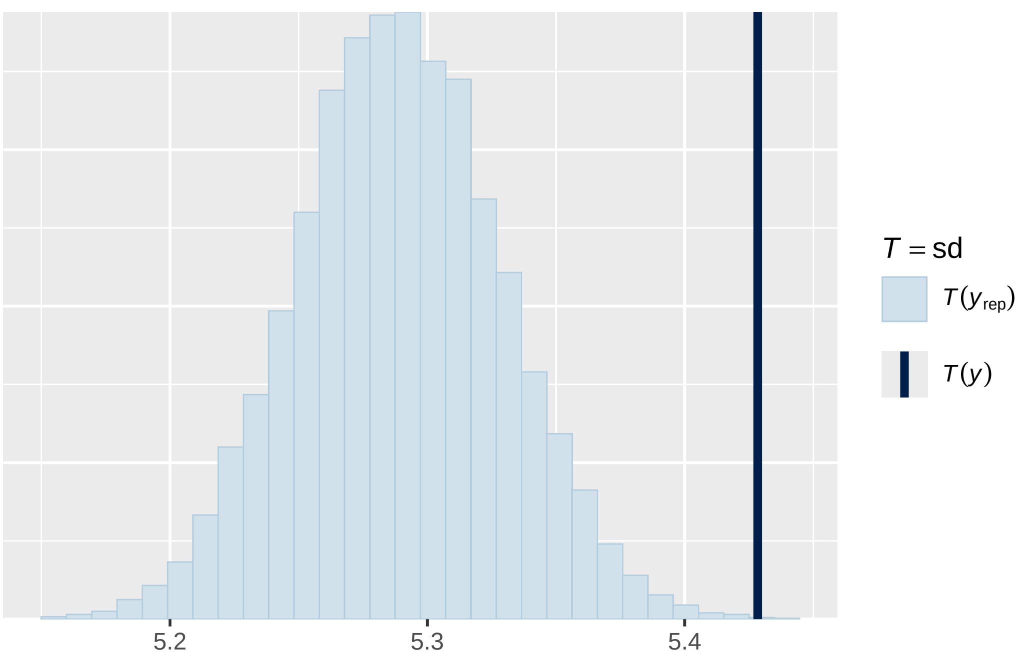



```
Using all posterior draws for ppc type 'stat' by default.
```

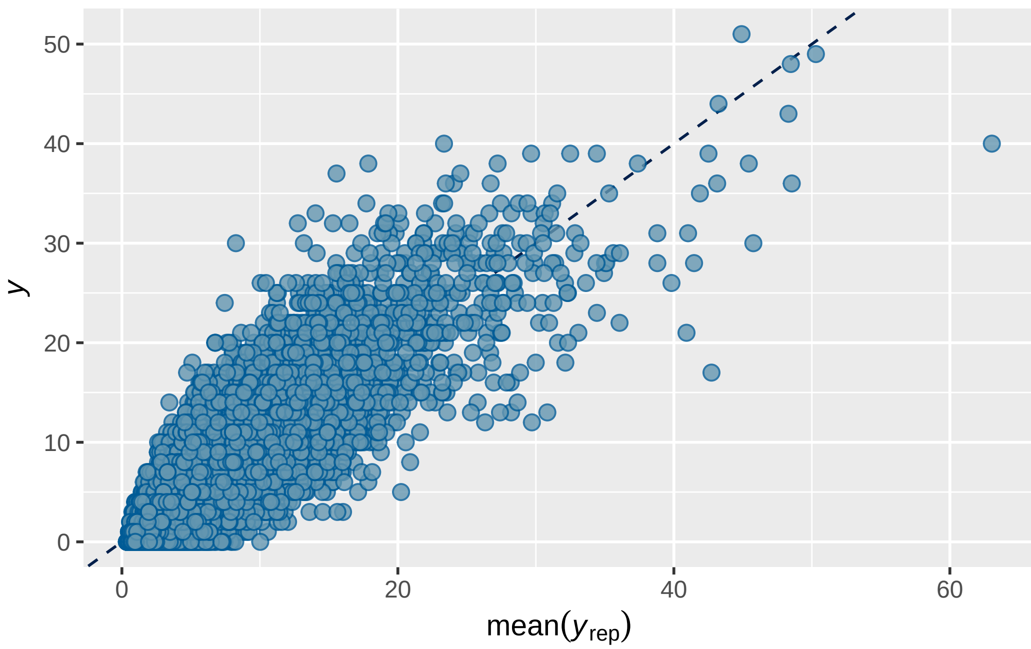
```
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 5 stimulant output

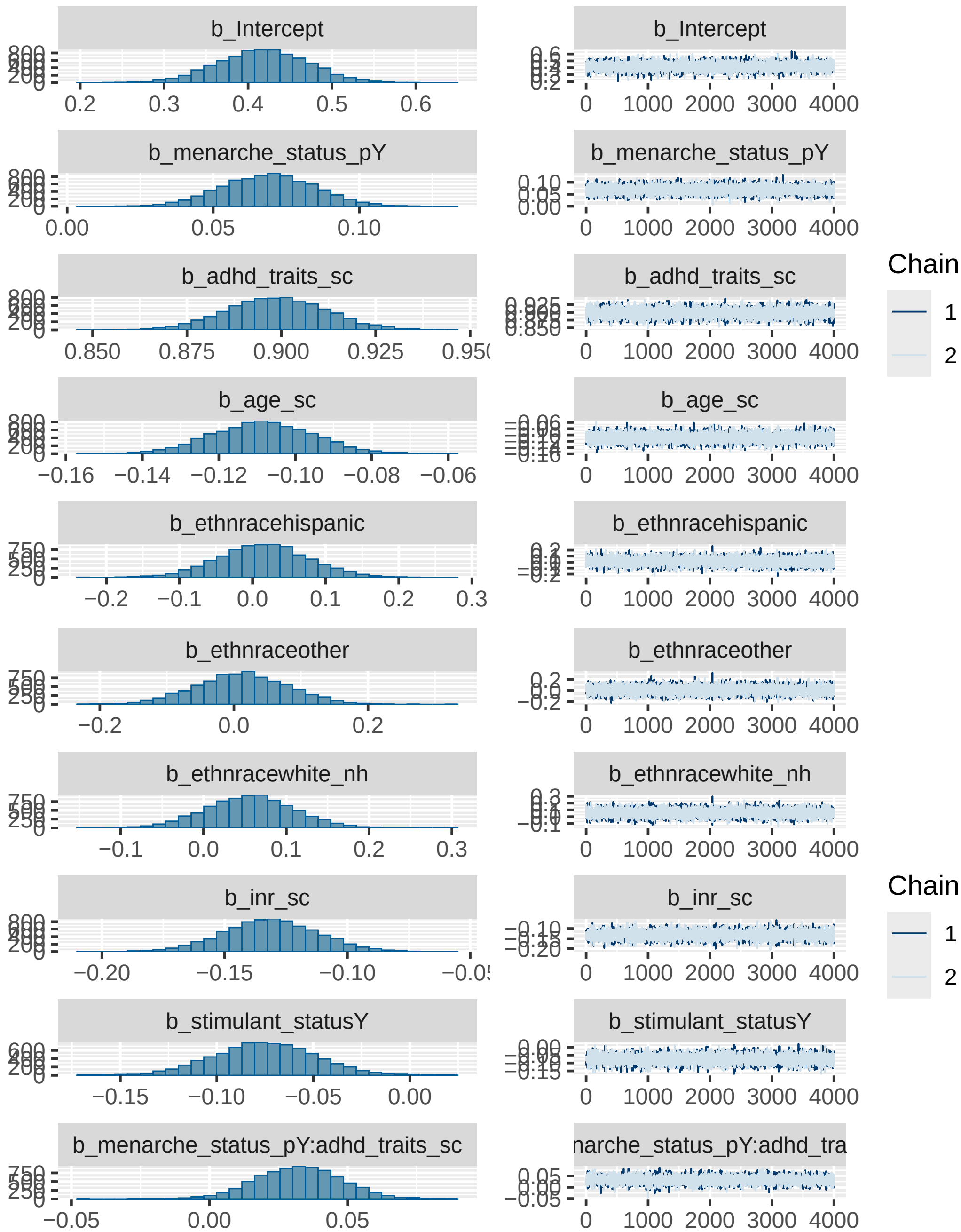
```
    Estimate  Est.Error      Q2.5     Q97.5
R2 0.7888422 0.00203214 0.7847808 0.7926695


 Family: poisson
  Links: mu = log
Formula: externalising ~ menarche_status_p * adhd_traits_sc + age_sc + ethnrace + inr_sc + st
   Data: analytic_df (Number of observations: 19381)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 4626)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.742     0.030    0.683    0.799 1.001      997     1777

~participant_id (Number of levels: 5297)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.756     0.027    0.704    0.811 1.003     1026     1704
```

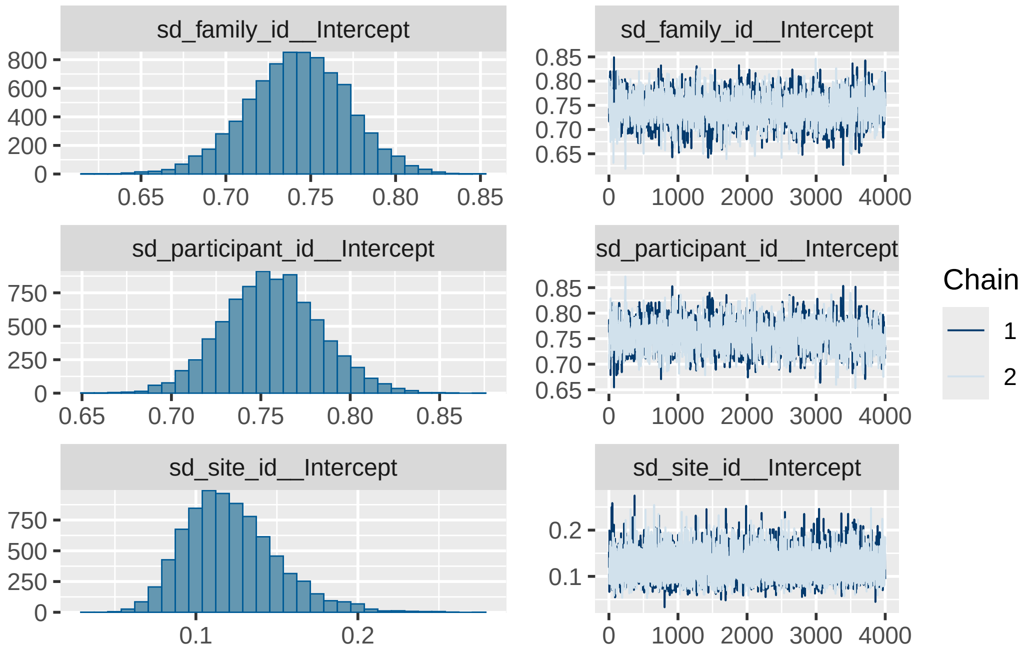
```
~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.122     0.029    0.075    0.189 1.000     7734     6102

Regression Coefficients:
                                  Estimate Est.Error l-95% CI u-95% CI  Rhat
Intercept                            0.418     0.054    0.312    0.523 1.000
menarche_status_pY                   0.069     0.016    0.038    0.100 1.000
adhd_traits_sc                       0.899     0.013    0.873    0.925 1.000
age_sc                              -0.108     0.013   -0.134   -0.083 1.000
ethnracehispanic                     0.015     0.061   -0.103    0.135 1.000
ethnraceother                        0.012     0.068   -0.120    0.143 1.000
ethnracewhite_nh                     0.056     0.053   -0.048    0.160 1.000
inr_sc                              -0.131     0.019   -0.169   -0.094 1.001
stimulant_statusY                   -0.075     0.027   -0.127   -0.023 1.000
menarche_status_pY:adhd_traits_sc    0.032     0.015    0.002    0.062 1.001
                                  Bulk_ESS Tail_ESS
Intercept                             6988     5541
menarche_status_pY                   14966     5962
adhd_traits_sc                       14865     5274
age_sc                               15313     6322
ethnracehispanic                      8316     6193
ethnraceother                         8483     6398
ethnracewhite_nh                      8022     6231
inr_sc                               18023     5441
stimulant_statusY                    17814     4897
menarche_status_pY:adhd_traits_sc    17067     6170

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

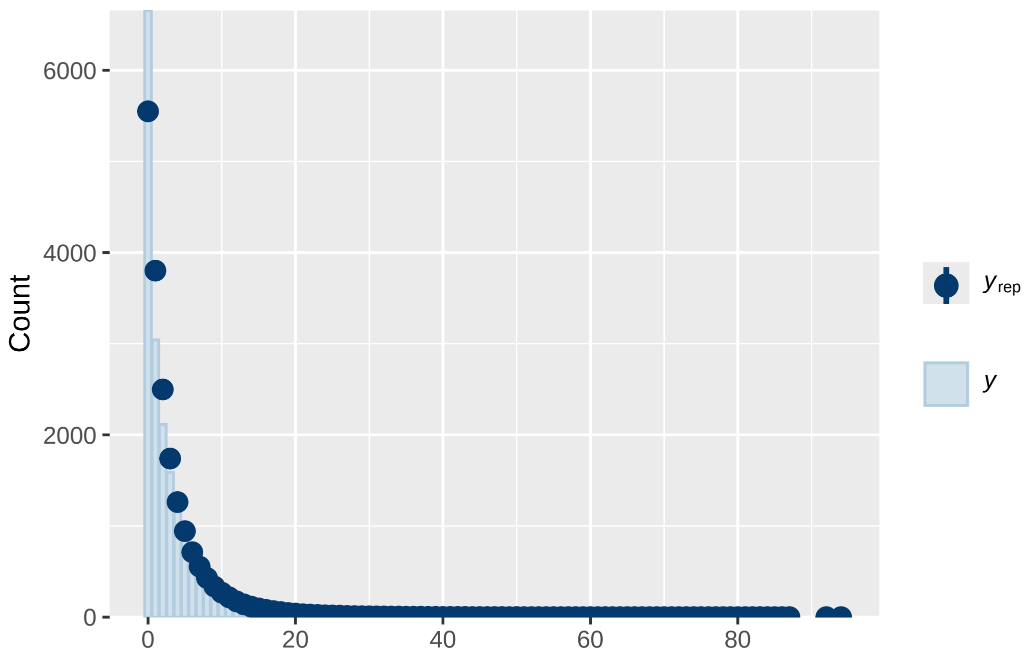
Model 5 stimulant diagnostics

```
       prior     class                              coef
normal(0, 1)         b
normal(0, 1)         b                    adhd_traits_sc
normal(0, 1)         b                            age_sc
normal(0, 1)         b                  ethnracehispanic
normal(0, 1)         b                     ethnraceother
normal(0, 1)         b                  ethnracewhite_nh
```

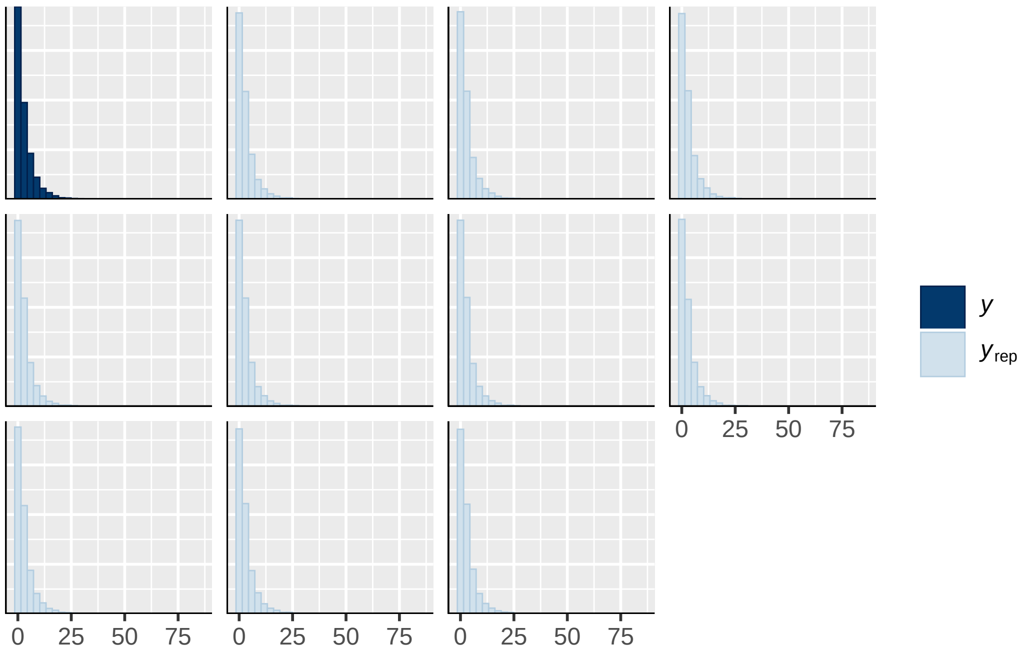
```
        normal(0, 1)         b                                 inr_sc
        normal(0, 1)         b                    menarche_status_pY
        normal(0, 1)         b menarche_status_pY:adhd_traits_sc
        normal(0, 1)         b                     stimulant_statusY
student_t(3, 0, 2.9) Intercept
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd                             Intercept
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd                             Intercept
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd                             Intercept
          group resp dpar nlpar lb ub tag       source
                                                  user
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                               default
                                0              default
      family_id                 0         (vectorized)
      family_id                 0         (vectorized)
 participant_id                 0         (vectorized)
 participant_id                 0         (vectorized)
        site_id                 0         (vectorized)
        site_id                 0         (vectorized)
```


b_Intercept
b_menarche_status_pY
b_adhd_traits_sc
b_age_sc
b_ethnracehispanic
b_ethnraceother
b_ethnracewhite_nh
b_inr_sc
b_stimulant_statusY
b_menarche_status_pY:adhd_traits_sc
narche_status_pY:adhd_tra
Chain
1
2

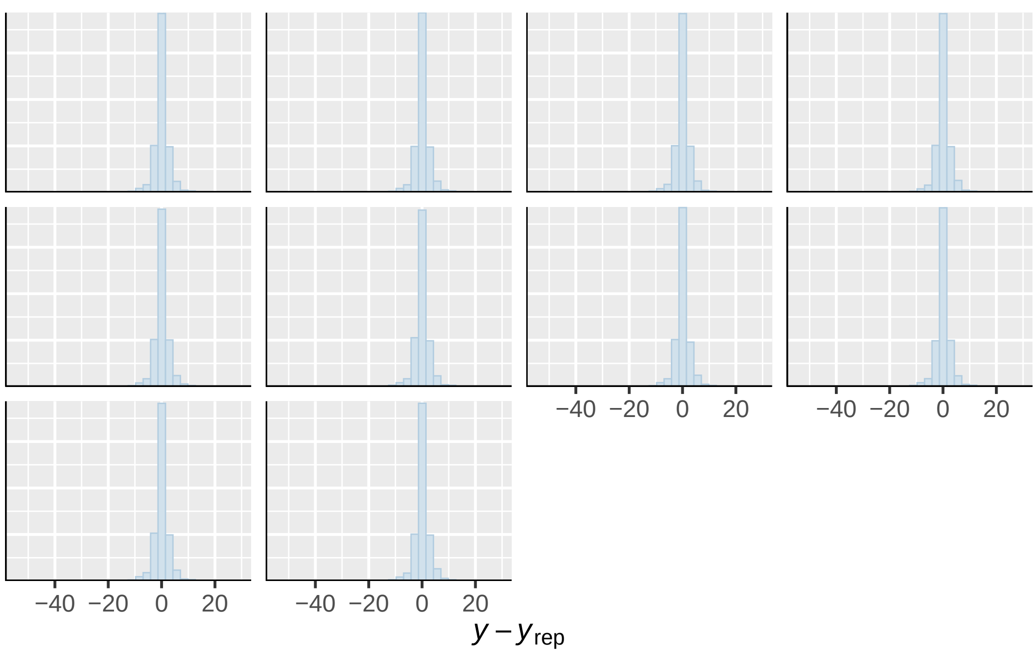




`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

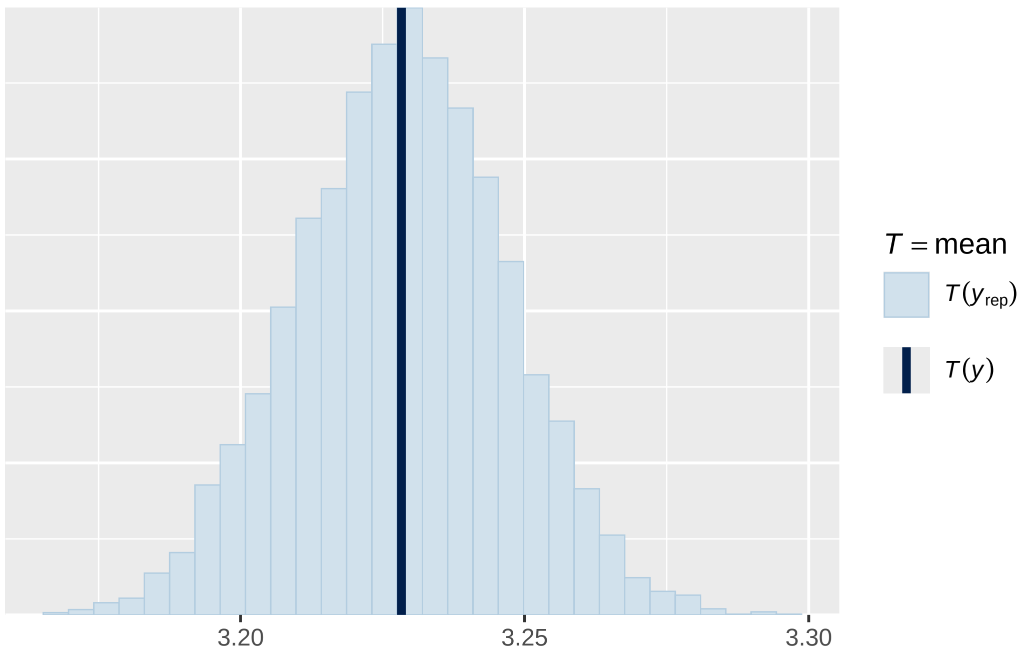



```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

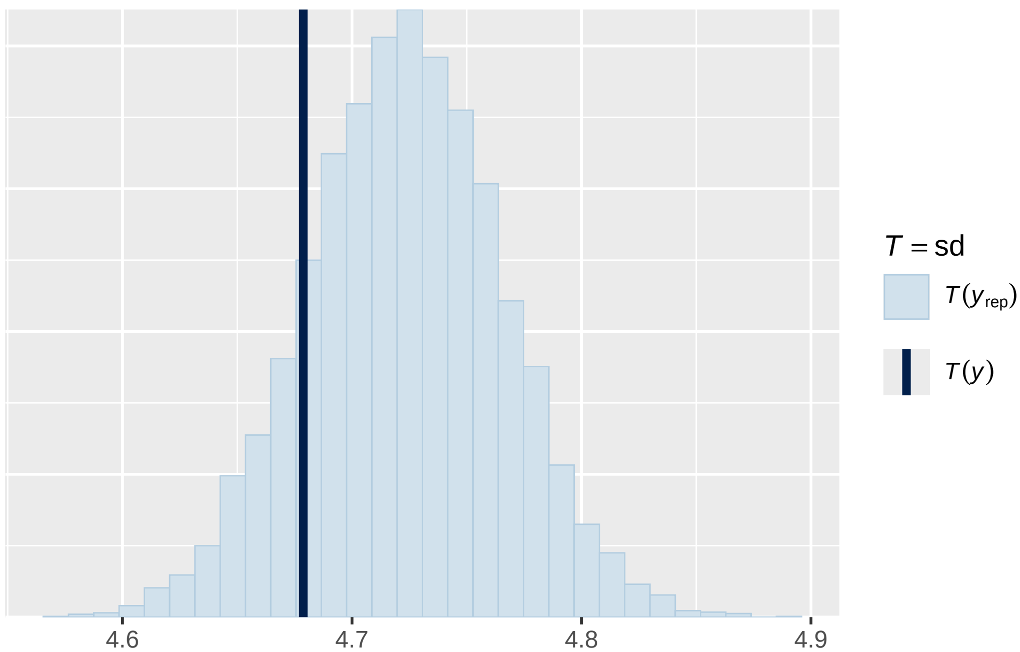



```
Using all posterior draws for ppc type 'stat' by default.
```

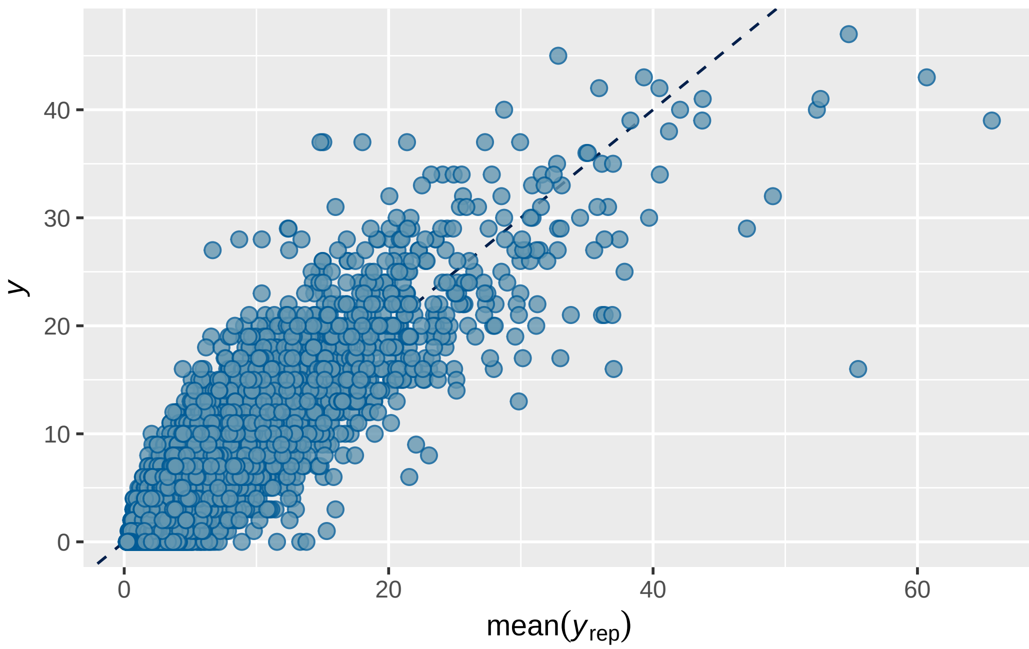
```
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



ABCD ADHD menarche analysis: Output for cross-sectional supplementary/sensitivity models

Model 1b prior 1 output

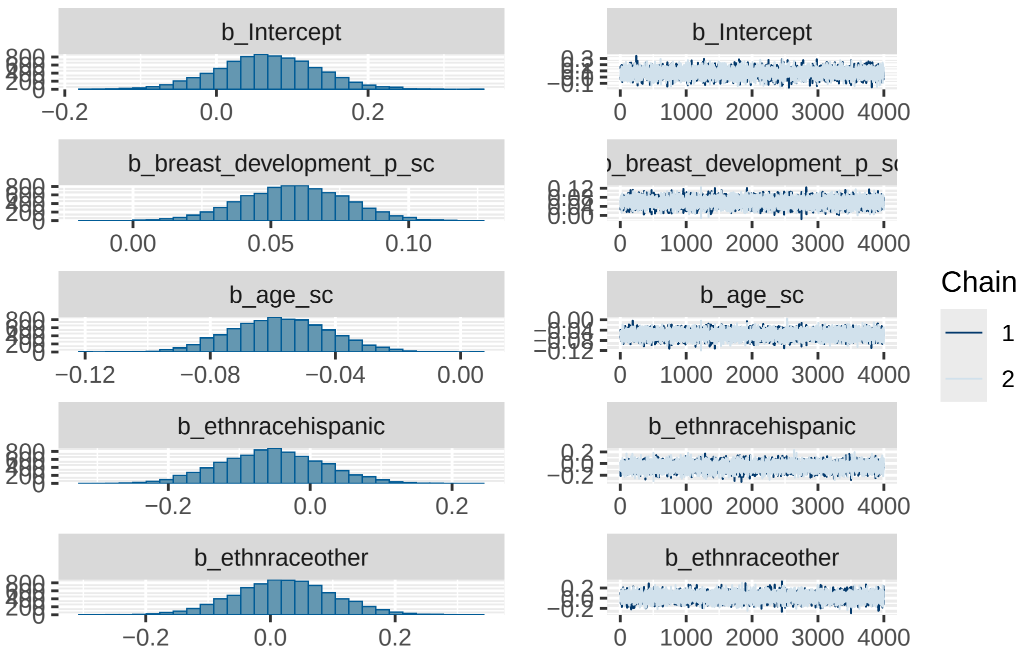
```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.7026681 0.003297907 0.6961068 0.7091473


 Family: poisson
  Links: mu = log
Formula: adhd_traits ~ breast_development_p_sc + age_sc + ethnrace + inr_sc + (1 | participan
   Data: analytic_df (Number of observations: 19364)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 4619)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.834     0.046    0.742    0.919 1.002      659     1369

~participant_id (Number of levels: 5289)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    1.036     0.036    0.966    1.107 1.001      773     1630

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.176     0.038    0.113    0.262 1.000     6842     5715

Regression Coefficients:
                        Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS
Intercept                  0.068     0.067   -0.064    0.198 1.001     6850
```

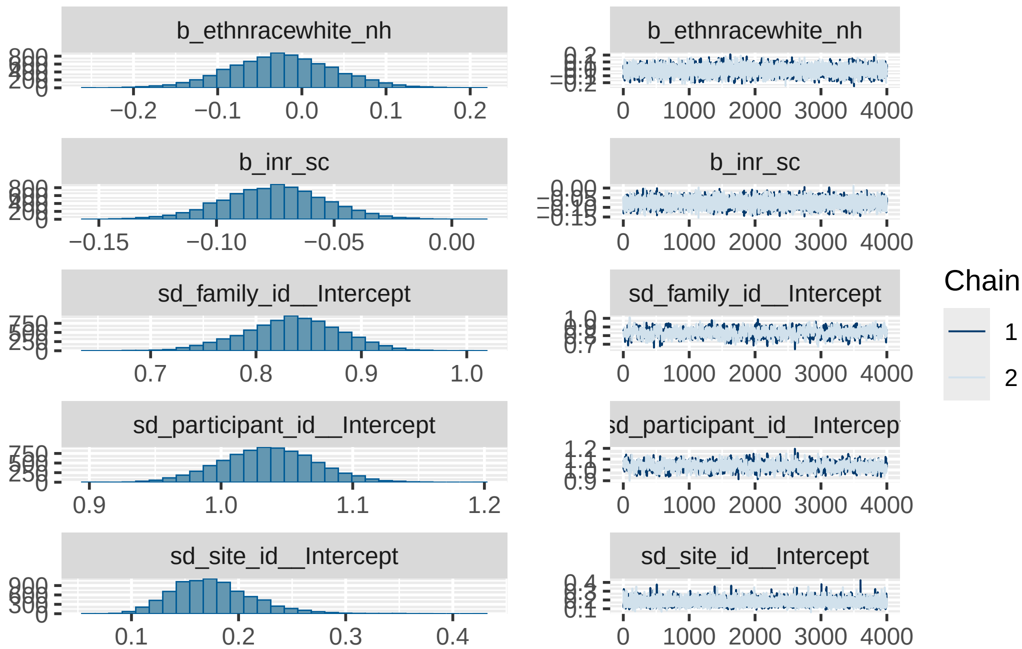
```
breast_development_p_sc    0.058     0.019    0.022    0.095 1.000    15511
age_sc                    -0.058     0.016   -0.087   -0.027 1.000    15136
ethnracehispanic          -0.054     0.072   -0.192    0.087 1.000     8366
ethnraceother              0.022     0.079   -0.131    0.177 1.000     8397
ethnracewhite_nh          -0.023     0.062   -0.146    0.099 1.000     7789
inr_sc                    -0.075     0.022   -0.119   -0.032 1.001    19389
                        Tail_ESS
Intercept                   5900
breast_development_p_sc     6276
age_sc                      6469
ethnracehispanic            5743
ethnraceother               5474
ethnracewhite_nh            6030
inr_sc                      6078

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

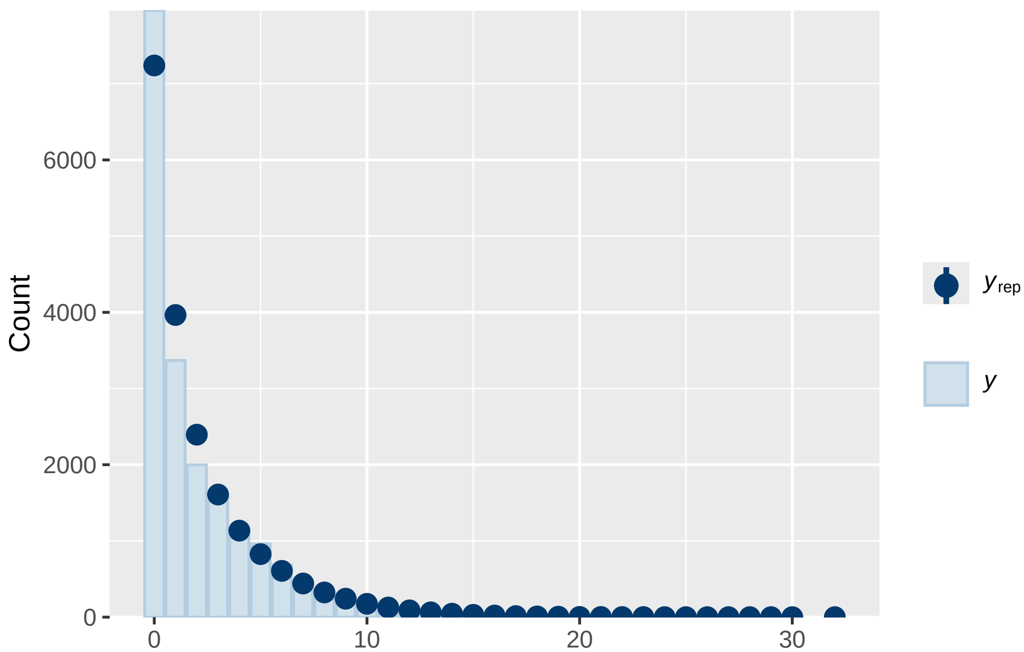
Model 1b prior 1 diagnostics

```
                prior     class                    coef          group resp
      normal(0, 0.25)         b
      normal(0, 0.25)         b                  age_sc
      normal(0, 0.25)         b breast_development_p_sc
      normal(0, 0.25)         b        ethnracehispanic
      normal(0, 0.25)         b           ethnraceother
      normal(0, 0.25)         b        ethnracewhite_nh
      normal(0, 0.25)         b                  inr_sc
 student_t(3, 0, 2.7) Intercept
 student_t(3, 0, 2.7)        sd
 student_t(3, 0, 2.7)        sd                              family_id
 student_t(3, 0, 2.7)        sd               Intercept      family_id
 student_t(3, 0, 2.7)        sd                         participant_id
 student_t(3, 0, 2.7)        sd               Intercept participant_id
 student_t(3, 0, 2.7)        sd                                site_id
 student_t(3, 0, 2.7)        sd               Intercept        site_id
 dpar nlpar lb ub tag       source
                              user
                      (vectorized)
                      (vectorized)
                      (vectorized)
```

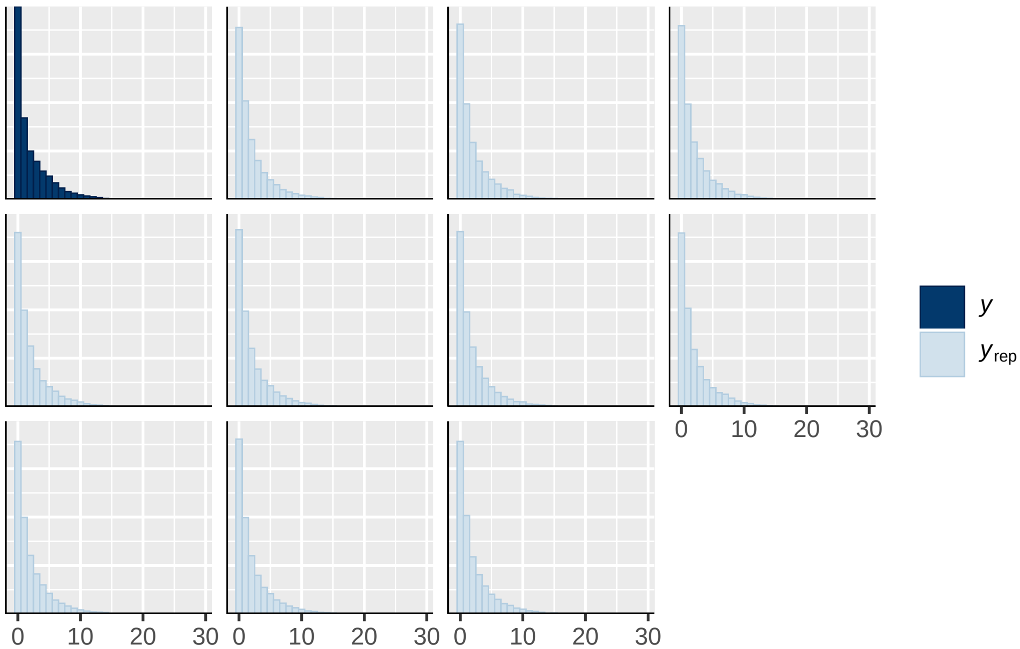
	(vectorized)
	(vectorized)
	(vectorized)
	default
0	default
0	(vectorized)
0	(vectorized)
0	(vectorized)
0	(vectorized)
0	(vectorized)
0	(vectorized)



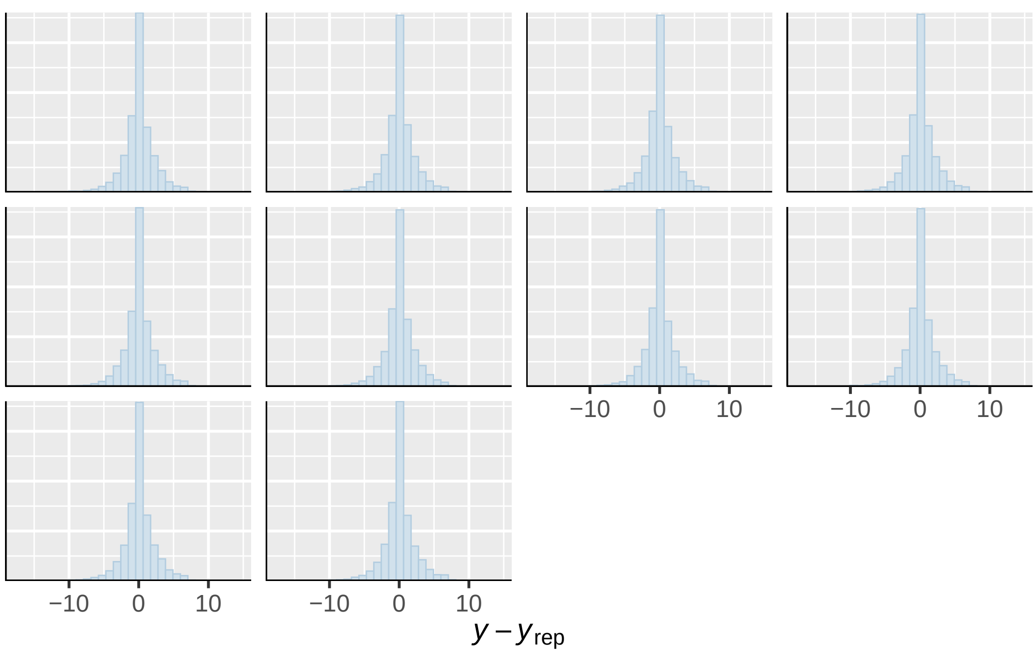





```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

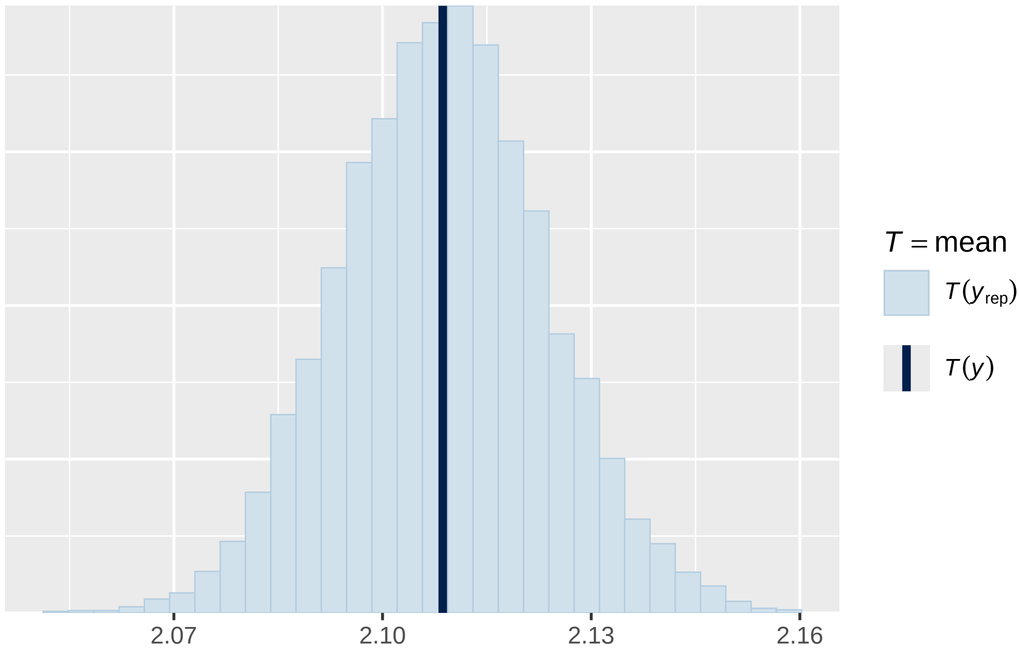



```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

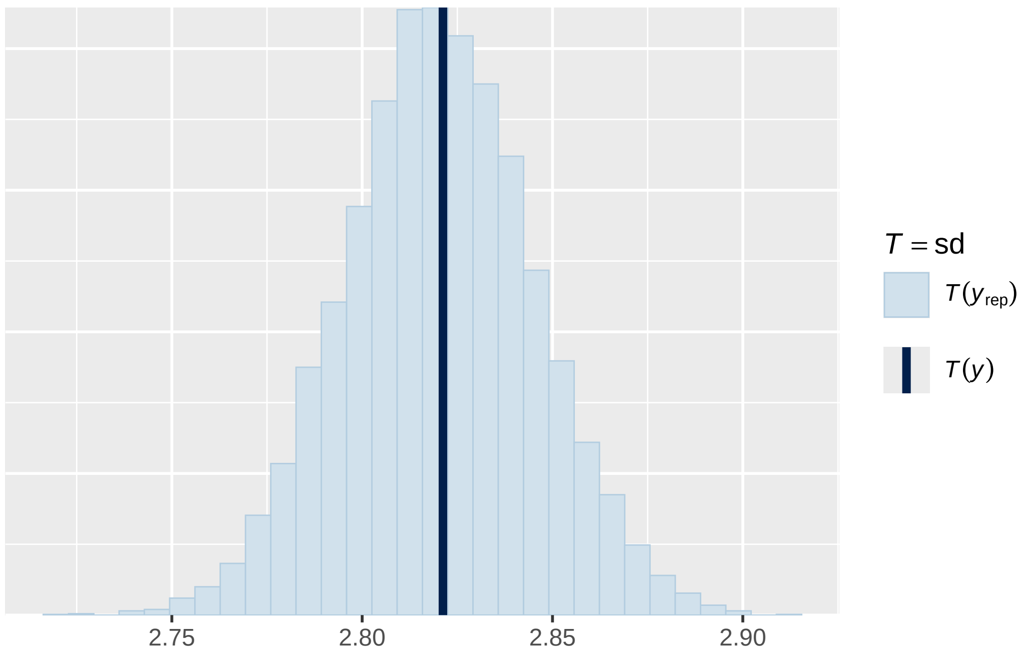



```
Using all posterior draws for ppc type 'stat' by default.
```

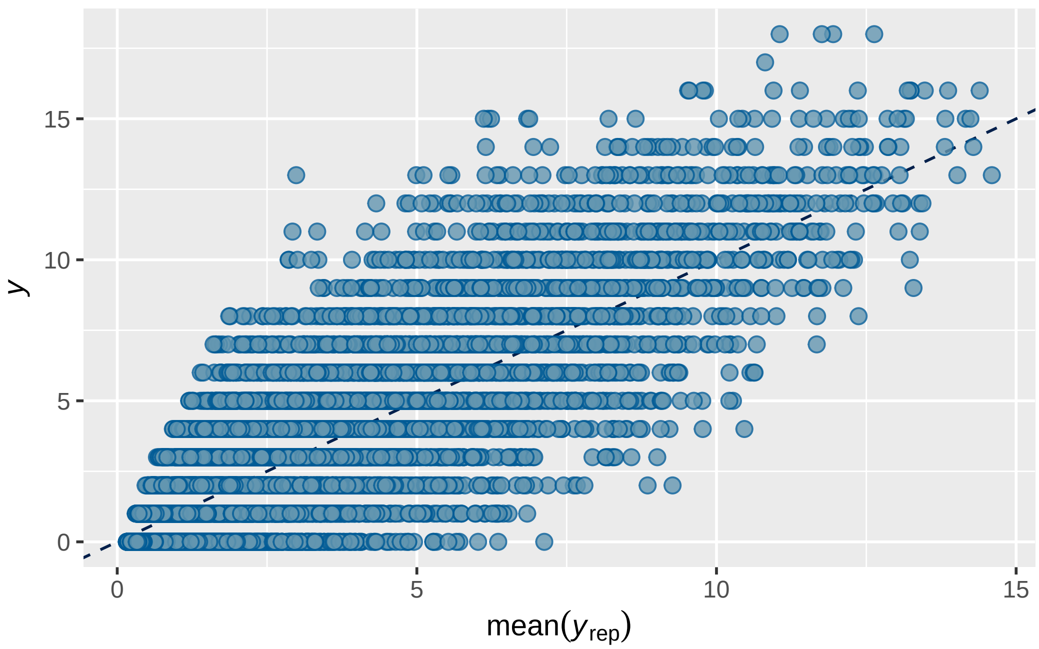
```
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 4 prior 1 output

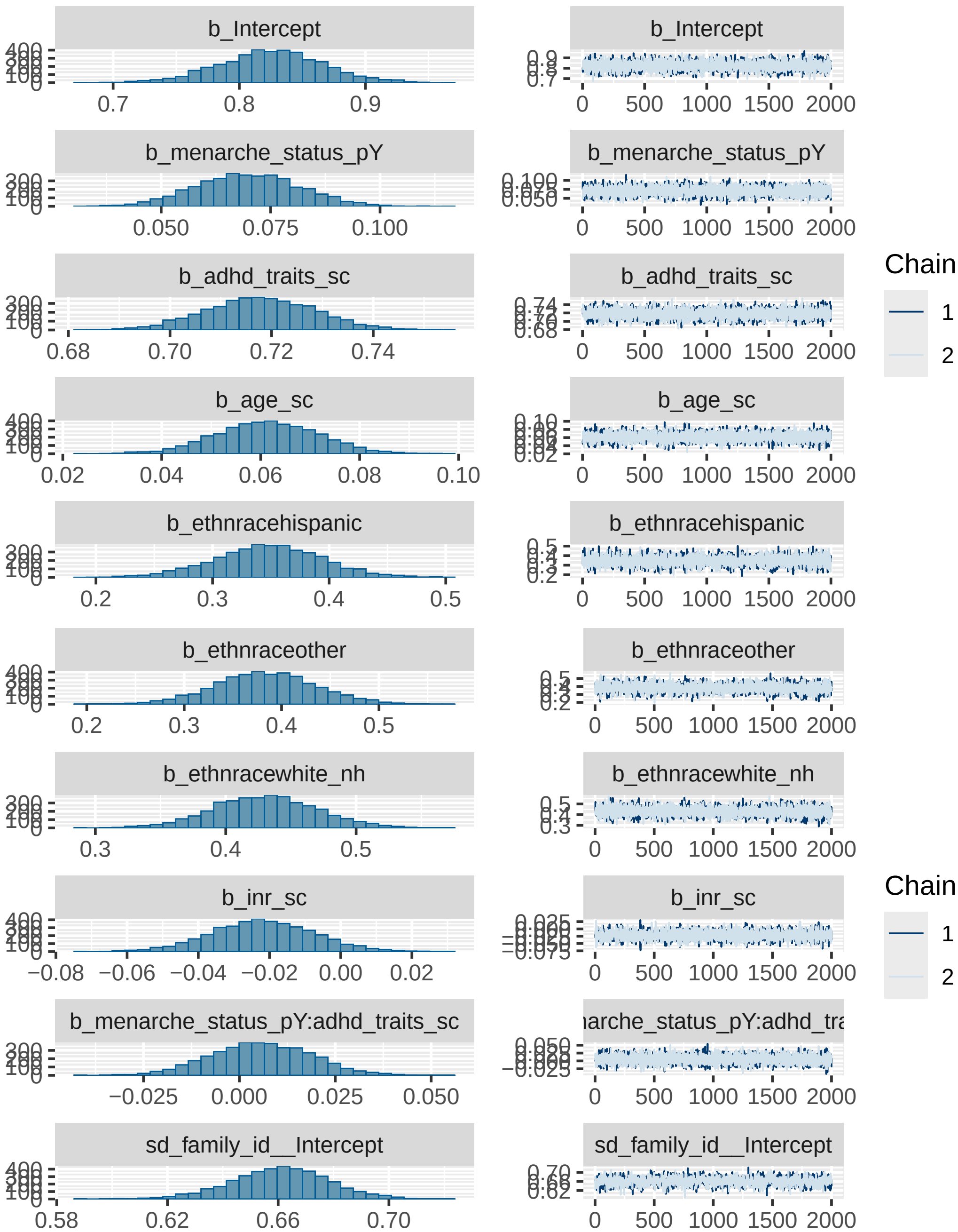
```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.7362667 0.002173641 0.7320194 0.7403498


 Family: poisson
  Links: mu = log
Formula: internalising ~ menarche_status_p * adhd_traits_sc + age_sc + ethnrace + inr_sc + (1
   Data: analytic_df (Number of observations: 19381)
  Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~family_id (Number of levels: 4626)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.661     0.017    0.625    0.695 1.000     1038     1831

~participant_id (Number of levels: 5297)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.488     0.018    0.453    0.526 1.001      652     1123
```

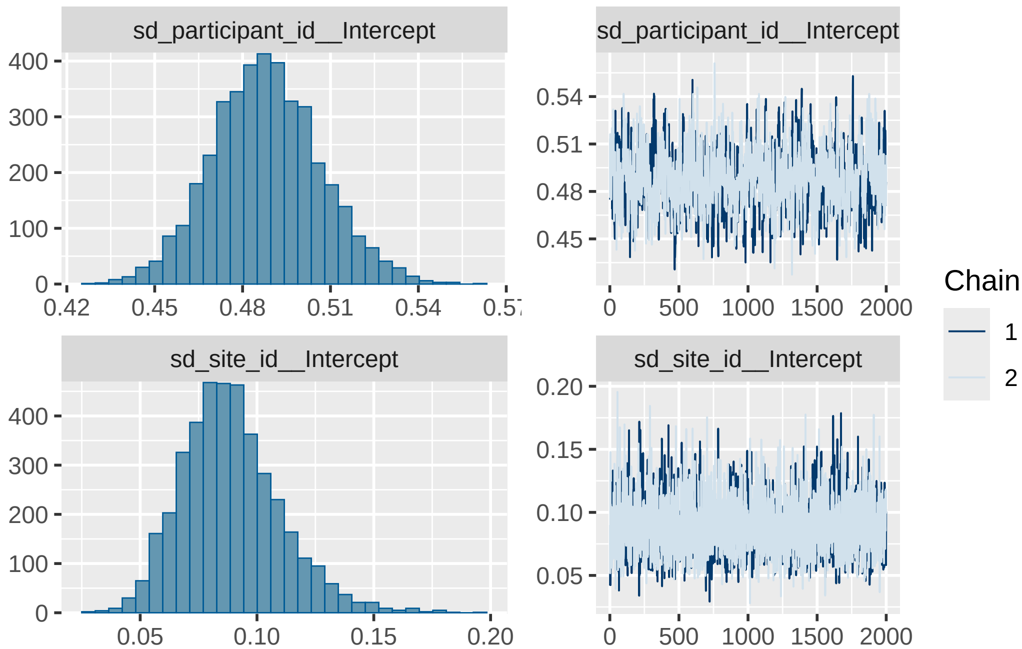
```
~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.089     0.021    0.053    0.136 1.001     3615     3561

Regression Coefficients:
                                  Estimate Est.Error l-95% CI u-95% CI  Rhat
Intercept                            0.826     0.041    0.744    0.908 1.000
menarche_status_pY                   0.070     0.012    0.047    0.093 1.001
adhd_traits_sc                       0.718     0.011    0.697    0.739 1.001
age_sc                               0.061     0.010    0.041    0.081 1.001
ethnracehispanic                     0.348     0.046    0.255    0.438 1.000
ethnraceother                        0.384     0.052    0.281    0.487 1.000
ethnracewhite_nh                     0.430     0.040    0.353    0.510 1.000
inr_sc                              -0.022     0.014   -0.050    0.007 1.002
menarche_status_pY:adhd_traits_sc    0.006     0.013   -0.020    0.031 1.000
                                  Bulk_ESS Tail_ESS
Intercept                             4472     3184
menarche_status_pY                    8541     3199
adhd_traits_sc                        8375     3403
age_sc                                8250     3161
ethnracehispanic                      4315     3109
ethnraceother                         4333     2600
ethnracewhite_nh                      5126     3151
inr_sc                                9537     2720
menarche_status_pY:adhd_traits_sc     6527     3341

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

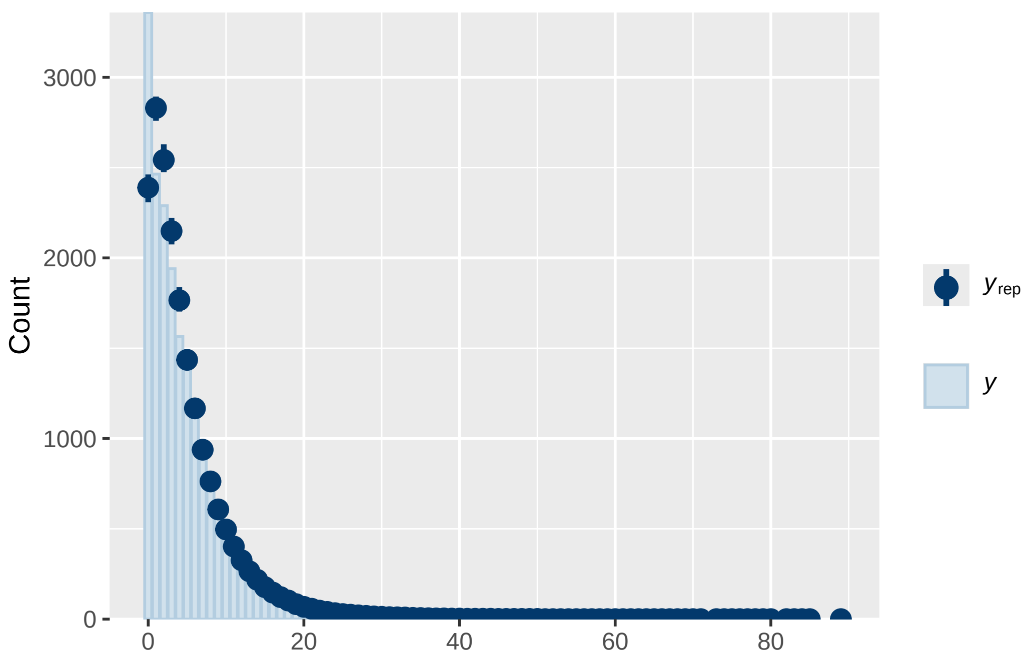
Model 4 prior 1 diagnostics

```
          prior     class                              coef
   normal(0, 1)         b
   normal(0, 1)         b                    adhd_traits_sc
   normal(0, 1)         b                            age_sc
   normal(0, 1)         b                  ethnracehispanic
   normal(0, 1)         b                     ethnraceother
   normal(0, 1)         b                  ethnracewhite_nh
   normal(0, 1)         b                            inr_sc
   normal(0, 1)         b                menarche_status_pY
```

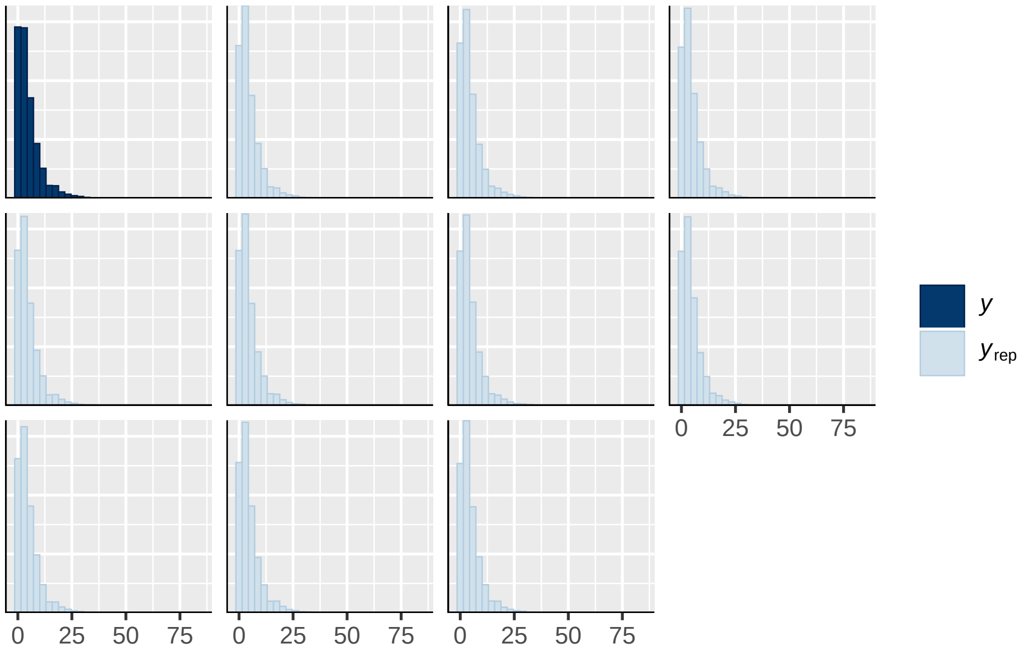
```
          normal(0, 1)         b menarche_status_pY:adhd_traits_sc
student_t(3, 1.1, 2.5) Intercept
  student_t(3, 0, 2.5)        sd
  student_t(3, 0, 2.5)        sd
  student_t(3, 0, 2.5)        sd                          Intercept
  student_t(3, 0, 2.5)        sd
  student_t(3, 0, 2.5)        sd                          Intercept
  student_t(3, 0, 2.5)        sd
  student_t(3, 0, 2.5)        sd                          Intercept
         group resp dpar nlpar lb ub tag       source
                                                 user
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                              default
                                0             default
     family_id                  0        (vectorized)
     family_id                  0        (vectorized)
participant_id                  0        (vectorized)
participant_id                  0        (vectorized)
       site_id                  0        (vectorized)
       site_id                  0        (vectorized)
```


b_Intercept
b_menarche_status_pY
b_adhd_traits_sc
b_age_sc
b_ethnracehispanic
b_ethnraceother
b_ethnracewhite_nh
b_inr_sc
b_menarche_status_pY:adhd_traits_sc
sd_family_id__Intercept
Chain
1
2

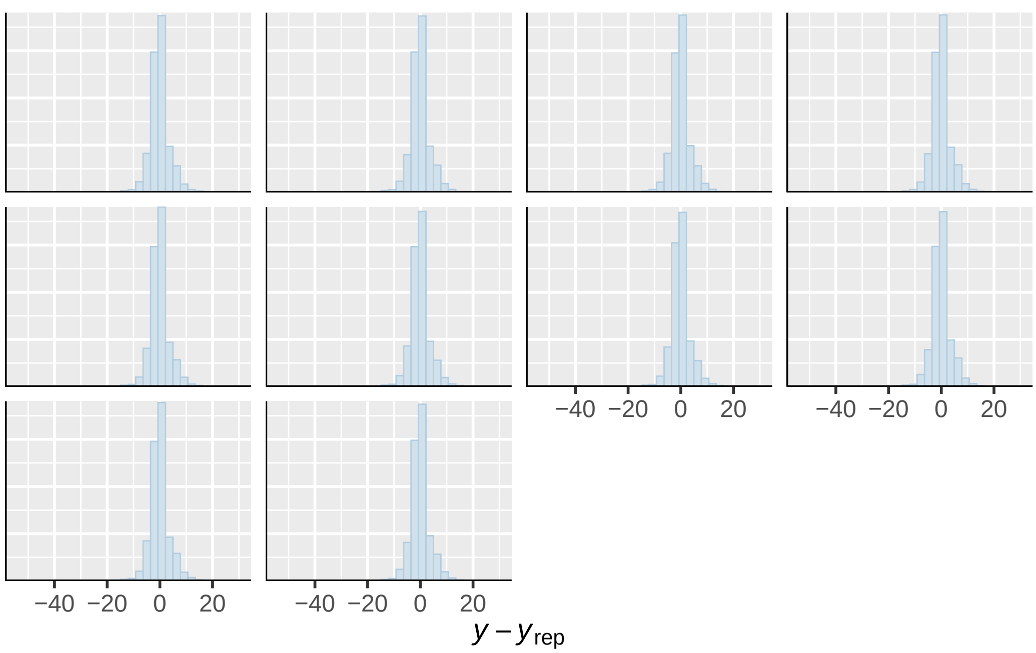





```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

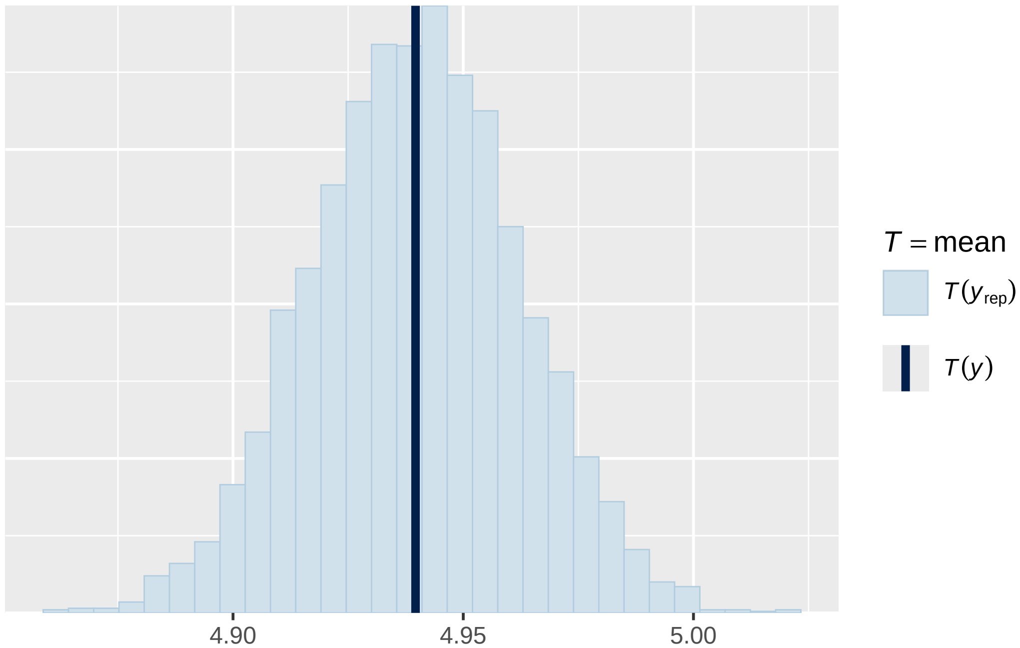



```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

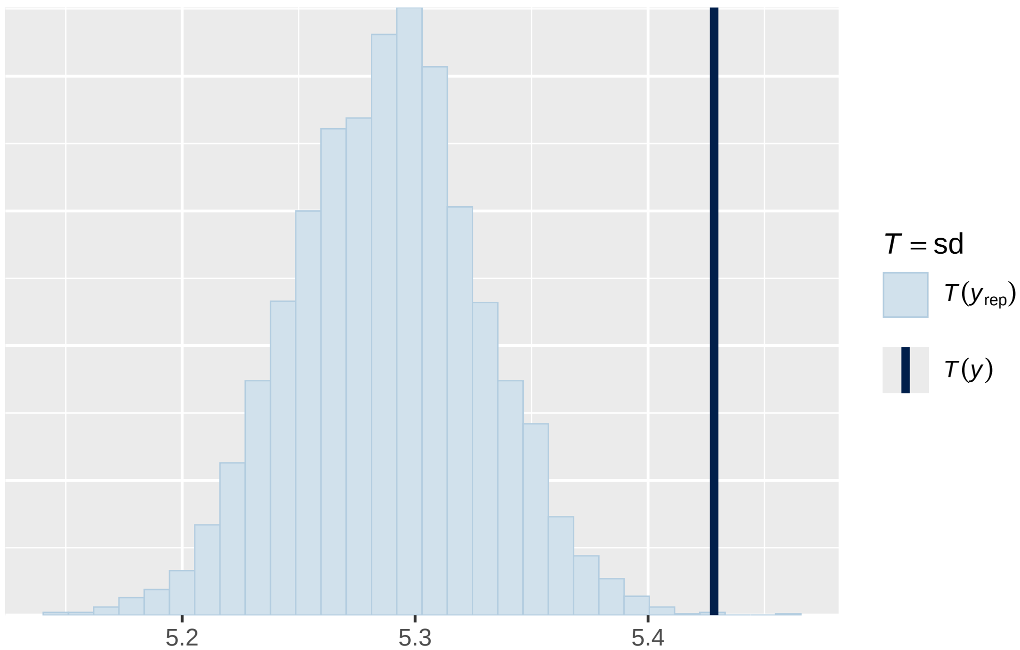



```
Using all posterior draws for ppc type 'stat' by default.
```

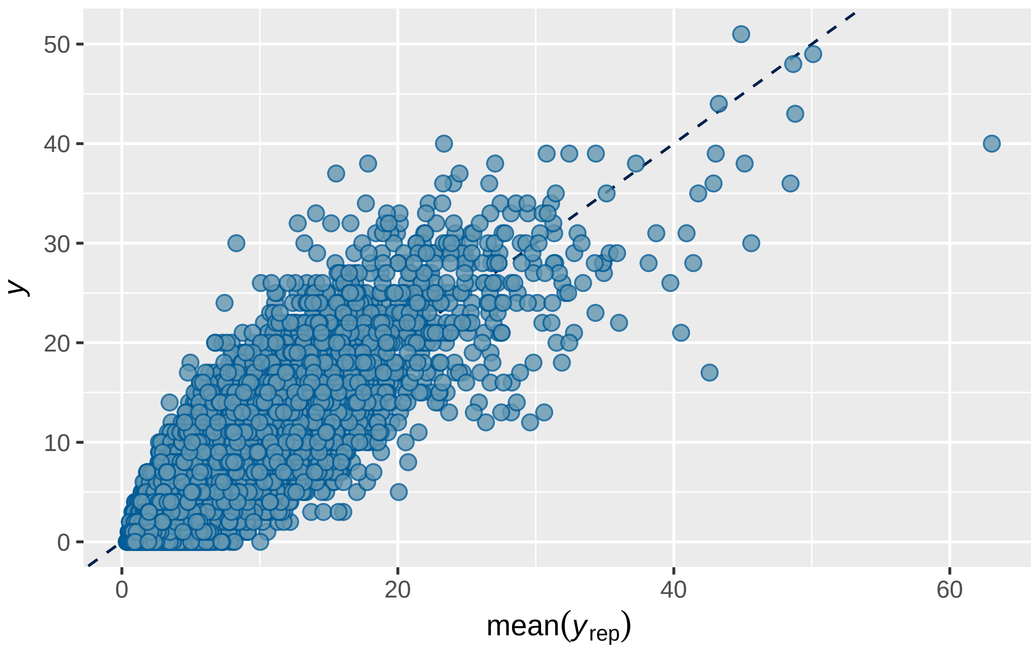
```
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 5 prior 1 output

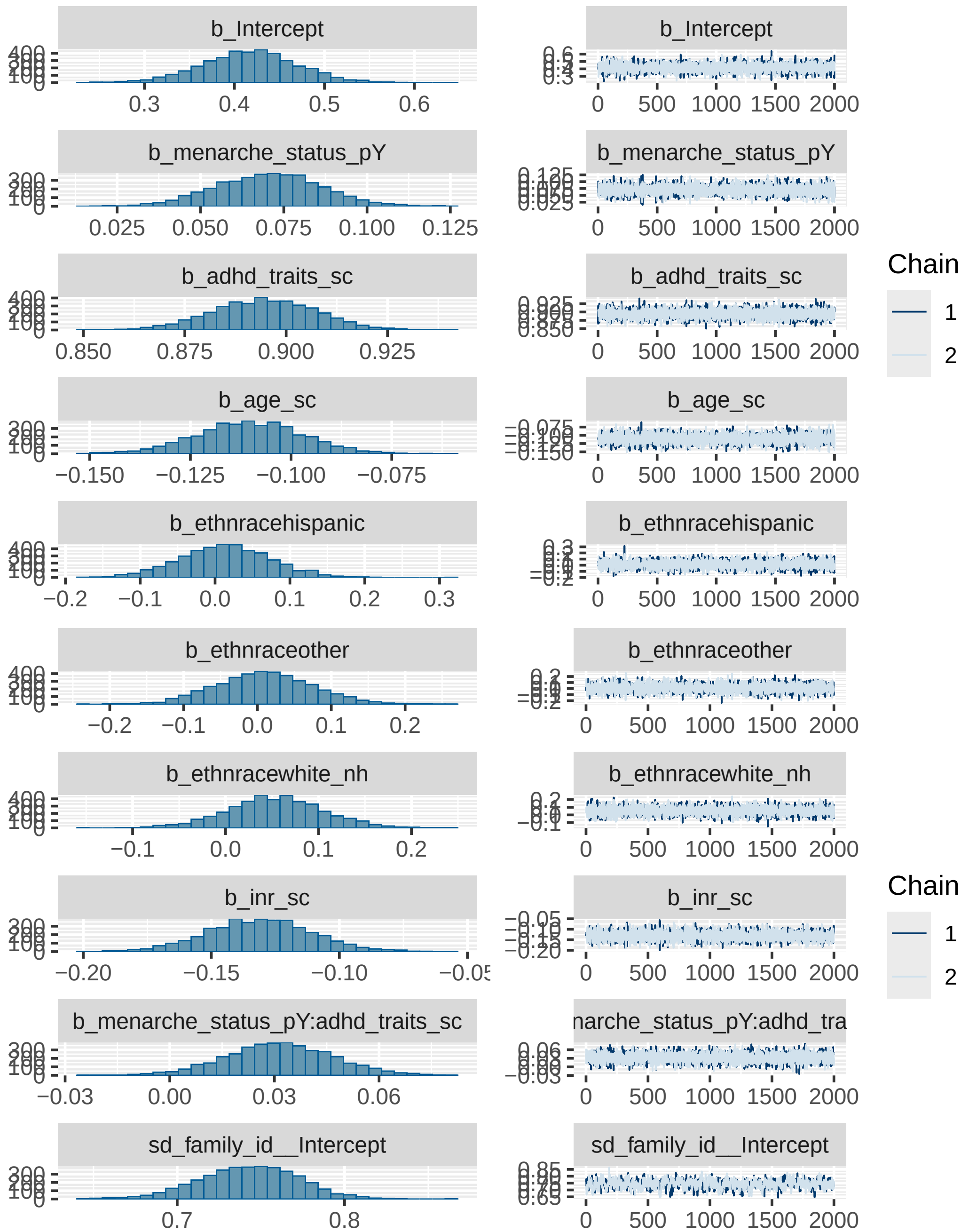
```
    Estimate   Est.Error      Q2.5    Q97.5
R2 0.7884055 0.001954511 0.7844898 0.792049


 Family: poisson
  Links: mu = log
Formula: externalising ~ menarche_status_p * adhd_traits_sc + age_sc + ethnrace + inr_sc + (1
   Data: analytic_df (Number of observations: 19381)
  Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~family_id (Number of levels: 4626)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.743     0.030    0.682    0.799 1.003      524      878

~participant_id (Number of levels: 5297)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.755     0.026    0.705    0.808 1.004      593      903
```

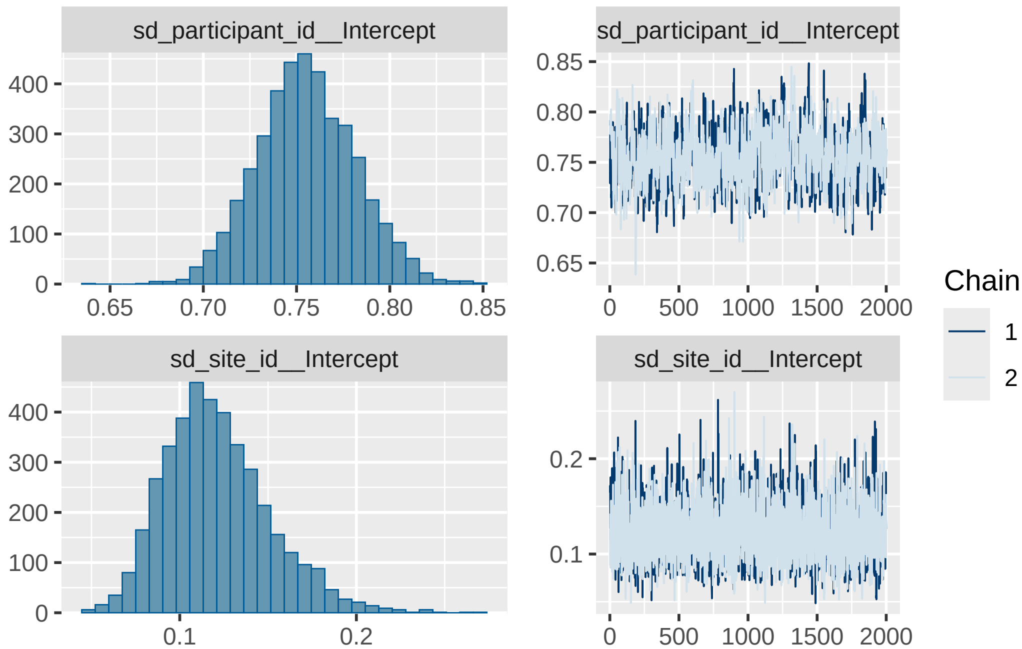
```
~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.121     0.030    0.073    0.186 1.001     3306     2787

Regression Coefficients:
                                  Estimate Est.Error l-95% CI u-95% CI  Rhat
Intercept                            0.417     0.053    0.311    0.519 1.001
menarche_status_pY                   0.070     0.016    0.040    0.102 1.001
adhd_traits_sc                       0.895     0.013    0.870    0.920 1.000
age_sc                              -0.110     0.013   -0.136   -0.084 1.000
ethnracehispanic                     0.014     0.060   -0.104    0.131 1.001
ethnraceother                        0.010     0.065   -0.116    0.139 1.002
ethnracewhite_nh                     0.054     0.051   -0.047    0.154 1.001
inr_sc                              -0.131     0.020   -0.171   -0.090 1.002
menarche_status_pY:adhd_traits_sc    0.032     0.015    0.001    0.062 1.000
                                  Bulk_ESS Tail_ESS
Intercept                             3364     2957
menarche_status_pY                    7027     3123
adhd_traits_sc                        7531     3248
age_sc                                7003     3334
ethnracehispanic                      4071     3161
ethnraceother                         4047     3136
ethnracewhite_nh                      3900     3122
inr_sc                                8985     2813
menarche_status_pY:adhd_traits_sc     6332     3130

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

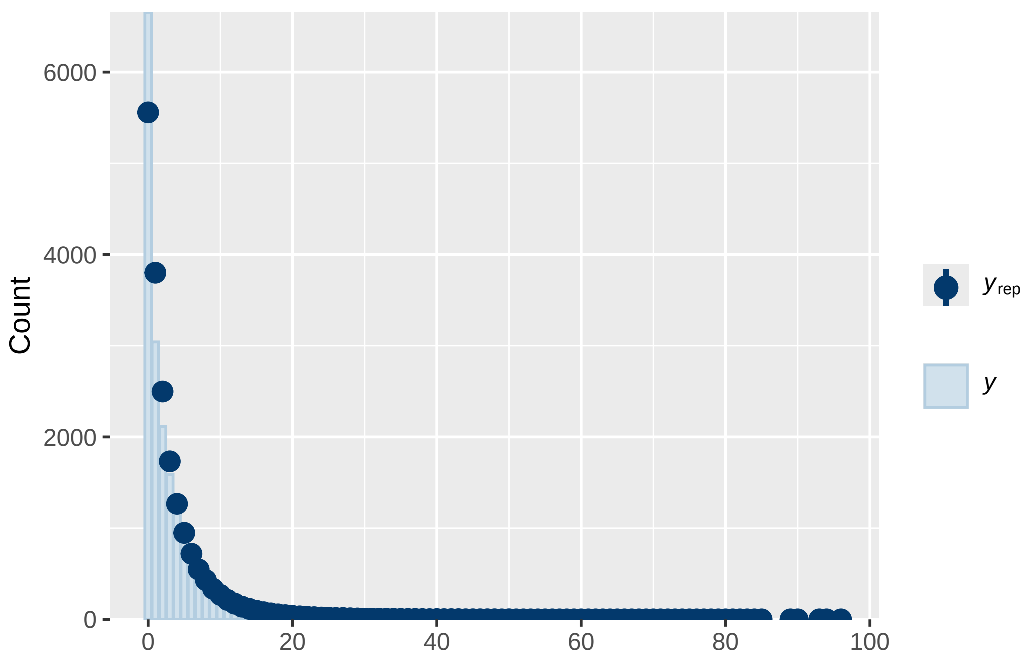
Model 5 prior 1 diagnostics

```
         prior     class                              coef
  normal(0, 1)         b
  normal(0, 1)         b                    adhd_traits_sc
  normal(0, 1)         b                            age_sc
  normal(0, 1)         b                  ethnracehispanic
  normal(0, 1)         b                     ethnraceother
  normal(0, 1)         b                  ethnracewhite_nh
  normal(0, 1)         b                            inr_sc
  normal(0, 1)         b                menarche_status_pY
```

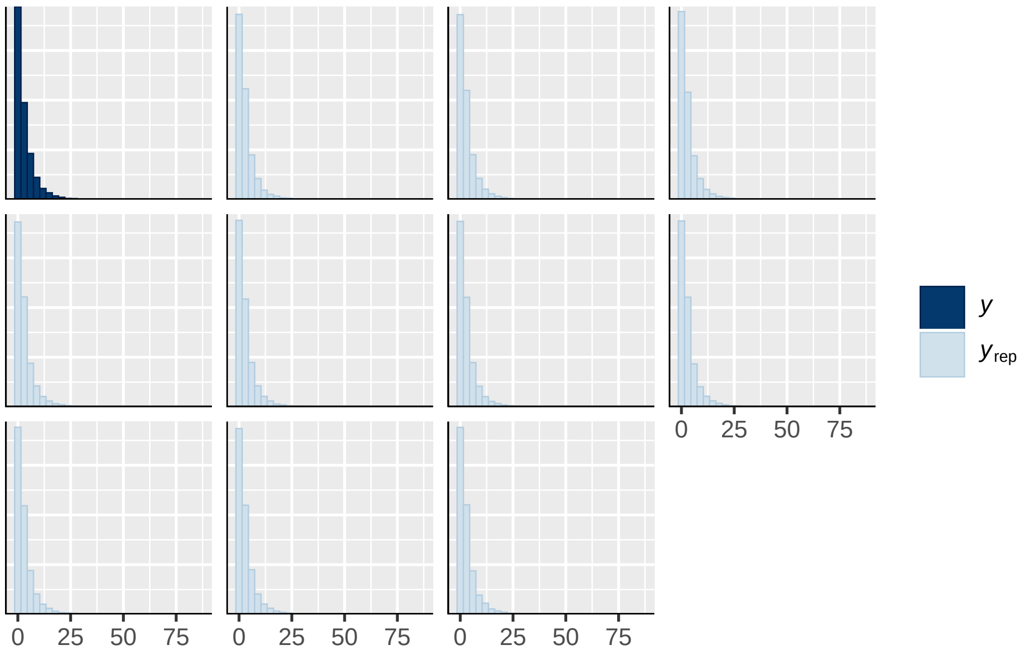
```
        normal(0, 1)         b menarche_status_pY:adhd_traits_sc
student_t(3, 0, 2.9) Intercept
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd                         Intercept
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd                         Intercept
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd                         Intercept
         group resp dpar nlpar lb ub tag       source
                                                 user
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                              default
                                0             default
     family_id                  0        (vectorized)
     family_id                  0        (vectorized)
participant_id                  0        (vectorized)
participant_id                  0        (vectorized)
       site_id                  0        (vectorized)
       site_id                  0        (vectorized)
```


b_Intercept
b_menarche_status_pY
b_adhd_traits_sc
b_age_sc
b_ethnracehispanic
b_ethnraceother
b_ethnracewhite_nh
b_inr_sc
b_menarche_status_pY:adhd_traits_sc
sd_family_id__Intercept
Chain
1
2

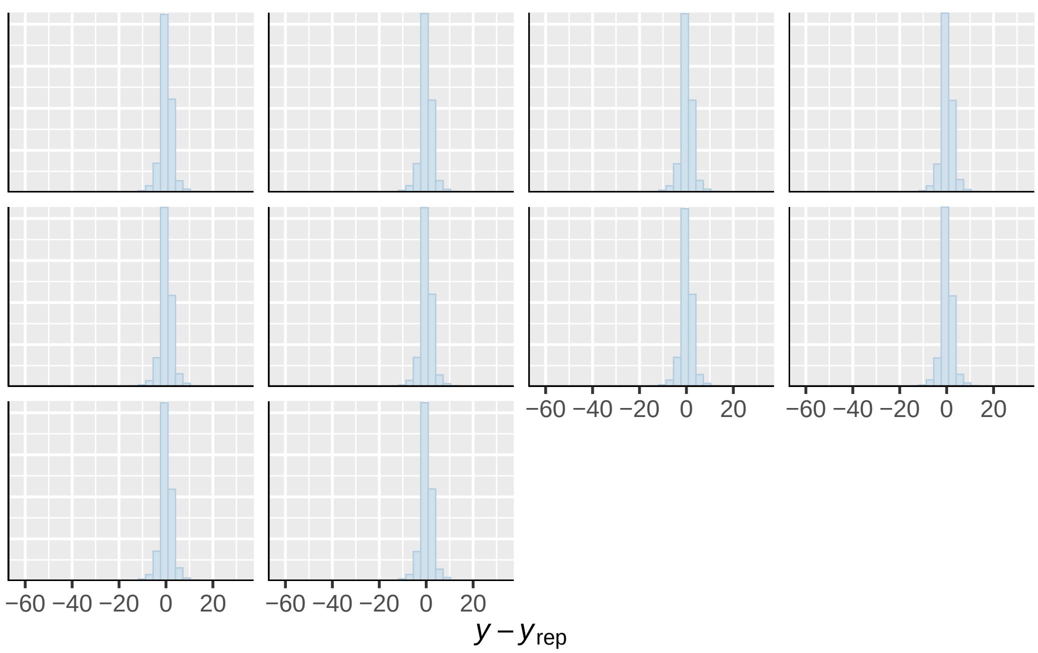





```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

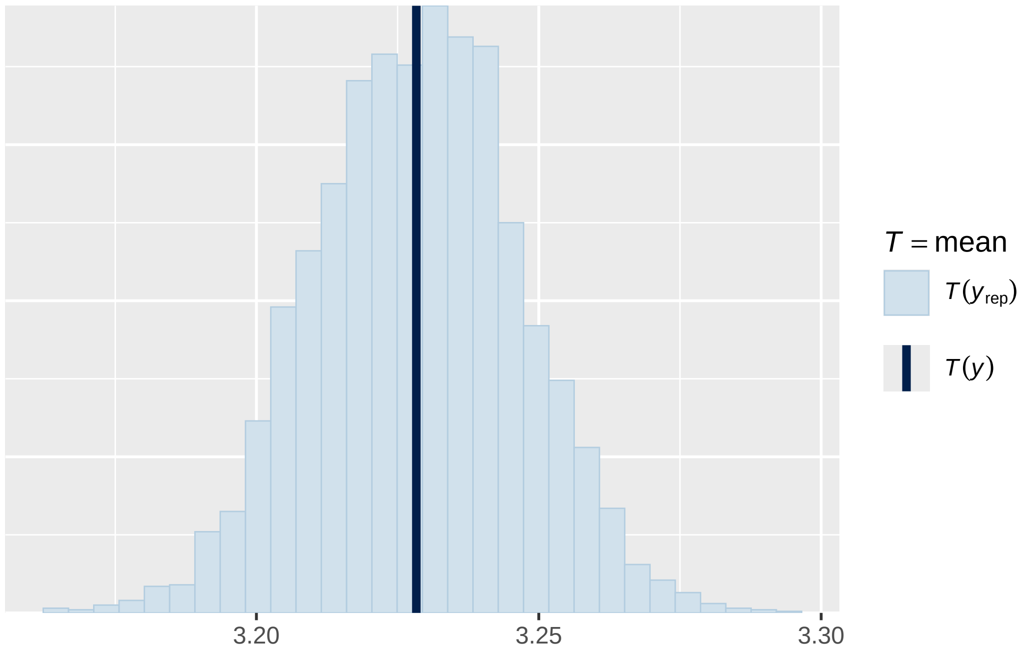



```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

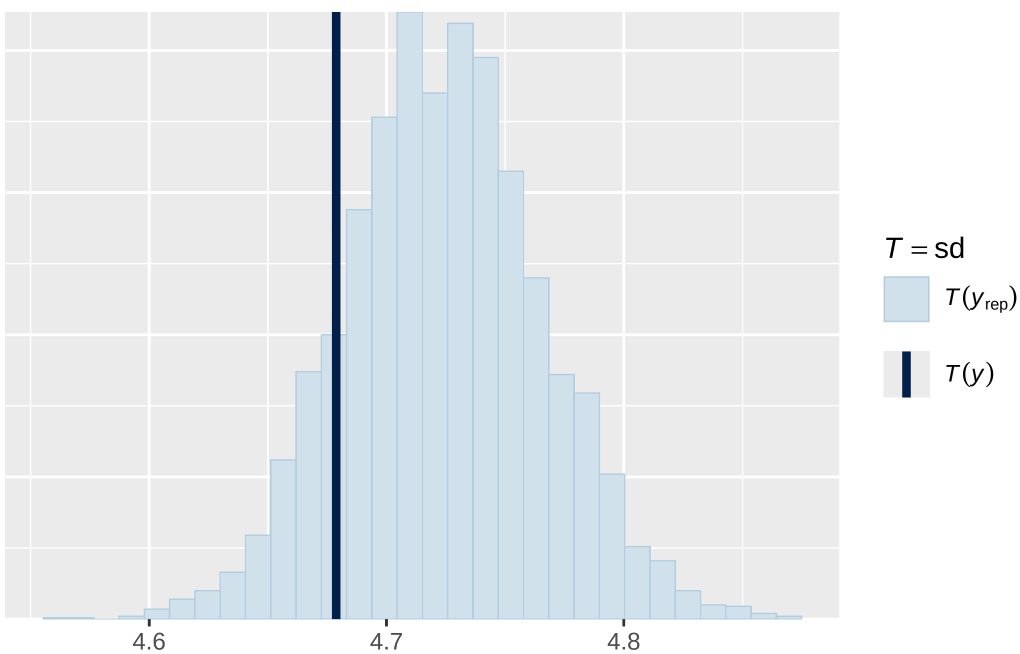



```
Using all posterior draws for ppc type 'stat' by default.
```

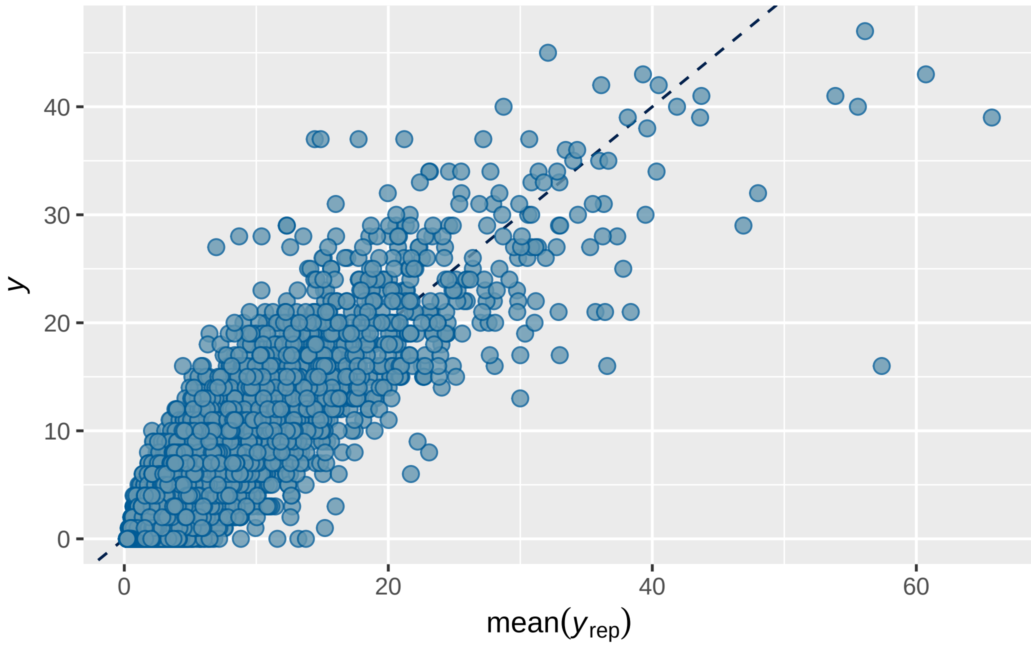
```
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 1b prior 2 output

```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.7026753 0.003300624 0.6961416 0.7091743
```

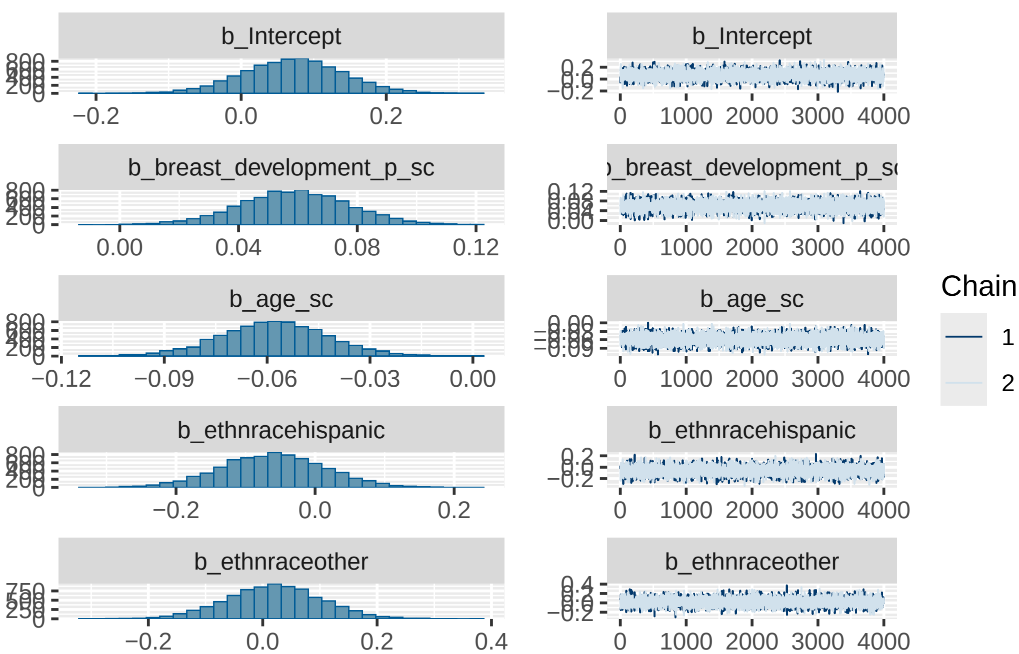
```
 Family: poisson
  Links: mu = log
Formula: adhd_traits ~ breast_development_p_sc + age_sc + ethnrace + inr_sc + (1 | participa
   Data: analytic_df (Number of observations: 19364)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 4619)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.837     0.046    0.744    0.924 1.001      698     1345

~participant_id (Number of levels: 5289)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    1.035     0.036    0.964    1.107 1.001      862     1602
```

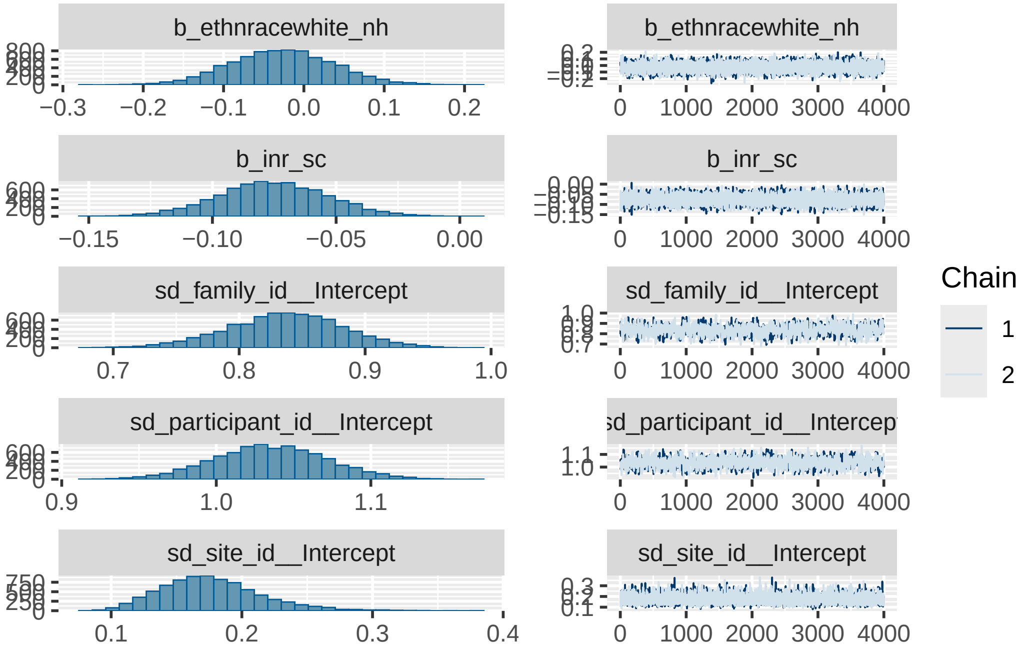
```
~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.177     0.038    0.113    0.262 1.000     6823     6294

Regression Coefficients:
                        Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
Intercept                  0.071     0.069   -0.068    0.207 1.000     6981
breast_development_p_sc    0.059     0.019    0.022    0.095 1.001    14782
age_sc                    -0.058     0.016   -0.089   -0.027 1.000    14253
ethnracehispanic          -0.061     0.075   -0.210    0.087 1.000     8214
ethnraceother              0.020     0.084   -0.142    0.184 1.000     8236
ethnracewhite_nh          -0.027     0.064   -0.150    0.098 1.000     7578
inr_sc                    -0.075     0.022   -0.119   -0.032 1.000    15782
                        Tail_ESS
Intercept                   6284
breast_development_p_sc     5778
age_sc                      6485
ethnracehispanic            6441
ethnraceother               6484
ethnracewhite_nh            6227
inr_sc                      6094

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

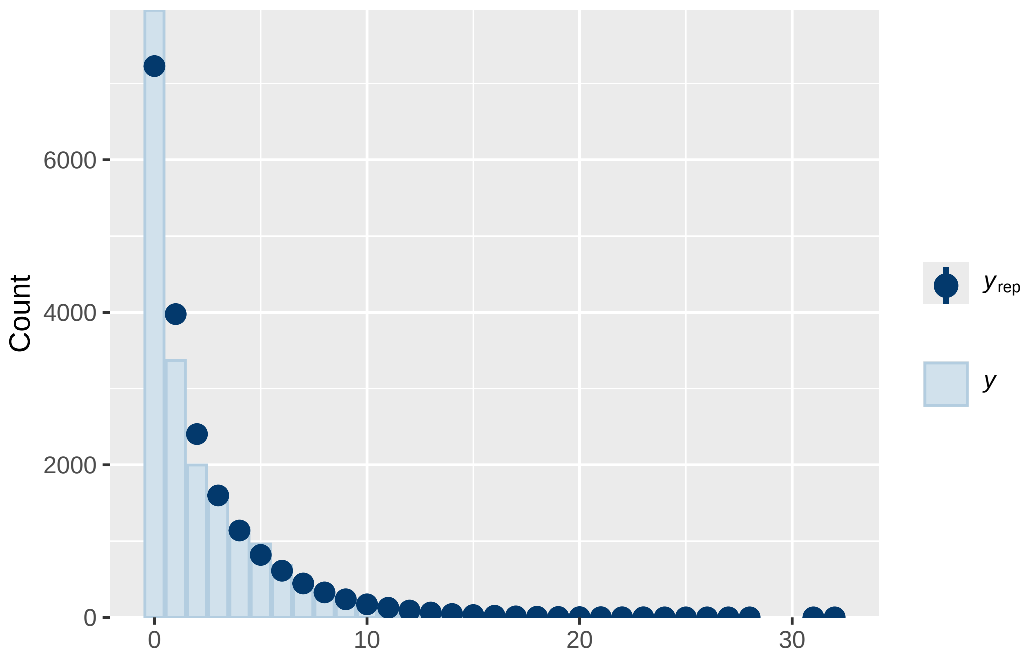
Model 1b prior 2 diagnostics

```
               prior     class                    coef          group resp
        normal(0, 10)         b
        normal(0, 10)         b                  age_sc
        normal(0, 10)         b breast_development_p_sc
        normal(0, 10)         b        ethnracehispanic
        normal(0, 10)         b           ethnraceother
        normal(0, 10)         b        ethnracewhite_nh
        normal(0, 10)         b                  inr_sc
 student_t(3, 0, 2.7) Intercept
 student_t(3, 0, 2.7)        sd
 student_t(3, 0, 2.7)        sd                              family_id
 student_t(3, 0, 2.7)        sd               Intercept      family_id
 student_t(3, 0, 2.7)        sd                         participant_id
```

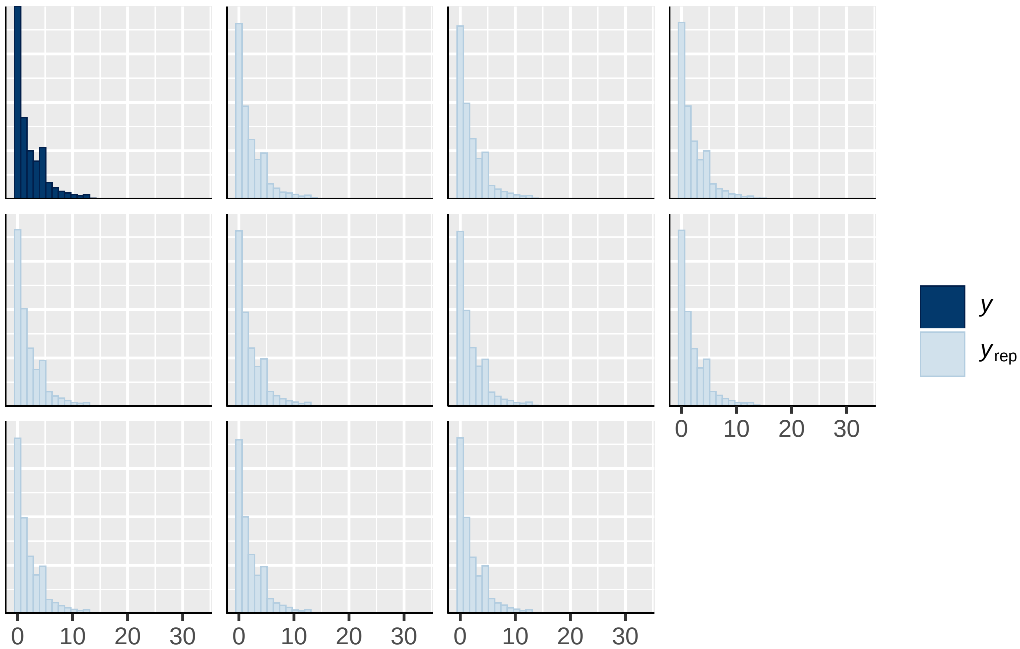
```
student_t(3, 0, 2.7)          sd            Intercept participant_id
student_t(3, 0, 2.7)          sd                             site_id
student_t(3, 0, 2.7)          sd            Intercept        site_id
dpar nlpar lb ub tag        source
                              user
                      (vectorized)
                      (vectorized)
                      (vectorized)
                      (vectorized)
                      (vectorized)
                      (vectorized)
                           default
            0              default
            0         (vectorized)
            0         (vectorized)
            0         (vectorized)
            0         (vectorized)
            0         (vectorized)
            0         (vectorized)
```



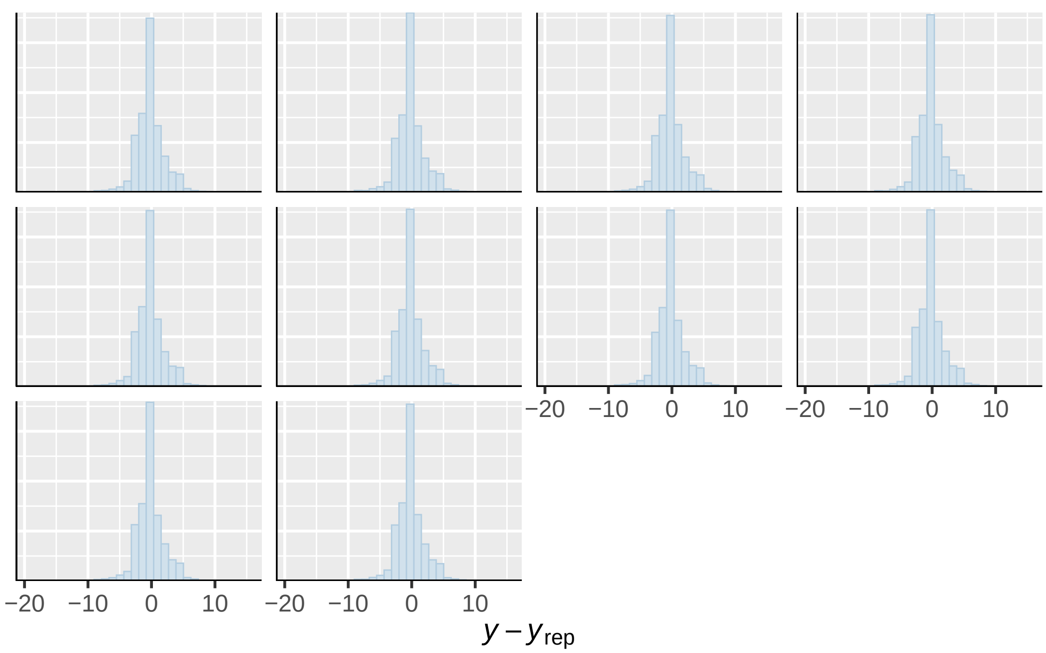





```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

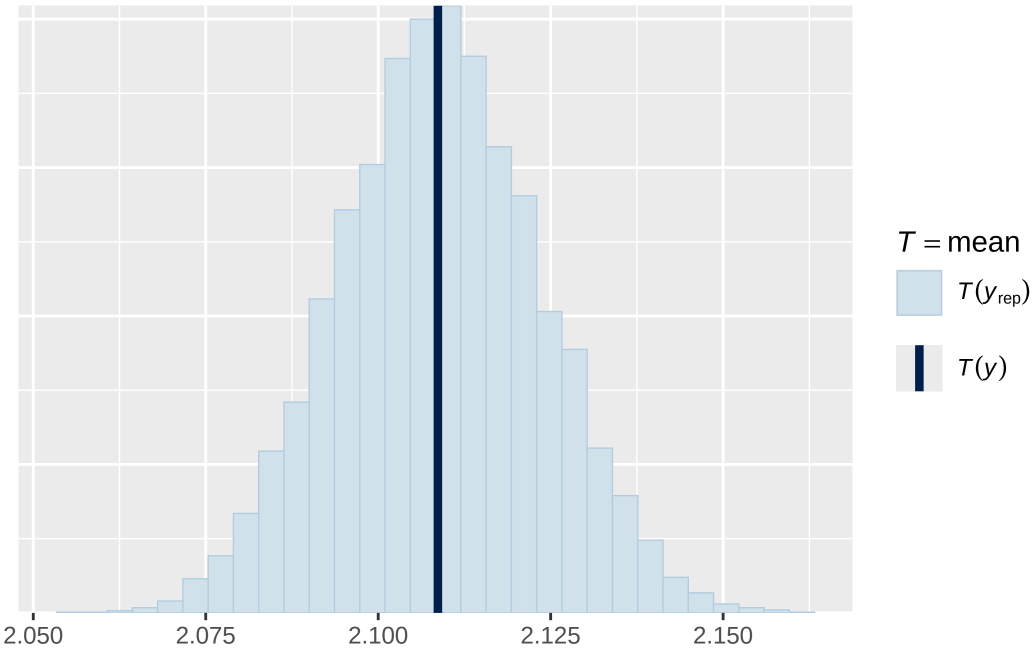


Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

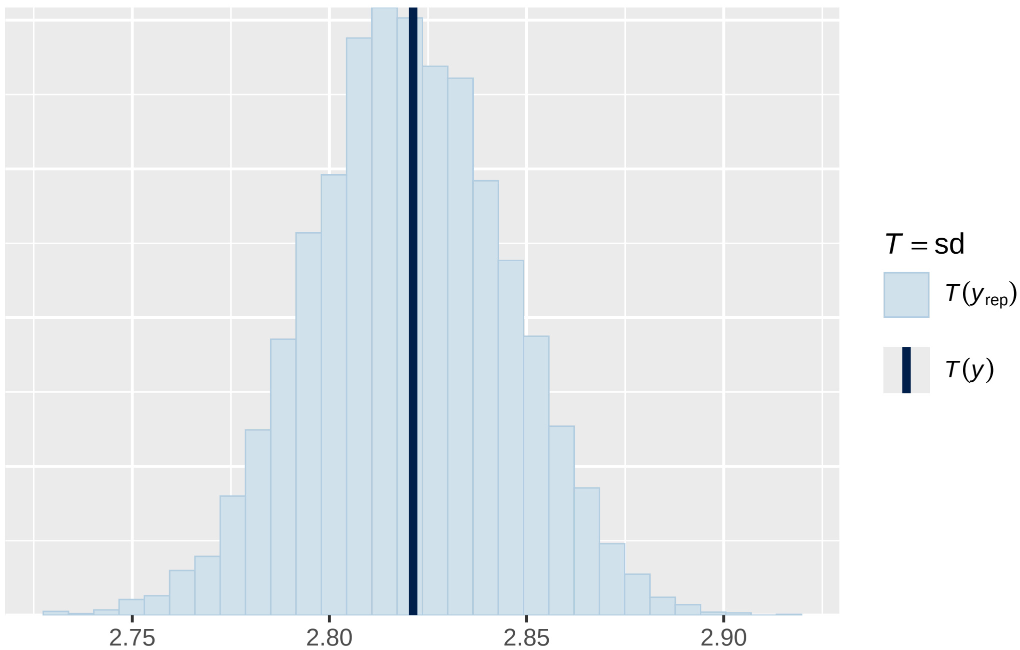


Using all posterior draws for ppc type 'stat' by default.

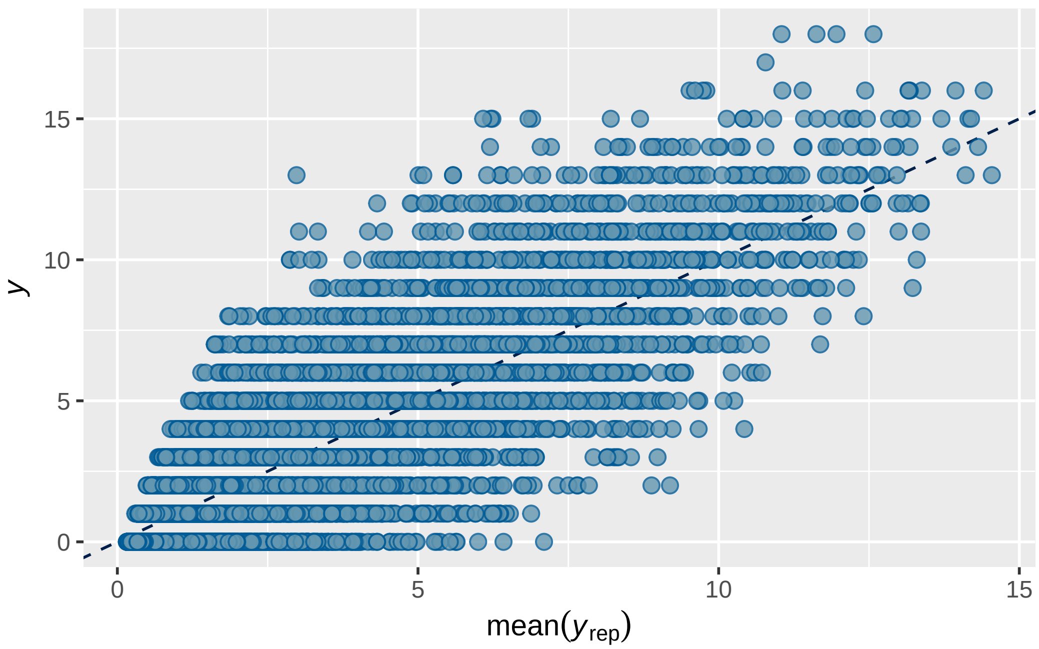
```
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 4 prior 2 output

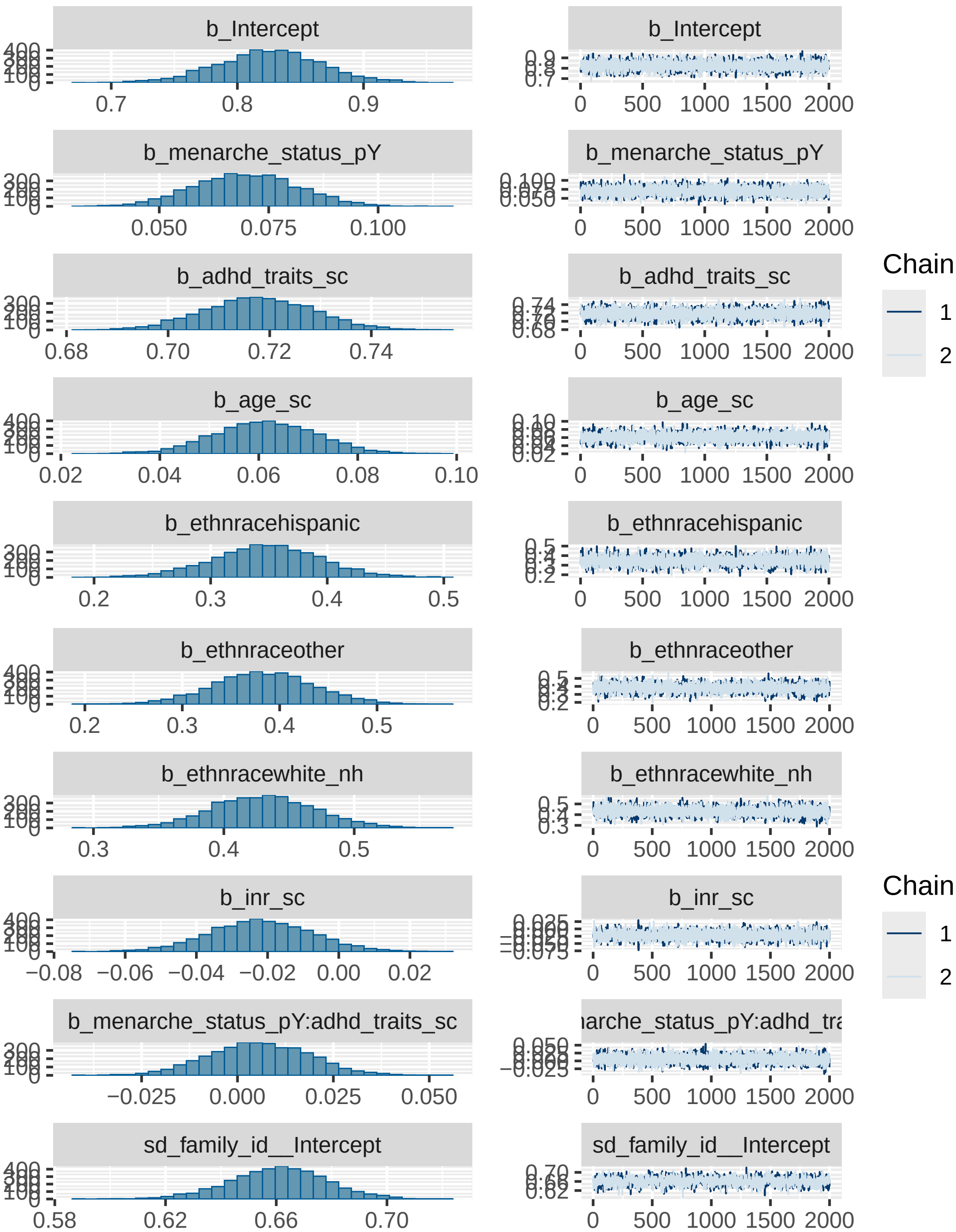
```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.7362667 0.002173641 0.7320194 0.7403498


 Family: poisson
  Links: mu = log
Formula: internalising ~ menarche_status_p * adhd_traits_sc + age_sc + ethnrace + inr_sc + (
   Data: analytic_df (Number of observations: 19381)
  Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~family_id (Number of levels: 4626)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.661     0.017    0.625    0.695 1.000     1038     1831

~participant_id (Number of levels: 5297)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.488     0.018    0.453    0.526 1.001      652     1123
```

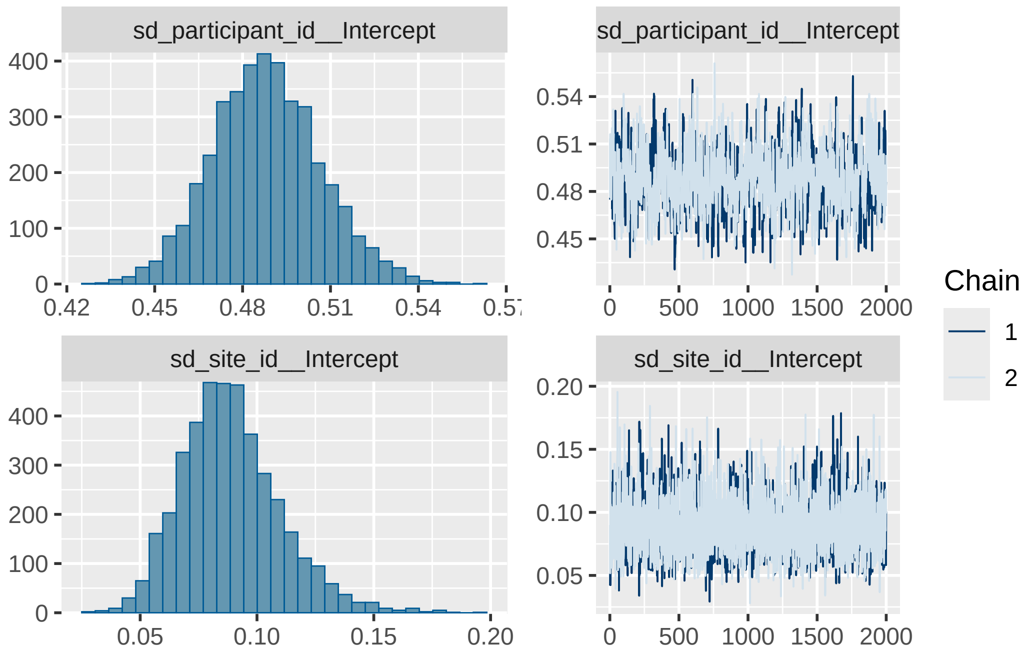
```
~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.089     0.021    0.053    0.136 1.001     3615     3561

Regression Coefficients:
                                  Estimate Est.Error l-95% CI u-95% CI  Rhat
Intercept                            0.826     0.041    0.744    0.908 1.000
menarche_status_pY                   0.070     0.012    0.047    0.093 1.001
adhd_traits_sc                       0.718     0.011    0.697    0.739 1.001
age_sc                               0.061     0.010    0.041    0.081 1.001
ethnracehispanic                     0.348     0.046    0.255    0.438 1.000
ethnraceother                        0.384     0.052    0.281    0.487 1.000
ethnracewhite_nh                     0.430     0.040    0.353    0.510 1.000
inr_sc                              -0.022     0.014   -0.050    0.007 1.002
menarche_status_pY:adhd_traits_sc    0.006     0.013   -0.020    0.031 1.000
                                  Bulk_ESS Tail_ESS
Intercept                             4472     3184
menarche_status_pY                    8541     3199
adhd_traits_sc                        8375     3403
age_sc                                8250     3161
ethnracehispanic                      4315     3109
ethnraceother                         4333     2600
ethnracewhite_nh                      5126     3151
inr_sc                                9537     2720
menarche_status_pY:adhd_traits_sc     6527     3341

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

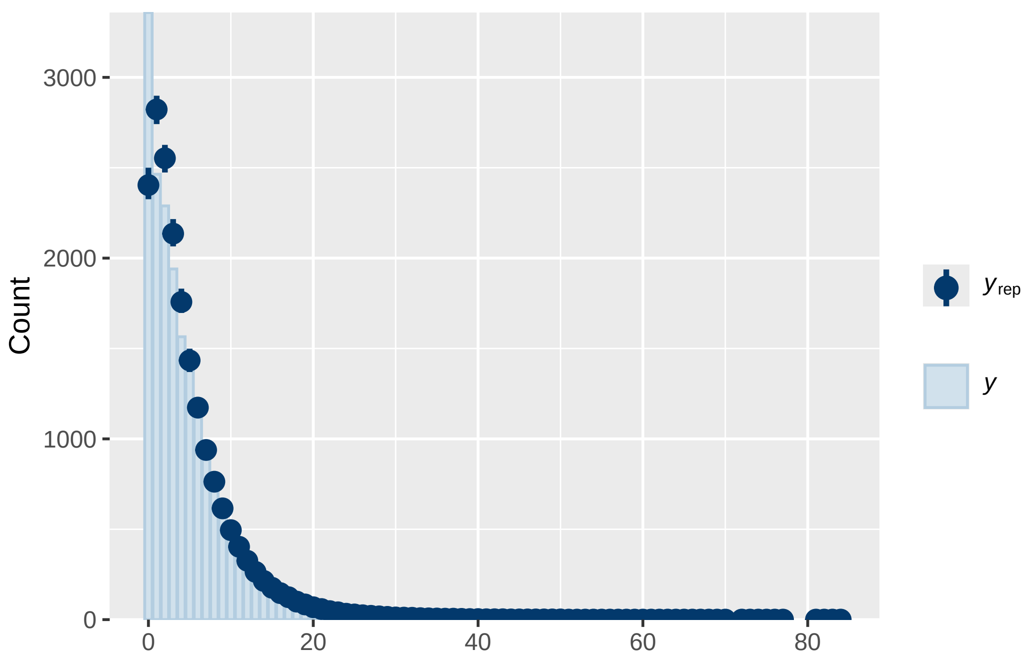
Model 4 prior 2 diagnostics

```
          prior     class                               coef
   normal(0, 1)         b
   normal(0, 1)         b                     adhd_traits_sc
   normal(0, 1)         b                             age_sc
   normal(0, 1)         b                   ethnracehispanic
   normal(0, 1)         b                      ethnraceother
   normal(0, 1)         b                   ethnracewhite_nh
   normal(0, 1)         b                             inr_sc
   normal(0, 1)         b                 menarche_status_pY
```

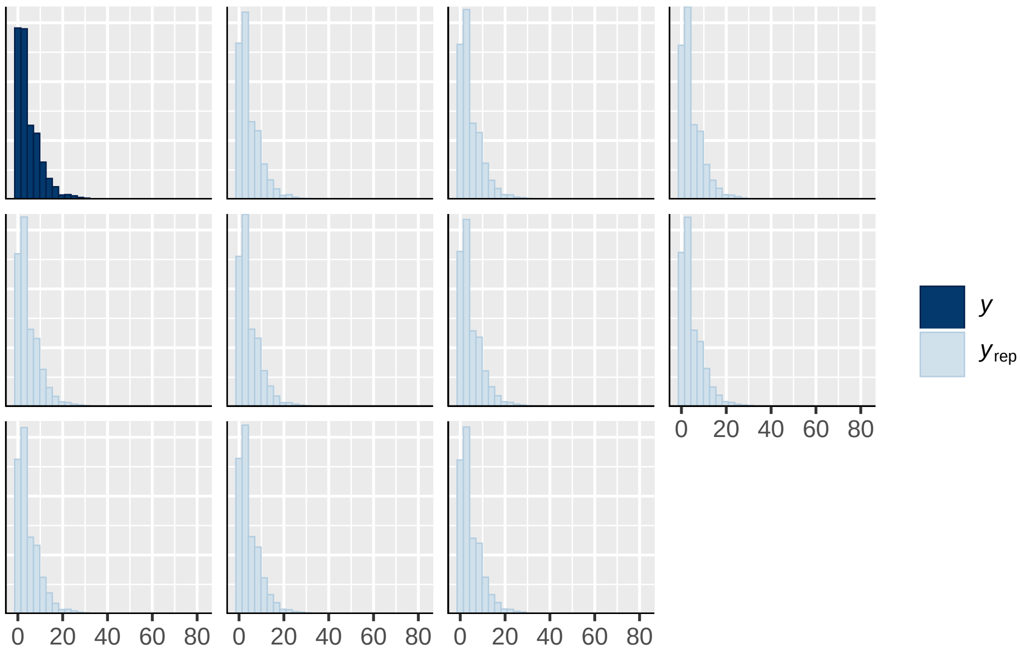
```
          normal(0, 1)         b menarche_status_pY:adhd_traits_sc
student_t(3, 1.1, 2.5) Intercept
  student_t(3, 0, 2.5)        sd
  student_t(3, 0, 2.5)        sd
  student_t(3, 0, 2.5)        sd                         Intercept
  student_t(3, 0, 2.5)        sd
  student_t(3, 0, 2.5)        sd                         Intercept
  student_t(3, 0, 2.5)        sd
  student_t(3, 0, 2.5)        sd                         Intercept
         group resp dpar nlpar lb ub tag       source
                                                 user
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                              default
                               0              default
     family_id                 0         (vectorized)
     family_id                 0         (vectorized)
participant_id                 0         (vectorized)
participant_id                 0         (vectorized)
       site_id                 0         (vectorized)
       site_id                 0         (vectorized)
```


b_Intercept
b_Intercept
b_menarche_status_pY
b_menarche_status_pY
b_adhd_traits_sc
b_adhd_traits_sc
b_age_sc
b_age_sc
b_ethnracehispanic
b_ethnracehispanic
b_ethnraceother
b_ethnraceother
b_ethnracewhite_nh
b_ethnracewhite_nh
b_inr_sc
b_inr_sc
b_menarche_status_pY:adhd_traits_sc
sd_family_id__Intercept
sd_family_id__Intercept
Chain
1
2
Chain
1
2

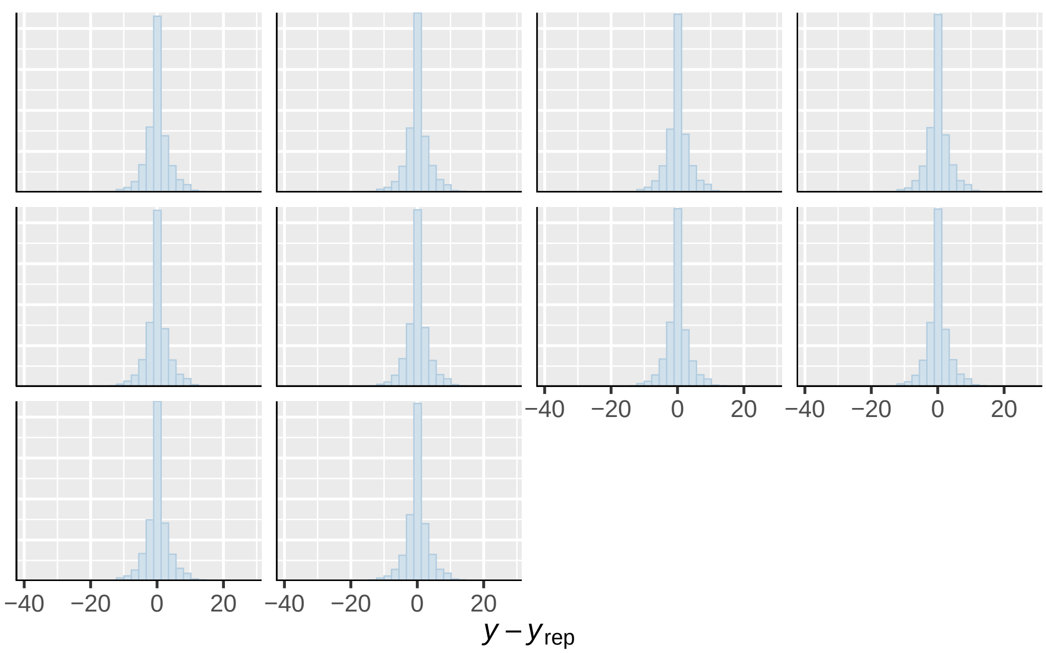





```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

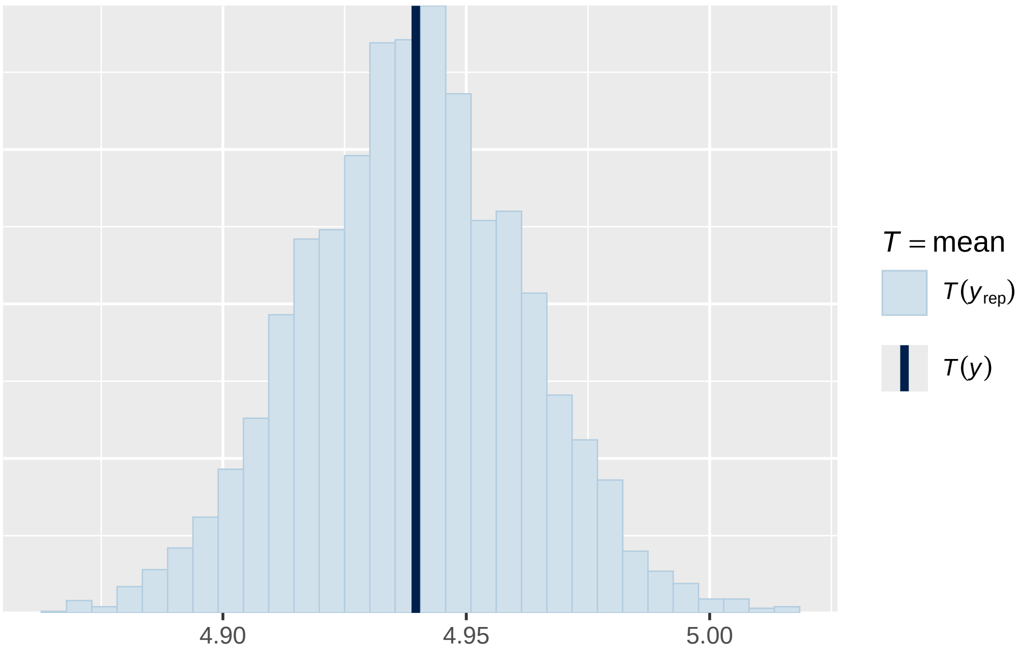



```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

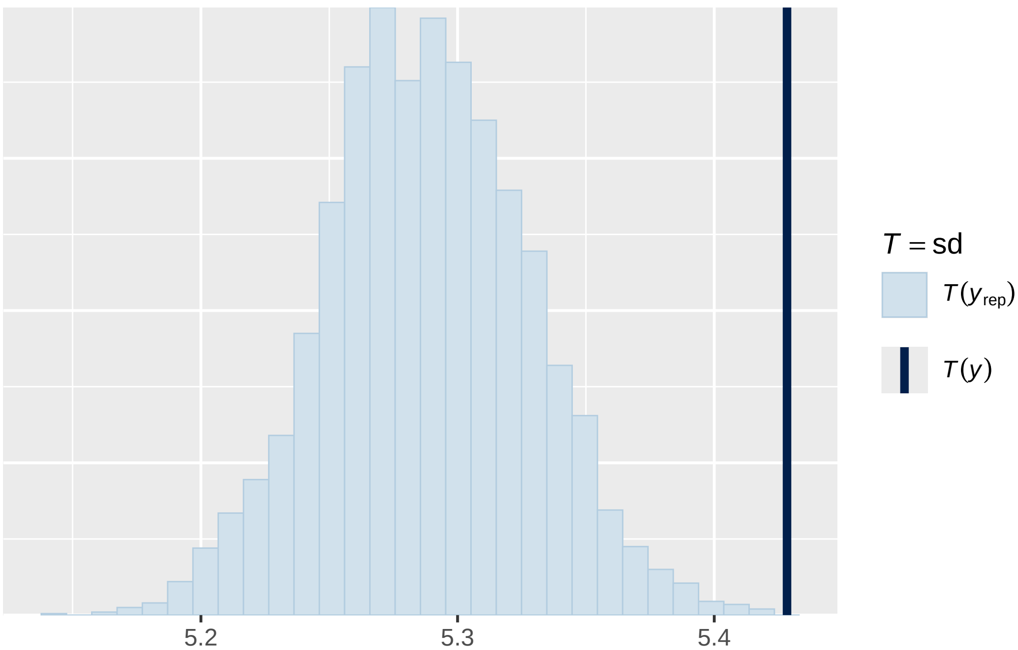



```
Using all posterior draws for ppc type 'stat' by default.
```

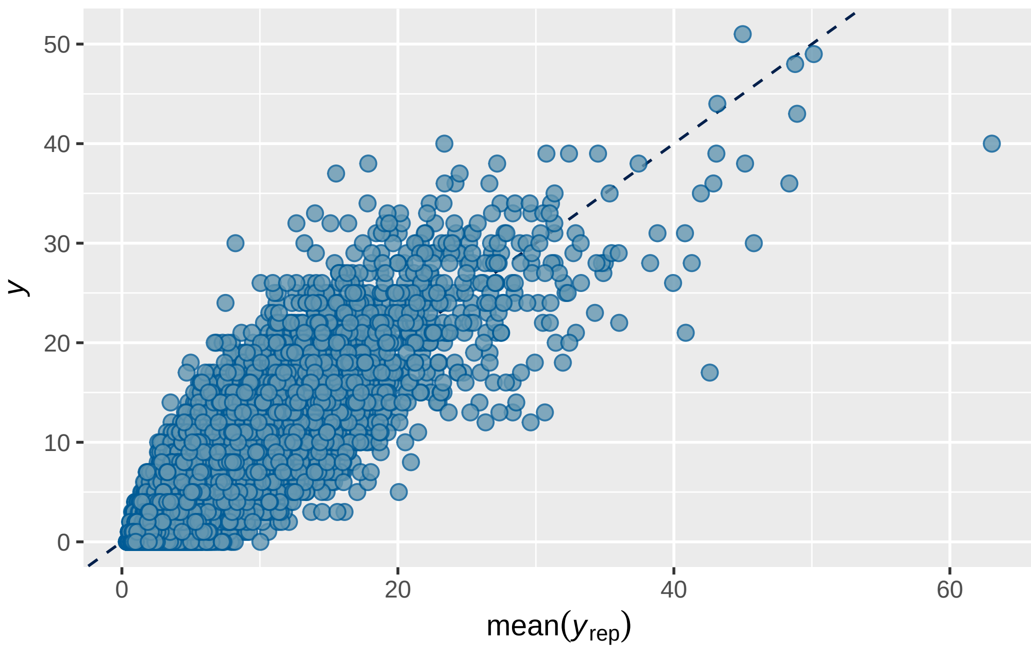
```
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 5 prior 2 output

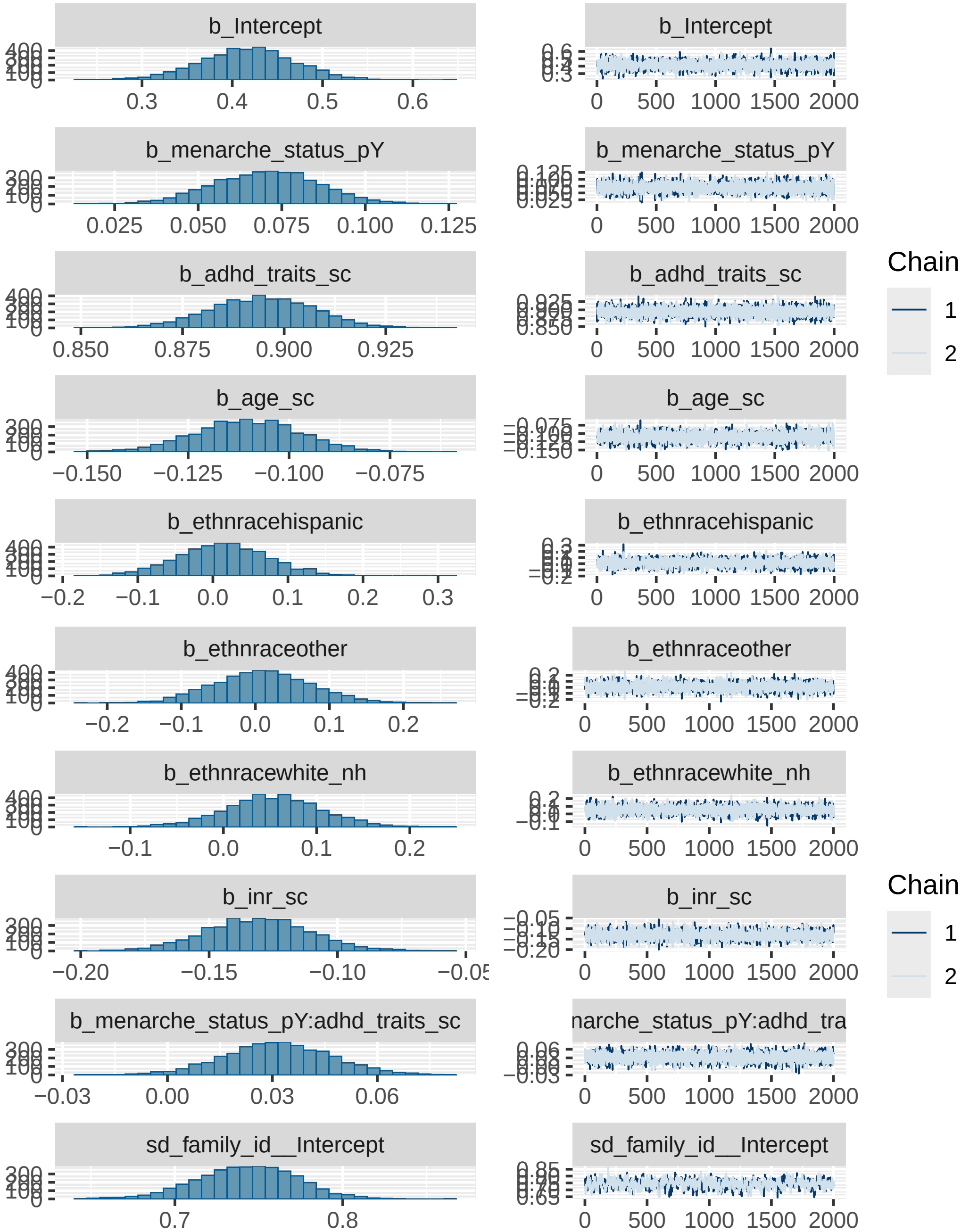
```
    Estimate   Est.Error      Q2.5    Q97.5
R2 0.7884055 0.001954511 0.7844898 0.792049


 Family: poisson
  Links: mu = log
Formula: externalising ~ menarche_status_p * adhd_traits_sc + age_sc + ethnrace + inr_sc + (
   Data: analytic_df (Number of observations: 19381)
  Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~family_id (Number of levels: 4626)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.743     0.030    0.682    0.799 1.003      524      878

~participant_id (Number of levels: 5297)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.755     0.026    0.705    0.808 1.004      593      903
```

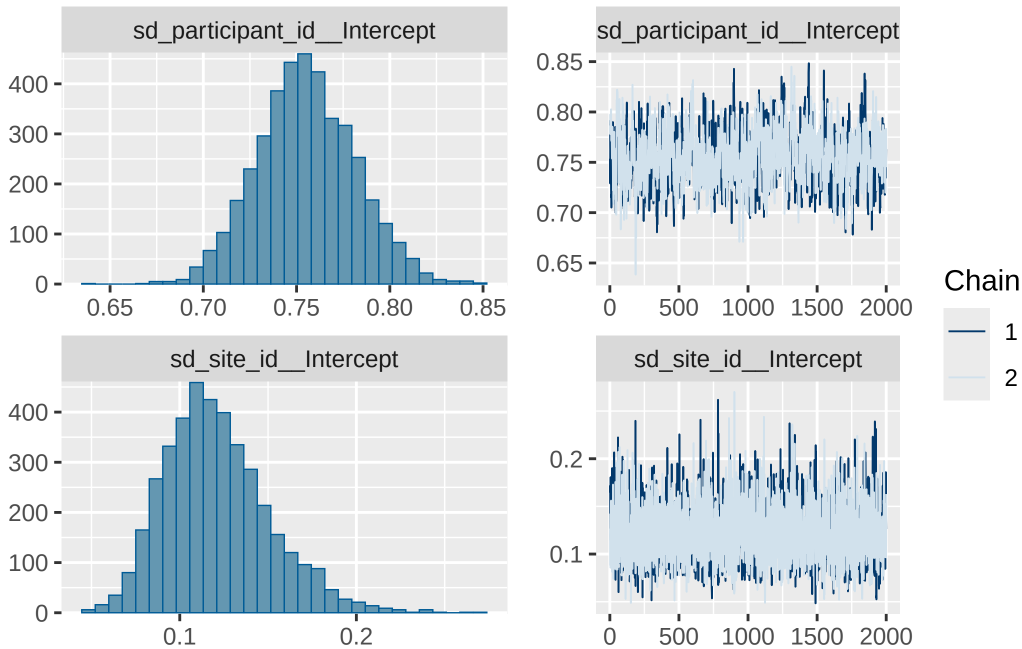
```
~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.121     0.030    0.073    0.186 1.001     3306     2787

Regression Coefficients:
                                  Estimate Est.Error l-95% CI u-95% CI  Rhat
Intercept                            0.417     0.053    0.311    0.519 1.001
menarche_status_pY                   0.070     0.016    0.040    0.102 1.001
adhd_traits_sc                       0.895     0.013    0.870    0.920 1.000
age_sc                              -0.110     0.013   -0.136   -0.084 1.000
ethnracehispanic                     0.014     0.060   -0.104    0.131 1.001
ethnraceother                        0.010     0.065   -0.116    0.139 1.002
ethnracewhite_nh                     0.054     0.051   -0.047    0.154 1.001
inr_sc                              -0.131     0.020   -0.171   -0.090 1.002
menarche_status_pY:adhd_traits_sc    0.032     0.015    0.001    0.062 1.000
                                  Bulk_ESS Tail_ESS
Intercept                             3364     2957
menarche_status_pY                    7027     3123
adhd_traits_sc                        7531     3248
age_sc                                7003     3334
ethnracehispanic                      4071     3161
ethnraceother                         4047     3136
ethnracewhite_nh                      3900     3122
inr_sc                                8985     2813
menarche_status_pY:adhd_traits_sc     6332     3130

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

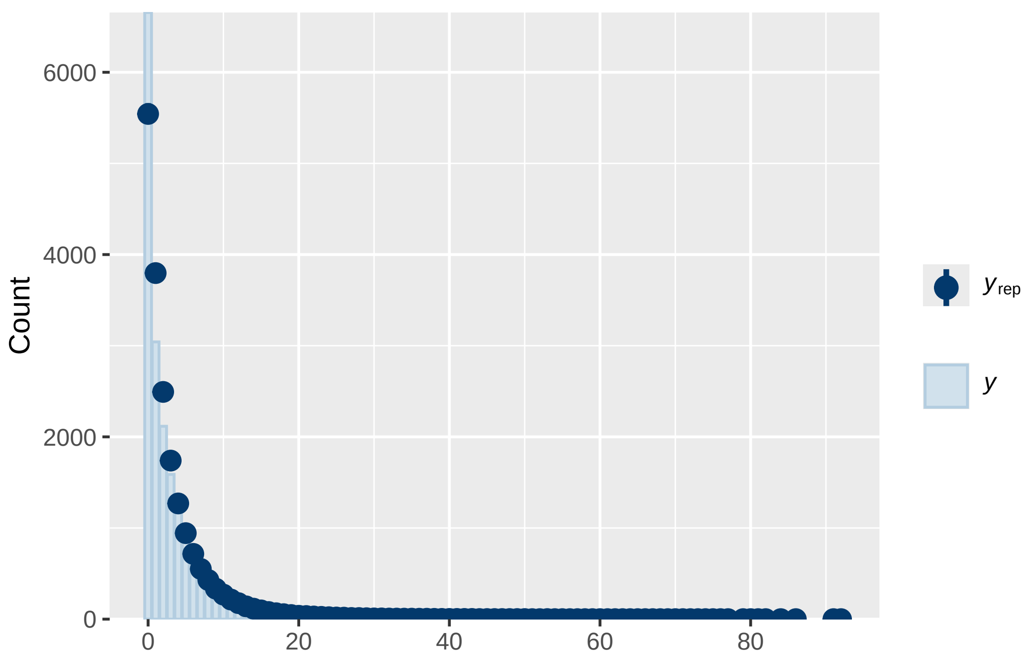
Model 5 prior 2 diagnostics

```
         prior     class                               coef
  normal(0, 1)         b
  normal(0, 1)         b                     adhd_traits_sc
  normal(0, 1)         b                             age_sc
  normal(0, 1)         b                   ethnracehispanic
  normal(0, 1)         b                      ethnraceother
  normal(0, 1)         b                   ethnracewhite_nh
  normal(0, 1)         b                             inr_sc
  normal(0, 1)         b                 menarche_status_pY
```

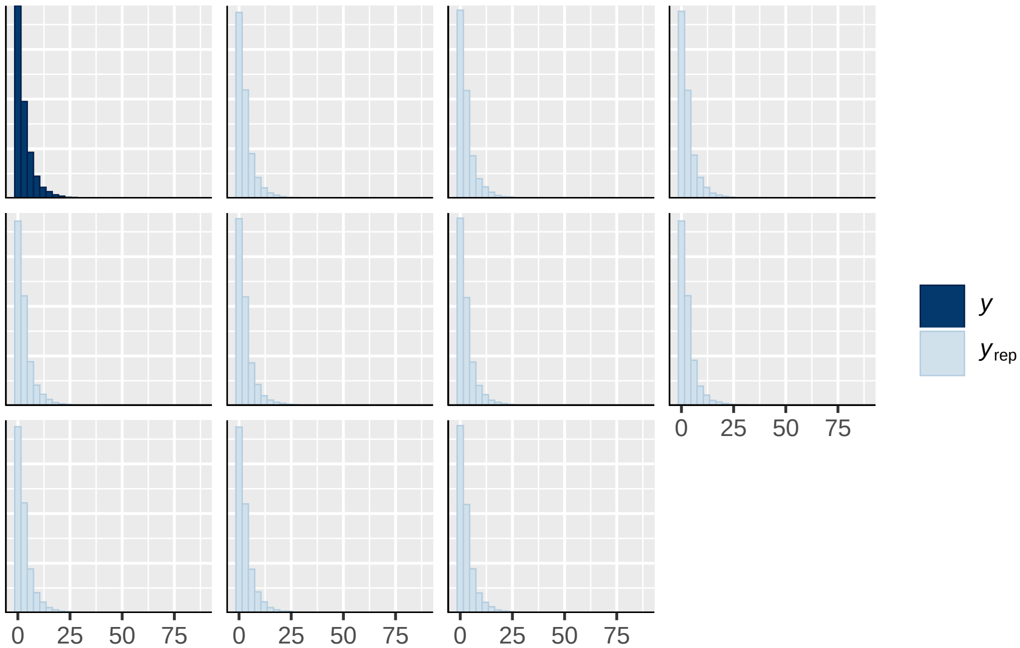
```
        normal(0, 1)         b menarche_status_pY:adhd_traits_sc
student_t(3, 0, 2.9) Intercept
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd                         Intercept
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd                         Intercept
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd                         Intercept
         group resp dpar nlpar lb ub tag       source
                                                 user
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                              default
                             0                default
     family_id               0           (vectorized)
     family_id               0           (vectorized)
participant_id               0           (vectorized)
participant_id               0           (vectorized)
       site_id               0           (vectorized)
       site_id               0           (vectorized)
```


b_Intercept
b_menarche_status_pY
b_adhd_traits_sc
b_age_sc
b_ethnracehispanic
b_ethnraceother
b_ethnracewhite_nh
b_inr_sc
b_menarche_status_pY:adhd_traits_sc
sd_family_id__Intercept
Chain
1
2

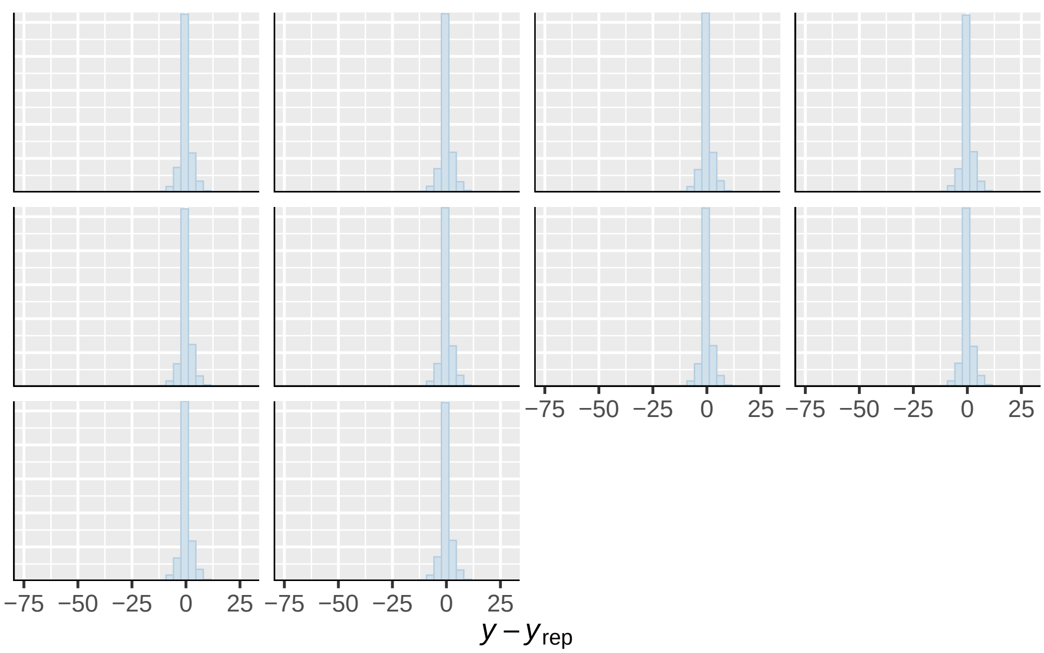




`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

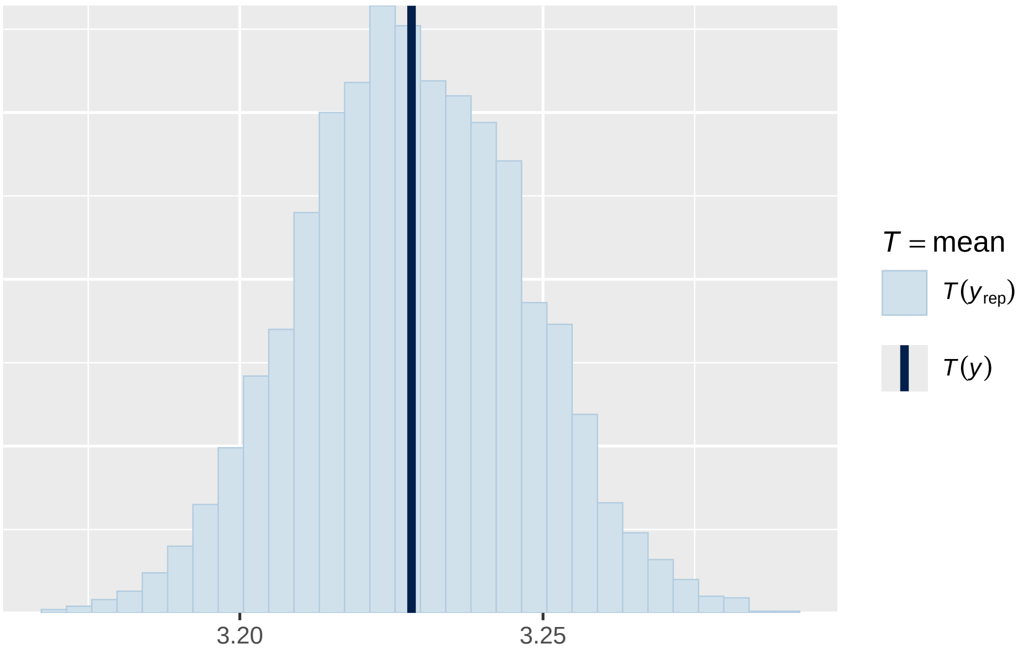



```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

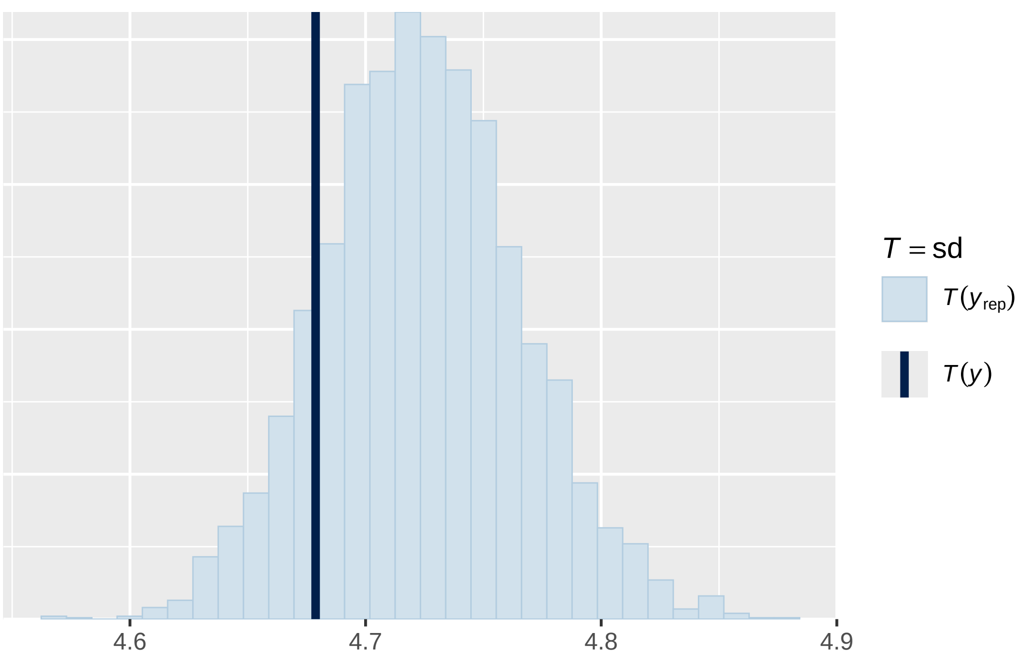



```
Using all posterior draws for ppc type 'stat' by default.
```

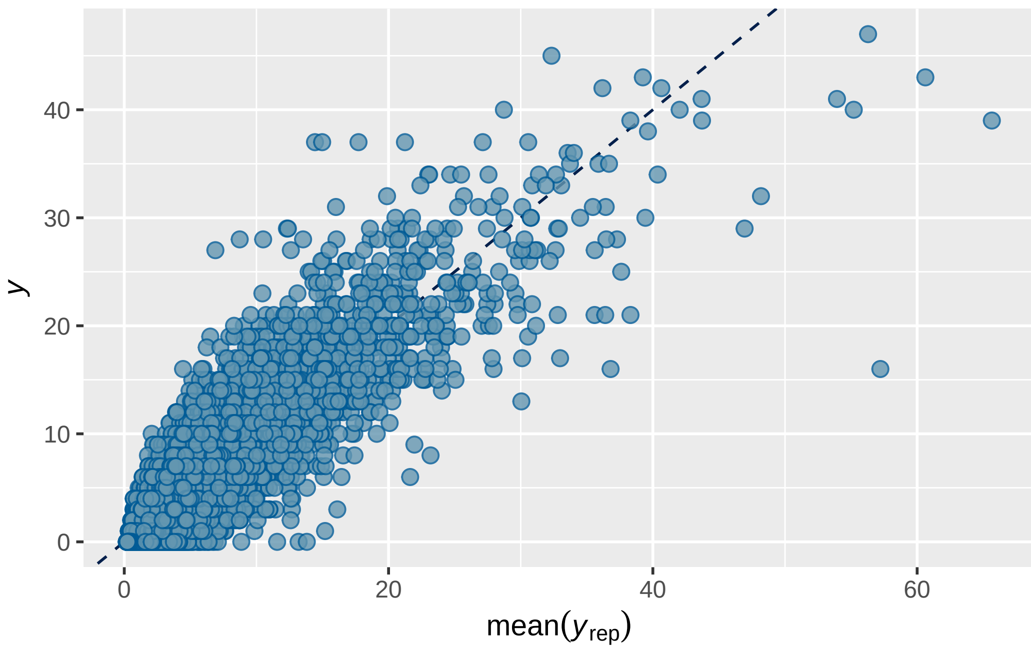
```
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



ABCD ADHD menarche analysis: Output for cross-sectional supplementary/sensitivity models

Model 1b observed output

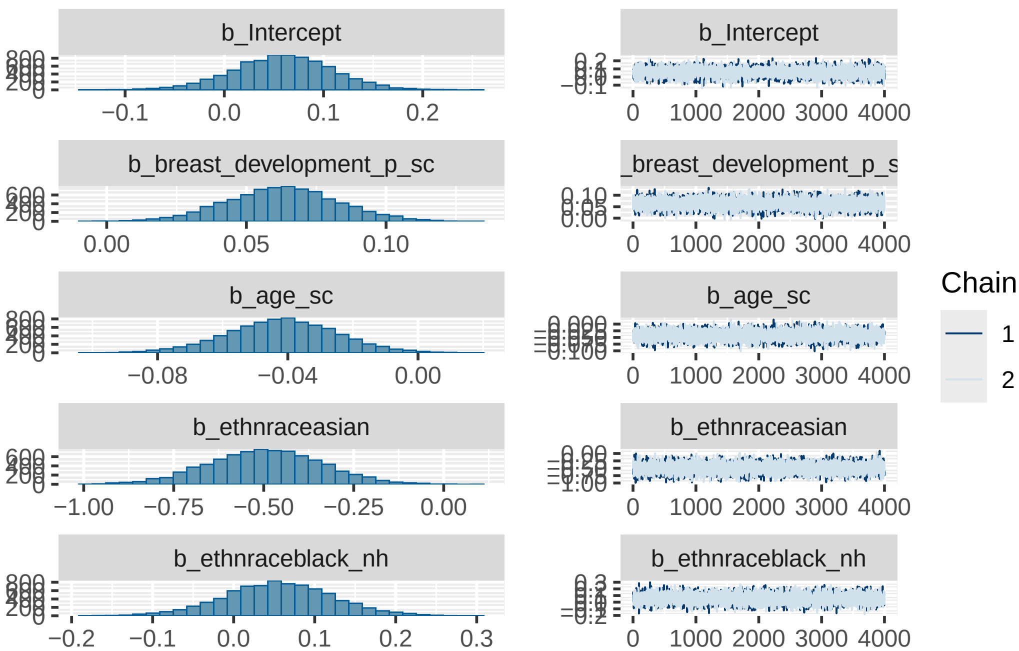
```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.7074451 0.003418142 0.7006664 0.7139525


 Family: poisson 
  Links: mu = log 
Formula: adhd_traits ~ breast_development_p_sc + age_sc + ethnrace + inr_sc + (1 | participan
   Data: obs_df (Number of observations: 17684) 
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 4483) 
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.807     0.048    0.707    0.897 1.003      504      954

~participant_id (Number of levels: 5137) 
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    1.037     0.036    0.966    1.110 1.001      764     1610

~site_id (Number of levels: 22) 
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.175     0.039    0.109    0.261 1.000     7157     6296

Regression Coefficients:
                        Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS
Intercept                  0.059     0.050   -0.043    0.157 1.000     7878
```

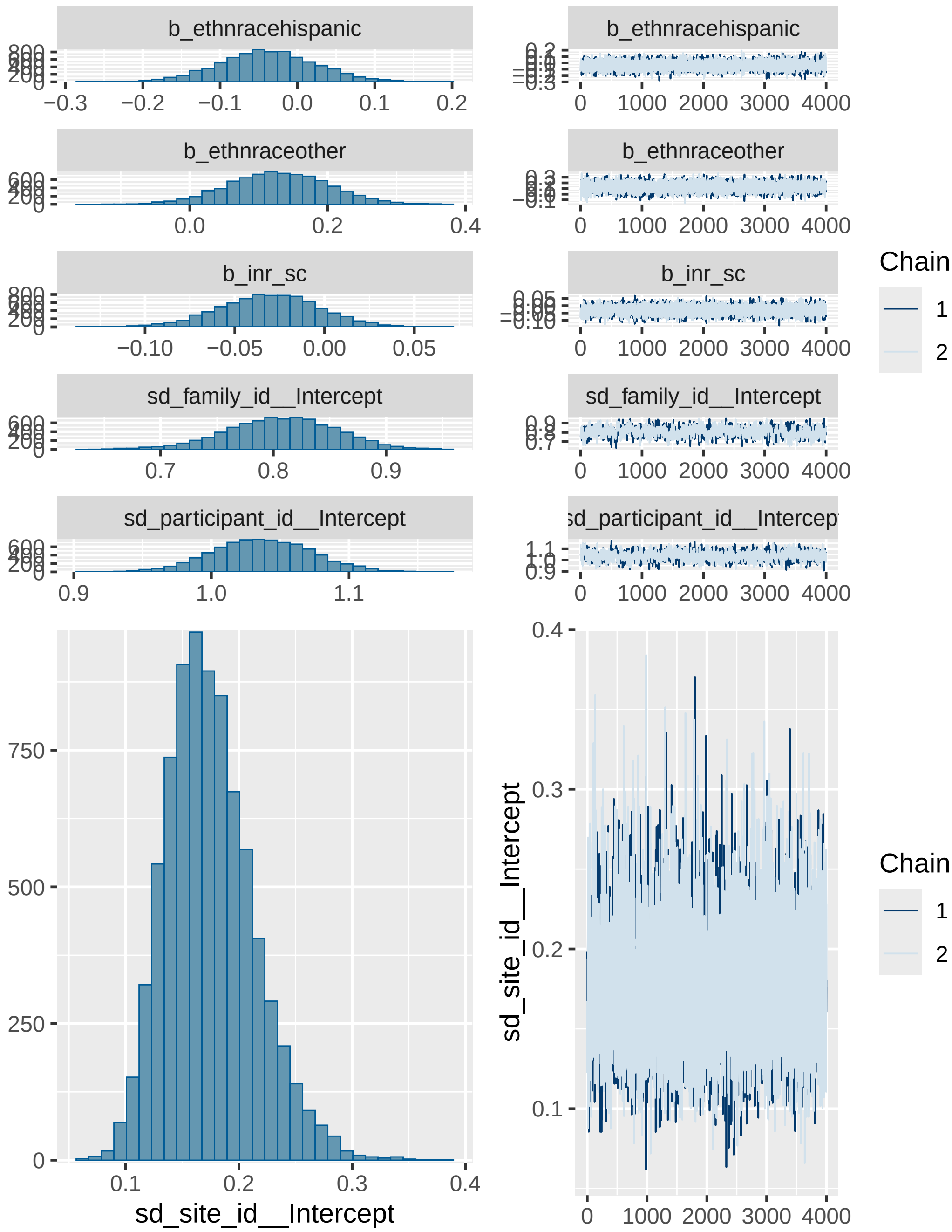
```
breast_development_p_sc    0.063     0.020    0.024    0.103 1.000    14577
age_sc                    -0.042     0.017   -0.076   -0.009 1.000    15123
ethnraceasian             -0.495     0.157   -0.806   -0.192 1.000    10381
ethnraceblack_nh           0.055     0.067   -0.079    0.189 1.000    10213
ethnracehispanic          -0.039     0.063   -0.161    0.086 1.000    11202
ethnraceother              0.129     0.073   -0.014    0.271 1.000    11871
inr_sc                    -0.031     0.027   -0.086    0.023 1.000    16084
                        Tail_ESS
Intercept                   5920
breast_development_p_sc     5750
age_sc                      5446
ethnraceasian               5816
ethnraceblack_nh            6150
ethnracehispanic            6106
ethnraceother               6273
inr_sc                      5933

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

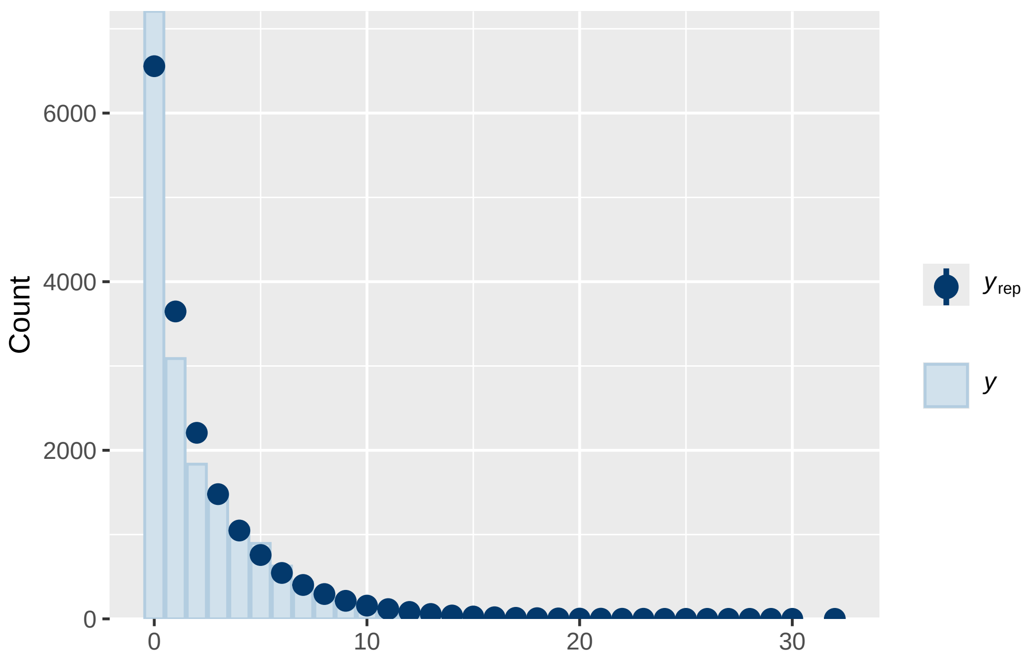
Model 1b observed diagnostics

```
                prior     class                    coef          group resp
         normal(0, 1)         b
         normal(0, 1)         b                  age_sc
         normal(0, 1)         b breast_development_p_sc
         normal(0, 1)         b           ethnraceasian
         normal(0, 1)         b        ethnraceblack_nh
         normal(0, 1)         b        ethnracehispanic
         normal(0, 1)         b           ethnraceother
         normal(0, 1)         b                  inr_sc
 student_t(3, 0, 2.7) Intercept
 student_t(3, 0, 2.7)        sd
 student_t(3, 0, 2.7)        sd                             family_id
 student_t(3, 0, 2.7)        sd               Intercept     family_id
 student_t(3, 0, 2.7)        sd                        participant_id
 student_t(3, 0, 2.7)        sd               Intercept participant_id
 student_t(3, 0, 2.7)        sd                               site_id
 student_t(3, 0, 2.7)        sd               Intercept       site_id
 dpar nlpar lb ub tag       source
                              user
```

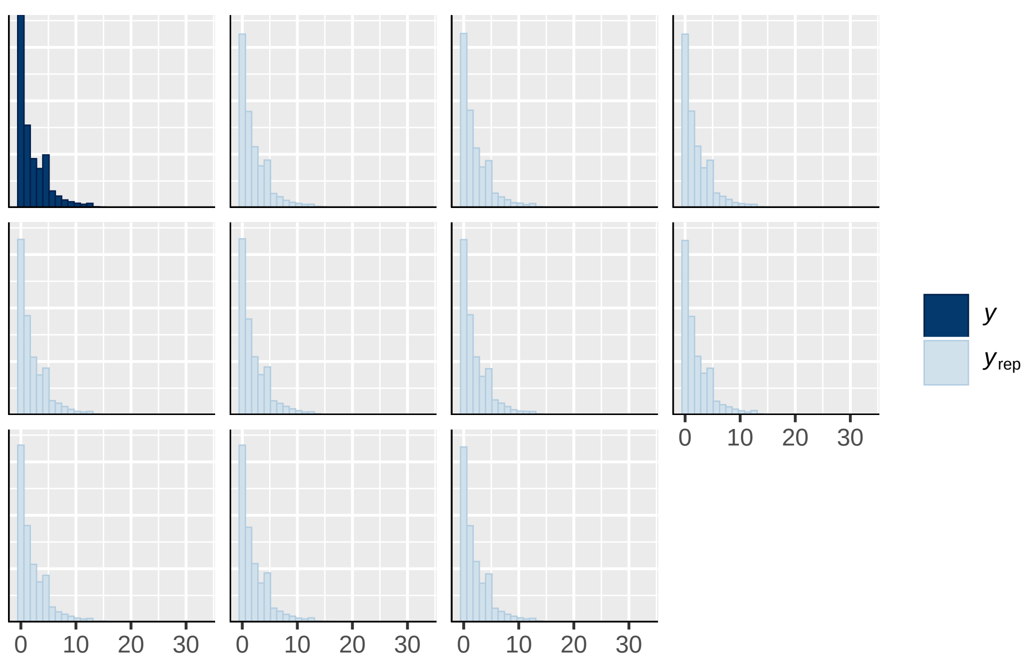
	(vectorized)
	(vectorized)
	(vectorized)
	(vectorized)
	(vectorized)
	(vectorized)
	(vectorized)
	default
0	default
0	(vectorized)
0	(vectorized)
0	(vectorized)
0	(vectorized)
0	(vectorized)
0	(vectorized)



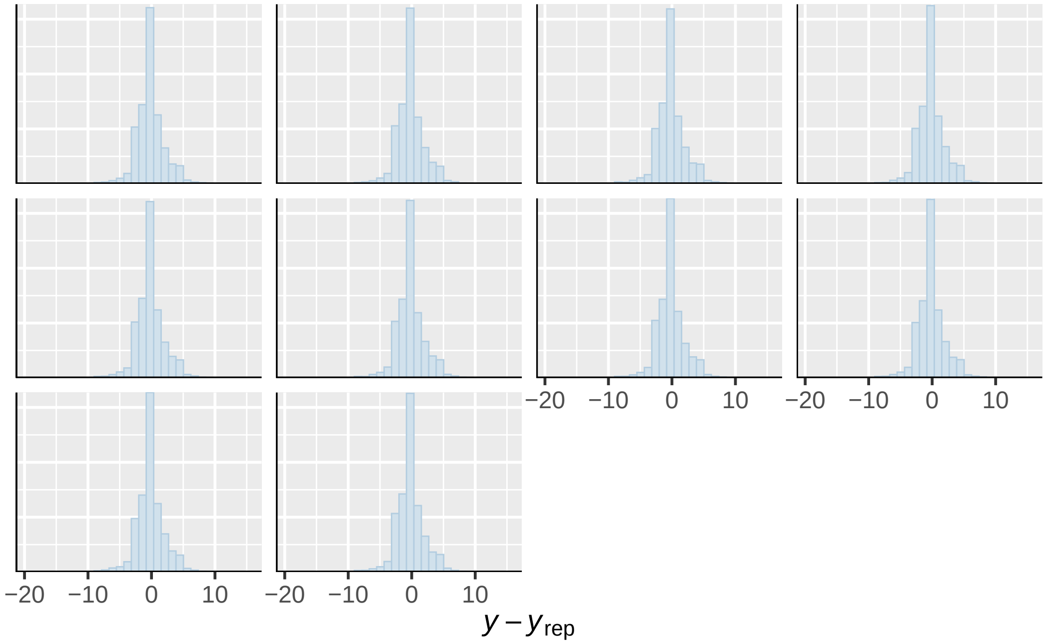

b_ethnracehispanic
−0.3
−0.2
−0.1
0.0
0.1
0.2
b_ethnraceother
0.0
0.2
0.4
b_inr_sc
−0.10
−0.05
0.00
0.05
sd_family_id__Intercept
0.7
0.8
0.9
sd_participant_id__Intercept
0.9
1.0
1.1
750
500
250
0
0.1
0.2
0.3
0.4
sd_site_id__Intercept
0
1000
2000
3000
4000
sd_site_id__Intercept
Chain
1
2



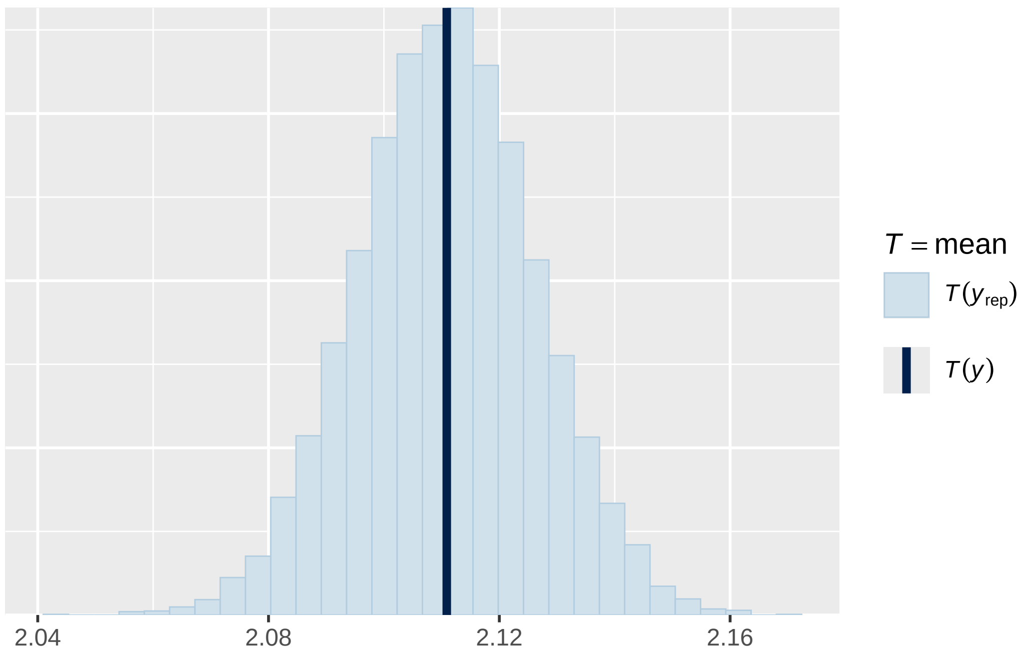

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



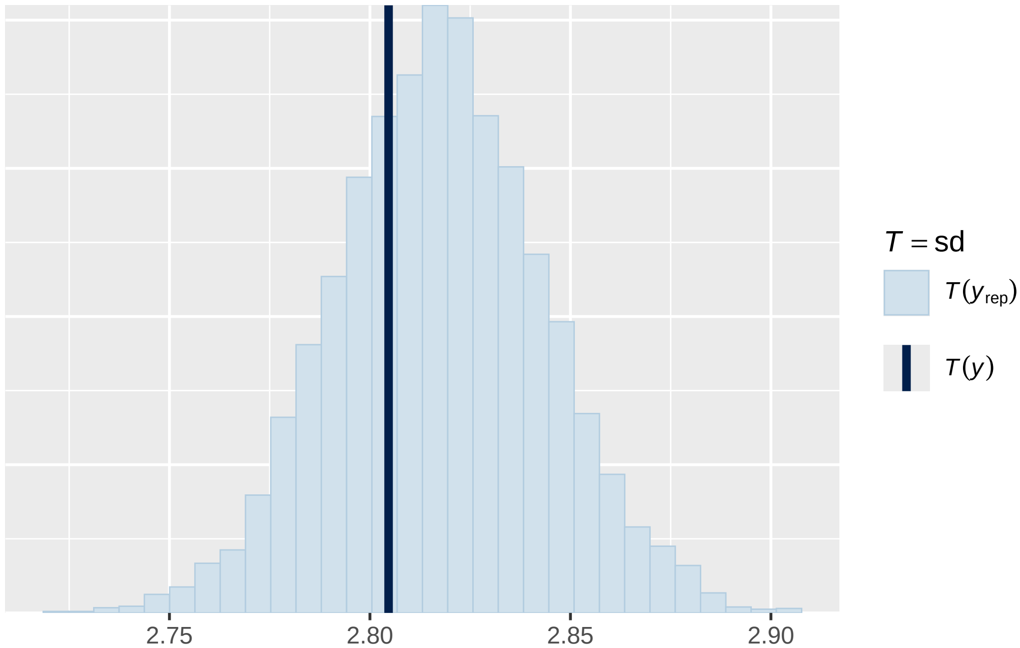

```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



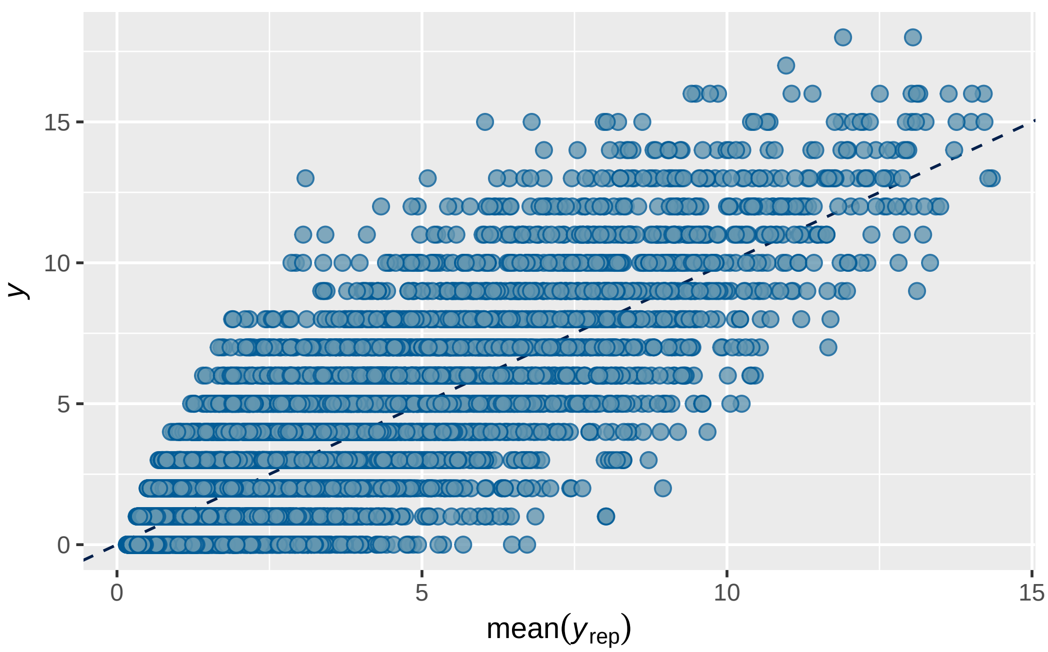

```
Using all posterior draws for ppc type 'stat' by default.
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 4 observed output

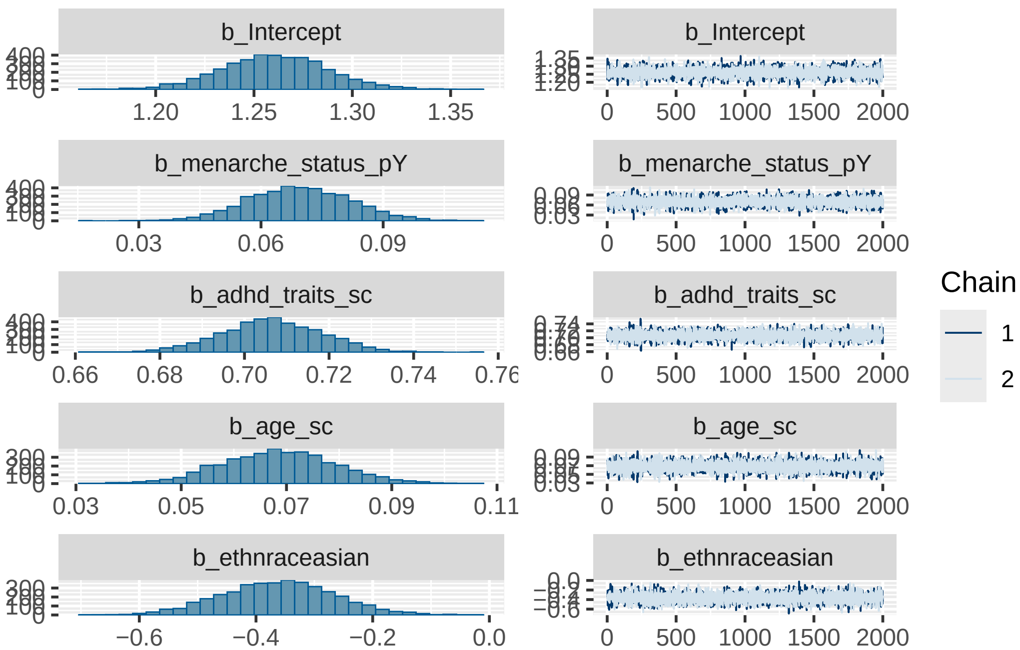
```
    Estimate   Est.Error      Q2.5    Q97.5
R2 0.7350095 0.002364186 0.7303298 0.739427


 Family: poisson
  Links: mu = log
Formula: internalising ~ menarche_status_p * adhd_traits_sc + age_sc + ethnrace + inr_sc + (
   Data: obs_df (Number of observations: 17646)
  Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~family_id (Number of levels: 4488)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.648     0.018    0.612    0.683 1.001     1047     1886

~participant_id (Number of levels: 5143)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.493     0.019    0.457    0.533 1.002      831     1586

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.095     0.023    0.057    0.145 1.000     3332     3185

Regression Coefficients:
                                  Estimate Est.Error l-95% CI u-95% CI  Rhat
Intercept                            1.261     0.027    1.207    1.314 1.000
menarche_status_pY                   0.070     0.013    0.046    0.094 1.001
adhd_traits_sc                       0.706     0.012    0.683    0.729 1.001
age_sc                               0.069     0.011    0.047    0.090 1.001
ethnraceasian                       -0.362     0.096   -0.552   -0.172 1.000
ethnraceblack_nh                    -0.399     0.041   -0.479   -0.317 1.001
ethnracehispanic                    -0.074     0.038   -0.146    0.002 1.000
ethnraceother                        0.013     0.043   -0.071    0.100 1.002
inr_sc                              -0.002     0.018   -0.036    0.032 1.000
menarche_status_pY:adhd_traits_sc    0.007     0.014   -0.020    0.034 1.000
                                  Bulk_ESS Tail_ESS
Intercept                             5913     3101
menarche_status_pY                    7446     3431
adhd_traits_sc                        8462     2860
age_sc                                8443     3198
```

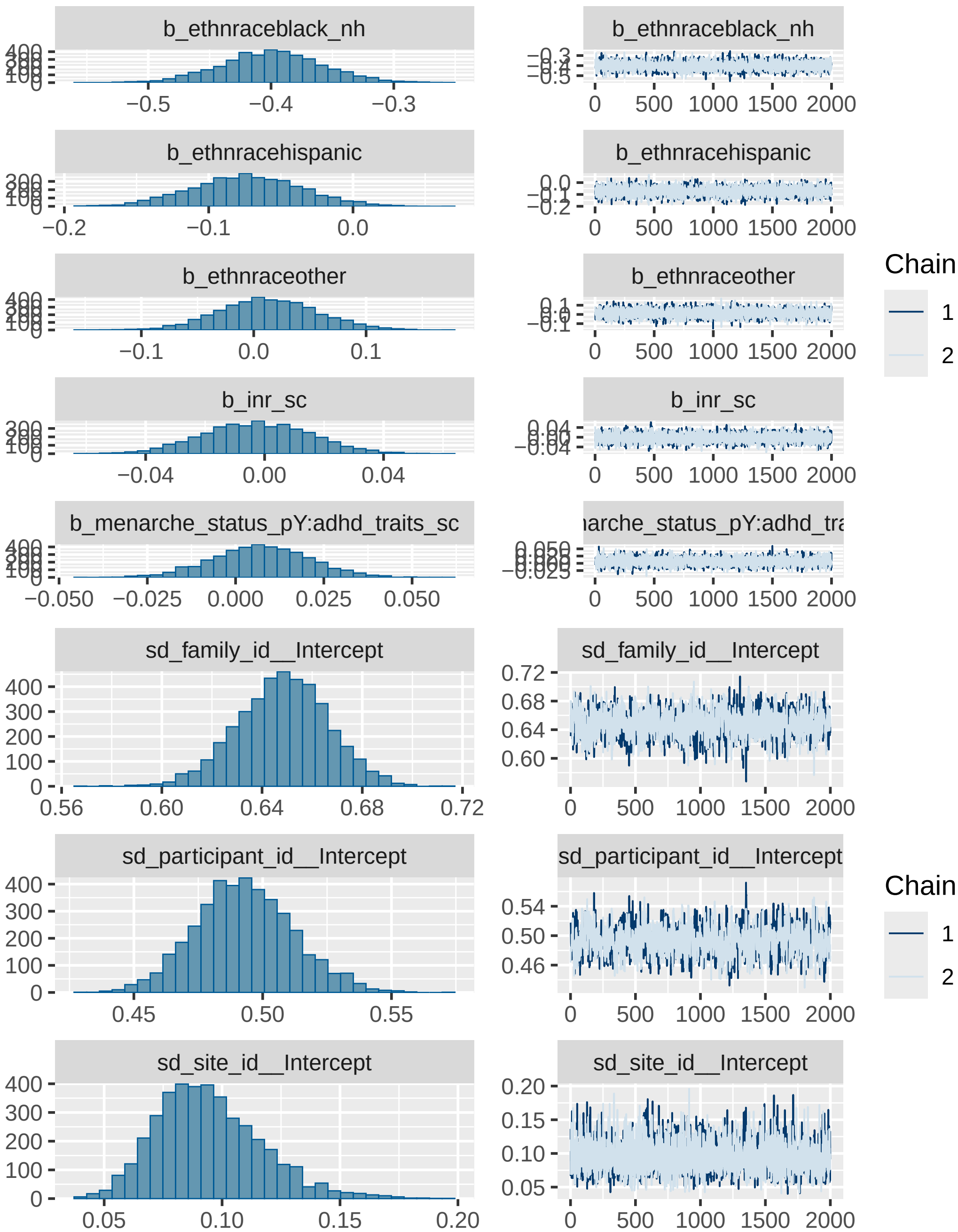
```
ethnraceasian                       8158     3153
ethnraceblack_nh                    6646     2703
ethnracehispanic                    7221     3210
ethnraceother                       6243     2852
inr_sc                              9607     2879
menarche_status_pY:adhd_traits_sc   6582     2877

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

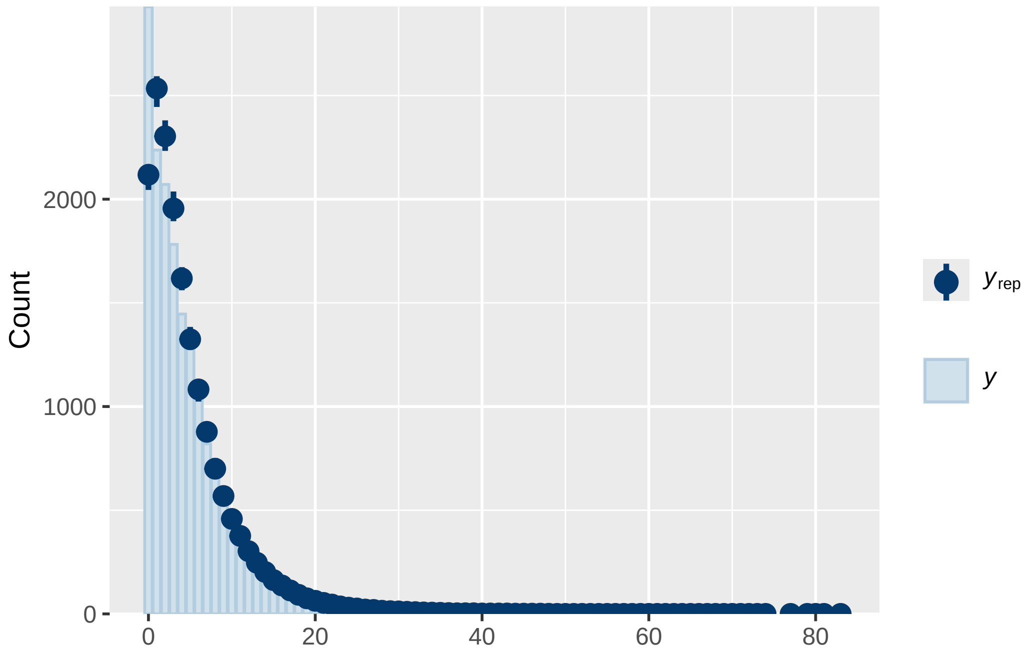
Model 4 observed diagnostics

```
                  prior     class                               coef
           normal(0, 1)         b
           normal(0, 1)         b                     adhd_traits_sc
           normal(0, 1)         b                             age_sc
           normal(0, 1)         b                      ethnraceasian
           normal(0, 1)         b                   ethnraceblack_nh
           normal(0, 1)         b                   ethnracehispanic
           normal(0, 1)         b                      ethnraceother
           normal(0, 1)         b                             inr_sc
           normal(0, 1)         b                 menarche_status_pY
           normal(0, 1)         b menarche_status_pY:adhd_traits_sc
 student_t(3, 1.1, 2.5) Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd                          Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd                          Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd                          Intercept
          group resp dpar nlpar lb ub tag       source
                                                  user
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
                                          (vectorized)
```

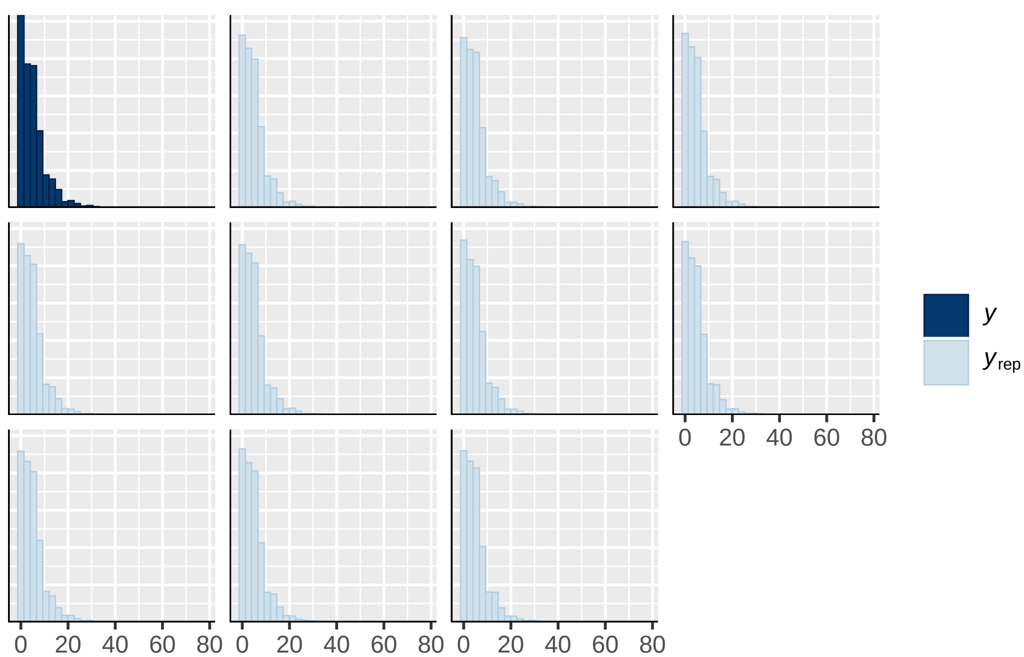
		(vectorized)
		default
	0	default
family_id	0	(vectorized)
family_id	0	(vectorized)
participant_id	0	(vectorized)
participant_id	0	(vectorized)
site_id	0	(vectorized)
site_id	0	(vectorized)



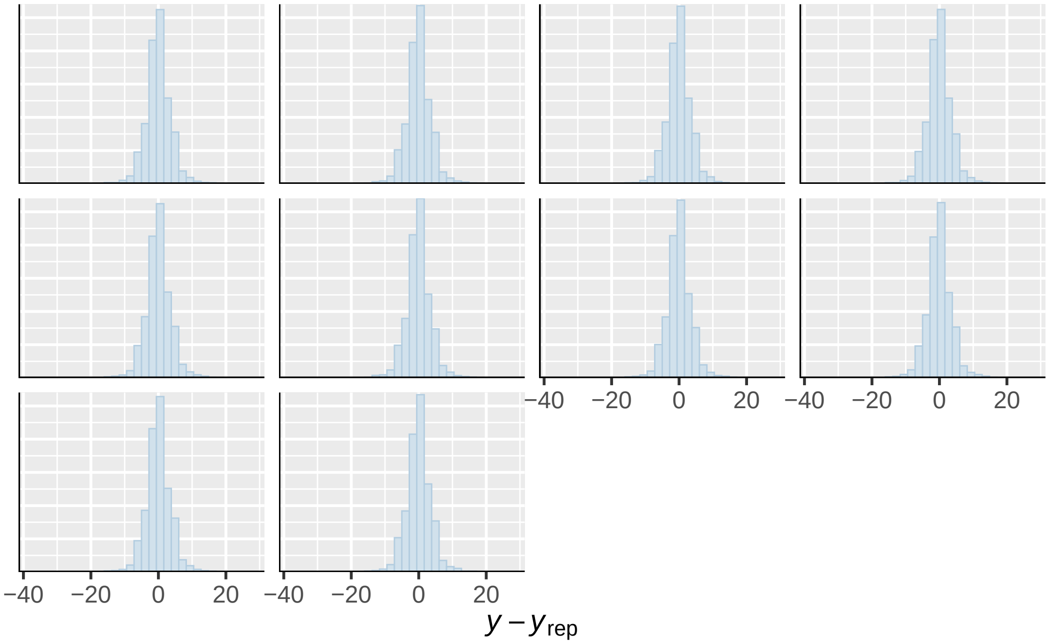

b_ethnraceblack_nh
b_ethnraceblack_nh
b_ethnracehispanic
b_ethnracehispanic
b_ethnraceother
b_ethnraceother
b_inr_sc
b_inr_sc
b_menarche_status_pY:adhd_traits_sc
iarche_status_pY:adhd_tra
sd_family_id__Intercept
sd_family_id__Intercept
sd_participant_id__Intercept
sd_participant_id__Intercept
sd_site_id__Intercept
sd_site_id__Intercept
Chain
1
2
Chain
1
2



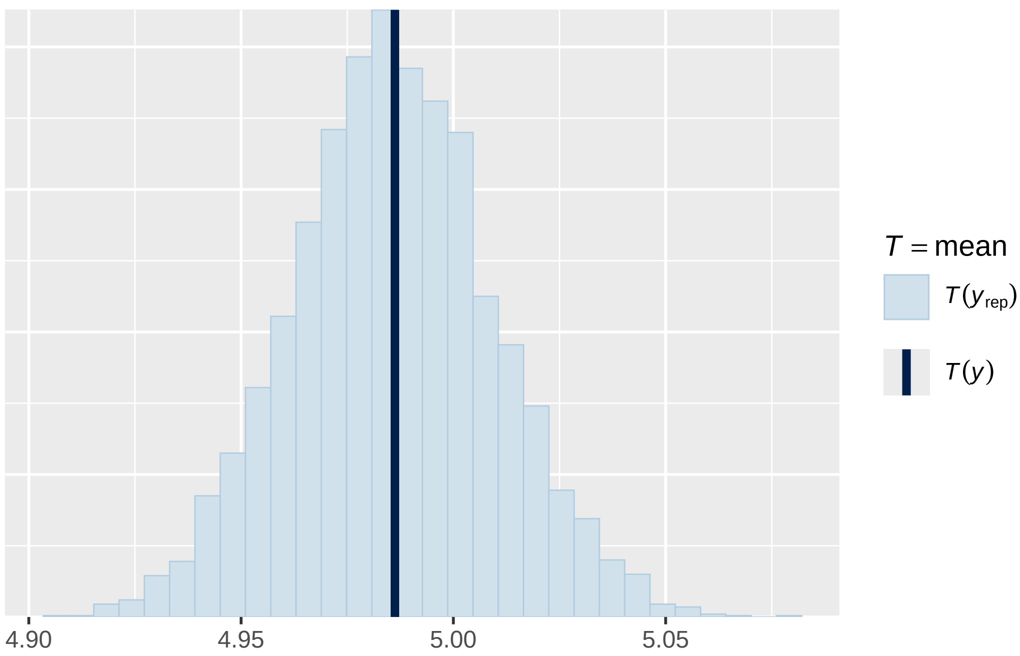
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.



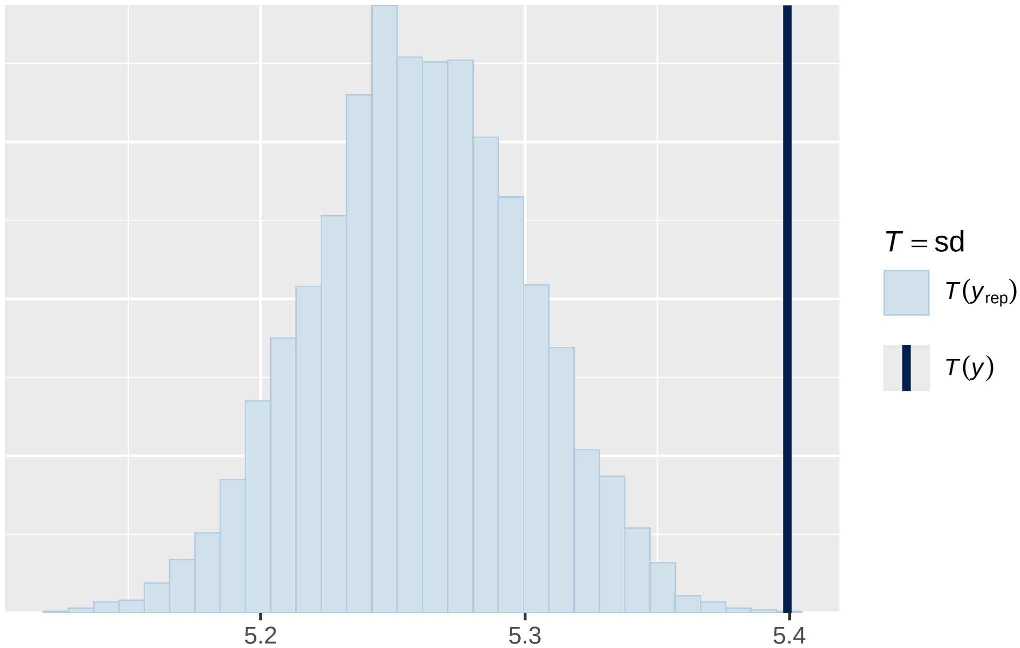
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.



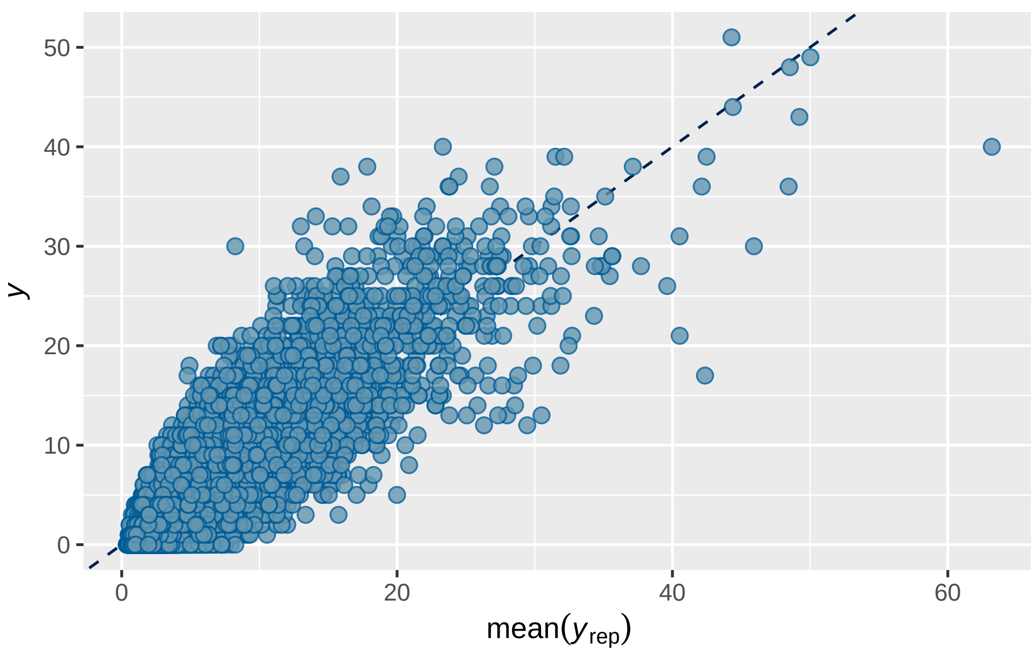

```
Using all posterior draws for ppc type 'stat' by default.
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 5 observed output

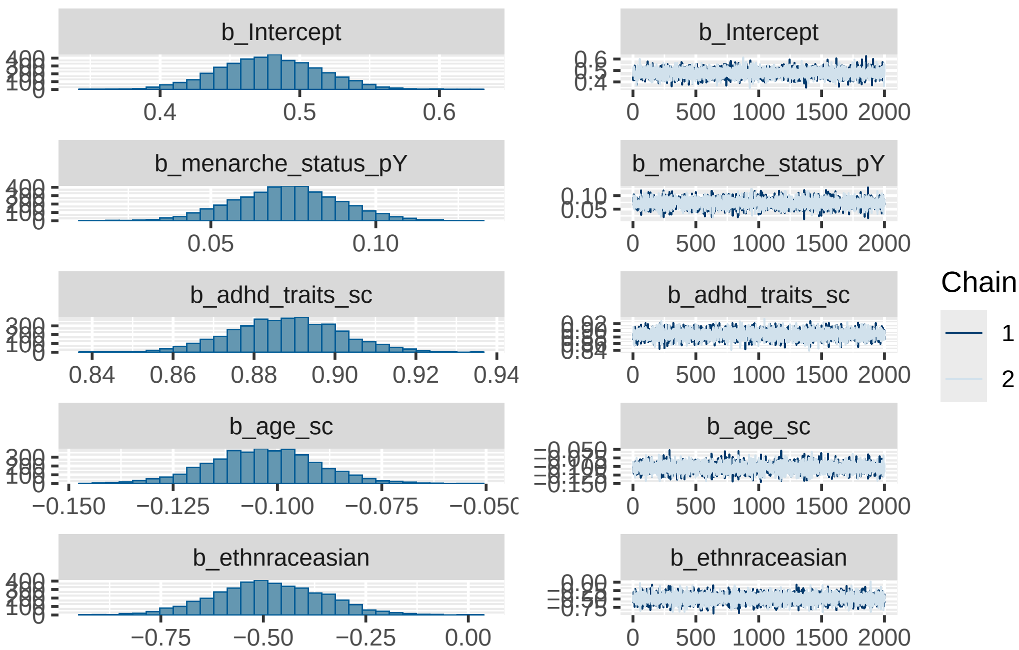
```
    Estimate   Est.Error     Q2.5     Q97.5
R2 0.7895093 0.002177434 0.785262 0.7936142


 Family: poisson
  Links: mu = log
Formula: externalising ~ menarche_status_p * adhd_traits_sc + age_sc + ethnrace + inr_sc + (
   Data: obs_df (Number of observations: 17646)
  Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~family_id (Number of levels: 4488)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.748     0.030    0.688    0.804 1.006      717     1444

~participant_id (Number of levels: 5143)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.753     0.027    0.701    0.807 1.004      693     1104

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.120     0.030    0.070    0.187 1.001     3797     3183

Regression Coefficients:
                                  Estimate Est.Error l-95% CI u-95% CI  Rhat
Intercept                            0.478     0.036    0.408    0.549 1.002
menarche_status_pY                   0.073     0.016    0.042    0.105 1.000
adhd_traits_sc                       0.888     0.013    0.862    0.914 1.000
age_sc                              -0.103     0.013   -0.130   -0.077 1.000
ethnraceasian                       -0.487     0.134   -0.747   -0.223 1.000
ethnraceblack_nh                    -0.051     0.054   -0.151    0.053 1.001
ethnracehispanic                    -0.051     0.048   -0.146    0.041 1.000
ethnraceother                        0.039     0.059   -0.072    0.154 1.000
inr_sc                              -0.137     0.024   -0.183   -0.090 1.000
menarche_status_pY:adhd_traits_sc    0.027     0.017   -0.005    0.061 1.000
                                  Bulk_ESS Tail_ESS
Intercept                             4690     3150
menarche_status_pY                    8393     3184
adhd_traits_sc                        7801     2895
age_sc                                8317     2523
```

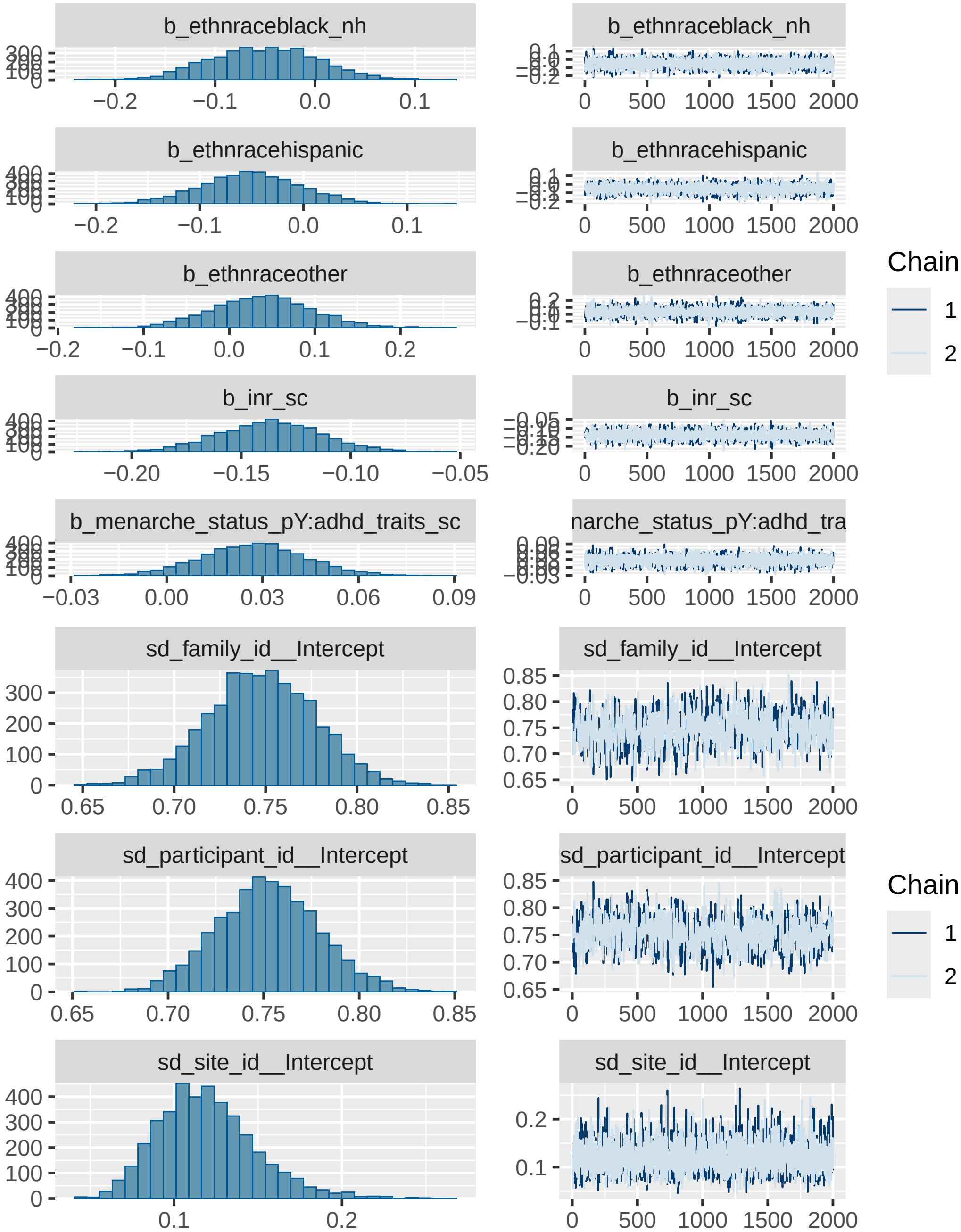
```
ethnraceasian                        7432     2898
ethnraceblack_nh                     6514     2585
ethnracehispanic                     6314     3010
ethnraceother                        6278     2861
inr_sc                               9253     3188
menarche_status_pY:adhd_traits_sc    7524     3252

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

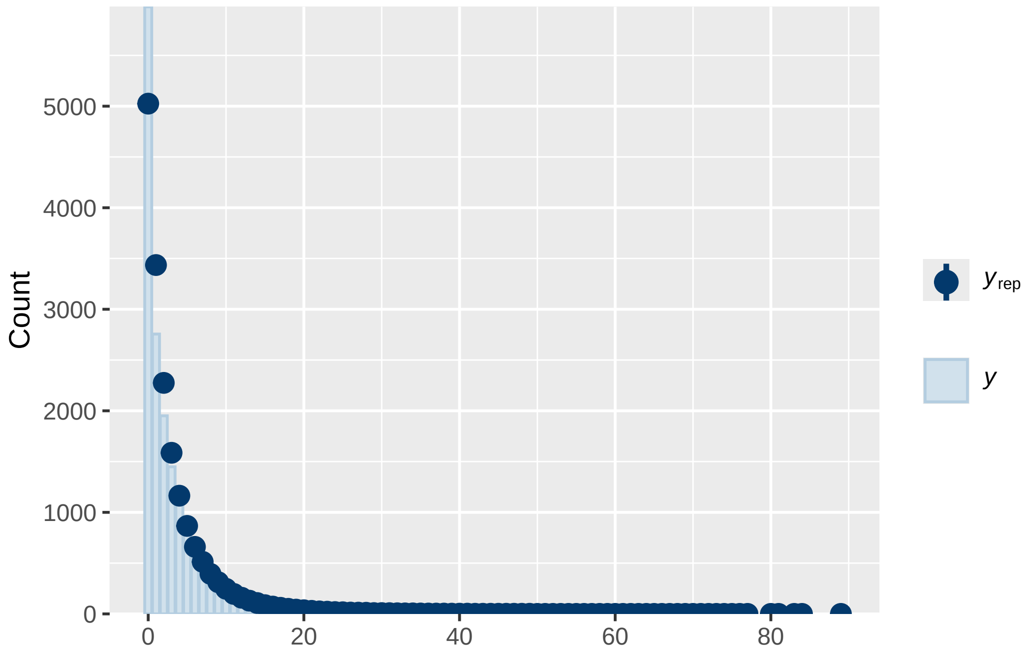
Model 5 observed diagnostics

```
                 prior     class                              coef
          normal(0, 1)         b
          normal(0, 1)         b                    adhd_traits_sc
          normal(0, 1)         b                            age_sc
          normal(0, 1)         b                     ethnraceasian
          normal(0, 1)         b                  ethnraceblack_nh
          normal(0, 1)         b                  ethnracehispanic
          normal(0, 1)         b                     ethnraceother
          normal(0, 1)         b                            inr_sc
          normal(0, 1)         b                menarche_status_pY
          normal(0, 1)         b menarche_status_pY:adhd_traits_sc
 student_t(3, 0.7, 2.5) Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd                         Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd                         Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd                         Intercept
         group resp dpar nlpar lb ub tag       source
                                                 user
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
                                         (vectorized)
```

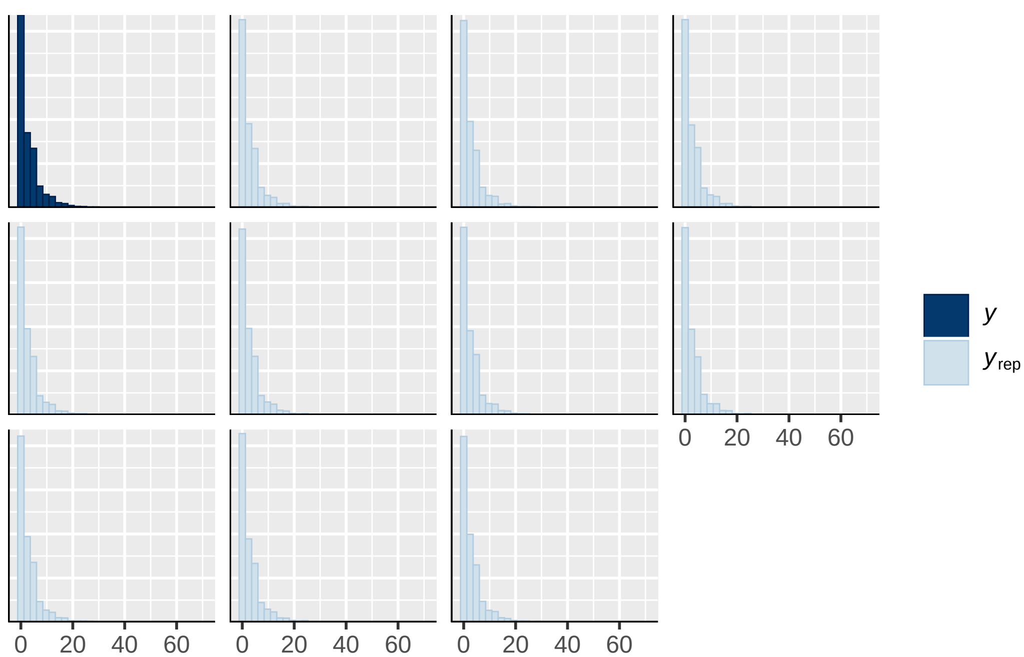
		(vectorized)
		default
	0	default
family_id	0	(vectorized)
family_id	0	(vectorized)
participant_id	0	(vectorized)
participant_id	0	(vectorized)
site_id	0	(vectorized)
site_id	0	(vectorized)



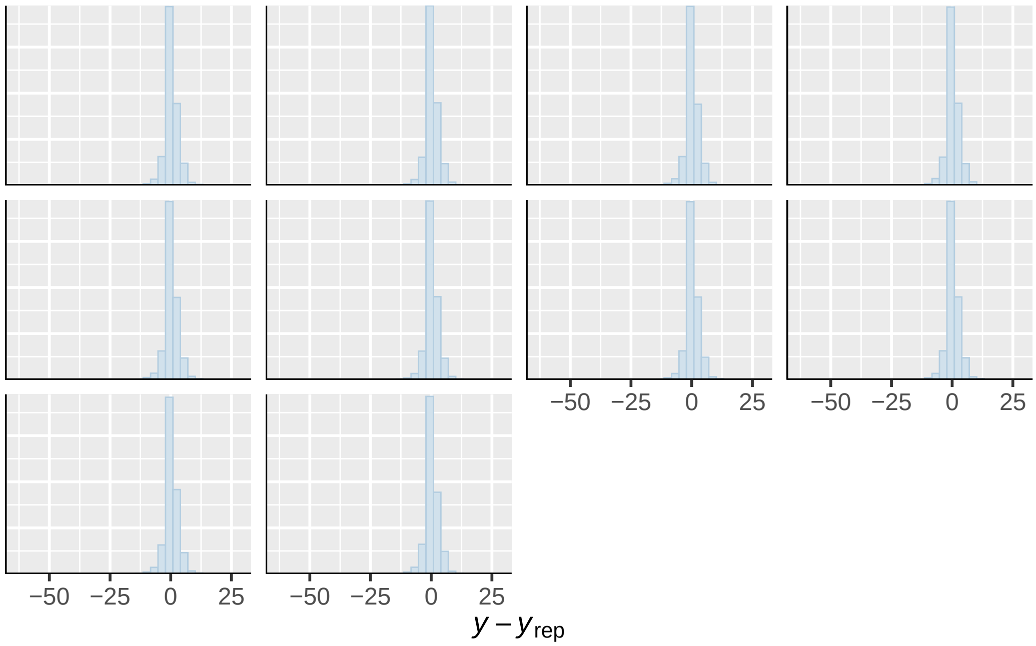

b_ethnraceblack_nh
b_ethnraceblack_nh
b_ethnracehispanic
b_ethnracehispanic
b_ethnraceother
b_ethnraceother
b_inr_sc
b_inr_sc
b_menarche_status_pY:adhd_traits_sc
narche_status_pY:adhd_tra
sd_family_id__Intercept
sd_family_id__Intercept
sd_participant_id__Intercept
sd_participant_id__Intercept
sd_site_id__Intercept
sd_site_id__Intercept
Chain
1
2
Chain
1
2



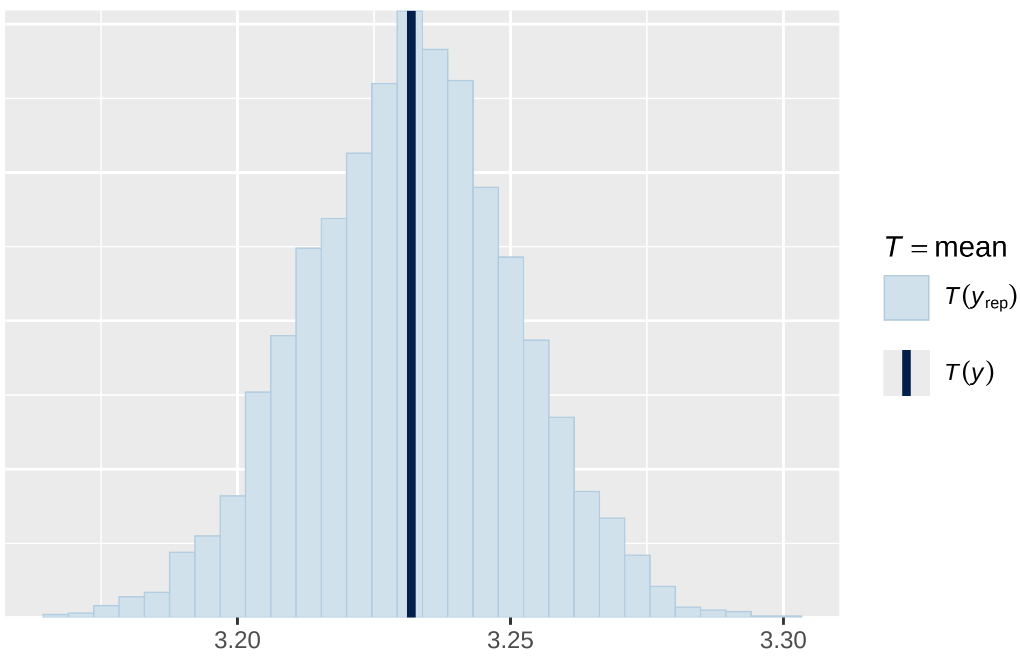
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.



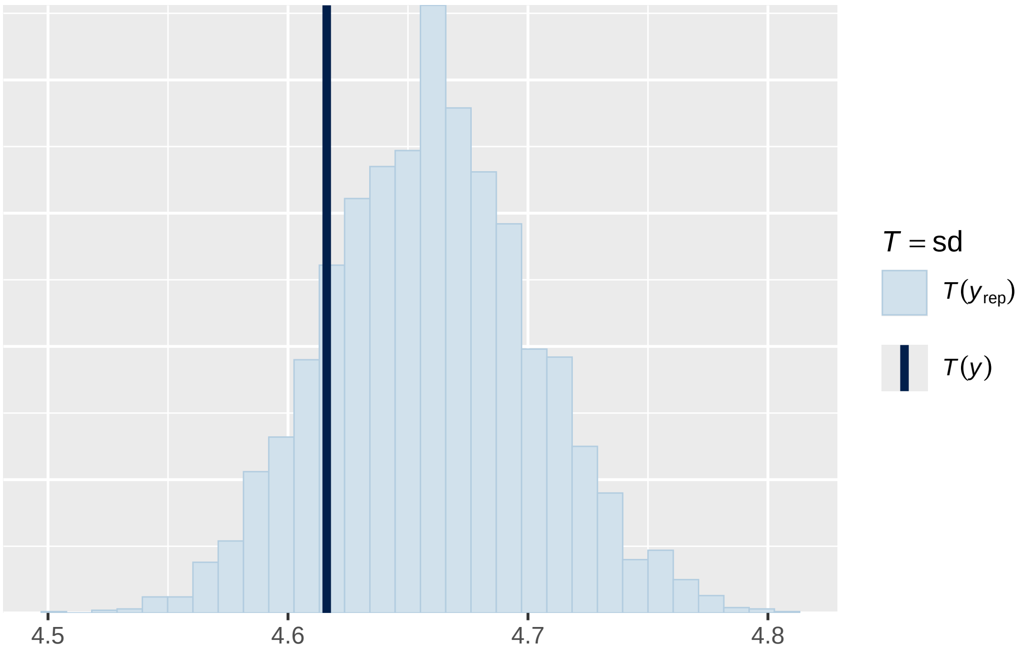
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.




```
Using all posterior draws for ppc type 'stat' by default.
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'scatter_avg' by default.
```